Notes

OPTM6423: POSTERIOR EYE THERAPEUTICS

WEEK 1: OPTOMETRIC PRESCRIBING FOR THE POSTERIOR EYE

LECTURE 1: Optometric Prescribing

How do Optometrists Manage Posterior Eye Conditions?

- Ocular disease screening, diagnosis
- Early detection of systemic disease
- Important role in triage (being able to recognise urgency of care patient needs), referral, coordination of care
- Patient education and counselling
- Glaucoma prescribing
- Collaborative care with ophthalmologists
- Role in rural and underserved communications
- Advocacy for expanded scope

Professional and Ethical Responsibilities

An entry-level optometrist must practise in a manner that adheres to ethical and professional principles consistent with legislative and regulatory requirements (AKA AHPRA guidelines and code of conduct). This complies with:

- **OptomBA (OBA) Standards, Guidelines and Code of Conduct** → National authority for registering and regulating optometrists in Australia
- **Relevant medicine prescribing competency frameworks and medicine safety guidance** e.g., Quality Use of Medicines
- **Jurisdictional medicines/poisons acts** → Refers to state law

Prescribing Principles + Basics

- Endorsed optometrists are expected to comply with the OBA guidelines, and must be familiar with TGA approved indications found in the product information. It is important to know that the listed drug indication that is TGA approved could be different to the reason that you intend to use it.
- Used as national evidence of appropriate professional conduct in legal proceedings.

POP QUIZ 💡

Regulatory Compliance Optometrists must know Therapeutic Goods Administration (TGA) approved indications for any medicine they prescribe ***NOTE:*** Can use TGA medicines different to how they are approved.	✅
Prescription requirements Schedule 4 medicine prescriptions must be handwritten or computer-generated and include the prescriber's signature	✅
Optometrists with general registration can prescribe S4 medicines (Rx only meds) for therapeutic use **To make it true**: Optometrists with scheduled medicine endorsements can prescribe S4 medicines for therapeutic use	❌

- There are 2 types of AHPRA registered optometrists:
  1. Optometrists with general registration who use scheduled medicines for diagnostic purposes, **and**
  2. Optometrists whose registration is endorsed for scheduled medicines (therapeutically endorsed), who are considered qualified and competent to
     - Obtain, possess, administer, prescribe, supply or use specified scheduled medicines, and
     - Use those medicines appropriately for the purposes of the practice of optometrists

SCHEDULE 1 MEDICINES	These are medicines that are not currently used.
SCHEDULE 2 MEDICINES **encouraged to write Rx*	On the shelf medicines.
SCHEDULE 3 MEDICINES **encouraged to write Rx*	Pharmacist ONLY medicines (kept behind the counter)
SCHEDULE 4 MEDICINES	Prescription ONLY medicines.

Appendix B

Board-approved list of topical schedule 2, 3 and 4 medicines that optometrists with a scheduled medicines endorsement are qualified to administer, obtain, possess, prescribe, supply or use for the purposes of the practice of optometry[1]

Schedule 2 Pharmacy Medicine

Anti-infectives
Dibromopropamidine
Propamidine

Decongestants/ anti-allergics
Antazoline
Azelastine
Ketotifen
Levocabastine
Lodoxamide
Naphazoline
Pheniramine
Sodium Cromoglycate

Miotics, mydriatics and cycloplegics
Phenylephrine ≤2.5%

Schedule 3 Pharmacist Only Medicine

Anti-infectives
Chloramphenicol

Schedule 4 Prescription Only Medicine

Anti-infectives
Aciclovir
Azithromycin
Bacitracin
Cephazolin
Ciprofloxacin
Framycetin
Ganciclovir
Gentamicin
Gramicidin
Neomycin
Ofloxacin
Polymyxin
Tetracycline
Tobramycin

Anti-inflammatories
Cyclosporin
Dexamethasone
Diclofenac
Fluorometholone
Flurbiprofen
Hydrocortisone
Ketorolac
Lifitegrast
Loteprednol
Prednisolone

Decongestants/ anti-allergics
Olopatadine

Anti-glaucomas
Apraclonidine
Betaxolol
Bimatoprost
Brimonidine
Brinzolamide
Dorzolamide
Latanoprost
Pilocarpine
Tafluprost
Timolol
Travoprost

Miotics, mydriatics and cycloplegics
Atropine
Cyclopentolate
Homatropine
Pilocarpine
Phenylephrine
Tropicamide

Local anaesthetics
Amethocaine
Lignocaine
Oxybuprocaine
Proxymetacaine

NOTE: Just because a drug is on this list, it does NOT mean it is PBS listed or even available in Australia e.g dibropropamidine, propamidine (Brolene) are not available in Australia

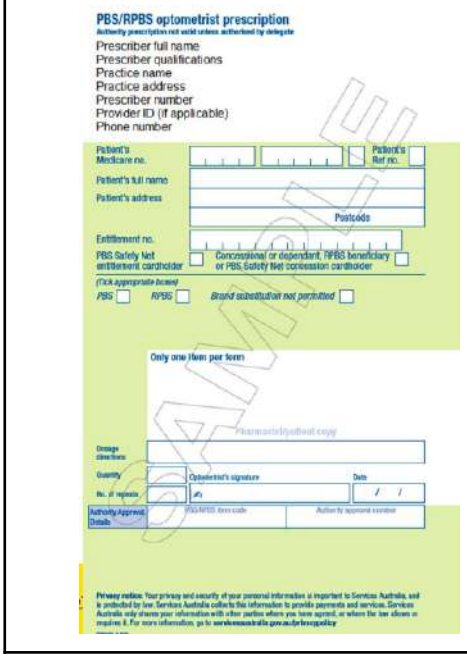
What is the PBS?

- **PBS =** Pharmaceutical Benefits Scheme

- Government subsidy program that provides timely, reliable and affordable access to necessary medicines for Australians.
- The PBS schedule lists the medicines available to be dispensed to patients at a government-subsidised price, can be visited online.
- Some of the medicines optometrists can prescribe are listed under the PBS schedule. Some are not PBS listed. Medicines not listed on the PBS can still be prescribed, but must be written as a **'private' non-PBS prescription.**

Prescription Requirements

- A prescription is a legal document. Features precise written instruction from a prescriber to a pharmacy for preparing + dispensing a medicine for a patient (must be legibly handwritten or printed)
- For S2 and S3 medicines, optometrists are encouraged to issue a prescription.
- S4 medicines prescriptions may be handwritten (must be legible!) or computer-generated and include;
  - ★ Date of issue
  - ★ Details of prescriber (name, address, prescriber number)
  - ★ Details of patient (name, address, DOB)
  - ★ Details of medicine (name by active ingredient, strength, quantity, what eye, course, quantity, # repeats, brand name)
  - ★ Precise directions + dosages
  - ★ Prescriber's signature

HANDWRITTEN PRESCRIPTION	COMPUTER-GENERATED PRESCRIPTION
	

POP QUIZ 💡

Prescribing Authority Phenylephrine at concentrations of 2.5% or less is approved for diagnostic use by optometrists with general registration	✅
Prescribing Authority Optometrists are authorised to prescribe Diamox (acetazolamide) in emergency situations To make it correct: Optometrists **CANNOT** prescribe Diamox without prior medical consultation with GP. Under Appendix C of OBA ➔ **Appendix C** states Optometrists whose registration is endorsed for scheduled medicines are **not** currently authorised under state and territory drugs and poisons legislation to prescribe, possess or supply acetazolamide (Diamox).	❌
Self-Prescribing Optometrists are advised against self-diagnosing and then self-prescribing S4 medicines	✅

Adverse Event Reporting

Quality Use of Medicines (QUM)

- QUM principles emphasise the importance of considering both pharmacological and non-pharmacological management.
- Involves:
  1. Selecting management options wisely
  2. Choosing suitable medicines (considering individual themselves, healthcare contexts, whether they can even access the drug)
  3. Use medicines safely and effectively to get the best possible results (implies adherence and getting patient to demonstrate how they are using their drops for e.g.)

Appendix A - Stresses what diagnostic agents optometrists can use	**Appendix B** (as shown above)

List of scheduled medicines for diagnostic use approved by the OBA for administration by optometrists holding general registration. The Optometry Board of Australia has approved the following diagnostic drugs for optometrists to administer in the course of their practice: • anaesthetics, local (synthetic cocaine substitutes) — when prepared and packed in the form of eye drops • tropicamide — when prepared and packed in the form of eye drops containing one (1) per cent or less of tropicamide • cyclopentolate hydrochloride — when prepared and packed in the form of eye drops containing one (1) per cent or less of cyclopentolate hydrochloride • atropine — when prepared and packed in the form of eye drops containing one (1) per cent or less of atropine sulphate • homatropine — when prepared and packed in the form of eye drops containing two (2) per cent or less of homatropine hydrobromide, and • pilocarpine nitrate — when prepared and packed in the form of eye drops containing two (2) per cent or less of pilocarpine nitrate • phenylephrine – when prepared and packed in the form of eye drops containing 2.5% or less of phenylephrine Registered optometrists should be familiar and comply with the current requirements of state and territory drugs and poisons legislation in the jurisdictions in which they practise.[16]	Board-approved list of topical schedule 2, 3 and 4 medicines that optometrists with a scheduled medicines endorsement are qualified to administer, obtain, possess, prescribe, supply or use for the purposes of the practice of optometry

POP QUIZ💡

Principles of Quality Use The principle of QUM requires optometrists to consider the cost of treatment not only to the patient but also to the broader health system.	✅
Considering Alternatives QUM encourages optometrists to consider non-pharmacological options before deciding to prescribe a medicine (sometimes a drug is NOT the best answer)	✅
Holistic Treatment Choices When prescribing, optometrists should select the most potent available medications to ensure rapid symptom resolution To make it true: When prescribing, optometrists should select the best available option taking into account several factors (i.e cost, form)	❌

Adverse Event Reporting

- The TGA collects reports of adverse events associated with medicines and medical devices. This allows them to investigate and take action on any medicine and safety issues.
- Endorsed optometrists must be familiar with TGA approved indications found in the product information
- Optometrists can help the TGA in safeguarding public health by reporting all suspected adverse events associated with medicines, particularly those associated with new products. It is only voluntary, not yet mandatory.

POP QUIZ💡

Responsibility in Reporting Adverse event reporting is a responsibility of pharmacists only To make correct : Adverse event reporting is a **shared** responsibility	❌
Mandatory Reporting by Optometrists NSW law requires that relevant authorities are notified of the loss or theft of a scheduled medicine (as they are powerful + potent drugs)	✅
Contribution to Medicine Safety Reporting adverse events helps improve medicine safety, supporting pharmacovigilance and protecting public health	✅

Practice Procedures

- Optometrists should ensure that:
  - Scheduled medicines are stored securely following manufacturer's recommendations and as required under the legislation of the relevant jurisdiction
  - Details of in-practice administration of scheduled medicines are recorded on the patient's clinical record
  - If required under the legislation of the jurisdiction, relevant authorities are notified of the loss or theft of a scheduled medicine
- As future therapeutically endorsed optometrists, you have a range of professional and ethical responsibilities, specified in the OBA guidelines for use of scheduled medicines
- Knowledge of ophthalmic drug delivery is foundational to understanding posterior eye therapeutics and holds practical relevance

Glaucoma Management

- When an initial diagnosis of chronic glaucoma is made, or a patient is at high risk of developing the disease, optometrists must:
  1. Refer the patient for specialist assessment and advice about confirmation of diagnosis and ongoing management, **OR**
  2. Develop a management plan that includes initiation of treatment and monitoring of the patient's response
  3. Have the equipment to measure and.or assess: IOP, Central corneal thickness (CCT), threshold visual fields, anterior chamber angle, ONH and RNFL
- The optometrist MUST provide a referral for ophthalmological assessment and advice after making an initial diagnosis and initiating treatment for chronic glaucoma IF:
  - The anti-glaucoma treatment **does not** stabilise the patient's condition

- A patient needs assessment by an ophthalmologist or ophthalmology service for possible surgical intervention or laser treatment, or
- A patient experiences side effects of initial treatment

- When you first initiate treatment for glaucoma (whether new or high risk), ensure there is a follow up plan in place, review appointment at least every 4-6 weeks, minimally. Checking response to therapy, if they are using drug, checking adherence, side effects etc.

NOTE: Must provide the patient with a referral to an ophthalmologist or ophthalmology service **WITHIN 4 MONTHS** of starting treatment for chronic glaucoma

POP QUIZ 💡

Referral Timelines Optometrists must provide patients with a referral to an ophthalmologist to ophthalmology services within four months of starting treatment for chronic glaucoma.	✅
Pilocarpine Usage Guidelines Pilocarpine is always used immediately in first aid for acute angle-closure (ALWAYS comes up in oral exams) To make correct : Pilocarpine is **not** always used immediately in acute angle closure, its use depends on IOP and angle closure mechanisms.	❌
Equipment for Ocular Assessment Proper equipment is essential for assessing ocular parameters to aid clinical decisions and referrals in glaucoma management.	✅

EMERGENCY MANAGEMENT OF ACUTE PRIMARY ANGLE CLOSURE 🚨

- Standard management of these cases is emergency referral to an ophthalmologist or hospital. HOWEVER, therapeutically qualified optometrists are able to (in collaboration with a medical practitioner), provide first aid.
- Definitive intervention is laser therapy (laser peripheral iridotomy)

First aid *unless* contraindicated:

1. **Oral acetazolamide 500mg** (2 tablets of **Diamox**) – If cannot keep down need to go to hospital for IV/intramuscular injection

NOTE: Pharmacists can also supply this drug via a telephone order. Acute angle closure causes a lot of pain , so you need to make sure the patient keeps the drug down.

2. **Topical medications** (allowing ~2mins between each drop):
    a. One drop **beta blocker** e.g., **timolol 0.5%**
    b. One drop **alpha agonist** e.g., **apraclonidine 0.5%**
    c. One drop **carbonic anhydrase inhibitor** e.g., **brinzolamide 1%**
    d. If the eye is inflamed, one drop **high-penetrance steroid** **prednisolone 1%** with a further three drops at regular intervals in the first hour, then hourly thereafter
3. ONLY After discussion with an ophthalmologist can we give one drop **pilocarpine 2%** (since there are some instances in which it is **contraindicated** e.g when IOP is > 40 mmHg as then pilocarpine will not have an effect).

NOTE: Pilocarpine should only be used for mechanisms involving phakic pupillary block or angle crowding. Use of pilocarpine is contraindicated for retrolenticular causes of angle closure (like thickened lens/cataract) as well as those involving aphakic or pseudophakic closure events.

LECTURE 2: Ophthalmic Drug Delivery

Why Does Ophthalmic Drug Delivery Matter?

- The pharmacotherapy of eye disease generally requires **high local concentrations of drug** at the ocular tissues, we need the drug at the site that it is treating.
- Treatment of uveitis, glaucoma, or retinal disease involves **therapeutic drug levels at appropriate target sites** deep within the globe. It can be hard logistically to get drugs into these areas, especially since we have to penetrate multiple tissues of various structures, anatomy and physiology .
- Many ophthalmic drug delivery systems have been developed; most suffer from lack of precision
- Intraocular drug delivery systems can lead to toxicity

(A) Topical Administration (MOST COMMON)

✅ Convenient, simple, non-invasive
✅ Self-adminstration possible
❌Primary source of drug loss is diffusion into the circulating blood via blood vessels of the conjunctiva, episcleral, intraocular vessels and drainage through the nasolacrimal system. Consequently, a lot of these drugs **do not typically penetrate in useful concentrations** to the posterior ocular structures and therefore are often of no therapeutic benefit for diseases of the posterior segment (e.g retinal tissue does not benefit from topical administration).

Eye drops: solutions and suspensions
(Most common for glaucoma)
NOTE: Suspensions need to be shaken to activate the drug.

WORKED EXAMPLE: Ocular hypotensive therapy for glaucoma typically packaged as a multidose dropper bottle designed to be discarded after 1 month of use.

Eye drop cap colour chart:

Turquoise: **Prostaglandin analogues**
- ★ Xalatan (latanoprost)
- ★ Lumingan (bimatoprost)
- ★ Travatan Z (travoprost)

Purple: **Alpha adrenergic agonists**
- ★ Bromindine (alphagan)

Orange: **Carbonic Anhydrase inhibitors**
- ★ Dorzolamide
- ★ Brinzolamide

Red: **Mydriatics and cycloplegics**

Tan: **Antibiotics**
- ★ Ciprofloxacin (Ciloxan)
- ★ Ofloxacin (Ocuflox)
- ★ Gatifloxacin (xymar, Zymaxid)
- ★ Moxifloxacin (Vigamox, Moxeza)

Dark blue: **Beta blockers combinations**
- ★ Combigan (brimonidine-timolol), Cosopt (dorzolamide-timolol)

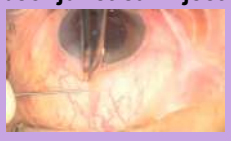
Yellow: **Beta blockers**
- ★ Timolol

Two unpreserved, single dose antiglaucoma options available: • **Lumigan** PF (**bimatoprost 0.03%**, 30x0.4mL unit doses) • **Ganfort** PF 0.3/5 (**bimatoprost 0.03%** + **timolol 0.5%** eye drops, 30 x 0.4 mL unit doses) – combo drug	★ Atropine, homatropine, cyclopentolate, tropicamide, phenylephrine ★ Levobunolol ★ Betaxolol - light blue top White: Topical anaesthetics and OTC products Direct patients to follow the storage instructions on the consumer medicine information. **Looking after your medicine** **Store the unopened bottle in its original container in a refrigerator (2° to 8° C), protected from light. Do not freeze.** **After opening, store the bottle in its box, protected from light and in a cool place where the temperature stays below 25°C. You do not need to keep it refrigerated.** For example, do not store it: • in the bathroom or near a sink, • in the car or on window sills, or • in the pockets of your clothes. Heat and dampness can destroy some medicines. **Keep it where young children cannot reach it.** **Looking after your medicine** **Store TIMOPTOL drops below 30◦C .** **Keep the drops away from direct light.** Follow the instructions in the carton on how to take care of your medicine properly. Store it in a cool dry place away from moisture, heat or sunlight; for example, do not store it: • in the bathroom or near a sink, or • in the car or on window sills. • In the pockets of your clothes **Keep it where young children cannot reach it.** **Do not leave the lid off the bottle for any length of time to avoid contaminating the eye drops.**
Sprays, ointments, lid scrubs	✅ Applicable to anterior eye segment ★ **Tearsagain** (soy lecithin 1% + tocopherol 0.002% + vitamin A palmitate 0.025% eye spray, 100 actuations) ★ **Hycor** (hydrocortisone acetate 1% eye ointment, 5 g) ★ **Tobrex** (tobramycin 0.3% eye ointment, 3.5 g) ★ **ViruPOS, XOROX** (aciclovir 3% eye ointment, 4.5 g) ★ Ocular lubricants
Gels	✅ **More sustained drug delivery** (than drop, suspension or solution) whilst minimising accompanying blurred vision typical of ointments E.g Optifresh Eye Gel, PAA, Viscotears (**Carbomer-980** 0.2% eye gel, 10 g) In situ-activated gel-forming systems are delivered as eyedrops the converted by temperature changes and ionic movement into a gel-like viscosity E.g., Timoptol XE (timolol 0.5% delivered using gellan gum) dosed once daily in the morning versus Timoptol (timolol 0.5% solution) once-twice a day ***NOTE:*** Standard dosing of beta blocker is 2 days but Timopotol does not need to be dosed twice a day.
Solid delivery devices + Continuous flow device	❌Not applicable to posterior eye disease Includes soft contact lenses, collagen shields, filter paper strips, cotton pledgets, conventional and continuous irrigation systems

(B) Periocular Administration

✅Useful when higher drug concentrations (especially corticosteroids + antibiotics) are required within the eye that cannot be delivered by topical administration, includes:

- Subconjunctical injection
- Sub-Tenon's injection
- Retrobulbar injection
- Peribulbar injection

Subconjunctical injection 	Involves passing the needle between the anterior conjunctiva and Tenon's capsule ✅ Useful in patients who are unreliable with topical medication ✅ Can be injected at the end of the surgery to avoid topical or systemic drug therapy ✅ Useful for severe corneal disease e.g antibiotics for bacterial ulcers ✅ Used as an initial supplement to systemic or intravitreal antibiotics for bacterial endopthalmitis ***NOTE:*** 5-fluorouracil (ANTIFIBROTIC) is sometimes injected after high-risk trabeculectomy. Used in high risk glaucoma surgeries where there is risk of fibrosis scarring closing up the trabeculectomy to prevent drainage failure and increase the chance of success.
Sub-Tenon's injection	❌Offers no significant advantages over subconjunctival injection ❌Delivers lower quantities of drug to the eye ❌Associate with a greater risk of perforating the globe E.g Sub-Tenon's injections of corticosteroids for severe uveitis, cystoid macular oedema following cataract surgery, diabetic macular oedema.
Retrobulbar injection + Peribulbar injection	Intentionally aims for the muscle cone, which contains vital structures (like the optic nerve) and has been superseded by peribulbar injection). ❌No longer offered for people having cataract surgery ❌Because of the risks associated with retrobulbar injections, peribulbar injection (injected around the

	globe but not directly into the muscle cone) is the primary alternative to topical or sub-tenon's anaesthesia for people having cataract surgery.

(C) Intracameral Administration

Describes drug injection directly into the A.C.

✅ Can be useful in cataract extractions e.g viscoelastic substances during cataract extraction and glaucoma filtering surgeries to protect against corneal endothelial cell loss and flat anterior chamber (as it broadens the AC, lowers the IOP).

(D) Intravitreal Administration

Describes drug injection directly into the vitreous.

✅ Needed because systemic drugs penetrate poorly into the eye.

Examples include:

- Intravitreal anti-VEGr injections for neovascular AMD
- Antibacterial and antifungal drugs for bacterial and fungal endophthalmitis
- Intravitreal triamcinolone to treat diffuse diabetic macular oedema

(E) Photodynamic Therapy (PDT)

It is a form of laser therapy. Tries to fill a gap because conventional laser can destroy adjacent healthy tissue so it offers the advantage of being a little more selective. Previously used to treat AMD but has been superseded by anti-VEGF injections for neovascular AMD. May still be used for type 1 aneurysmal neovascularisation (PCV), chronic/recurrent CSCR (AKA pachychoroid disease entities)

1. Involves **intravenous administration of photosensitive dye**, verteporfin (Visudyne), for 10 minutes. Travels to the abnormal vessels
2. Drug attaches to molecules that are commonly found in rapidly growing cells i.e in wet AMD BV
3. **Low-power laser non-thermal light** to **abnormal tissues** which activates drug leading to selective cell death

Case Studies

CASE STUDY 1: A 58-year-old patient with poorly controlled type 2 diabetes presents with blurred central vision. OCT shows macular thickening

Condition = Diabetic macular oedema (most likely centre involving as patient mentioned central blurred vision) . Caused by fluid accumulation in macula due to leaky retinal blood vessels from breakdown of inner BRB.

As an optometrist, what is your role in identifying, monitoring, and referring patients with DMO?	How do you decide when therapeutic intervention is needed versus referral?
• Understanding limitations of skills, competency and experience and knowing when another professional can be called if it is beyond our scope. • Not topical management here (but optometrists can only prescribe topical) • Basically need to refer to ophthalmologist (in most cases, will refer outside your practice) • Educating the patient about the disease to make sure they visit the ophthal • Reinforce the importance of managing diabetes + consulting with an endocrinologist • Document your findings • Report back to their GP	• Grade the severity of the DR: + Monitor progression with time (better/worse/stay the same) + Dictates urgency, allows us to triage the condition and figure out management + Allows us to predict the risk of progression • Assess the patient's signs and symptoms • Educate patients on lifestyle choices/change, the role of other members in healthcare (e.g GP, endocrinologists). • ALWAYS document findings + send referral off to GP, even if nil findings • Think about the tests that you want to do in house VS tests you want the specialist to do • Think about ongoing monitoring schedule (so when DMO dries up, what schedule of care will they receive – thinking about cost, adherence).

CASE STUDY 2: A 76-year-old px reports sudden, painless vision loss in one eye. Fundus exam reveals a pale, swollen optic disc.

Condition = Condition = Arteritic anterior ischaemic optic neuropathy (AAION), Giant cell arteritis, Temporal cell arteritis

- Caused by obstruction of optic nerve blood supply
- Ischaemia causes swelling of RNFL, resulting in pale swelling of ONH (***NOTE:*** if hyperaemic optic nerve, would be NAION)
- One of the red flag in ocular emergencies
- It is a systemic disease , it is the temporal artery that is inflamed.

What is your responsibility in detecting early signs of arteritic AION, educating the patient about the risk of bilateral vision loss, and facilitating urgent referral for corticosteroid therapy? What might happen if these responsibilities are not fulfilled?	How do you support long-term care and systemic monitoring to prevent further harm?
If not managed, the patient may pass away (from stroke/aneurysm),	• Not good enough to write a standard letter and send

may sue or have permanent vision loss. Representatives may even sue.	them on their way, so instead: + Emergency department of hospitals + Sydney Eye Hospital + Pick up the phone + contact your connections

Treatment: ~ 2 years of IV corticosteroids (long-term) initially then modifying to oral.

CASE STUDY 3: A 64-year-old patient presents for a routine eye exam. They are asymptomatic but have a family history of glaucoma. On examination, their IOP is 23 mmHg OU, optic nerve head shows mild cupping (C/D ratio 0.6), and OCT reveals early RNFL thinning. Visual fields show a subtle nasal step in one eye.

Condition = Glaucoma

What is your role as an optometrist in diagnosing and managing early glaucoma?
How do you decide when to initiate treatment, refer, or monitor?
What are your responsibilities in educating the patient and ensuring long-term adherence to therapy?

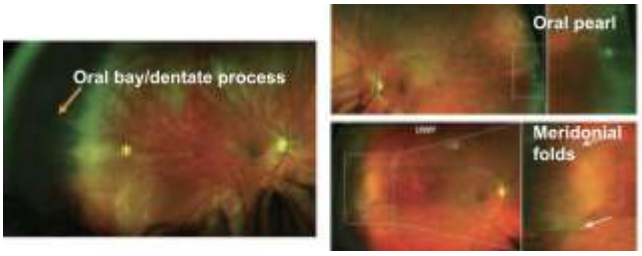
- Optometrists are often the first point of contact
- Can start prostaglandin analogue
- Monitor side effects, compliance, and continue follow-ups + see them on going.
- Mandatory referral to ophthalmologist in 4 months

WEEK 2: MACULAR AND RETINAL DISEASE

LECTURE 3: Macular and Retinal Disease

Prioritise

- **Red Flags =** Symptoms that signal high risk of serious underlying pathology, or a significant medical emergency, requiring further investigation or specialist referral to avoid potential harm or life-altering consequences.



Peripheral retinal and related conditions

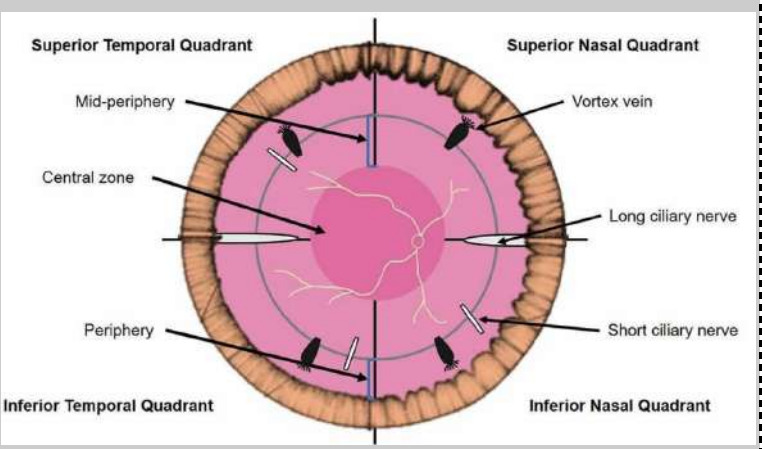
- Peripheral retinal conditions are one of the highest reasons why optometrists are being sued for 'malpractice' (for missing detachment).

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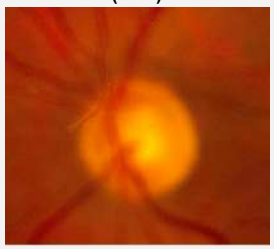
Retinal periphery: Zone from the **equator** out to the **ora serrata**, typically spans the view of a 20D BIO lens (where most diseases/signs will present). Indicated by 4x vortex veins. The central zone is indicated by darker pink area (AKA posterior pole) .

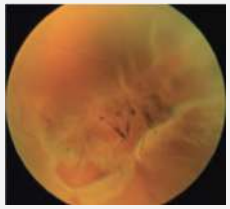
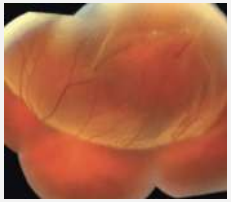
Imaging Techniques (relevant to workup)

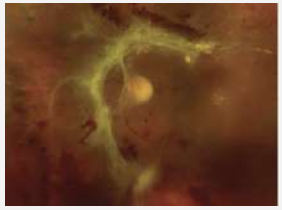
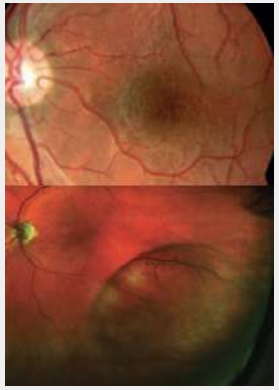
- ★ DFE using BIO + noncontact/contact funduscopy and/or scleral indentation
- ★ Ultrawide field imaging (OPTOS, Clarus)
- ★ OCT
- ★ Ultrasonography (B-scan) – qualitative view of the globe, helps us view media opacities, dense floaters.

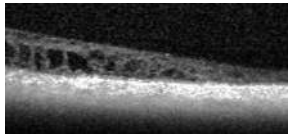
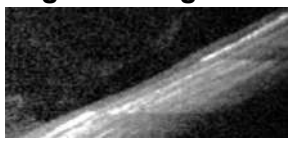
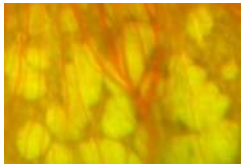
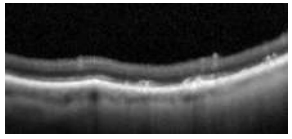
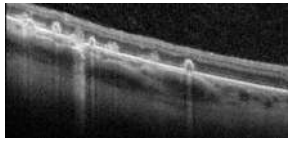
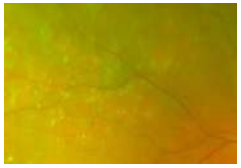
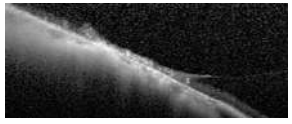
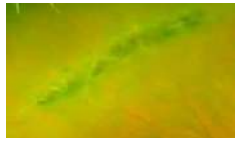


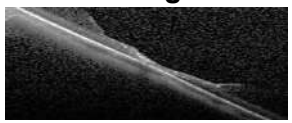
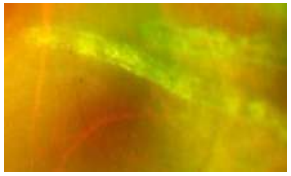
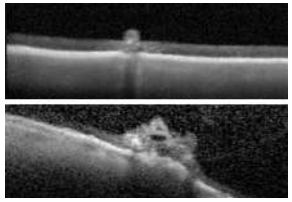
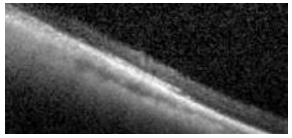
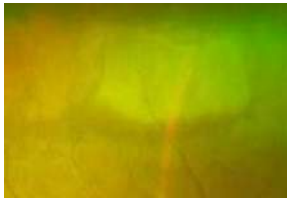
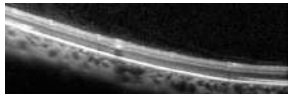
- PVD is the singular most important event of the aging eye. Monitor for complications: Most cases are benign, but patients should be educated to report new floaters or flashes, which may signal retinal tears or detachment (retinal detachment = RD).
- **RD is an ophthalmic emergency requiring** prompt surgical intervention (e.g., vitrectomy, scleral buckle, or pneumatic retinopexy)
- Laser photocoagulation or cryotherapy are often used to seal retinal breaks and prevent progression to detachment

CONDITION	SIGNS/SYMPTOMS + DIFFERENTIALS	WORKUP	MANAGEMENT	FOLLOW- UP
Posterior vitreous detachment (PVD)  ***Weiss Ring:*** Semi-translucent film in front of ON The vitreous is more strongly attached to the disc, peripheral retina (vitreous base – strongest) and BVs. It is more weakly attached to the macula.	**Signs:**  • **Weiss Ring:** One or more discrete near-translucent or light grey vitreous opacities, one often in shape of a ring or broken ring suspended over the disc. • Retinal break/retinal tear • Retinal detachment • Vitreous haemorrhage • Peripheral retinal and disc margin haemorrhages • Released retinal pigment epithelial cells in the anterior vitreous ("tabacco dust"/"Schaffer's sign"). **Symptoms:** Floaters, blurred vision, and/or flashes of light which are more common in dim illumination or with eye movement. Symptoms usually present acutely and progress over hours-days. **DDx:** 💬 **Uveitis** → In vitritis, vitreous cells may be found in both the posterior and anterior vitreous, the condition may be bilateral and the cells are not typically pigmented. Many uveitis (particularly white dot syndromes) will present with floaters and white photopsias. 💬 **Migraine** → Multicoloured photopsias in a zig-zag pattern that obstructs vision lasts ~ 20 minutes. A headache may or may not follow and symptoms may be bilateral. Normal fundus examination.	• Distinguish retinal photopsia from the visual distortion of migraines ***NOTE:*** Visual aura is described more as a zigzag pattern that advances across the visual field as time passes. Retinal photopsia is described as a very quick sudden flash, like a lightning flash in the periphery, often temporal. • Enquire about risk factors ★ Age 46-65 year olds (if a younger onset than it is more likely to be a retinal tear) ★ Associated symptoms ★ Myopic (especially if high) ★ Trauma/surgeries (already existing vulnerability) ★ FOHx of tears/ detachments **Mandatory:** DFE (ruling out RD/tears) and scleral depression/indentation BOTH eyes • If you do not do scleral indentation, you will miss ~11% of retinal breaks/detachments. • If a break has occurred will have seen pigmented **Schaeffer's sign** (**tobacco dust).** Need the patient to look around to see it when performing fundoscopy (because pigment settles by gravity) Dilate both eyes to **see fellow eye** – if PVD has occurred in one eye, PVD is imminent in the other eye (assess its status/whether it has occurred or not) **Optional:** • Contact funduscopy (3x mirror lens with coupling fluid on the flange of the lens - similar to gonio) • OCT • B-scan ultrasonography	No treatment is indicated for PVD unless an acute retinal break or dense vitreous hemorrhage is found; In the presence of **symptomatic PVD**, chronic retinal breaks or lattice degeneration usually warrant Tx. **Same day/ within 24 hours referral** to a SURGICAL retinal specialist if you suspect a retinal break. ***NOTE:*** If a patient is getting flashes of light, this indicates that the patient is getting some sort of traction happening, so it is best to seal those off.	• **4-6 weeks** for repeat DFE to: ***1.*** Ensure that we did not miss anything initially ***2.*** PVD is an event that occurs on a continuum (still a critical period with risk of hole developing ~ 5% risk of retinal break developing) • Advise patient of RD symptoms (a significant increase in flashes/floaters, worsening vision, persistent curtain or shadow in field of vision) – IMMEDIATE VISIT • **PVD symptoms usually diminish** over time (sometimes months) • Symptoms of RD, explain **likelihood in the fellow eye** • If no retinal break is found but significant VH or anterior pigmented vitreous cells are present, repeat examination should be performed within 24 hours by a retinal specialist (high risk of retinal break).
Retinal break (tear) 	**Signs:** A full thickness retinal defect, usually seen in the periphery. **Symptoms:** **Acute =** Flashes of light, floaters ("cobwebs,hair, or film" that changes position with eye movement) with or without VA changes. Can be identical to PVD symptoms but more profound. **Chronic =** Usually asymptomatic Predisposing conditions: • Lattice degeneration • High myopia • Aphakia/Pseudophakia • Age-related retinoschisis • Vitreoretinal tufts • Meridional folds • History of previous retinal break or detachment in the fellow eye • FOHx of retinal break or detachment • Collagen vascular disorders • Trauma.	**SAME AS ABOVE!** Complete ocular examination with a slit lamp and indirect ophthalmoscopy of both eyes with scleral depression. After trauma, scleral depression may be gently performed once open globe injury has been ruled out. B-scan US may be helpful when the retina is not visible (e.g., vitreous hemorrhage, dense cataract, etc.)	**Laser photocoagulation or cryotherapy** required **within 24hrs** especially for: • Acute symptomatic break/new onset • Acute traumatic break • Acute symptoms and lattice degeneration **Asymptomatic** cases are not treated unless it is **superior and large** (since gravity will pull it down). Can also consider treating in absence of PVD and if associated with other risk factors like pseudophakia (cataract surgery), high myopia. Treatment may be less urgent for chronic breaks. ENSURE that when you see a retinal break/tear document the absence or presence of a PVD. Can also document the area of rub-retinal fluid or detachment to see if it is advancing over time.	**Non-treated** retinal breaks may be reviewed at 3 months then every 6-12 months if stable. **Treated breaks** are reviewed at 2 weeks (as scars take that long to form), 6 weeks, 3 months, then every 6-12 months if stable. RD symptoms (a dramatic increase in floaters or flashing lights, worsening visual acuity, or the appearance of a curtain, shadow, or bubble anywhere in the field of vision) are explained and patients are told to return immediately if these symptoms develop.

RETINAL DETACHMENTS There are 3 distinct types of RD:				
<u>Rhegmatogenous</u> retinal detachment (RRD)  [PIC] Lattice degeneration – hyperpigmentation, spindle-shaped, causing associated detachment + sub-retinal fluid. 	**Signs:** Elevation of the retina from the RPE by fluid in the subretinal space due to an accompanying full-thickness retinal break/breaks. **Symptoms:** Flashes of light, floaters, a curtain or shadow moving over the field of vision, peripheral or central visual loss, or both. Posterior vitreous detachment (PVD) can cause retinal breaks and rhegmatogenous retinal detachment (RRD). While PVD itself is often harmless, the vitreous gel pulling away from the retina can create tears or holes in the retina. If this tear allows fluid to seep behind the retina, it can cause a more serious rhegmatogenous retinal detachment. **Etiology:** A retinal break allows fluid to move through the hole and separate the overlying retina from the RPE. ***<u>NOTE:</u>*** A chronic rhegmatogenous retinal detachment (RRD) often shows a pigmented demarcation line at the posterior extent of the RD, intraretinal cysts, fixed folds, and/or subretinal precipitates or a combination of these with a relative visual field defect. It should be differentiated from retinoschisis, which is typically smooth and dome-shaped, translucent, and produces an absolute visual field defect. Underlying choroidal vasculature appears normal (unlike RRD, where it is obscured).	**SAME AS ABOVE!** 1. Slit lamp examination to assess for signs of uveitis, lens status, presence of vitreous pigment/hemorrhage, and PVD. 2. Indirect ophthalmoscopy with scleral depression of both eyes. Slit lamp examination of the periphery with a 90-diopter or widefield lens may help in finding small breaks. 3. B-scan US may be helpful if media opacities are present.	**Macula on but threatened:** Should have URGENT surgical repair (scleral buckle, vitrectomy or combination) performed urgently. **Macula OFF:** Should be repaired but not necessarily urgently, as it has been shown there is no difference in visual outcomes if performed within 7-10 days.	Follow up after surgery is typically 1 day, 1 weeks, 1 month, 2-3 months then every 6-12 months thereafter.
<u>Exudative</u>/Serous retinal detachment 	**Signs:** Serous elevation of the retina with shifting SRF with patient positioning. While sitting, the SRF accumulates inferiorly, detaching the inferior retina; while in the supine position, the fluid accumulates in the posterior pole, detaching the macula. There is no retinal break; fluid accumulation is due to breakdown of the normal blood-retinal barrier. The detachment typically does not extend to the ora serrata. **Symptoms:** Visual field defect with varying degrees of vision loss; visual changes may vary with changes in head position. The detached retina is smooth and may become quite bullous. A mild RAPD may be present in large RDs. <u>Common causes</u>: • Choroidal neovascularisation (leaky vessels that have broken through RPE) AKA SRF • Inflammation • Tumours • **Neoplastic:** Choroidal malignant melanoma, metastasis, choroidal hemangioma, multiple myeloma, retinal capillary hemangioblastoma, • **Vascular:** Choroidal neovascularization (CNV), Coats disease, malignant hypertension (HTN), preeclampsia, and familial exudative vitreoretinopathy AKA anything that can exude fluid	**Mandatory**: DFE **Optional**: FFA (dye study - trying to understand where fluid is coming from), OCT, B-scan US (qualitative investigation to see shape of globe + see any media opacities), systemic workup to identify SRF source **1.** Intravenous fluorescein angiography (IVFA) may demonstrate leakage or pooling and identify the SRF source. **2.** OCT may help identify the source of SRF (e.g., CNV). **3.** B-scan US may help delineate the underlying cause. **4.** Systemic workup to rule out aforementioned causes.	Treat underlying condition	Depends on cause and individual patient factors

<u>Tractional</u> retinal detachment Occurs when **<u>traction</u>** caused by proliferative, fibrous membranes present over the retinal surface or vitreous **pulling neurosensory retina from RPE**  [PIC] Fibrosis + scar formation pulling the retina off	**MOA –** Scar formation pulling the retina off. In trauma/PVD you have a break and then fluid creeps under there. Mechanism is different for tractional RD. **Symptoms:** Visual loss or visual field defect; may be asymptomatic. **Sign:** A traction retinal detachment (TRD) appears concave with a smooth surface; cellular and vitreous membranes exerting traction on the retina are present; retinal striae extending from these areas may also be seen. Detachment may become a convex RRD if a tractional RT develops (combined RRD/TRD). The retina is immobile, and the detachment rarely extends to the ora serrata. A mild RAPD may be present in large RDs. <u>Causes</u>: • Fibrosis • Scar formation • ERM • RoP (retinopathy of prematurity) • **Sickle cell retinopathy** • **Proliferative DR (MOST COMMON)** **Etiology:** Fibrocellular bands in the vitreous (e.g., resulting from proliferative diabetic retinopathy [PDR], sickle cell retinopathy, retinopathy of prematurity, FEVR, toxocariasis, trauma, proliferative vitreoretinopathy) contract and detach the retina. Also important to assess lens status to see status of any other retinal vascular disease e.g NVI, PVD.	**SAME AS PVD** 1. Slit lamp examination to assess lens status, sequelae of retinal vascular disease (e.g., neovascularization of the iris), and PVD. 2. Indirect ophthalmoscopy with scleral depression of both eyes. Slit lamp examination of the periphery with a 90-diopter or widefield lens may help in finding small breaks. 3. B-scan US may be helpful if media opacities are present. 4. OCT is useful in identifying tractional membranes and can be useful in differentiating tractional membranes from detached retina.	May or may not require treatment depending on aetiology, fellow eye, extent/location of the traction	Depends on cause and patient factors.
X-linked *(JUVENILLE)* **and age-related retinoschisis** *(MORE COMMON)*  **TOP** – X-linked **BOTTOM** – age-related	If we have a schisis and we have lifting and there are vessels within that, then it may be prone to breaking and causing a vitreous haemorrhage. Can also get formation of inner + outer retinal holes. **X-LINKED** **Symptoms:** Decreased vision due to macular involvement. Sometimes VH. Can be asymptomatic. The condition is congenital but may not be detected at birth if an examination is not performed. A family history may or may not be elicited (X-linked recessive). **Signs:** Foveal schisis seen as stellate maculopathy: Cystoid foveal changes with retinal folds that radiate from the center of the fovea (petaloid pattern). Unlike the cysts of cystoid macular edema (CME), they do not stain or leak on IVFA, but can be seen with indocyanine green angiography (ICGA) and on OCT. The macular appearance changes in adulthood and the petaloid pattern may disappear. **AGE-RELATED** **Symptoms:** Usually asymptomatic; may have decreased vision. **Signs:** The schisis cavity is dome-shaped with a smooth surface and is usually located temporally, typically inferotemporally. The findings are usually bilateral and may show sheathing of retinal vessels and "snowflakes" or "frosting" (persistent Mueller fibers) on the elevated inner wall of the schisis cavity. Splitting usually occurs at the level of the outer plexiform layer. The area of schisis is not mobile and there is no associated RPE pigmentation in contrast to an RD, which may have corrugations and a pigmented demarcation line.	**Mandatory**: DFE **Optional**: OCT, VF (to see scotoma), FFA (to differentiate from exudative cause), FAF (for contrast to show that part of the retina is above the typical retinal plane), ERG Will see an **ABSOLUTE scotoma** **X-LINKED:** 1. Family history. 2. Dilated retinal examination with scleral depression to rule out retinal break or detachment. 3. OCT can help determine the layer of schisis and help to differentiate schisis from an RD. 4. IVFA will show no leakage. 5. Fundus autofluorescence (FAF) may help delineate areas of schisis. 6. Electroretinography (ERG) is not necessary for diagnosis but can show a reduced b-wave with a preserved a-wave **AGE-RELATED:** 1. Slit lamp evaluation for anterior chamber inflammation and pigmented anterior vitreous cells; neither should be present in isolated retinoschisis. 2. Dilated retinal examination with scleral depression to rule out a concomitant RD or an outer layer retinal hole, which may lead to an RD. 3. A slit lamp examination using a 90-diopter lens or fundus contact lens as needed to aid in recognizing outer layer retinal breaks. 4. OCT can help determine which layer of the retina is split. 5. Visual field testing will reveal an absolute scotoma in the area of schisis.	• **No treatment required for age-related retinoschisis** • **No <u>definitive</u> Tx for stellate maculopathy** (AKA X-linked) • Topical CAIs have been shown to decrease foveal thickness and improve VA for juveniles. • Manage associated vitreous haemorrhage if present, RD or amblyopia	• Every 6 months or more frequently for amblyopia • Advise Px of Sx of RD

Condition	Workup	Management	Followup	Other considerations
Microcystoid (peripheral cystoid) degeneration 		Multiple small intraretinal microcystoid cavities Appears as a hazy grey area of thickened retina extending from the ora serrata	Routine Review	Coalescence of cysts with accompanying splitting of retina is thought to lead to development of retinoschisis
Pavingstone degeneration 		Circumscribed areas of retinal thinning from loss of RPE and photoreceptors, which allows increased visualisation of the choroidal vasculature.	Routine Review	present in 4%–28% of patients; often bilateral; similar prevalence in men and women; increasingly common with age
Reticular (honeycomb) degeneration 		It appears as a circumferential band of granular pigment between the ora serrata and equator. The pigment can take on a reticular or bone spicule appearance.	Age related and innocuous Routine Review	Due to pigment scattered throughout the sensory retina from degenerating RPE Usually bilateral and often accompanied by peripheral drusen
Peripheral Drusen 		Extracellular protein and fat deposits between the RPE and Bruch's membrane, caused by degeneration of RPE cells	Routine Review	Commonly found in people aged ≥40 years It is not associated with a risk for central vision loss in the absence of changes within the macula, however there is evidence that peripheral drusen are part of the spectrum of AMD.
Pars Plana Cyst	Normal finding, near the ora serrata		Routine Review	
Lattice Degeneration 		Retinal thinning with loss of neurosensory layer; vitreoretinal adhesions at the margin of lesion Associated with RPE hyperplasia, yellow glistening flecks, sclerosed vessel formation and atrophic holes. Most common in the superior and inferior retina and usually in a circumferential orientation around the eye.	Annual Review Risk for retinal detachment and rhegmatogenous retinal detachment due to vitreous traction Retinal detachment seen in 14%–35% of cases	Found in 6%–8% of patients

Snailtrack Degeneration 		Shiny bands of retina due to numerous glistening yellow-white dots on the inner retinal surface.	Annual Review May lead to retinal tears in up to 54% of patients; retinal holes may also be seen	A variant of lattice degeneration Found in 10% of the general population; more commonly seen in myopic eyes (40%)
Vitreoretinal Tuft 	 	Area of retinal degeneration caused by attachment to and pulling of the vitreous **Non-cystic**: Short (<0.1mm), thin projections of altered retinal and glial tissue **Cystic:** Round/oval-shaped, grey-white nodules of varying size with internal microcysts and often with RPE hyperplasia near/within the tuft.	**Non-cystic**:Not associated with retinal breaks, routine review **Cystic:** May predispose to a retinal tear/operculated hole or detachment with PVD. Annual review, if asymptomatic	Noncystic tufts are present in up to 72% of adults, cystic tufts are present in up to 5% of adults
White with pressure		Findings: The retina appears normal without depression Distinctive milky white or opalescent appearance of the peripheral retina that is observed in many normal eyes when examined with scleral depression	Benign condition Routine Review	Seen in around 30%–35% of eyes examined with scleral depression; incidence increases with age; no sex predilection; more commonly observed in myopic eyes during the second decade of life
White without pressure 		Distinctive white appearance of the peripheral retina without indentation and without mechanical stimulus	There is a low risk for association with retinal detachment, so patients are watched and examined every 1–2 years	Found in up to 30% of normal eyes; often bilateral; more frequently diagnosed in younger patients; may be associated with longer axial length
Dark without pressure 		Discrete, flat patches of darker retina, often with well-defined scalloped edges. OCT findings show hyporeflectivity of the ellipsoid zone; border corresponds to the site where the ellipsoid zone faded or disappeared	Routine Review	More common in darkly pigmented fundi.

Rate conditions from **least** to **most** likely to require treatment:

1. PVD (assuming no retinal break, tear or detachment – just a clean break)
2. Age-related retinoschisis
3. X-linked retinoschisis
4. Exudative/serous retinal detachment
5. Tractional retinal detachment
6. Retinal break (tear)
7. Rhegmatogenous retinal detachment (red flag – within 24 hours)

Your turn: It ain't rare if it's in your chair! You encounter a patient with juvenile x-linked retinoschisis and look into the peer-reviewed evidence for some management options. You find a systematic review which finds that "Yang et al. 's was the only one of six studies in this group to report no improvement with carbonic anhydrase inhibitors. The other five studies all reported significant improvement in VA" The parents and child would like to consider the use of topical carbonic anhydrase inhibitors to manage the maculopathy. ***How will you proceed?***	Inform them about the risks and benefits of that specific treatment. In this, it is off-label treatment. "Off-label" – not approved by the TGA for its drug indication. Refer them to a retinal specialist to co-manage the condition

Detriments of ADR

Adverse drug reactions: An unintended, harmful outcome from taking a medicine under normal conditions of use

- ★ Dosage
- ★ Pharmacology
- ★ Pharmacokinetics: absorption, metabolism and excretion
- ★ Route of administration
- ★ Patient characteristics: age, gender, presence of kidney or liver disease
- ★ Drug interaction

World Health Organization Definitions—Causality Assessment of Suspected Adverse Drug Reactions

Assessment of Suspected Adverse Drug Reactions	Definition
"Certain"	A clinical event noted temporally to be related to the administration of a drug that cannot be explained otherwise by concurrent disease or other drugs or chemicals. Dechallenge (drug withdrawal) and rechallenge (drug reintroduction causing recurrence of the effect) should be definitive.
"Probable/likely"	A clinical event occurs within a reasonable time to drug introduction, which is unlikely to be attributed to concurrent disease or other drugs or chemicals. The drug dechallenge is clinically reasonable. Rechallenge corroboration is not available or required for this definition.
"Possible"	A clinical event occurs with a reasonable time relationship to drug initiation but that could also be explained by concurrent disease or other drugs or chemicals. Dechallenge data may be unavailable or unclear.
"Unlikely"	A clinical event not necessarily related to drug initiation, such that a causal relationship seems improbable, and in which other drugs, chemicals, or underlying disease provide plausible explanations.
"Conditional/unclassified"	A clinical event reported as an adverse reaction, about which more data are essential for a proper assessment or the additional data are being processed.
"Unable to assess/ unclassifiable"	An unverifiable report suggesting an adverse reaction but with insufficient or contradictory information.

Detecting ocular ADRs (OADR):

- **Detailed drug history** includes prescription drugs, OTC drugs, recreational drugs, nutritional and herbal medicines
- Establish a temporal relationship between **drug use and ocular signs or symptoms**
- Familiarise yourself with the **possible ocular effects of the drug** the patient is taking
- Review the literature for new reports, systemic reviews

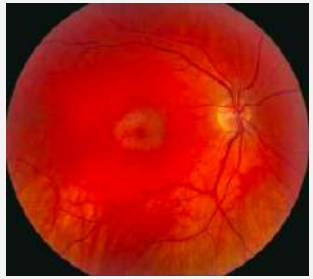
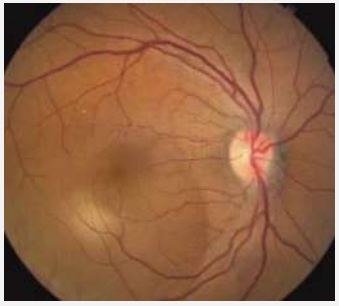
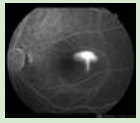
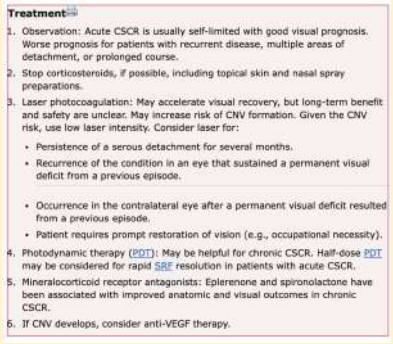
Assessment + management of OADRs:

- Causality + risk assessment:
  - + What's the likelihood it is associated with the drug?
  - + What's the significance?
- Report to the medicine prescribers, GP, and specialists
- Manage the OADR
  - + Optometric management: e.g. artificial tears for dry eyes
  - + Refer to an ophthalmologist if appropriate e.g. cataract removal
  - + Alter drug dose or discontinue drug
  - + Need to balance between the significance of an OADR against the physiologic need for the drug treatment
- Consider reporting to TGA

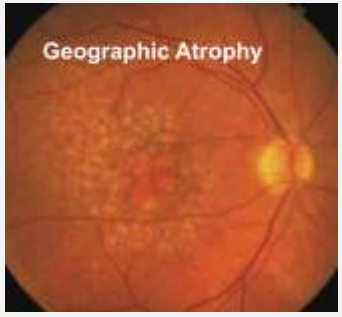
Macular Disease

- Macular disease makes a real difference to a patient's QOL as they impact central vision.

What is the approximate diameter of the macula? 5-6mm or 2 from the centre AKA 4 disc diameters	**What are the parts of the macula?** • Foveola (impacts vision the most – as affects fixation) • Fovea (impacts vision the most – fixation, is 1DD) • Parafocea • Perifovea 
Which retinal layer contains the PRCs? Outer nuclear layer (ONL) – is a relatively large space where the PRCs are.	**What is the role of the blood-retinal barrier and how does its breakdown contribute to retinal disease?** Replaces blood with fluid 2x blood retinal barriers: 1. **INNER BRB** – formed by tight junctions and the vascular endothelium of retinal vessels (within the capillaries of the endothelial cells) Weakness in the tight junctions can cause leaky blood retinal barriers resulting in (vasculopathic causes): • Exudates • Oedema (such as in diabetic retinopathy) 2. **OUTER BRB** – formed by tight junctions in the RPE, purpose is to prevent fluid from creeping into sub-retinal fluid. When impaired, will get neovascularisation and central-serous chorioretinopathy.

CONDITION	SIGNS/SYMPTOMS + DIFFERENTIALS	WORKUP	MANAGEMENT	FOLLOW- UP
Cystoid Macular Oedema 	**Signs:** • Irregularity and blunting of the foveal light reflex, macular thickening with or without small intraretinal cysts in the foveal region • Vitreous cells • Optic nerve swelling • Dot haemorrhages **Symptoms:** Decreased vision **Etiology:** Post-op (any ocular surgery including laser photocoagulation and cryotherapy), DR, CRVO, BRVO, uveitis , RP, AMD, Behcet syndrome, sarcoidosis, RD, systemic HTN	Any of these can cause CMO (take history):: • Recent intraocular surgery? • Diabetes? HTN? • Uveitis or ocular inflammation? • Uveitis or ocular inflammation? • Night blindness or FOHx (like RP)? • Medicines including topical epinephrine, dipivefrin or prostaglandin analogs (especially in patients that have undergone cataract surgery) DFE useful **Optional:** OCT (can be seen without an OCT), FFA **(1)** Complete ocular examination, including gonioscopy to rule out the presence of retained lens fragments following cataract surgery and haptic malposition of implanted anterior chamber intraocular lens. **(2)** IVFA shows early leakage of perifoveal capillaries and late macular staining, classically in a petaloid or spoke-wheel pattern. Optic nerve head leakage is sometimes observed (Irvine-Gass syndrome). Fluorescein leakage does not occur in select cases of pseudo-CME (see above). **(3)** OCT can be utilized to document the presence of CME and demonstrate the efficacy of therapy. OCT outlines the loss of foveal contour resulting from enlarged cystic spaces within the retina and thickening of the macula. **(4)** Other diagnostic tests when indicated: Fasting blood sugar and hemoglobin A1c, ERG, others.	• Treat the underlying cause: DR, CRVO, BRVO, uveitis, RP, retinal vasculitis, etc BASICALLY anything that **causes breakdown in the blood retinal barrier** (BRB). • Consider discontinuing associated medications (consult with prescribing physician) • Postsurgical CMO is **not** an emergency condition (post cataract surgery) • Other forms of CMO may benefit from early treatment	• **4-6 weeks** to determine response to therapy (want to see swelling reduce) Postsurgical CME patients should be started on a topical NSAID and a topical steroid with follow up in 4-6 weeks. Other forms of CMO should be followed in a similar time frame to monitor response to initial therapy.
Central Serous Chorioretinopathy (CSCR)  Part of pachychoroid spectrum disease. TYPE A people	Fluid collects under the macula (the central part of the retina), causing blurry or distorted central vision, often described as a wavy appearance of straight lines. Can happen at macula (central) but ALSO peripherally.  "Smoke stack" appearance on ICGA, caused by breakdown of the outer blood retinal barrier. It looks like a mushroom or a cloud plume. **Etiology:** • **Idiopathic:** Most common in young adult to middle-aged men. In women, CSCR has an association with pregnancy. Increased incidence in patients with lupus. • **Increased endogenous cortisol:** This might explain a possible association with psychological or physiologic stress (type A personality). Rare cases exist with cortisol producing adrenal adenomas or Cushing syndrome. • **Exogenous cortisol:** Corticosteroid use, including nasal corticosteroid sprays and topical creams.	**History**: Ask about **corticosteroid use** (topical cream or nasal spray), sleep patterns, **stress**, **anxiety**, **pregnancy**, systemic lupus erythematosus (SLE), h.pylori (stomach gut ulcers). • Observe for ***Cushing syndrome*** (is hypercortisolism – extra stress hormone in the body) • **Amsler grid/VF** – to identify visual function deficit (will see scotoma and distortion of straight lines) • **DFE** to rule out concomitant CNV or other causes of RD • FFA (fluorescein fundus angiography) or ICGA (indocyanine green angiography) if diagnosis is uncertain or atypical ***NOTE:*** Different dyes used in FFA (blue light) vs ICGA (infrared viewing). ICGA goes deeper to see choroid. **Optional**: OCT with enhanced depth imaging (to see into choroid with increased contrast)	• **Laser photocoagulation** or **photodynamic therapy (PDT)** to accelerate visual recovery especially in chronic, recurrent, or bilateral cases requiring prompt visual restoration • Manage associated risk factors: Refer to GP, stop corticosteroids if possible • **Anti-VEGF for CNV** • **Emerging Tx**: Mineralocorticoid receptor antagonists e.g., **Eplerenone**  **Treatment** 1. Observation: Acute CSCR is usually self-limited with good visual prognosis. Worse prognosis for patients with recurrent disease, multiple areas of detachment, or prolonged course. 2. Stop corticosteroids, if possible, including topical skin and nasal spray preparations. 3. Laser photocoagulation: May accelerate visual recovery, but long-term benefit and safety are unclear. May increase risk of CNV formation. Given the CNV risk, use low laser intensity. Consider laser for: • Persistence of a serous detachment for several months. • Recurrence of the condition in an eye that sustained a permanent visual deficit from a previous episode. • Occurrence in the contralateral eye after a permanent visual deficit resulted from a previous episode. • Patient requires prompt restoration of vision (e.g., occupational necessity). 4. Photodynamic therapy (PDT): May be helpful for chronic CSCR. Half-dose PDT may be considered for rapid SRF resolution in patients with acute CSCR. 5. Mineralocorticoid receptor antagonists: Eplerenone and spironolactone have been associated with improved anatomic and visual outcomes in chronic CSCR. 6. If CNV develops, consider anti-VEGF therapy.	Observation every 6-8 weeks until resolved.

Non-Exudative Age-related Macular Degeneration (MOST COMMON)



Will see signs rather than symptoms first in early-intermediate stages.

Symptoms: Gradual loss of central vision, Amsler grid changes; may be asymptomatic.

Signs:
- Macular drusen, clumps of pigment in the outer retina, and RPE atrophy, almost always in both eyes.
- Other. Confluent retinal and choriocapillaris atrophy (e.g., geographic atrophy), dystrophic calcification.

History: Enquire about associated risk factors like FOHx, smoking, HTN, CVD, BMI, poor diet, lack of exercise

Mandatory:
- Amsler grid (in early-intermediate no defect, in later stages will complain of poor light/dark adaptation, in GA will get corresponding scotomas).
- DFE to stage and assess risk of conversion + look at periphery (may see reticular pigmentary degeneration or paving stone degeneration).
- Baseline and periodic FAF may be useful to monitor for progression, particularly geographic atrophy.
- IVFA or OCT when exudative AMD is suspected based on subjective or objective change in vision or examination findings. Drusen and RPE atrophy are often more visible on IVFA and FAF. OCTA is also a potentially useful diagnostic test as a noninvasive substitute for IVFA or if IVFA is inconclusive, especially with masquerade conditions such as pattern dystrophy or CSCR.

NOTE: For late AMD: Low vision care, Review driving status,

Optional: FAF, FFA, OCT, OCTA

Management can depend on the stage of AMD:

1. Early
2. Intermediate
3. Late

AS SEEN BELOW:

Review regularly
- **Early:** Review regularly or every 12 months
- **Intermediate:** 6 months - 12 months
- **Late:** 3-6 months

EARLY AMD, SMOKERS AND PATIENTS WITH RISK FACTORS

❌No smoking, supplements ***NOT*** recommended (can't do anything if patient has FOHx of AMD)

Diet and lifestyle changes recommended, aimed at reducing risk of progression or primary prevention.

✅Modify nutrient intake
- *Antioxidant rich foods*: carotenoids (lutein, zeaxanthin or beta-carotene), vitamin C (orange, carrots) and vitamin E in **green leafy vegetables**
- *Omega-3* (found in Ph outer segment membranes) in **cold water fish and nuts**

✅Modify other lifestyle factors
- Cigarette smoking
- Obesity, cardiovascular health (exercise)
- Some weak evidence regarding sunlight exposure

INTERMEDIATE AMD

Diet and lifestyle, Relatives, Use and amsler grid, Smoking cessation, Eye tests, Nutritional supplements = **DRUSEN**

Nutritional supplements: AREDS2 type supplements indications
- **Intermediate AMD** (MOST COMMON)
  - \+ Intermediate AMD one eye, fellow eye with early AMD
  - \+ Intermediate AMD OU
  - \+ Intermediate AMD one eye, fellow eye with advanced (GA or MNV)
- **Non-central GA:** One eye, fellow eye with GA or MNV
- **Non-central GA:** OU

AREDS 1 AND 2

Focus more on diet + lifestyle changes rather than supplementation.
- Antioxidants only established in intermediate to high-risk disease
- **No good evidence to support routine supplementation in healthy adults** to delay or prevent AMD i.e. family history of AMD or only small drusen
- Discuss more about habits + lifestyle
- No point initiating treatment
- Recommended replacement of betacarotene with zeaxanthin and lutein because of risk of lung cancer with betacarotene
- Addition of Omega-3 found no benefit to fish oil

LATE AMD (GEOGRAPHIC ATROPHY)
- Applicable strategies for iAMD
- Low vision care
- Review driving status (especially if centre involving GA), good conversation about this, record this conversation
- Amsler grid or other self- monitoring
- Other aspects of the patient's overall eye/vision needs (good refraction!)
- Review every 3-6 months
- **Emerging treatment:** Intravitreal complement inhibitors (**Syfovre** + **ISAVA**)

NEOVASCULAR

Signs: Drusen and SRF, ME or RPE detachment associated with CNV.

Symptoms: Variable onset of central visual loss, central or paracentral scotoma, metamorphopsia, photopsias in the central visual field. Subretinal or intraretinal blood. Retinal exudates, subretinal fibrosis (disciform scar). Retinal angiomatous proliferation (RAP) is an intraretinal variant of neovascular AMD and is characterized by focal telangiectatic retinal vessels with an adjacent superficial retinal hemorrhage and

Mandatory: DFE, OCT
Optional: OCTA, FFA, ICGA

Indications for referral

Key symptoms: Sudden onset central blur/vision loss/distortions, Blood, fluid lipids are also indications for referral.

(1) Slit lamp biomicroscopy with a 60-, 90-diopter, or fundus contact lens to detect CNV and associated exudation. Must examine both eyes.
(2) Perform IVFA or OCTA if CNV is suspected. IVFA is useful to confirm neovascular AMD size, type, and location. OCTA is useful as a noninvasive alternative to IVFA or when IVFA is inconclusive such as in pattern dystrophy or central CSCR. OCTA is also useful if there is an

15 years ago AMD was largely untreatable, until the rise of anti-VEGF therapy. BUT the problem with VEGF is that:
1. Anti VEGF blocks permeability but does not cause the MNV to regress so when treatment is stopped the MNV often becomes active again
2. Patients therefore need ongoing therapy which is a huge burden

Usual treatment regimen is typically monthly until MNV is inactive based on FFA and/or OCT
- PRN = As needed
- Fixed monthly
- Treat and extend (MOST common treatment regimen used in Australia) – warn these patients about signs and symptoms of

[Exudative] Age-Related Macular Degeneration



associated intraretinal edema and RPE detachment. Some neovascular AMD patients may present with VH.

allergy to fluorescein dye or in pregnancy when it is best to avoid dye use.

(3) OCT is helpful in determining retinal thickness, CNV thickness, location, and extent of ME, SRF, and RPE detachment. OCT is the primary modality for following response to treatment.

(4) ICGA may help delineate the borders of certain obscured occult CNV, particularly with subretinal blood or exudation. It also shows RAP and IPCV lesions better than IVFA.

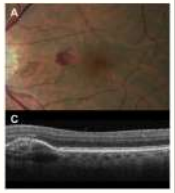
Exudative features of MNV	**Lesion components in addition to MNV**
• Leakage (intra-cellular oedema, intraretinal oedema) • Subretinal fluid • Lipid (hard exudates) • Subretinal hyperreflective exudative material (more obvious on OCT)	• RPE detachment • Haemorrhage • Fibrosis • RPE rip or tear

X3 subtypes of MNV (Macular neovascular membranes):

TYPE 1: (**Occult**) – Neovas **beneath the RPE**

Sub-RPE version, looking for pigment epithelial detachments + mixed internal reflectivity. Can see bleeding on fundoscopy + hard exudates. Neovas looks grey/green. Nothing going on in sub-retinal space, just bump beneath the RPE. **STAGE 1 NOT THRU RPE.**

Ill-defined, stippled, flat, or elevated subtle late leakage on IVFA and located in the sub-RPE location by OCT.



TYPE 2: (**Classic**) – Can see a break in the RPE + subretinal fluid. BVs gone through RPE and in sub-retinal space. **STAGE 2 IS JUST CHOROIDAL NEOVAS.**

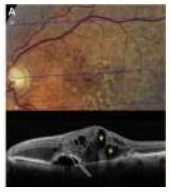
Early-phase IVFA demonstrates a well-delineated area of lacy hyperfluorescence with prominent leakage in later phases and located in the sub-neurosensory retinal location by OCT.



TYPE 3: (**Retinal angiomatous proliferation**)

Growth of new vessels in the retina growing down to meet new vessels in the choroid. More intra-retinal disturbance (AKA intraretinal oedema – looking for fluid at any level). Gone through RPE, retinal vessels from top and choroidal vessels from below join together.

Focal intraretinal hyperfluorescence on IVFA and ICGA. High-speed ICGA is particularly sensitive and may show characteristic "hair pin loop" with retinal feeder and draining vessels. OCTA shows this focal form of neovascularization to be intraretinal.



It is important to consider other aspects of a patient's overall eye and vision needs (in collaboration with an ophthalmologist).

- LV care, reviewing driving status
- Amsler grid or other self monitoring to help detect progression of disease.

Management options

- ★ **Ranibizumab** 0.5 mg (Lucentis)
- ★ **Aflibercept** 2 mg (Eylea)
- ★ **Bevacizumab** (Avastin) "off label"
- ★ **Brolucizumab** (Beovu) – extended dosing (3 monthly)
- ★ **Faricimab** (Vabysmo) – extended dosing (3 monthly)

A lot of these drugs (when compared), none have shown a significant improvement ; they are non-inferior.

- Guided by RCTs; however, most RCTs have not been designed for head-to-head comparison
- Efficacy: No consistent meaningful difference has been demonstrated
- Side effects: Safety is so good with existing drugs that new drugs need to be just as safe
- Systemic side effects: meta-analyses suggest little difference
- Treatment frequency: No difference between Eylea and Lucentis when used monthly or treat and extend

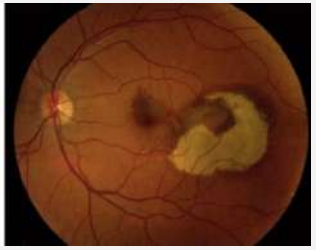
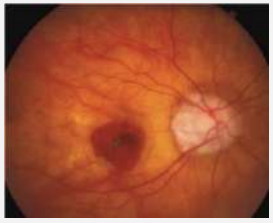
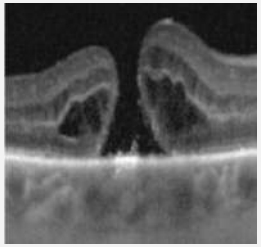
Ultimately, drug of choice comes down to individual doctor preference.

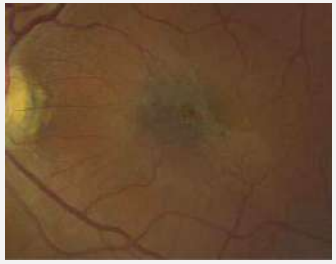
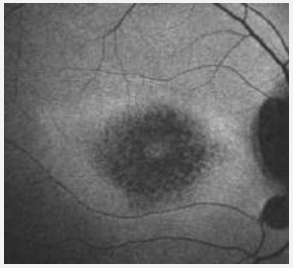
endophthalmitis and RD.

REFERRAL PERIOD

Refer to ophthalmologist (from Optometry Australia guideline)

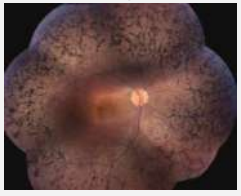
- Within 1 week - if symptomatic
- Within 2 week - if asymptomatic

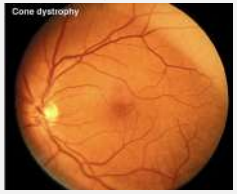
[Aneurysmal type 1 neovascularisation] AKA Idiopathic polypoidal choroidal vasculopathy (IPCV) 	Bloody and fluid type of neovascularisation, most chronic version of pachychoroid disease (long-lasting standing phase 4) This form of neovascularisation is more confined + focal than neovascular AMD (which is generally more devastating to vision). **Symptoms:** Decreased central vision; may be sudden or gradual. **Signs:** Subretinal red-orange, polyp-like lesions of the choroidal vasculature. Can be macular (more symptomatic) or peripapillary. **Other:** Bilateral subretinal and/or sub-RPE blood, VH, circinate subretinal exudates, subretinal fibrosis (disciform scar), SRF, atypical CNV, and multiple serous PEDs. **Risk Factors:** More common in females, individuals of African or Asian descent, and in patients with HTN. Can occur at a younger age compared to neovascular AMD, but usually without significant drusen or geographic atrophy.	• DFE, ICGA (deeper + uses infrared so we see choroid better) – **GOLD STANDARD** for diagnosing PCV • **Optional:** OCT, OCTA, FFA **(1)** Slit lamp biomicroscopy with a 60-, 90-diopter, or fundus contact lens to detect signs of exudation. **(2)** ICGA is the gold standard for diagnosis. ICGA is used to confirm the presence of a branching network of vessels arising from the inner choroidal circulation with terminal aneurysmal dilations (popcorn lesions). Unlike occult CNV, the IPCV lesions do not stain late unless active leakage is present. **(3)** IVFA is performed to evaluate for other causes of CNV. **(4)** OCT is used to assess for ME, SRF, and PEDs and can detect polyps in some eyes.	• Typically just observe • PDT, Anti-VEGF mono- or combination therapy for those that are visually threatening ***NOTE:*** EVEREST-II and PLANET trials supports anti-VEGF mono or combination therapy Refer within a week for anti-VEGF (especially for new onset PCV) or for combination therapy.	Review every **1-2 months** with repeat OCT, FFA and ICGA as needed
Pathologic Myopia  **Hallmark =** Posterior staphyloma [PIC] Coin shaped haemorrhage	More confined, focal, coin-shaped haemorrhage. Very common and is often associated with glaucoma. X3 different scales of pathologic myopia: ***1.*** Can present with **neovascular** development, tesselated fundus, lacquer cracks that cause breaks in Bruch's membrane that allows the CNV to grow. ***2.*** **Tractional myopic maculopathy** causing macula hole from pulling ***3.*** **Atrophic maculopathy** from focal or diffuse chorioretinal atrophy occurring in myopia	• Baseline glaucoma assessment • DFE of the macula and peripheral retina • **Optional:** OCT (see if there is foveal schesis or full thickness macular hole), FFA	• Treatment depends on the specific presentation • RD prophylaxis IF neovascular = anti-VEGF within a week IF RD/retinal breaks /lattice = Prophylaxis	Review every **6-12 months** with repeat DFE
Vitreomacular adhesion, traction and macular hole  [PIC] looks tractional	Adhesion is OK – normal event Traction is less OK – getting some tugging Hole – can be predisposed to from traction **Symptoms:** Variable decreased vision (typically around 20/200 level for a full-thickness hole, better for a partial-thickness hole), metamorphopsia, or central scotoma. Three times more likely in women; usually occurs in sixth to eighth decade. 10% bilateral. **Signs:** A full-thickness macular hole appears as a round, red spot in the center of the macula, usually from one-third to two-thirds of a disc diameter in size; may be surrounded by a gray halo/cuff of SRF.	**History**: • **Photopsia, metamorphopsia** – experiencing symptoms of the traction or visual loss/distortion from the hole/traction • POHx trauma, eye surgery, myopia • Sun-gazing, poppers (nitric oxide drugs to get high) – can burn a hole and create similar presentation **DFE**: Check for PVD and rule out peripheral retinal breaks **Watzke-Allen test** • Requires very thin beam on slit lamp • Ask patient to look directly at it with one eye occluded • If the line of the beam appears broken to the patient (i.e gap in beam), confirms a present full thickness macular hole **OCT** **Optional**: FFA	***NOTE:*** 50% of cases resolve spontaneously It is okay to watch, but if progressing significantly + lots of risk factors (maybe hole in fellow eye) **Pars plana vitrectomy** with membrane peel within 6 months of onset (GOLD STANDARD)→ *Removes vitreous gel from posterior eye* Can consider: **Ocriplasmin** for symptomatic VMA, VMT, and macular hole → *Enzyme medication injected into eye (recombinant version of plasmin → breaks down protein matrix responsible for abnormal adhesions between vitreous gel and retina)*	• Depends on Sx and Tx administered • Patients with high myopia 6 months • Advise Px of risk of RD and risk to fellow eye

	Vitreomacular traction (VMT) demonstrates loss of the normal foveolar depression and often a yellow spot or ring in the center of the macula.			
Epiretinal membrane (ERM) *SO COMMON!* 	Avascular,semi-translucent, fibrocellular membranes that form on the inner surface of the retina. Most often caused by normal ageing process where the vitreous jelly inside the eye separates from the retina, leading to scar tissue formation. They can also develop secondary to other eye conditions such as diabetes, retinal tears or detachments, inflammation, trauma or previous eye surgery. **Etiology:** ● PVD ● Retinal break, RRD ● Idiopathic ● After retinal cryotherapy or photocoagulation. ● After intraocular surgery, trauma, or vitreous hemorrhage. ● Uveitis ● Other retinal vascular disease (diabetic retinopathy, vein occlusion, etc.).	**History**: POHx eye surgery or other conditions including diabetes (in this case would be secondary not primary) **DFE**: Check for PVD (MOST COMMON CAUSE), peripheral retina, uveitis and other retinal vascular disease ● ERM may have developed secondary to other conditions so important to check peripheral retina. **Optional**: OCT (to help categorise + stage the ERM)	● Treat the underlying disorder ● Surgical peeling of the membrane	**Not an emergent condition,** annual observation or referral Mostly chronic + stable, will not progress.
Chloroquine/ hydroxychloroquine toxicity (BULL'S EYE MACULOPATHY)  ***NOTE:*** Hydroxychloroquine (Plaquenil) is a medication commonly used in the treatment of various rheumatological diseases including lupus and rheumatoid arthritis. Both chloroquine and hydroxychloroquine can result in irreversible macular toxicity and classically disrupts the outer retina parafoveally. When macular toxicity is evident clinically and on OCT, advanced disease is already present. Both OCT and multi-focal ERG are recommended screenings to rule out toxicity.	Looking for characteristic bull's eye pattern and risk factors associated with it. **Symptoms:** Often asymptomatic early, then decreased vision, abnormal color vision, reduced dark adaptation. **Signs:** Bull's eye macula (ring of depigmentation surrounded by a ring of increased pigmentation), loss of foveal reflex.	**Enquire about risk factors:** Daily dosage, weight, duration of use, total cumulative dose ● **Chloroquine** >300g, >2.3mg/kg ● **Hydroxychloroquine** >1000g, >5mg/kg ● Duration of use > 5years ● Renal disease, tamoxifen use, comorbid macular disease Comprehensive baseline evaluation within the first year of starting the medication including: ● BCVA ● SLE (slit lamp exam) ● DFE ● CFP (colour fundus photograph) ● 10-2 standard automated perimetry (SAP) ● One of mfERG, FAF or SD-OCT	**Discontinue medication** in conjunction with the prescribing physician	**Annual screening 5 years** after medication use or sooner for high risk patients

Crystalline retinopathy 	Deposits of reflective crystals form in the retina, particularly the macula.	• **PMHx**: IV drug use, cardiovascular disease, oral tanning agents, anaesthesia in patients with renal failure (increases risk for developing retinopathy) • Examine Px for iv drug abuse signs • DFE **Optional:** OCT, FFA, OCTA	**Discontinue associated drug use** Consider PRP or anti-VEGF for peripheral non-perfusion or NV	Follow up depends on underlying aetiology
Optic disc pit 	Rare congenital excavation of the optic disc, resulting from malformation during development of the eye. Can also be caused by normal tension glaucoma.	**Mandatory**: VFs, IOP, DFE **Optional**: OCT, FFA	• **No Tx required for isolated cases** • Associated serous macular detachment may be treated by vitrectomy and intravitreal gas, or laser photocoagulation at the temporal disc margin There is no clear consensus regarding treatment of optic pit maculopathy. There is no evidence for anti-VEGF therapy for treatment of optic pits. Conservative management would not be appropriate given the vision of 20/200 and the associated risk of amblyopia in the pediatric population. The most commonly used intervention for the treatment of optic pit maculopathy is pars plana vitrectomy (PPV).	Follow up annually, with Amsler grid
Solar retinopathy 	Damage to the eye's retina, particularly the macula, that comes from prolonged exposure to bright light or solar radiation. **Signs:** Acute findings include a yellow-white spot in the fovea with or without surrounding granular gray pigmentation. Classic late finding is a red, sharply demarcated lesion in the fovea. **Etiology:** Unprotected solar eclipse viewing, sungazing (e.g., related to religious rituals, psychiatric illnesses, hallucinogenic drugs), sunbathing, vocational exposure (e.g., aviation, military service, astronomy, arc welding).	• Enquire about former eclipse viewing, sungazing, work exposure like welding , recreational drug use, photosensitizing medications • **Mandatory**: Amsler grid testing, DFE • **Optional**: FFA, OCT	Observation and patient counselling	

Retinal Dystrophies

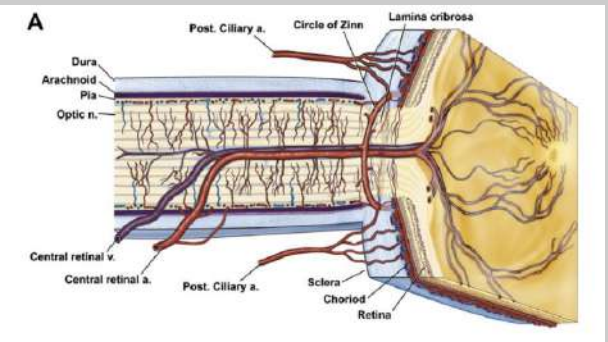
CONDITION	SIGNS/SYMPTOMS	WORKUP	MANAGEMENT	FOLLOW- UP
Retinitis pigmentosa 	**Symptoms:** Decreased night vision (often night blindness) and loss of peripheral vision. Decrease in central vision can occur early or late in the disease process depending on whether rod or cone involvement is predominant. The same is true for color vision. **Signs:** Classically, vitreous cells (most consistent sign), clumps of pigment dispersed throughout the peripheral retina in a perivascular pattern, often assuming a "bone spicule" arrangement (though bone spicules may be absent), areas of RPE	VA, VFs, electrophysiology (absent a & b wave)	• Some evidence to support high dose Vitamin A supplementation (15,000 IU/day) and lutein (12mg/day) to slow midperipheral field loss in nonsmokers • Where applicable: + Genetic counselling + Low vision, retinal implants + Gene therapy (Luxturna for RPE65)	Monitor liver function and vitA if vitA Tx is initiated

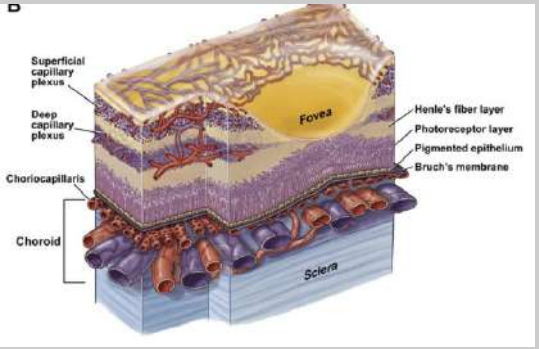
	depigmentation or atrophy, arteriolar narrowing, and, later, waxy optic disc pallor. Progressive visual field loss, usually a ring scotoma, which progresses to a small central field. ERG usually moderately to markedly reduced.			
Cone dystrophies 	**Symptoms:** Slowly progressive visual loss, photophobia, and poor color vision, with onset during the first 3 decades of life. Vision is worse in bright than dim light. **Signs:** • **Early:** Essentially normal fundus examination, even with poor visual acuity. Abnormal cone function on ERG (e.g., a reduced single-flash photopic response and a reduced flicker response). • **Late:** Bull's eye macular appearance or central geographic atrophy of the RPE and choriocapillaris.	• FOHx retinal dystrophies, blindness • PMHx chloroquine/HCQ use • VA, Appropriate colour vision testing, VFs • DFE • Full field ERG • **Optional:** OCT, FAF	✅Pallative • Tinted lenses • Miotic drops for photophobia • Genetic counselling • Low vision	Yearly
Stargardt disease (Fundus Flavimaculatus) 	**Symptoms:** Decreased vision in childhood or young adulthood. Early in the disease, the decrease in vision is often out of proportion to the clinical ophthalmoscopic appearance; therefore, care must be taken not to label the child a malingerer. **Signs:** • A relatively normal-appearing fundus except for slight granularity in the foveola. • Yellow or yellow-white, fleck-like deposits at the level of the RPE, often in a pisciform (fish-tail) configuration. • Atrophic macular degeneration: May have a bull's eye appearance as a result of atrophy of the RPE around a normal central core of RPE, a "beaten-metal" appearance, pigment clumping, or marked geographic atrophy.	• Enquire about age of onset, medications, FOHx retinal dystrophies, blindness • VA, DFE, VFs • **Optional:** FFA, FAF, ERG and EOG, consider genetic testing	✅Palliative, as with cone dystrophy ❌Avoid VitA supplementation	
Best disease (Vitelliform macular dystrophy) 	**Symptoms:** Decreased vision, scotoma, metamorphopsia, or asymptomatic. **Signs:** Yellow, round, subretinal lesion(s) likened to an egg yolk (lipofuscin) or in some cases to a pseudohypopyon. Typically bilateral and located in the fovea, measuring approximately one to two disc areas in size. Likely present at birth, though may not be detected until examination is performed. Ten percent of lesions are multiple and extrafoveal. ERG is typically normal, while EOG is abnormal, showing severe loss of the light response.	• Enquire about FOHx • VA, DFE, Amsler/VFs • **Optional:** EOG highly specific, FFA, OCT for MNV, genetic testing	Consider prompt Tx for CNV – anti-VEGF, PDT, focal laser	Annual DFE, Amsler grid monitoring

LECTURE 4: Retinal Vascular Disease and Glaucoma

Retinal Vascular Diseases

REFRESHER





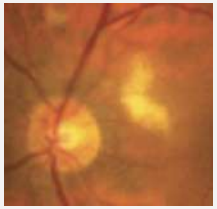
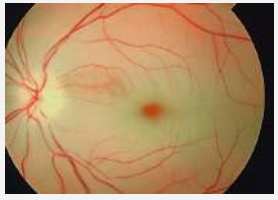
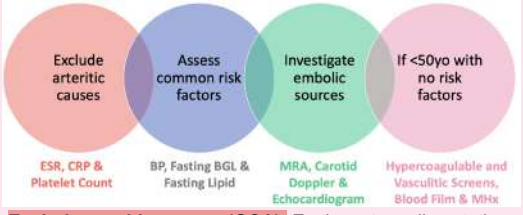
Why is the fovea considered avascular?	***Which two circulations supply the retina?***
• It is a natural bending of all the inner retinal layers • Minimise light scattering for best VA (given there is no vessels in the central fovea) • In dye studies (fluorescein angiography, FAF) will see an absence of vessels, including the small capillaries centrally. • FAF, FFA, OCTA shows absence of vessels at fovea	• Long and short ciliary artery → Choroid • Inner retina: Retinal arteries (supplies inner 2/3) • Outer retina: Choroidal vessel circulation (supplies posterior 1/3) The choroid supplies the outer one-third of the retina, while the central retinal artery supplies the inner two-thirds of the retina. There are retinal vessels arranged in different layers, including a superficial capillary plexus and a deep capillary plexus, which lies on either side of the inner nuclear layer.

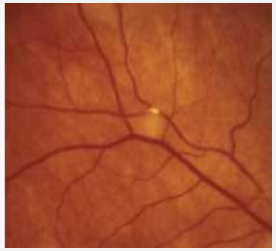
Retinal Vascular Occlusions

Can have both arteriolar (inflow disruption) VS venous (outflow disruption) occlusions. They differ in their pathophysiology and systemic implications. Often studied together and can be mixed up.

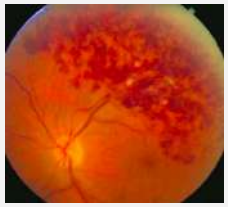
Common feature is that they are:

- Usually >50 years old
- Carry associated CV risk factors

CONDITION	SIGNS/SYMPTOMS + DIFFERENTIALS	WORKUP	MANAGEMENT	FOLLOW- UP
Cotton Wool Spot 	**Symptoms:** Visual acuity usually normal. Often asymptomatic. **Signs:** Localized whitening in the superficial retinal NFL with fluffy appearance to margins ***NOTE:*** The presence of even a single cotton-wool spot (CWS) is not normal. In a patient without diabetes mellitus, acute changes in blood pressure (most commonly hypertension), or a retinal vein occlusion, a workup for an underlying systemic condition should be performed. **Etiology:** Thought to be an acute obstruction of a precapillary retinal arteriole causing blockage of axoplasmic flow and subsequent buildup of axoplasmic debris in the NFL. Common causes include HTN, DM, retinal vein occlusion, retinal emboli, GCA	**(1)** History: Diabetes or HTN? Prior ocular or periocular radiation? GCA symptoms in appropriate age group? Symptoms of collagen vascular disease including joint pain, rashes, etc.? HIV risk factors? Hematologic abnormalities? **(2)** Complete ocular examination, including dilated retinal examination with a slit lamp and a 60- or 90-diopter lens and indirect ophthalmoscopy. Look for concurrent hemorrhages, vascular occlusion, vasculitis, hard exudates. **(3)** Check blood pressure. **(4)** Check fasting blood sugar and hemoglobin A1c. **(5)** Consider erythrocyte sedimentation rate (ESR), C-reactive protein (CRP), and platelets if GCA suspected. **(6)** Consider blood and urine cultures, chest x-ray, carotid and orbital Doppler examination, chest computed tomography (CT), and echocardiography if emboli are suspected. **(7)** Consider HIV testing. **(8)** Fluorescein angiography is generally not helpful for an isolated CWS without concomitant pathology. IVFA reveals areas of capillary nonperfusion adjacent to CWS location.	Identify and treat underlying etiology.	Depends on underlying etiology. If concern for infectious process, serial dilated examinations are recommended. CWSs typically fade in 5 to 7 weeks but can remain longer if associated with diabetic retinopathy.
Central Retinal Artery Occlusion   ***CHERRY RED SPOT!***	Has a very classical kind of appearance with a cherry red spot and superficial whitening/opacification of the retina itself. Central retinal artery occlusion (CRAO) is monocular because the central retinal artery is the sole supplier of blood to the inner layers of the retina for one eye. Most common causes are emboli (cholesterol, calcium and platelet-fibrin) or thrombi. Other causes include: • GCA • SLE • Thrombosis • Trauma **Symptoms:** Unilateral, painless, acute vision loss (counting fingers to light perception in 94% of eyes) occurring over seconds; may have a history of transient visual loss (amaurosis fugax). **Signs:** Superficial opacification or whitening of the retina in the posterior pole and a cherry-red spot in the center of the macula (may be subtle). **Other:** • Marked RAPD. • Narrowed retinal arterioles; boxcarring or segmentation of the blood column in the arterioles. • Occasionally, retinal arteriolar emboli or cilioretinal artery sparing of the foveola is evident. • If visual acuity is light perception or worse, strongly suspect ophthalmic artery occlusion.	Should be treated as an acute stroke. • Enquire about amaurosis fugax (phrase as "Any history of transient vision loss, any recent loss of vision last seconds or more, in one or both eyes?" • **Pupils: RAPD** • VA: Lowered • DFE • **Optional:** OCT, ERG, FFA POSSIBLE work up with GP (aim is to **reveal underlying aetiology**), liaise with GP to find the underlying cause.  • **Exclude arteritic causes (GCA):** Erythrocyte sedimentation rate (ESR - will be elevated in GCA), CRP & **Platelet** Count + Want to exclude this quickly as it is an ocular emergency. • **Assess common risk factors:** BP, Fasting BGL & Fasting Lipids • **Investigate embolic sources:** MRA (magnetic resonance angiography), Carotid Doppler & Echocardiogram (heart ultrasound) • **If <50yo with no risk factors:** Hypercoagulable and Vasculitic Screens, Blood Film & MHx	**REFER IMMEDIATELY** TO ED FOR STROKE MANAGEMENT IF **GCA = OCULAR EMERGENCY** Look for underlying causes with GP (stoke centre) Anecdotal reports of improvement if instituted within **90-120mins** of the occlusive event *(Not standard care)* ★ Ocular massage ★ Anterior chamber paracentesis ★ **Acetazolamide** 500mg iv or two 250mg tablets po ★ Topical **betablocker** daily or bid	As directed by managing internist/neurologist. Repeat DFE in 1-4 weeks checking for NVI, NVD, NVA,NVE, which develops in up to 20% (a fifth) of causes 4 weeks after onset.

Branch Retinal Artery Occlusion 	**Signs:** • Superficial opacification or whitening along the distribution of a branch retinal artery. The affected retina becomes edematous. • Narrowed branch retinal artery • Boxcarring (here blood in a retinal artery appears segmented, like boxcars on a train, with visible columns of blood alternating with empty sections)  • Emboli (cholesterol emboli appear as bright, reflective crystals usually at vessel bifurcation, CWS may appear in the involved area). **Symptoms:** Unilateral, painless, abrupt change in vision, usually partial visual field loss; may have a history of transient visual loss (amaurosis fugax).	**SAME AS ABOVE BUT pupils + ERG are less helpful** (as only branch involved or sectorial, less widespread, not as helpful)	• As with CRAO • No proven therapy • Immediately treat any underlying medical problem **1.** The AAO 2018 guidelines suggest that all these patients should be sent immediately to an emergency department, preferably affiliated with a stroke center, for evaluation and workup. **2.** No ocular therapy of proven value is available. **3.** Treat any underlying medical problem.	Patients need immediate evaluation to treat any underlying disorders (especially GCA). Follow up every 3-6 months, NV is rare
Central Retinal Vein Occlusion  	**Signs:** • Diffuse retinal hemorrhages in **all four quadrants** of the retina; dilated, tortuous retinal veins. • CWS • Disc edema and hemorrhages • Macular edema (ME) • Optociliary collateral vessels on the disc (later finding); NVD, NVI, NVA, and NVE. **Symptoms:** Painless loss of vision, usually unilateral. **Etiology:** • Atherosclerosis of the adjacent central retinal artery: The artery compresses the central retinal vein in the region of the lamina cribrosa, secondarily inducing thrombosis in the vein lumen. • HTN (MOST COMMON) • Glaucoma (MOST COMMON) • Optic disc drusen • Drugs: Oral contraceptives, diuretics, and others. Types: **1.** **Ischemic CRVO:** Vision typically worse (<20/200) with RAPD and visual field defects. Extensive retinal hemorrhage, CWSs, venous tortuosity, and widespread capillary nonperfusion on IVFA (often >10 disc diameters). ERG shows decreased b-wave amplitude. Higher risk of neovascularization. **2.** **Nonischemic CRVO:** Vision often better than 20/200, mild or no RAPD, mild fundus changes. Lower risk of neovascularization.	**Ocular Workup:** **(1)** Complete ocular examination, including IOP measurement, careful slit lamp examination and gonioscopy to rule out NVI and NVA (both of which are best observed before dilation), and dilated fundus examination. **(2)** IVFA: Risk of neovascularization proportional to degree of capillary nonperfusion. **(3)** OCT: Used to help detect presence and extent of ME as well as to monitor response to therapy. **(4)** If the diagnosis is uncertain, oculopneumo-plethysmography or ophthalmodynamometry may help to distinguish CRVO from carotid disease (but are infrequently performed). Ophthalmic artery pressure is low in carotid disease but is normal to increased in CRVO. VA, pupils, raised IOP (more relevant, raised IOP increases risk of CRVO) , SLE and gonio to rule out NVI and NVA prior to DFE DFE to distinguish **ischemic** (non-perfused) vs **nonischemic** (perfused) **Systemic Workup:** Enquire about risk factors: • PMHx (esp HTN, diabetes) • Medications (oral contraceptives, diuretics) • Eye diseases **Optional**: ERG, FFA, OCT 	• Discontinue related medications if possible - plan B, contraceptives, diuretics, diabetic medications, HTN medications • **Reduce IOP** if increased • Tx underlying medical conditions • Prompt PRP (panretinal photocoagulation) for any new vessels • **Aspirin** 81 to 325 mg po **daily** often recommended but no clinical trials have demonstrated efficacy and it may increase risk of haemorrhage **<u>Management of related macular oedema:</u>** • **Intravitreal anti-VEGF** or corticosteroids • May be combined with **laser photocoagulation**	• Every month initially, gradual interval taper based on vision, presence/absence of macular oedema and response to treatment. • Advise patient of 8-10% risk of fellow eye i/CRVO

Branch Retinal Vein Occlusion



Signs: Superficial hemorrhages in a sector of the retina along a retinal vein. The hemorrhages usually do not cross the horizontal raphe (midline).

CWSs, retinal edema, a dilated and tortuous retinal vein, narrowing and sheathing of the adjacent artery, retinal neovascularization, VH.

Symptoms: Blind spot in the visual field or loss of vision, usually unilateral.

Etiology: Disease of the adjacent arterial wall (usually secondary to HTN, arteriosclerosis, or diabetes) compresses the venous wall at a crossing point.

AS WITH CRVO, FFA is obtained after the haemorrhages clear or sooner if NV is suspected.

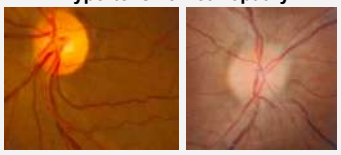
1. History: Systemic disease, particularly HTN or diabetes?
2. Complete ocular examination, including dilated retinal examination with indirect ophthalmoscopy to look for retinal neovascularization and ME.
3. OCT: Used to help detect presence and extent of ME as well as monitor response to therapy.
4. Check blood pressure.
5. Blood tests: Fasting blood sugar and hemoglobin A1c, lipid profile, CBC with differential and platelets, PT/PTT. If clinically indicated, consider a more comprehensive workup.
6. Medical examination: Performed by an internist to check for cardiovascular disease.
7. An IVFA is obtained after the hemorrhages clear or sooner if neovascularization is suspected.

SAME AS ABOVE + may be combined with focal retinal laser photocoagulation (no PRP or widespread treatment).

SAME AS ABOVE

In general, every month initially, with gradual interval taper based on vision, presence of ME, and treatment response. At each visit, the patient should be checked for neovascularization and ME.

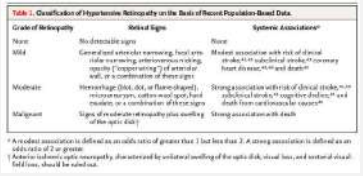
Hypertensive Retinopathy



VERY COMMON!

Risk profile:
- Age (>60)
- Smoking
- Overweight
- Diet: High salt / saturated fat
- Lack of physical exercise
- Alcohol (>1-2/day)
- Stress
- Family history
- Chronic kidney disease
- Pregnancy

Symptoms:
- None / "feel fine"(Majority)
- Headaches
  - Early morning
  - "like a 'pressure' in the head / body"
- Shortness of breath / tired easily
- Nose bleeds
- Dizzy spells
- Chest pains
- Nausea, vomiting
- Fits / black outs
- Vision changes



- Enquire about risk factors: PMHx HTN, diabetes, pregnancy (pre-eclampsia/eclampsia)
- Check blood pressure (consider doing in-house)
- DFE to grade severity (chronic, acute, **mild, moderate, malignant**)

- Refer to GP
- Refer urgently to ED if symptomatic, malignant signs, or if BP ≥180/110 (severe HTN)
- Control the HTN, as per the internist.

- Review every 2-3 months
- Review 6-12 monthly thereafter

Signs:
- **Chronic HTN:** Arteriovenous crossing changes ("AV nicking"), retinal arteriolar sclerosis ("copper" or "silver" wiring), CWSs, flame-shaped hemorrhages, arterial macroaneurysms, central or branch occlusion of an artery or vein. Rarely, neovascular complications can develop.
- **Acute ("malignant") HTN or accelerated HTN:** Hard exudates often in a "macular star" configuration, retinal edema, CWSs, flame-shaped hemorrhages, optic nerve head edema. Rarely serous RD or VH. Areas of focal chorioretinal atrophy (from previous choroidal infarcts [Elschnig spots]) are a sign of past episodes of acute HTN.

Symptoms: Usually asymptomatic, although may have decreased vision.

HTN defined according to WHO – condition in which the BVs have persistently raised blood pressure. According to Joint National Committee report, HTN is diagnosed if BP is ≥140/90 mmHg.

It is an incredibly common condition, and can increase risk of:
- Heart, brain, kidney and other disease (end stage organ disease)
- Premature death

MILD hypertension = Only when chronic signs are present	**MODERATE hypertension** = When there are signs of acute changes	**MALIGNANT hypertension** =
• Arteriolar narrowing • AV nicking • Copper/silver wiring	• Retinal haems(blot, dot, flame) • Microaneuryms • Cotton wool spots • Hard exudates	Signs of hypertensive retinopathy AND elevated optic disc oedema (swelling)

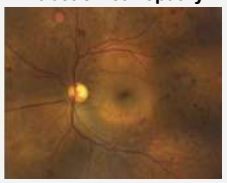
Table 2.1 Classification of clinic blood pressure levels in adults

Diagnostic category*	Systolic (mmHg)		Diastolic (mmHg)
Optimal	<120	and	<80
Normal	120–129	and/or	80–84
High-normal	130–139	and/or	85–89
Grade 1 (mild) hypertension	140–159	and/or	90–99
Grade 2 (moderate) hypertension	160–179	and/or	100–109
Grade 3 (severe) hypertension	≥180	and/or	≥110
Isolated systolic hypertension	>140	and	<90

HTN Classification:

1. **Hypertensive urgency:** Severe BP elevations ≥180/110 that are not immediately life threatening but the patient does have associated new symptoms. Needs treatment with oral antihypertensives within 24-72 hrs
   - AKA New symptoms but no organ damage
2. **Hypertensive <u>emergency:</u>** BP is very high (>220/140) AND new onset end organ damage or dysfunction (optic disc oedema, heart failure, pulmonary oedema, myocardial infarct, or acute focal neurology) is present. Needs hospitalisation and parenteral antihypertensive drug treatment
   - AKA new symptoms AND onset of organ damage
   - **Ask for:** Chest pain, difficult breathing, headaches

Diabetic Retinopathy



Can be difficult to distinguish between HTN retinopathy and DR (as can co-exist).

Classification Scheme:

	Classification	Findings Observable on Dilated Ophthalmoscopy*
Diabetic retinopathy	No apparent DR	No abnormalities
	Mild NPDR	Microaneurysms only
	Moderate NPDR	Microaneurysms and other signs (e.g., dot and blot hemorrhages, hard exudates, cotton wool spots), but less than severe nonproliferative DR
	Severe NPDR	Moderate nonproliferative DR with any of the following: intraretinal hemorrhages (≥20 in each quadrant); definite venous beading (in 2 quadrants); intraretinal microvascular abnormalities (in 1 quadrant); and no signs of proliferative retinopathy
	Proliferative DR	Severe nonproliferative DR and 1 or more of the following: neovascularization, vitreous/preretinal hemorrhage
Diabetic macular oedema	No DMO	No retinal thickening or hard exudates† in the macula
	Non-center-involving DMO	Retinal thickening in the macula that does not involve the central subfield zone that is 1 mm in diameter
	Center-involving DMO	Retinal thickening in the macula that does involve the central subfield zone that is 1 mm in diameter

- Early detection of disease & correct grading is important to prevent advanced disease and treatment side effects.
- Enquire about risk factors: Control of blood sugar, lipids and blood pressure, duration, other complications (like neuropathy or associated kidney disease), adherence to care, first diagnosis of DM.
- SLE and gonio to rule out NV prior to DFE
- DFE to subtype severity and search for NV and DMO

(1) Check fasting blood sugar, hemoglobin A1c, and lipid panel.
(2) Check blood pressure.
(3) Consider IVFA to determine areas of perfusion abnormalities, foveal ischemia, microaneurysms, and subclinical neovascularization, especially if considering focal macular laser therapy.
(4) Consider OCT to evaluate for presence and extent of DME. OCT angiography (OCTA) can be useful to check for presence of significant central macular ischemia.

Optional: FFA, OCT, OCTA

DR AND PREGNANCY

Pregnancy in females with diabetes may accelerate the development and progression of DR. Pregnant women should:

1. Have a DFE **prior to conception** if planning pregnancy
2. Have a DFE in the first trimester
   - Again at **28 weeks** (third trimester) if first trimester results are normal.
   - Again at **16-20 weeks** (second trimester) if mild-moderate DR is found, consider routine referral to an ophthalmologist, *Tx unlikely (NOT URGENT)*
   - Monthly thereafter if Severe NPDR or PDR is found. Refer PDR for PRP
3. Have a DFE **6-12 weeks post-partum** (~ 3 months after)

NOTE: Women with **gestational diabetes** during the course of pregnancy **do not need ophthalmic surveillance** during or after pregnancy, unless diabetes persists.

Tx:

- PRP:fpr severe NPDR (esp. T2DM) significant reduced the risk of severe vision loss
- Pars plana vitrectomy (within 3 months) : include severe PDR
- Medical therapies: ACE inhibitors, Protein kinase C inhibitors (slow progression of DR), intravitreal depot steroid therapy
-

Classification	One year risk of PDR	Review / referral timeframe*	Ophthamologist referral
No apparent DR, no DMO	N/A	1–2 yrs	Not required
Mild NPDR	1-5%	6–9 mos	Not required
Moderate NPDR	12-27%	4 mos (3–6 mos)	Required
Severe NPDR	~52%	4 weeks (<3 mos)	Required
Proliferative DR		Within 1 week (<1 mo)	Required
Non-center-involving DMO		3 mos	Required
Center-involving DMO		1 mo	Required

NOTE: Follow up varies with pregnancy.

GENERAL MANAGEMENT OF DR (ESPECIALLY MILD + MODERATE NPDR):
Indications for **Fenofibrate** now include the reduction in the progression of DR in patients with T2DM and existing DR (kind of like the supplements for AMD) – especially those with **mild + moderate NPDR.**

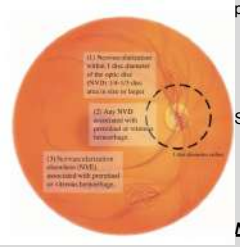
NOTE: No high quality data on the effect of fenofibrate in severe NPDR, PDR or DMO, patients less than 40. Using **Lipidil.**

Reasonable to continue or commence fenofibrate in these populations with caution.

MANAGEMENT OF DMO (DIABETIC MACULAR OEDEMA): Refer to ophthalmologist within 4 weeks especially if VA is reduced (6/9.5 or worse). Will typically receive:

- Intravitreal anti-VEGF
- Intravitreal corticosteroids
- Focal or grid laser

MANAGEMENT OF PDR: Will use **panretinal photocoagulation laser** (PRP - shrinks abnormal vessels) can be +/- with intravitreal Anti-VEGF and or Pars plana vitrectomy. Indicated IF:



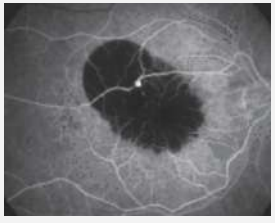
- NVD > ¼ disc area
- Any NVD with pre-retinal or overlying VH (vitreous haemorrhage)
- NVE > ½ disc area with preretinal or VH
- Any NVI or NVA

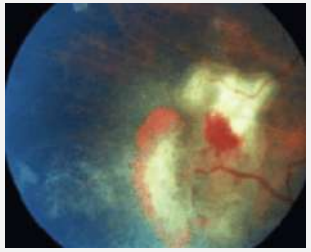
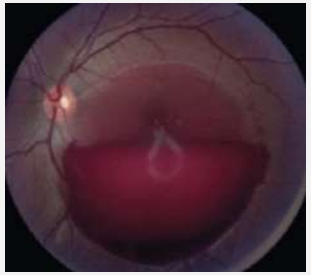
Suspected role of PRP is:

- ★ Reduction in O2 demand from the outer retina
- ★ Increase in O2 transfer to inner retina via scar
- ★ Decrease in VEGF signal
- ★ Decrease in oedema

NOTE: PRP is not without side effects. It is not a benign treatment, and will cause dead retina.

DRUG NAME	MOA	INDICATIONS/CONTRAINDICATIONS	SAFETY PROFILE (PRECAUTIONS/INTERACTIONS)	ADVERSE EFFECTS
Active Ingredient: Fenofibrate **Trade Names: Lipidil, Apo-Fenofibrate** **ORAL** ❌On the OBA **(MUST BE PRESCRIBED BY GP)** **Dosage:** Oral, 145 mg once daily.	**MOA:** Fenofibrate, a fibric acid derivative, is a peroxisome proliferator activated-receptor alpha (PPARa) agonist. Lowers free fatty acids by upregulating the synthesis of proteins responsible for fatty acid transport and ß-oxidation, which inhibits the formation of triglycerides and very low density lipoproteins. Activation of PPARa is thought to also increase high-density lipoprotein (HDL)-cholesterol. Fibrates have been shown to be clinically effective in lowering triglyceride levels and improving HDL-cholesterol levels. **PROPOSED MECHANISMS OF ACTION IN DR:** • Increase the levels of circulating apolipoprotein A-I, an independent protective factor for DR development. • May also be important in regulating intra-retinal lipid metabolism and reducing lipid deposition and lipotoxicity . • Other non-lipid mechanisms: anti-apoptotic, anti-oxidant, anti-inflammatory activity and anti-angiogenic activity, and protective effects on the blood-retinal barrier breakdown.	✅Severe hypertriglyceridaemia ✅Dyslipidaemia associated with type 2 diabetes (second line) ✅Reduction in progression of existing diabetic retinopathy associated with type 2 diabetes (with appropriate BP, glycaemic and lipid control) in mild + moderate to slow risk of progression to more advanced disease. ✅Hypercholesterolaemia (rarely used) ***NOTE:*** Does not prevent presence of DR (like prevent DR if you have DM). ❌Photosensitivity from ketoprofen (discontinued) ❌Pancreatitis, unless due to hypertriglyceridaemia ❌Severe renal impairment ❌Contraindicated when hepatic impairment, primary biliary cirrhosis, gallstones or gall bladder disease are present.	**Precautions:** • Severe diabetic retinopathy in type 2 diabetes—no data on effect of fenofibrate, as these people were excluded from the clinical trials. 💊**Interactions: Ezetimibe** as may further increase risk of gall bladder disease. Seek specialist advice with children Avoid with breastfeeding ❌Avoid risk with pregnant women if possible, except on specialist advice in cases of severe hypertriglyceridaemia with risk of pancreatitis	**Common (>1%):** GI disturbances (eg dyspepsia, abdominal pain), increased creatinine concentration (reversible) **Infrequent (0.1–1%):** Gallstones, pancreatitis (may be rare with gemfibrozil), photosensitivity (may be rare with gemfibrozil) **Rare (<0.1%):** Cholestatic jaundice, anaemia, leucopenia, myopathy, rhabdomyolysis, hypersensitivity reactions (eg angioedema, anaphylaxis, exfoliative dermatitis)

CONDITION	SIGNS/SYMPTOMS	WORKUP	MANAGEMENT	FOLLOW- UP
Ocular Ischaemic Syndrome/Carotid Occlusive Disease  Mimics BRVO	Carotid artery has stenosis (narrow). **Symptoms:** Decreased vision, ocular or periorbital pain, afterimages or prolonged recovery of vision after exposure to bright light, may have a history of transient monocular visual loss (amaurosis fugax). Usually unilateral, although up to 20% of cases can be bilateral. Typically occurs in patients aged 50 to 80 years. Men outnumber women 2:1. **Signs:** Although retinal veins are dilated and irregular in caliber, they are typically not tortuous. The retinal arterioles are narrowed. Associated findings include midperipheral retinal hemorrhages (80%), iris neovascularization (66%), neovascularization of the disc (35%), and neovascularization of the retina (8%). **Etiology:** ● Ipsilateral carotid artery disease: Usually ≥90% stenosis. ● Ipsilateral ophthalmic artery disease: Less common. ● Ipsilateral central retinal artery obstruction: Rare. ● Giant cell arteritis: Rare.	**(1)** History: Previous episodes of transient monocular visual loss? Cold hands or arm muscle spasms with exercise? **(2)** Complete ocular examination: Search carefully for anterior chamber flare, asymmetric cataract, and NVI/NVA/NVD/NVE. **(3)** Medical examination: Evaluate for HTN, diabetes, and atherosclerotic disease. Check pulses. Cardiac and carotid auscultation. **(4)** Laboratory workup for GCA in the appropriate settings. **(5)** Consider IVFA for diagnostic purposes. **(6)** Noninvasive carotid artery evaluation: Duplex Doppler US, oculoplethysmography, magnetic resonance angiography, others. **(7)** Consider orbital color Doppler US. **(8)** Consider ophthalmodynamometry if CRVO diagnosis cannot be excluded. **(9)** Carotid arteriography is reserved for patients in whom surgery is to be performed. **(10)** Consider cardiology consultation, given the high association with cardiac disease.	**1.** Carotid endarterectomy for significant stenosis. Refer to neurovascular surgeon. **2.** Consider PRP and anti-VEGF agents in the presence of neovascularization. **3.** Manage glaucoma if present. **4.** Control HTN, diabetes, and cholesterol. Refer to internist. **5.** Lifestyle modification (e.g., smoking cessation).	Depends on the age, general health of the patient, and the symptoms and signs of disease. Surgical candidates should be evaluated urgently.
Vitreous Haemorrhage 	**Symptoms:** Sudden, painless loss of vision or sudden appearance of black spots, cobwebs, or haze in the vision. **Signs:** In severe VH, the red fundus reflex may be absent, and there may be limited or no view to the fundus. Red blood cells may be seen in the anterior vitreous (or anterior chamber). In mild VH, there may be a partially obscured view to the fundus. Chronic VH has a yellow ocher appearance from hemoglobin breakdown. **Etiology:** ● DR ● PVD ● Retinal break/ retinal detachment ● Retinal vein occlusion (usually BRVO) ● Exudative AMD ● Sickle cell disease	**(1)** History: Any ocular or systemic diseases? Trauma? **(2)** Complete ocular examination, including slit lamp examination with undilated pupils to check for iris neovascularization, IOP measurement, and dilated fundus examination of both eyes by using indirect ophthalmoscopy. In cases of spontaneous VH, scleral depression is performed if a retinal view can be obtained. **(3)** When no retinal view can be obtained, B-scan US is performed to detect an associated RD or intraocular tumor. Flap RTs may be detected with scleral depression and may be seen on B-scan US (elevated flap). **(4)** IVFA may aid in defining the etiology, although the quality of the angiogram depends on the density of the hemorrhage. Additionally, it may be useful to highlight abnormalities in the contralateral eye.	If etiology unknown = Observation OR Vitrectomy. ● No heavy lifting/straining/bending ● Keep head of bed elevated This reduces the chance of recurrent bleeding and allows blood to settle inferiorly, permitting a view of the superior peripheral fundus, a common site for possible retinal breaks. ● No Aspirin, NSAIDs or other anticlotting agents unless medically necessary. The underlying etiology is treated as soon as possible (e.g., retinal breaks are sealed with cryotherapy or laser photocoagulation, detached retinas are repaired, and proliferative retinal vascular diseases are treated with anti-VEGF therapy or laser photocoagulation).	If observation is elected, the patient is evaluated daily for the first 2 to 3 days. If a total, dense VH persists, and the etiology remains unknown, vitrectomy should be considered.
Retinal Arterial Macroanerysm 	**Symptoms:** Decreased vision; history of systemic HTN. Usually unilateral, but 10% bilateral. **Signs:** Acute hemorrhages in multiple layers of the retina (subretinal, intraretinal, preretinal) possibly with VH; often with a white or yellow spot in the middle of the retinal arterial macroaneurysm (RAM). Chronic leakage may cause a ring of hard exudates and retinal edema around the aneurysm resulting in decreased vision if the macula is involved.	**(1)** History: Systemic disease, particularly HTN or diabetes? **(2)** Complete ocular examination with dilated retinal examination with a 60- or 90-diopter lens and indirect ophthalmoscopy. Look for concurrent retinal venous obstruction (present in one-third of cases) and signs of hypertensive retinopathy (visible in fellow eye as well). **(3)** Check blood pressure. **(4)** Consider checking lipid panel as well as fasting or random blood sugar and hemoglobin A1c. **(5)** IVFA may demonstrate early hyperfluorescence if there is no blockage from hemorrhage. Late frames may show leakage or staining of vessel wall. **(6)** OCT is helpful in demonstrating and following any ME.	Consider laser treatment if edema and/or exudate threatens central vision. Caution must be taken when treating arterioles that supply the central macula since distal thrombosis and obstruction with resultant ischemia can occur. Laser can also cause aneurysmal rupture resulting in retinal and vitreous hemorrhage. Anti-VEGF agents may be beneficial in patients with macroaneurysm-associated ME. Dense or nonclearing vitreous hemorrhage, sub-internal limiting membrane (ILM) hemorrhage, or thick submacular hemorrhage may benefit from vitrectomy.	Frequency based on the amount and location of exudate and hemorrhage.

Sickle Cell Retinopathy 	**Symptoms:** Usually without ocular symptoms. Floaters, flashing lights, or loss of vision with advanced disease. Systemically, patients often have painful crises with severe abdominal or musculoskeletal pain. Patients are typically of African or Mediterranean descent. Sickle cell patients with temporal macular atrophy can be asymptomatic, but can also present with scotomas, impaired color vision and loss of contrast sensitivity. Sickle cell maculopathy occurs more commonly in patients with HbSS > HbSC. **Signs:** Peripheral retinal neovascularization in the shape of a fan ("sea fan sign"), sclerosed peripheral retinal vessels, or a dull gray peripheral fundus background color as a result of peripheral arteriolar occlusions and ischemia. **Staging:** • **Stage 1:** Peripheral arteriolar occlusions. • **Stage 2:** Peripheral arteriovenous anastomoses. • **Stage 3:** Neovascular proliferation. • **Stage 4:** VH. • **Stage 5:** RD.	**Workup:** **(1)** Medical history and family history: Sickle cell disease, diabetes, or known medical problems? Intravenous drug abuse? **(2)** Dilated fundus examination using indirect ophthalmoscopy. **(3)** Sickledex, sickle cell preparation, and hemoglobin electrophoresis.		**1.** No retinopathy: Annual dilated fundus examinations. **2.** Retinopathy present: Repeat dilated fundus examination every 3 to 6 months, depending on severity.
Angioid Streaks 	**Symptoms:** Usually asymptomatic. Decreased vision may result from CNV. **Signs:** Bilateral reddish-brown or gray bands located deep to the retina, due to breaks within Bruch membrane, usually radiating in an irregular or spoke-like pattern from the optic disc. CNV may occur. **Etiology:** Fifty percent of cases are associated with systemic diseases; the rest are idiopathic.	**(1)** History: Any known systemic disorders? Previous ocular trauma? **(2)** Complete ocular examination: Look carefully at the macula with a slit lamp using a 60-, 90-diopter, or fundus contact lens to detect CNV. **(3)** FAF if diagnosis uncertain or IVFA or OCTA if CNV suspected. **(4)** Physical examination: Look for clinical signs of etiologic diseases. **(5)** Serum alkaline phosphatase and urine calcium levels if Paget disease of bone is suspected. **(6)** Sickle cell preparation and hemoglobin electrophoresis in patients of African descent. **(7)** Skin or scar biopsy if PXE is suspected. **(8)** CBC if hematologic etiology suspected.	**1.** Anti-VEGF therapy is now used for angioid streak-associated CNV, as focal laser photocoagulation and PDT have discouraging outcomes. **2.** Management of any underlying systemic disease by an internist. **3.** Recommend wearing one-piece polycarbonate safety glasses for sports due to an increased risk of subretinal hemorrhage and choroidal rupture from minor trauma.	Fundus examination every 6 months, monitoring for CNV. Instruct patient to check Amsler grid daily and return immediately if changes are noted.
Valsalva Retinopathy 	Valsalva causes sudden increase in intraocular venous pressure leading to rupture of superficial capillaries in macula or elsewhere in the retina. May be associated with anticoagulant therapy. **Symptoms:** Decreased vision or asymptomatic. History of Valsalva maneuver (forceful exhalation against a closed glottis), which may occur during heavy lifting, coughing, vomiting, or straining during bowel movement. Sometimes, no history of Valsalva can be elicited. **Symptoms:** Single or multiple hemorrhages under the ILM in the area of the macula. Can be unilateral or bilateral. Blood may turn yellow after a few days. Other signs include vitreous, intraretinal, subretinal, and subconjunctival hemorrhage can occur.	**(1)** History: History of Valsalva including any recent heavy lifting, straining during bowel movement, coughing, sneezing, vomiting, etc.? The patient may not remember the incident. **(2)** Complete ocular examination, including dilated retinal examination with a slit lamp and a 60- or 90-diopter lens, and indirect ophthalmoscopy. Look for findings suggestive of a different etiology including microaneurysms, dot-blot hemorrhages, CWSs, RT, PVD. **(3)** If dense VH is present, perform a B-scan US to rule out RT or RD. **(4)** IVFA may be helpful to rule out other causes including RAM or diabetic retinopathy.	Prognosis is excellent. Most patients are observed, as sub-ILM hemorrhage usually resolves after a few days to weeks. Occasionally laser is used to permit the blood to drain into the vitreous cavity, thereby uncovering the macula. Vitrectomy rarely considered, typically only for nonclearing VH.	May follow up every 2 weeks for the initial visits to monitor for resolution, then follow up routinely.

Glaucoma

Group of disordered, which have in common:
- Potentially progressive and characteristic optic neuropathy
- Associated, progressive VF loss
- IOP as a key modifying factor

Classifications:
1. **Congenital** (developmental) OR **acquired** (more common)
2. **Open-angle** OR **closed-angle**
3. **Primary** OR **secondary**
   a. Primary = There is no evident cause of intraocular pressure elevation/development of glaucoma
   b. Secondary = Cause of intraocular pressure elevation can be identified, can be caused by other ocular causes (pseudoexfoliation i.e pseudoexfoliative material in AC clogging up the TM, pigment dispersion syndrome), other disease processes, injury or side effects of medical treatment.

Aetiological factors:
- ★ Mechanical
- ★ Ischaemic → Vascular risks (related to H&S questions)
- ★ Neurodegenerative → Increased prevalence as patient ages
- ★ Autoimmune
- ★ Prevalence of open-angle glaucoma is 1.7% in over 40 year olds

Statistics:
- ★ Prevalence increases with age; 10% in the over 80 age group
- ★ Glaucoma affects 2–3% of people over the age of 40 years
- ★ 50% of glaucoma cases are undiagnosed
- ★ Currently ~80 million people have glaucoma worldwide
- ★ Lots of diagnostic ambiguity, can make a name for yourself as a glaucoma specialist.

(1) Refresher on the foundations of Glaucoma

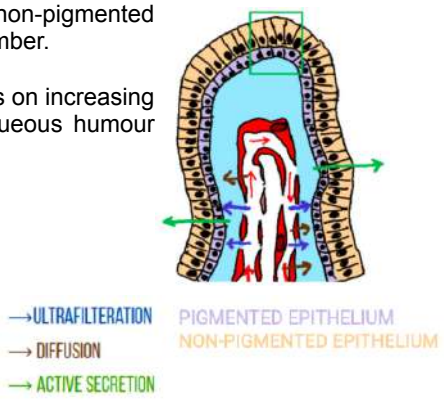
Aqueous Humour

Production

Aqueous humour = Produced from plasma by the ciliary processes of the non-pigmented epithelium of the ciliary body (pars plicata) and secreted into the posterior chamber.

Secretion is subject to the influence of the sympathetic nervous system. Takes on increasing relevance when we think about the therapies for reducing IOP through aqueous humour production.
- Beta-2 receptors **increase** secretion
- Alpha-2 receptors **decrease** secretion
- Enzymatic action from carbonic anhydrase plays a key role

Any imbalance in AH production or outflow does increase IOP , which therefore increases risk of glaucoma.

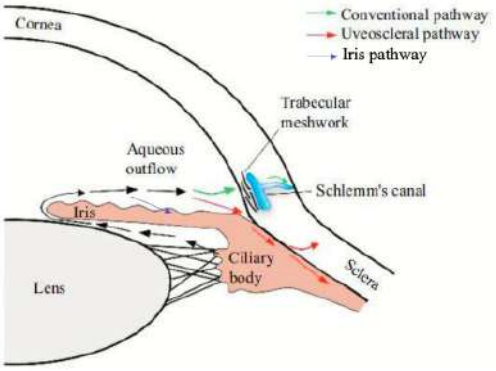


Outflow

Trabecular meshwork = sieve-like structure at the angle of anterior chamber with 3x components:
1. **Uveal** meshwork
2. **Corneoscleral** meshwork
3. **Juxtacanalicular** meshwork

Schlemm's canal = Circumferential channel within the perilimbal sclera + has collector channels draining it off. Lies adjacent to the TM.

Collector channels leave Schelmm's canal at oblique angles + connect directly/indirectly with episcleral veins (collector channels drain it out)



Aqueous flows from the POSTERIOR chamber to ANTERIOR chamber, exiting 3x ways:
1. **Trabecular outflow (90%) – CONVENTIONAL PATHWAY :** Trabecular meshwork → Schlemm canal → collector channels → Episcleral veins (bulk flow pressure-sensitive route so that increasing IOP will increase outflow)
2. **Uveoscleral drainage (10%):** Aqueous passes across the face of the ciliary body into the suprachoroidal space, and is drained by the venous circulation in the ciliary body, choroid and sclera.
3. **Iris (very small amount)**

Ocular Hypertension (OHT) VS Glaucoma

Ocular hypertension: State of elevated IOP without evidence of structural or functional damage by standard clinical tests. Patient must:
- Have open angles
- Have no evidence of ocular or systemic cause of elevated IOP
- No evidence of glaucomatous damage YET, no secondary risk factors

NOTE: 4-10% of the population >40 have **IOP >21mmHg**

Treatment guided by Ocular Hypertension Treatment Study (OHTS). **All** should be followed for structural and functional damage suggestive of glaucoma.

(2) Types of Glaucoma

- It is difficult to define glaucoma precisely, partly because the term encompasses a diverse group of disorders. All forms of the disease have in common a characteristic potentially progressive optic neuropathy that is associated with visual field loss as damage progresses, and in which IOP is a key modifiable factor.

A. Pre-perimetric glaucoma

Defined as: Glaucomatous structural damage (suspicious disc and/or RNFL defects) **without** any visual field defect

NOTE: Although VF loss is associated with the disease it is not always measurable in the clinic. Wherever there is a structural defect, there should be a functional defect.

B. POAG – Primary Open Angle Glaucoma (MOST COMMON)

Defined as: Glaucomatous optic neuropathy in the presence of an open angle and no secondary mechanism

- **IOP >21mmHg** or has been at some stage
- Characteristic, progressive visual field loss
- Occurs frequently in white, Hispanic/Latino and black individuals
- Myopia increases the risk

Typically **BILATERAL**, but Aymmetrical

C. NTG – Normal Tension Glaucoma

Defined as:

- High prevalence in Japan, 20-30% of glaucoma approx in Australia
- Cupping and visual field loss characteristic of OAG, often more **paracentral** field loss
- **IOPs within the 'normal' range** + absence of secondary features otherwise similar to POAG
- Essentially an arbitrary division of POAG that may not have significant clinical value – some authorities suggest that the vascular ischemic mechanism of disease has a greater role in NTG (e.g more disc haem, more paracentral field loss)
- IOP independent factors may have a greater role in pathogenesis:
  - + OSA (obstructive sleep apnea)
  - + Reduced perfusion: Low blood pressure or reduced ocular blood flow (suggests more vascular ischemic role in NTG)
  - + Migraine or Raynaud's phenomenon (greater role in development or progression)
  - + Disc haemorrhage
  - + Myopia
  - + Low cerebrospinal fluid pressure

D. PACG – Primary Angle Closure Glaucoma

Features:

- PACG is more visually destructive form of disease than POAG PACG is responsible for half of all glaucoma blindness worldwide
- It is more unstable and needs to be managed accordingly.
- Particularly high prevalence in individuals of Asian descent
- Most angle closure is asymptomatic
- "Angle closure" is occlusion of the TM by the peripheral iris (iridotrabecular contact - ITC) obstructing aqueous outflow

Risk Factors:

- Age
- Women
- Family history
- Hypermetropia (hyperopia)
- Chinese
- Thicker lens (as lens ages, gets thicker = narrowing angle)
- More anteriorly positioned lens
- Shorter axial length

Table 5:
Grading scheme for angle closure spectrum disease[1,12]

Grade	Definition
Open angles	Posterior trabecular meshwork or deeper visible in all quadrants*
Primary angle closure suspect	Posterior trabecular meshwork not visible in three or more quadrants
Primary angle closure	Posterior trabecular meshwork not visible in three or more quadrants potentially with raised intraocular pressure (>21 mmHg) AND/OR presence of synechiae
Primary angle closure glaucoma	Primary angle closure glaucoma: posterior trabecular meshwork not visible in three or more quadrants AND glaucomatous optic nerve AND/OR visual field changes

Glaucoma = Structural + Functional loss

Primary Angle Closure (PAC)

- X3 subtypes: Acute, intermittent/resolved or chronic
- 3 can overlap and coexist in the one eye
- Gonioscopy important for this
- Chronic PAC is the most prevalent form of PAC
- Chronic PAC may develop after acute PAC or more commonly following gradual asymptomatic closure of the angle
- IOP control depends on the amount of TM damage and extent of closure
- Reopening of the angle prior to ultrastructural changes should restore the natural outflow pathway (if detected can be treated easily)

NOTE: On a worldwide basis, primary angle closure (PAC) constitutes up to half of cases, and has a particularly high prevalence in individuals of Asian descent, although with improved assessment such as the routine performance of gonioscopy in a darkened rather than a bright environment, PAC is known to be more prevalent in Caucasian individuals than previously realized.

E. Secondary Glaucoma

Defined as: Glaucomatous optic neuropathy together with signs of other pathological processes.

- Often less stable with higher/more variable IOP (HENCE should not be managed solo due to instability)
- PXF
- PDS
- Fuch's Heretchromic Iridocyclitis
- Possner-Schlossman Syndrome

Pseudoexfoliative Glaucoma	Pigment Dispersion Glaucoma
• Secondary open angle glaucoma • White deposits within the anterior lens capsule, ciliary body, iris, cornea and trabecular meshwork • 25% of patients with PXF develop high IOP; 5-10x risk of glaucoma if PXF • Incidence increases with age (thinking of an older patient, Scandinavian) • More common in Caucasians: Norway, Ireland, Greece, Saudi Arabia • Systemic associations: TIA, stroke, MI, Alzheimer's, hearing loss	• Secondary open angle glaucoma • Pigment dispersion syndrome - no glaucoma evident • Younger than POAG • Peak prevalence of PG – between 3rd and 4th decade • Almost always Caucasian, mild-moderate myopia, mostly bilateral • Conversion to PDG more often in males (male, gym junkie) • Midperipheral radial iris transillumination defects, Krukenberg spindle, pigment on lens zonules • Deep pigmented angle on gonioscopy (broad open angle) • 6-7% retinal detachment – remember to dilate!

(3) Glaucoma Clinical Examination

- Pupils to rule out other optic neuropathies
- Pachymetry
- Slit lamp examination for signs of PDS, PXF, iritis, trauma/angle recession, cataract (may impact fields)
- Gonioscopy
- DFE
- CV (red cap test – if concerned about other optic neuropathies)

History taking

Current medications: (Don't want to double dose the medication if patient is currently taking some)
- Steroids (increases IOP)
- Beta blockers (reduces IOP) meds with "olo" at the end
- OCP
- E.g "*Are you currently taking any medications?*"

FOHx of glaucoma:
- Immediate family risk with siblings and parents
- E.g "*Does anyone in your family have glaucoma or at risk of glaucoma?*"

Enquiring about previous symptoms of previous, intermittent acute angle attacks in the past
- Reduced VA
- Hyperaemia (associated red eyes)
- Dilated pupils (although unlikely that patient will notice)
- Sudden severe eye pain
- Eg. severe eye pain while watching movie in dark

Blood pressure
- High (typical risk factor for glaucoma)
- Low (increased risk of reduced perfusion i.e. NTG)

Standard glaucoma workup should include a thorough case history:

POHx	• Previous high IOP and treatment • Rule out secondary causes: trauma, known PXF/PS, uveitis, cataracts (as it gets worse it narrows the AC), diabetes, CRVO • Myopia, previous intraocular surgery (e.g. cataracts)
PMHx	Focused on risk factors • HTN, hypotension, diabetes, thyroid disease • Vasoplastic disorders (Raynauds, migraines), OSA, smoking
Medications	Topical/systemic corticosteroids or Beta blockers, OCP
FOHx	Immediate history of glaucoma

Tonometry

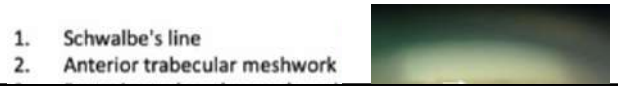
Measures IOP (***NOTE:*** Glaucoma can develop at any IOP)
- Several methods used clinically e.g. NCT, iCare, GAT – GAT is gold standard)
- A single reading may be misleading: Result varies with time of day, heartbeat, blood pressure and respiration (capture diurnal since POAG patients have more unstable IOP)
  - + Thicker cornea = IOP overestimated
  - + Thinner cornea = IOP underestimated
- Water drinking test is an option as well
- **Higher fluctuations have been associated with greater likelihood of disease progression** – role of IOP phasing and the water drinking test (to see fluctuations)

Gonioscopy

Essential on **all patients** with or at risk of glaucoma.
- Allows for: Correct diagnosis + Targeted therapy
- Pathological findings:
  - + Peripheral anterior synechiae (PAS)
  - + Pigmentation
  - + Abnormal blood vessels, blood in Schlemm's
  - + Angle recession

In a healthy eye:
1. Schwalbe's line
2. Anterior trabecular meshwork
3. Posterior trabecular meshwork
4. Scleral spur
5. Ciliary body band





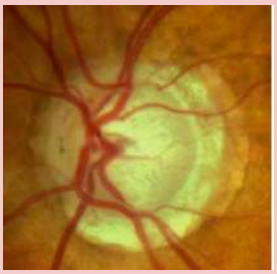
DFE: ONH

NOTE:

- Iris processes can be mistaken as synechiae
- Blood vessels can be mistaken as neovascularisation (recap that new vessels are weak!!). Tend to be chunkier than new vessels which are more fragile (any BV that passes the scleral spur is pathological).
- Will not see Schlemm's canal unless filled with blood

The 5 Rs:

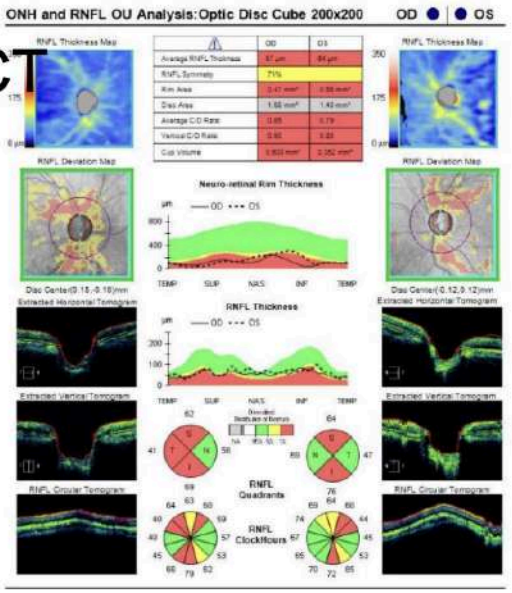
1. Observe the **scleral Ring** to identify the **limits of the optic disc** and evaluate its size, note asymmetries
2. Identify **focal** or **generalised thinning** of the **Rim**
3. Examine the **Retinal nerve fibre layer** (RNFL)
4. Examine the **Region outside the optic disc** for **parapapillary atrophy** (PPA)
5. Watch for **Retinal and optic disc haemorrhages**

CASE 1: FOCAL ISCHEMIC	CASE 2: MYOPIC DISC	CASE 3: SCLEROTIC DISC (old shallow saucer like cup)	CASE 4:CONCENTRICALLY ENLARGED
			
• Glaucomatous ON • Infero-temporal notching • Associated disc haemorrhage • Associated RNFL dropout	Differentiated with myopic cups • Inferior rim looks thin • Temporally sloped cups • Beta PPA • Tesselated fundus (difficult to see RNFL "*reduced visibility of RNFL from tesselated fundus*"	No rim (superiorly + inferiorly) ***NOTE:*** Healthy colour of the disc comes from a full rim	Thick rims Wide cups

RNFL assessment via OCT

RNFL assessment via OCT

Measurement	Matched to Normal Based On	Gray	White	Green	Yellow	Red
RNFL						
Average RNFL Thickness, RNFL Symmetry, RNFL Clock Hours, RNFL Quadrants, RNFL Thickness (graph)	Age	Gray shading does not apply to RNFL measurements	The thickest 5% of measurements fall in the white area (white > 95%).	90% of measurements fall in the green area (5%<green<95%).	The thinnest 5% of measurements fall in the yellow area or below (1%<yellow<5%, suspect).	The thinnest 1% of measurements: Measurements in red are considered outside normal limits (red < 1%, outside normal limits).
Optic Nerve Head						
Rim Area and Neuroretinal Rim Thickness (graph)	Disc Area and Age	ONH Normative Database is not applicable if: 1) The disc area is larger than 2.5mm² or smaller than 1.33mm², or 2) The Average or Vertical C/D Ratio is below 0.25, or 3) The ONH Normative Database license has not been activated.	The largest 5% of measurements fall in the white area (white > 95%).	90% of measurements fall in the green area (5%<green<95%).	The smallest 5% of measurements fall in the yellow area or below (1%<yellow<5%, suspect).	The smallest 1% of measurements: Measurements in red are considered outside normal limits (red < 1%, outside normal limits).
Average C/D Ratio, Vertical C/D Ratio, Cup Volume			The smallest 5% of measurements fall in the white area (white > 95%).	90% of measurements fall in the green area (5%<green<95%).	The largest 5% of measurements fall in the yellow area or below (1%<yellow<5%, suspect).	The largest 1% of measurements: Measurements in red are considered outside normal limits (red < 1%, outside normal limits).



- Can look for corresponding changes on GC analysis (observing cell bodies vs axons) if one structure is affected so will the others.
- OCT can be used OCT to track structural progression event vs trend analysis (can look at downward trent)
- GPA (guided progression analysis) can be used to help guide diagnosis (used to track progression over time)

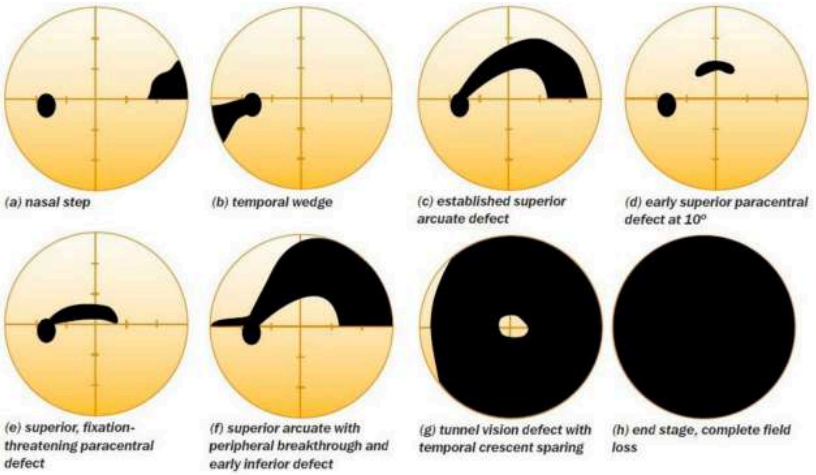
Perimetry

Prior testing: refraction and VA

Threshold SAP (standard automated perimetry) is the method recommended by NHRMC

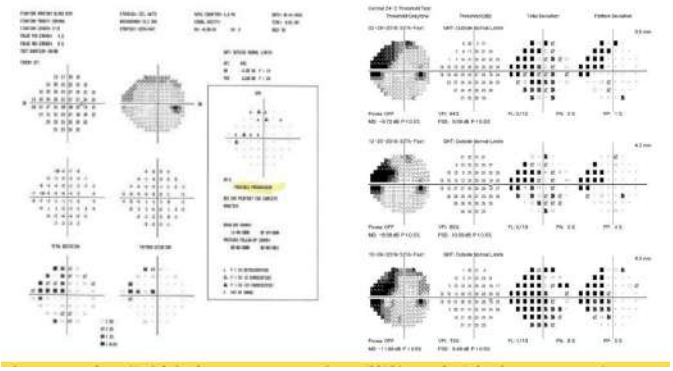
- Includes: Humphrey Visual Field Analyser, Octopus, Medmont, Henson
- **24-2 for glaucoma**
- **30-2 for neurological conditions**
- Consider:

+ **Frontloading** → Used for baseline for new patients (ideally same day)
+ **Repeat VFs** → Used for baseline for new patients or unreliable patients (ideally same day)
+ **10-2** → Used for **advanced glaucoma** where there is only central field left , + think about low vision support



Nerve fibre bundle abnormalities specifics (from optic neuritis trial):

- **Temporal wedge:** Small VF temporal to the blind pot
- **Enlarged blindspot:** Nerve fibre bundle region that involves at least 2 points and is contiguous with the BS
- **Nasal step:** Limited field loss adjacent to nasal horizontal meridian with at least 1 abnormal point at or outside 15* on the meridian. Cannot include more than 1 sig point in the NFB region on the temporal side
- **Paraentral:** Not continuous with the BS or nasal meridian. Does not involve points outside 15* that are adjacent to the nasal meridian . Can affect macular bundle (can be partially arcuate, central-coecal or central VF)
- **Partial arcuate:** extends incompletely from the BS to the nasal meridian. Generally contiguous with either the BS or nasal meridian and must include at least 1 abnormal location in the temporal VF
- **Arcuate:** Significant VF loss extending across contiguous abnormal points form the BS to at least 1 point outside 15* adjacent to the nasal meridian



AKA stages of VF loss in glaucoma:

1. **Nasal step** or **paracentral** defect (paracentral defects can be fixation-threatening)
2. Defects enlargens ⇒ **arcuate** defects
3. Spreads to other hemifield ⇒ **central vision loss**

AKA nasal step/paracentral defect ⇒ arcuate scotoma/defect ⇒ central scotoma/defect

OBA Requirements

To comply with OBA guidelines for glaucoma care, endorsed optometrists must:

1. Remain cognisant of the 4 mandatory referral triggers (section 7.1)
2. Communicate with patient's other health practitioners (section 7.2)
3. Have all equipment specified in section 7.3 (refer to NHMRC)
4. Refer to NHMRC guidelines when setting target IOP and deciding review frequency

Optometrists must have the equipment to measure and/or assess a patient's:

- **Intraocular pressure** → Tonometry (ideally applanation)
- **Central corneal thickness** → Pachymetry
- **Threshold visual fields** → Standard automated perimetry (SAP)/visual fields
- **Anterior chamber angle** → Gonioscopy
- **Optic nerve head** and **retinal nerve fibre layer** → DFE, OCT

Level	Hierarchy of Glaucoma Examination
Very good level	IOP*, Gonioscopy, threshold VF†, Optic Disc Images AND RNFL images
Good	IOP*, Gonioscopy, threshold VF†, Optic Disc Images OR RNFL images
Satisfactory	IOP*, Gonioscopy, threshold VF†, clinical optic disc examination
Insufficient	IOP*

Barrier to diagnosis + management

❌**Wrong equipment:** NCT, FDT, absence of a gonioscope/coupling fluid/anaesthetic e.g not the right type of tonometer

❌Mis-matched equipment for collaborative care with the local ophthalmologist/ophthalmology service: Medmont vs HVFA

❌Lack of practice, time

❌Lack of compensation

(4) Glaucoma Management

Glaucoma Diagnosis

- After a glaucoma workup, there are 4 possible outcomes:

1. The patient is normal, no further testing or follow up is required
2. The results are equivocal, and the patient has OHT or is a glaucoma suspect
3. The patient shows consistent structural and functional signs of glaucoma → classify the cause/type/mechanism and severity of glaucoma (as this will influence treatment plan)
4. The patient shows evidence of another mimicking condition

Table 6: Stages of glaucoma from the American Academy Preferred Practice Pattern 2021: one example of a staging system

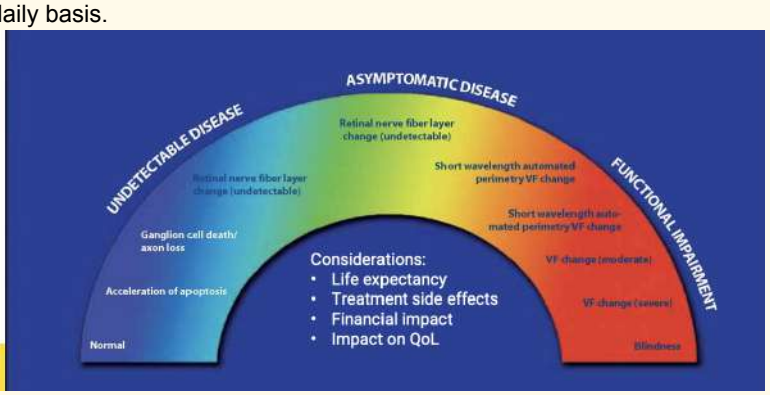
Stage of glaucoma	Structural changes	Functional changes
Mild or 'pre-perimetric'	Definite optic disc, nerve fibre layer or macular imaging abnormalities consistent with glaucomatous damage	Normal visual field
Moderate	Definite optic disc, nerve fibre layer or macular imaging abnormalities consistent with glaucomatous damage	Visual field abnormalities in one hemifield that are not within 5 degrees of fixation
Severe	Definite optic disc, nerve fibre layer or macular imaging abnormalities consistent with glaucomatous damage	Visual field abnormalities in both hemifields and/or loss within 5 degrees of fixation in at least one hemifield
Indeterminate	Definite optic disc, nerve fibre layer or macular imaging abnormalities consistent with glaucomatous damage	Inability of patient to perform visual field testing, unreliable/uninterpretable visual field test results, or visual fields not performed yet

WHEN TO TREAT GLAUCOMA?

GOAL: Sustainability for px and society, preserve visual function, QOL.

- Preserving visual function
- Thinking about QOL implications
- (ABOVE) at a sustainable cost for the patient and society

Recognise that we are trying to treat an asymptomatic disease that the patient may not know about + asking them to take medications that will cause them symptoms on a daily basis.



Glaucoma spans a spectrum and it is not until we identify a full-frank progression that we might choose to initiate treatment.

Optic nerve changes consistent with glaucoma

- Either observed on fundoscopy or OCT
- With or without **visual field loss**
- With or without **elevated IOP**

Optic nerve questionable/normal status w elevated IOP

- **Glaucoma suspect**
- Ocular hypertension **21-30 mmHg**
- Significant risk factors

Signs of Progression – Consider is a patient actually going to go blind in lifetime?

- Confirmed **new defect** OR worsening of prev ambiguous VF defect
- Progressive **OD cupping, notching or rim thinning**
- Progressive **thinning of circumpapillary RNFL or macular ganglion cell layer consistent** with glaucomatous progression

Management Principles

1. **Educate** the patient about the disease, the severity of their disease, and rationale for treatment (otherwise no point , want to bring patient on board with you , trying to boost adherence)
2. Set **target pressure**
3. Select **therapy type**: Topical therapy, laser or surgery i.e what is the safest and simplest medication that the patient agrees to use
   a. First choice is typically a **PGA** (prostaglandin analogues), **β-blocker** or **SLT** (selective laser trabeculoplasty)
      - **Prostaglandin analogues** – Enhances outflow of AQH through uveoscleral pathway (potent, once-daily efficacy)
      - **B-blocker** – decreases AQH production (lowers IOP when prostaglandins are insufficient/intensive treatment)
      - **SLT** – Uses lasers to selectively target pigmented cells in the trabecular meshwork, increasing AQH outflow
   b. Achieve IOP decrease, best safety profile, most affordable, simplest regimen, lowest effective concentration, preferably once daily administration
4. **Maximise adherence**:
   a. Discuss the therapy being recommended including technique and timing of administration, potential adverse effects
   b. Strategies for reducing systemic absorption if topical
5. **Monitor –** chronic and progressive condition

Target IOP

- Target IOP is defined when there is no progression with field, ONH or imaging. Inherent to this definition is that pre-treatment IOP was at a level that damaged it in the first place.
- Target IOP is ~ 25% reduction from baseline, further 20% if progression occurs at follow up so aiming for greater reduction in IOP) + DOCUMENT TARGET IOP.
- Lower IOP (compared to baseline) if patient has more severe disease or more risk factors (e.g diurnal variation)

Stage	Target IOP	Specified level
High-risk suspects	20%	≤24mmHg
Early-moderate, low-risk	20%	≤19mmHg
Moderate, high-risk	30%	≤18mmHg
Advanced	30-50%	≤14mmHg

Select Therapy Type

- First choice is typically a PGA, Beta blocker or SLT (selective laser trabeculoplasty).
- In terms of medical therapy – looking at what is the safest and simplest medication or treatment that the patient agrees to use.
- Also aiming to use the lowest effective concentration (where possible also once daily - PGA)


Table 7: IOP medications available in Australia for the management of glaucoma

Preparations by Class	Mechanism of action	Efficacy	Daily dosage	Order of treatment choices	Ocular side effects	Systemic side effects	Contraindications
Prostaglandin analogues • Latanoprost 0.005% (e.g. Xalatan) • Travoprost 0.004% (e.g. Travatan) • Bimatoprost 0.03% (e.g. Lumigan*) • Tafluprost 0.0015% (e.g. Saflutan*)	Increase aqueous outflow	25-35% Maximum effect: 8-12 hours	Once daily (night)	First	• Increase in iris pigmentation • Darkening • Thickening & lengthening of eyelashes • Conjunctival hyperaemia • Periorbital pigmentation	Uncommon – may cause respiratory symptoms in susceptible individuals	No contraindications **Precautions:** • Intraocular inflammation (iritis, uveitis) • History of herpetic keratitis • Aphakia or pseudophakia (potential for macular oedema)
Beta-blockers Non-selective agents: • Timolol 0.25%, 0.5%, 1% (e.g. Timoptol, Nyogel, Timoptic*) • Selective agents: • Betaxolol 0.25%, 0.5% (e.g. Betoptic)	Decrease aqueous production	20-25% Maximum effect: 2 hours	One to two times daily	First	• Transient ocular discomfort • Blurred vision • Increased lacrimation • Foreign body (FB) sensation	• Headache • Bradycardia • Decreased libido • Bronchospasm • Nausea	• Sinus bradycardia • Overt cardiac failure history • Cardiogenic shock **Precautions:** • Asthma • Severe COPD (selective agents, i.e. betaxolol preferred)
Alpha2-agonists • Brimonidine 0.2%, 0.15% (e.g. Alphagan) • Apraclonidine* 0.5% (e.g. Iopidine)	Increase aqueous outflow and decrease aqueous production	10-20% Maximum effect: 1-4 hours	Two to three times daily	Second	• Allergic reactions • Hyperaemia • Burning/stinging • Foreign body (FB) sensation • Blurring	• Headache • Bradycardia • Decreased libido • Bronchospasm • Nausea	Patients receiving MAOIs **Precautions:** • Severe cardiovascular disease • May have loss of effect over time
Carbonic anhydrase inhibitors Topical: • Dorzolamide 2% (e.g. Trusopt) • Brinzolamide 1% (e.g. Azopt)	Decrease aqueous production	15-20% Maximum effect: 2 hours	Two to three times daily	Second	• Allergic reactions • Burning/stinging	• Headache • Bitter taste • Dry mouth • Nausea • Fatigue	• Allergy to sulfonamides • Severe renal impairment **Precautions:** • Corneal grafts • Endothelial dystrophy (may cause corneal oedema)
Systemic: • Acetazolamide 250mg (e.g. Diamox)	Decrease aqueous production	25-50%	Two to four times daily	Third		• CNS depression • Lactic acidosis • (N.B. Up to 50% of patients do not tolerate acetazolamide)	• Adrenal or respiratory failure • Sodium or potassium depletion
Cholinergics (miotics) • Pilocarpine 1%, 2% (e.g. Isopto Carpine, Pilocarpine minims*)	Increase aqueous outflow	15-20% Maximum effect: 5-4 hours	Three to four times daily	Third	• Eye ache/pain • Blurred vision • Myopic shift • Miosis • Rare - retinal detachment	• Headache • Nausea • Dizziness	• Uveitis • Iritis • Secondary glaucoma
Combination therapies • Combigan (brimonidine 0.2%/timolol 0.5%) • Cosopt (dorzolamide 2%/timolol 0.5%) • DuoTrav (travoprost 0.004%/timolol 0.5%) • Xalacom (latanoprost 0.005%/timolol 0.5%) • Ganfort (bimatoprost 0.03%/timolol 0.5%) • Azarga (brinzolamide 1%/timolol 0.5%) • Simbrinza (brinzolamide 1%/brimonidine 0.2%)	As for individual components	20-30%	Combigan: Twice daily Cosopt: Twice daily DuoTrav: Once daily Xalacom: Once daily Ganfort: Once daily Azarga: Twice daily Simbrinza: Twice daily	Second	As for individual components		As for individual components

*Preservative-free option
†Currently not available on the PBS

1. NHMRC Guideline for the Screening, Prognosis, Diagnosis, Management and Prevention of Glaucoma. Canberra, Australia, 2010.

LiGHT trial (2019): RCT treatment naïve patients. Compared SLT vs drops as first line treatment for OHT and glaucoma + MEASURED Health related QoL, cost, clinical effectiveness, safety

- At 3 years, 74% of SLT patients required no drops to maintain target IOP
- Glaucoma surgery none in SLT group vs 11 in drops group
- SLT more cost effective + can be repeated

Conclusion – SLT should be offered as a first-line treatment (not recommended for everyone, for patients who refuse invasive treatment)

Class	Efficacy By Class
Prostaglandins	Diurnal: 18 – 31% Nocturnal: 8.5 – 17%
Beta-Blockers	Diurnal: 20 -27% Nocturnal: Little or none
Alpha Agonists	Diurnal: 12.5 – 29% Nocturnal: Little or none
Carbonic Anhydrase Inhibitors	Diurnal: 13.2 – 22% 1 Nocturnal: Modest efficacy *Brinzolamide has additive nocturnal effect with Latanoprost*
Miotics (pilocarpine)	Diurnal:? Nocturnal:?

- Initiate **one** medication at a time
- Patient returns to assess efficacy: **4-6 weeks**
  - ★ Prostaglandins: 2-4 weeks
  - ★ Timolol: 4 weeks
  - ★ Alpha agonists & Topical CAI: 2 week

Consider eye one trial (although glaucoma is a bilateral condition even though asymmetric)
- Questions raised on validity of monocular trials
- Monocular trial significantly improves precision of the estimation of treatment effect

Modify medication ONLY IF: IOP too high/variable + not reaching target/worried it will increase risk of progression, ON progression, VF progression

Follow Up Examination

Comprehensive history at follow-up	Recommended	Communication
• How are **your eyes and visi**on? • Are you managing to **take your medication as discussed**? If not, what are the problems or difficulties that you face? • Is there anything about your **condition or your treatment plan** that you would like **explained**? • Do you have **other medical conditions**? If yes, have they been exacerbated recently? • Are you taking **any prescription or OTC**, if so what? • Do you have **plans to conceive**/are you already **pregnant**? If yes, do you plan to breastfeed? • **When did you last attend an eye examination** or have your condition monitored?	Interested in adherence + understanding of disease, not just response to drugs • Check **patient drop instillation** technique and adherence • Enquire about **tolerance and side effects** • **Visual acuity** measurement • **Applanation** tonometry (review against target IOP, adjust if indicated) • **Do not change the medication at first review unless** essential ALSO, consider: • Anterior segment examination • Gonioscopy, if indicated • Assessment of ONH/RNFL (volk, photos, OCT) • SAP – Visual field assessment	• Educate patient on their diagnosis, prognosis and treatment options for shared decision making • If treating with medication, advise patient of: + Costs + Side effects + Dosage + Instillation • Letter to GP • Referral to ophthal **within 4 months of initiating therapy**

Managing Glaucoma Subtypes

OBA legally requires optometrists:

- **Minimum:** IOP, Gonioscopy (AC angle), Threshold VF, DFE (OD examination)
- **Optional:** OD imaging & RNFL images

<table>
<tr><th>CONDITION</th><th>WORKUP</th><th>PROGNOSIS/MANAGEMENT</th></tr>
<tr>
<td>Ocular Hypertension (OHT)</td>
<td>A OHT patient will have high IOP but no structural damage (which means no functional VF defect either).

History and examination should be carried out as for glaucoma . Of particular note, consideration should be given to whether any systemic medication is being taken that might be influencing IOP, either upwards (e.g. steroids) or downwards (e.g. beta-blockers). Look for risk factors of developing glaucoma (from someone who already has OHT):
● Older age
● Central corneal thickness (CCT) → The risk is greater in eyes with low CCT and lower in eyes with higher CCT.
● Cup/disc (C/D) ratio → The greater the C/D ratio the higher the risk.
● African American race
● Males
● Heart disease
● Myopia
● Diabetes
● Family Hx of glaucoma
● Steroid</td>
<td colspan="2">When deciding on whether to start treatment it is important to take into account that it will be necessary to treat a large number of patients in order to prevent the development of glaucoma in a single individual. In general, only those at higher risk should be treated, although patient preference may be a decisive factor.

● Treatment options are the same as for POAG, although a less aggressive pressure-lowering approach is frequently taken, e.g. <u>alternate day</u> prostaglandin dosing and low-intensity selective laser trabeculoplasty have been proposed; filtration surgery is only occasionally indicated. Cataract surgery commonly results in a significant IOP reduction.
● Careful monitoring is a reasonable alternative in many circumstances: baseline visual fields and RNFL/disc imaging should be performed.

<u>NOTE:</u> <u>Must treat all IOPs ≥ 30 mmHg as there is a >40% 5-year risk of glaucoma</u>

<u>COMPLICATIONS:</u> OHT almost certainly increases the risk of retinal venous occlusion , an additional point to take into account when considering whether to start treatment.

Increased risk of CRVO → BVs more vulnerable, easier to break due to the pressure constantly changing.

<u>Follow up:</u>
When you <u>commence medical treatment</u>, follow up in 4-6 weeks (checking IOP response, side effects, adherence). If you are happy with the patient, then progress every 3-6 months. Advise patients to return sooner if they experience side effects that are out of the norms (we have to explain the side effects associated with each drug that we prescribe). This short review period allows if there has been little or no response the initial drug is withdrawn and another substituted. If there has been an apparently incomplete response another drug may be added or a fixed combination substituted. But keep in mind that sometimes it may be worthwhile to allow a further month or two of treatment before altering a regimen, as response may improve over time.

If NOT using drops, monitor in 6-12 months. If unhappy GOMO</td>
</tr>
<tr>
<td>Primary Open Angle Glaucoma

<u>Risk Factors:</u>
● High IOP
● Older age
● Race (X4 common in black individuals)
● Family Hx of glaucoma
● DM
● Myopia
● Contraceptive pill
● Vascular disease
● Translaminar pressure gradient
● Large optic disc area
● Ocular perfusion pressure

Prognosis: 5-10% lifetime chance of blindness (varies significantly)

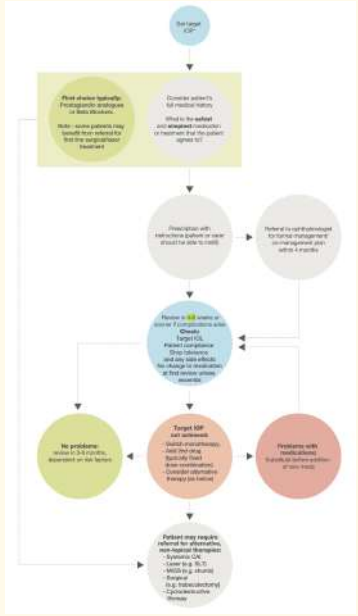
Poor prognostic factors include:
● Advanced disease at baseline
● Non-adherence to treatment
● African ethnicity</td>
<td><u>History</u>
● Visual symptoms will usually be absent, unless damage is advanced.
● Previous ophthalmic history (e.g myopia, trauma, previous inflammation, surgery)
● FOHx of POAG or related conditions like OHT
● Past medical history:
+ Asthma, heart failure or block, peripheral vascular disease: contraindications to the use of beta-blockers.
+ Head injury, intracranial pathology including stroke: may cause optic atrophy or visual field defects.
+ Vasospasm: migraine and Raynaud phenomenon.
+ Diabetes, systemic hypertension and cardiovascular disease may increase the risk of POAG.
+ Oral contraceptive pill for several years may be associated with an increased risk of glaucoma.
● Current medications:
+ Steroids including skin cream and inhalants.
+ Oral beta-blockers may lower IOP
● Allergies (see if allergic to sulfanomides)

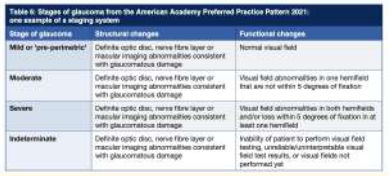
<u>Examination</u>
(1) Visual acuity is likely to be normal except in advanced glaucoma.
(2) Pupils → Exclude a relative afferent pupillary defect (RAPD); if initially absent but develops later, this constitutes an indicator of substantial progression.
(3) Colour vision assessment such as FM100/D-15 if there is any suggestion of an optic neuropathy other than glaucoma.</td>
<td colspan="2">Once a diagnosis of glaucoma is made, the severity of glaucoma should be determined by the degree of structural optic nerve damage and/or functional loss. Once the severity of glaucoma is established, this helps in setting the appropriate target IOP, selecting the best form of treatment (topical, laser or surgical), and guiding the follow-up and/or referral plan.

A reasonable initial treatment in a primary open angle glaucoma patient is to reduce IOP by 20-30% below baseline and to adjust up or down as indicated by disease course and severity (it is something that you tailor).

<table>
<tr><th>Stage</th><th>Target IOP</th><th>Specified level</th></tr>
<tr><td>High-risk suspects</td><td>20%</td><td>≤24mmHg</td></tr>
<tr><td>Early-moderate, low-risk</td><td>20%</td><td>≤19mmHg</td></tr>
<tr><td>Moderate, high-risk</td><td>30%</td><td>≤18mmHg</td></tr>
<tr><td>Advanced</td><td>30-50%</td><td>≤14mmHg</td></tr>
</table>

<table>
<tr><th>Higher percentage reduction in IOP</th><th>A lower percentage reduction in IOP may be chosen if:</th></tr>
<tr><td>● There is more severe optic nerve head damage
● Optic nerve head damage is progressing rapidly
● There are other risk factors present (e.g. family history, age, disc haemorrhage)</td><td>● The risk of aggressive treatment outweighs the benefit (e.g. if the patient does not tolerate medical/laser treatment or has high risks to surgical intervention)
● Patient has very mild disease/ if the patient's life expectancy is limited</td></tr>
</table>

 
Management = Drug (PGA, beta blockers etc) +</td>
</tr>
</table>

	Expect it to be normal. Can use red cap (if it shows you something NOT glaucoma as glaucoma B/Y). (4) Slit lamp examination →Exclude features of secondary glaucomas such as pigmentary and pseudoexfoliative. (5) Tonometry prior to pachymetry, noting the time of day → Esp. Asymmetry of 4mmHg or more (6) Gonioscopy. (7) Optic disc examination for glaucomatous changes should always be performed with the pupils dilated, provided gonioscopy does not show critically narrow angles. Red-free light can be used to detect RNFL defects. (8) Pachymetry for CCT. (9) Perimetry should usually be performed prior to clinical examination. (10) Imaging of the optic disc, peripapillary RNFL and/or ganglion cell complex, e.g. red-free photography, stereo disc photography, OCT, confocal scanning laser ophthalmoscopy and/or scanning laser polarimetry.	No practitioner can know the true target IOP for any particular patient prior to initiating glaucoma therapy. The appropriateness of the target is only revealed once structural and function stability is assessed over a number of years. If the patient continues to progress even at their target pressure, the target IOP needs to be lowered further until stability of structure and function occurs. Target IOP should be reviewed at follow up visits and reset and documented if required. **SLT process** In glaucoma, TM drainage is partly blocked hence pressure builds up. SLT uses a gentle laser to "wake up" or stimulate the drainage cells. This makes the eye's own drain work better, so fluid flows out more easily so pressure goes down. The laser doesn't burn or cut — it targets only pigmented cells, leaving the rest untouched (hence selective).	SLT  Table 6: Stages of glaucoma from the American Academy Preferred Practice Pattern 2021: one example of a staging system Stage of glaucoma / Structural changes / Functional changes Mild or 'pre-perimetric' / Definite optic disc, nerve fibre layer or macular imaging abnormalities consistent with glaucomatous damage / Normal visual field Moderate / Definite optic disc, nerve fibre layer or macular imaging abnormalities consistent with glaucomatous damage / Visual field abnormalities in one hemifield that are not within 5 degrees of fixation Severe / Definite optic disc, nerve fibre layer or macular imaging abnormalities consistent with glaucomatous damage / Visual field abnormalities in both hemifields and/or loss within 5 degrees of fixation in at least one hemifield Indeterminate / Definite optic disc, nerve fibre layer or macular imaging abnormalities consistent with glaucomatous damage / Inability of patient to perform visual field testing, unreliable/uninterpretable visual field test results, or visual fields not performed yet
Normal Tension Glaucoma Risk Factors: ● Age (patients tend to be older than the POAG) ● Gender (females) ● Race (NTG occurs more frequently in people of Japanese origin than in European or North American Caucasians) ● CCT is lower in patients with NTG than POAG. ● Family history. The prevalence of POAG is greater in families of patients with NTG than in the normal population. ● Systemic hypotension including nocturnal blood pressure dips of >20%, particularly in those on oral hypotensive ● medication. ● Obstructive sleep apnoea syndrome may be associated, perhaps via an effect on ocular perfusion. ● Autoantibody levels have been found to be higher in some groups of NTG patients by some investigators. ● Translaminar pressure gradient. This may on average be larger than in POAG. ● Ocular perfusion pressure may be relatively lower than in POAG. ● Myopia is associated with a greater likelihood of glaucoma and of its progression. ● Thyroid disease may be more common.	History and examination are essentially the same as for POAG but specific points warrant attention. ● Migraine and Raynaud phenomenon (reduced blood flow). ● Episodes of shock. ● Head or eye injury. ● Headache and other neurological symptoms (intracranial lesion). ● Medication, e.g. systemic steroids, beta-blockers. Notes of difference: (1) **IOP** is usually in the high teens,but may rarely be in the low teens. In asymmetrical disease the more damaged disc typically corresponds to the eye with the higher IOP. (2) The optic nerve head may be larger on average in NTG than in POAG. The pattern of cupping is similar, but acquired optic disc pits and focal nerve fibre layer defects may be more common. May also see drance haemorrhage (3) Visual field defects are essentially the same as in POAG although there is some evidence that they tend to be closer to fixation, deeper (greater extent of loss), steeper and more localized. In probably more than half of patients, field changes are non-progressive over a period of 5 years or more without treatment. (4) Other investigations are as for POAG although in selected patients the following can be considered. ● Assessment of systemic vascular risk factors. ● Blood pressure measurement can be used to calculate ocular perfusion pressure; ● Blood tests for other causes of non-glaucomatous optic neuropathy such as vitamin B 12, red cell folate, full blood count, erythrocyte sedimentation rate/C-reactive protein, treponemal serology including Lyme disease, serum angiotensin-converting enzyme level, plasma proteiN electrophoresis and autoantibody screen. ● Cranial MRI. ● Carotid Duplex imaging. ● Ocular blood flow assessment (e.g. laser flowmetry) may have useful clinical potential.	Further lowering of IOP is effective in reducing progression in many or most patients. However, as a large proportion of untreated patients will not deteriorate (approximately 50% at 5 years), in many cases progression should be demonstrated before commencing treatment. Exceptions include **advanced glaucomatous damage**, particularly if threatening central vision, and young age. Regular assessment including perimetry should be **performed at 4–6 monthly intervals initially.** Treatment Options: 1. **Medical treatment:** The alpha-2 agonist brimonidine may have a neuroprotective effect on the retina and optic nerve in addition to its IOP-lowering effect and may be superior to beta-blockers. Carbonic anhydrase inhibitors, particularly dorzolamide, may improve ocular perfusion. Prostaglandin derivatives tend to have a greater ocular hypotensive effect, which may be an over-riding consideration. Topical beta-blockers can have a dramatic effect on BP in a minority, and may contribute to nocturnal dips, though selective blockade (e.g. betaxolol) may actually have a beneficial effect on optic nerve perfusion. 2. **Laser trabeculoplasty**, particularly SLT, is a reasonable option to achieve IOP targets. 3. **Surgery** may be considered if progression occurs despite IOP in the low teens; antimetabolite enhancement of trabeculectomy is likely to be indicated in order to achieve a satisfactorily low pressure. 4. **Control of systemic vascular disease** such as diabetes, hypertension and hyperlipidaemia may be important, in order theoretically to optimize optic nerve perfusion. 5. **Systemic calcium-channel blockers** to address vasospasm have been advocated by some authorities. 6. **Antihypotensive measures.** If significant nocturnal dips in BP are detected, it may be necessary to reduce antihypertensive medication, especially if taken at bedtime. Non-selective topical beta-blockers in particular may cause a profound drop in systemic blood pressure in some individuals. Selected patients might be encouraged to increase their salted food intake, in consultation with the patient's cardiovascular physician. 7. **Neuroprotective agents** of proven benefit are not yet available; memantine is used to retard neuronal death in some CNS disorders, and its use has been adopted in glaucoma by some practitioners. **Ginkgo biloba** (40 mg three times daily) or an antiplatelet agent may confer some benefit in selected cases.
Primary Angle Closure Suspect	(1) Anterior segment OCT, ultrasound biomicroscopy or Scheimpflug photography may be useful to supplement gonioscopic findings and for patient education. (2) Gonioscopy (3) Anterior chamber depth measurement is helpful in some	Management: ● Consider provocative testing (trying to make them have a closed angle, sit in dark, lie flat forward) — when in dark pupil dilates which crowds angle, when u lie flat lens falls forward and that presses the angle) → Check IOP after ● Lens extraction may be considered for px with symptomatic cataract ● **Laser iridotomy** followed by observation	

 PAC – can be chronic, intermittent or even asymptomatic. Acute angle closure attack – is an event/clinical presentation.	cases. (4) Biometry if lens extraction is considered. (5) Posterior segment ultrasonography in atypical cases to exclude causes of secondary angle closure. (6) Provocative testing like dark room/prone provocative test like stimulating angle closure	+ If significant ITC persists after iridotomy, options include observation (most), laser iridoplasty, and long-term pilocarpine prophylaxis, e.g. 1% twice daily; provocative testing may be helpful in some patients. If symptomatic cataract is present, lens extraction usually definitively opens the angle. If IOP is elevated, then by definition PAC is present. • Laser iridoplasty • Long-term pilocarpine prophylaxis - 1% 2x/day <u>Follow up:</u> • 4-6 months → repeat perimetry <u>Prognosis:</u> • Risk of PACG at 5 years → 30%
Primary Angle Closure	(1) Anterior segment OCT, ultrasound biomicroscopy or Scheimpflug photography may be useful to supplement gonioscopic findings and for patient education. (2) Gonioscopy (3) Anterior chamber depth measurement is helpful in some cases. (4) Biometry if lens extraction is considered. (5) Posterior segment ultrasonography in atypical cases to exclude causes of secondary angle closure. (6) Provocative testing like dark room/prone provocative test like stimulating angle closure	• Management is as for PACS, but with a lower threshold for further intervention if angle widening is inadequate after iridotomy, particularly if IOP remains elevated. • Urgency and intensity of treatment, and frequency of review is tailored to the individual patient, taking into account IOP, extent of angle closure and glaucomatous damage, if present. • Medical treatment as for POAG may be required for eyes with substantial synechial closure or with persistently elevated IOP despite an opened angle. ***NOTE:*** In the context of PAC with IOP > 30 mmHg or PACG with IOP > 21 mmHg and no visually significant cataract, early LE (lens extraction) results in greater overall quality of life, lower IOP, and fewer medications or ongoing glaucoma interventions when compared to laser iridotomy, iridoplasty, and medical treatment.
Primary Angle Closure Glaucoma	With PACG, you can have 2 types: 1. **Appositionally close** → Reversible type of closure. Can be reversed with: • Laser peripheral iridotomy • Sometimes laser iridoplasty (if plateau iris is present) 2. **Synechially closed** → Iris is scarred to TM, laser will not peel it off. Goal is NOT to open the angle BUT: • Lower IOP • Prevent further PAS forming This can be done with medications, laser peripheral iridotomy, lens extraction and surgery (trabeculectomy).	***NOTE:*** In the context of PAC with IOP > 30 mmHg or PACG with IOP > 21 mmHg and no visually significant cataract, early LE results in greater overall quality of life, lower IOP, and fewer medications or ongoing glaucoma interventions when compared to laser iridotomy, iridoplasty, and medical treatment.
Acute primary angle closure	(1) Quick H&S, looking for: • Extreme pain • Blurry vision/sudden vision loss • Halos around lights • Nausea, vomiting • Headache (2) VA → Will be reduced from corneal oedema (3) SLE → Corneal oedemaoedema, shallow AC angle (4) IOP → Will be very high (5) Gonioscopy (will see closed angle – no view TM) MOA: pupillary block occurs when the iris is pushed against the lens, blocking the flow of aqueous humor from the posterior to the anterior chamber, which increases pressure in the posterior chamber and pushes the iris forward to block the trabecular meshwork. Plateau iris is caused by an abnormally positioned ciliary body that makes the peripheral iris shallow and prone to closure.	**EMERGENCY MANAGEMENT OF ACUTE PRIMARY ANGLE CLOSURE** 🚨 • Standard management of these cases is emergency referral to an ophthalmologist or hospital. HOWEVER, therapeutically qualified optometrists are able to (in collaboration with a medical practitioner), provide first aid. • Definitive intervention is laser therapy (laser peripheral iridonomy) First aid *unless* contraindicated: 4. **Oral acetazolamide 500mg** (2 tablets of **Diamox**) – If cannot keep down need to go to hospital for IV/intramuscular injection ***NOTE:*** Pharmacists can also supply this drug via a telephone order. Acute angle closure causes a lot of pain , so you need to make sure the patient keeps the drug down. 5. **Topical medications** (allowing ~2mins between each drop): a. One drop **beta blocker** e.g., **timolol 0.5%** b. One drop **alpha agonist** e.g., **apraclonidine 0.5%** c. One drop **carbonic anhydrase inhibitor** e.g., **brinzolamide 1%** d. If the eye is inflamed, one drop **high-penetrance steroid** **prednisolone 1%** with a further three drops at regular intervals in the first hour, then hourly thereafter 6. ONLY After discussion with an ophthalmologist can we give one drop **pilocarpine 2%** (since there are some instances in which it is contraindicated e.g when IOP is > 40 mmHg as then pilocarpine will not have an effect). ***NOTE:*** Pilocarpine should only be used for mechanisms involving phakic pupillary block or angle crowding. Use of pilocarpine is contraindicated for retrolenticular causes of angle closure (like thickened lens/cataract) as well as those involving aphakic or pseudophakic closure events.
Pseudoexfoliation glaucoma	**SAME AS POAG**	<u>Management:</u>

Patchy TM (as fibular material)	(1) Visual acuity is likely to be **normal** except in advanced glaucoma. (2) Pupils → Exclude a relative afferent pupillary defect (RAPD); if initially absent but develops later, this constitutes an indicator of substantial progression. (3) Colour vision assessment such as Ishihara chart testing if there is any suggestion of an optic neuropathy other than glaucoma. Expect it to be normal. Can use red cap. (4) Slit lamp examination →Exclude features of secondary glaucomas such as pigmentary and pseudoexfoliative. (5) Tonometry prior to pachymetry, noting the time of day → Esp. Asymmetry of 4mmHg or more (6) Gonioscopy. (7) Optic disc examination for glaucomatous changes should always be performed with the pupils dilated, provided gonioscopy does not show critically narrow angles. Red-free light can be used to detect RNFL defects. (8) Pachymetry for CCT. (9) Perimetry should usually be performed prior to clinical examination. (10) Imaging of the optic disc, peripapillary RNFL and/or ganglion cell complex, e.g. red-free photography, stereo disc photography, OCT, confocal scanning laser ophthalmoscopy and/or scanning laser polarimetry.	• Similar to POAG, but failure is common • Required laser trabeculoplasty, phacoemulsification, filtration surgery. • Annually gonioscopy <u>Prognosis:</u> • Worse than POAG • Likely to have ON damage, worse VF <u>Follow up:</u> • Within 3-6 months • Refer to opthal within 1 month
Pigment dispersion glaucoma On gonio will see heavily pigmented TM	**SAME AS POAG** (1) Visual acuity is likely to be **normal** except in advanced glaucoma. (2) Pupils → Exclude a relative afferent pupillary defect (RAPD); if initially absent but develops later, this constitutes an indicator of substantial progression. (3) Colour vision assessment such as Ishihara chart testing if there is any suggestion of an optic neuropathy other than glaucoma. Expect it to be normal. Can use red cap. (4) Slit lamp examination →Exclude features of secondary glaucomas such as pigmentary and pseudoexfoliative. (5) Tonometry prior to pachymetry, noting the time of day → Esp. Asymmetry of 4mmHg or more (6) Gonioscopy. (7) Optic disc examination for glaucomatous changes should always be performed with the pupils dilated, provided gonioscopy does not show critically narrow angles. Red-free light can be used to detect RNFL defects. (8) Pachymetry for CCT. (9) Perimetry should usually be performed prior to clinical examination. (10) Imaging of the optic disc, peripapillary RNFL and/or ganglion cell complex, e.g. red-free photography, stereo disc photography, OCT, confocal scanning laser ophthalmoscopy and/or scanning laser polarimetry.	<u>Management:</u> • Lifestyle change → avoid jolting forms of exercises (usually it could cause iris-lens rubbing that leading to the releasing of pigment) • Similar to Pseudoexfoliative glaucoma • Miotics may be beneficial → esp due to reduced iridozonular contact • But may exacerbate myopia, precipitating increased risk of RD <u>Follow up:</u> • Annual review → if low risk pigment dispersion syndrome (Without glaucoma • Refer to opthal within 1 month
Neovascular Glaucoma	**Other investigations:** FA may be helpful in confirming aetiology and delineating ischaemia. B-scan ultrasonography will help to exclude potential causes such as RD when the posterior segment view is impaired. Anterior segment OCT has been proposed as a useful tool for angle assessment. (1) SLE → Looking at AC, iris surface, pupillary margins (2) Gonioscopy → Looking for NV (3) IOP → May be normal early in the disease process, but is frequently extremely high later on. The anterior segment will often be congested once progression to elevated IOP has occurred.	(1) Review. Frequent review during high-risk periods is critical: the first few months following an ischaemic CRVO, and the first few weeks following diabetic vitrectomy. (2) Medical treatment of elevated IOP is as for POAG but **miotics should be avoided**, and prostaglandin derivatives used with relative caution due to their inflammation-promoting potential. • Topical atropine 1% once or twice daily will resist posterior synechiae and PAS formation, and topical steroids should be given if significant inflammation is present, watching for secondary raised IOP. • Steroids and atropine alone may be adequate if there is no visual potential. • Topical apraclonidine and oral acetazolamide may be useful temporizing measures; acetazolamide can be associated with renal dysfunction in diabetes, especially type 1, and should be used with caution in these patients. (3) Panretinal photocoagulation (PRP) (4) Goniophotocoagulation.

<table>
<tr>
<td>Inflammatory Glaucoma</td>
<td>Types of inflammatory glaucoma:
1. Angle-closure glaucoma with pupillary block:
2. Angle-closure glaucoma without pupillary block
3. Open angle glaucoma

Diagnosis:
- SLT: selusio pupillae, iris bombe and shallow AC
- Gonioscopy: angle closure from iridotrabecular contact, indentation may see</td>
<td>(1) Medical Tx: if the angle is completely open
(a) Beta-blocker is usually the 1st line treatment
(b) Anti-glaucoma drug can be used to lower IOP (Notes: some px may be sensitive to topical CAI)
(c) Long term steroid should be used with caution
(d) Contraindication: Miotic (as it increase vascular permeability and promote formation of posterior synechiae)
(2) Laser iridotomy (hole in iris to promote aqueous flow)
(a) Aims to prevent further pupil block
(3) Surgery
(a) Pre-operative preparation: control chronic uveitis for minimum of 3 months before surgery, preoperative topical steroid should be used as prophylaxis (oral prednisolone 0.5mg/kg/day)
(b) Trabeculectomy with mitomycin C enhancement/glaucoma drainage device implantation</td>
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<tr>
<td>Traumatic Glaucoma</td>
<td>For ANGLE RECESSION:
(1) Slit lamp examination may show signs of previous blunt trauma; these may be mild, such as a small sphincter rupture.
(2) Gonioscopy may initially reveal irregular widening of the ciliary body face</td>
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<tr>
<td>Hyphema</td>
<td>General:
• A coagulation abnormality, particularly a haemoglobinopathy, should be excluded.
• Any current anticoagulant medication should be discontinued after liaison with a general physician to assess the risk; NSAIDs should not be used for analgesia. Likewise, specialist advice should be sought regarding the management of a patient with a haemoglobinopathy, particularly before administering high-risk medication
• Hospital admission may be required for a large hyphaema.
• Strict bed rest is probably unnecessary, but substantially limiting activity is prudent, and the patient should remain in a sitting or semi-upright posture, including during sleep.

Medical:
• A beta-blocker and/or a topical or systemic CAI is administered, depending on the IOP; CAI are avoided in sickle haemoglobinopathies if possible. Miotics should also be avoided as they may increase pupillary block and disrupt the blood–aqueous barrier, and prostaglandins as they may promote inflammation. Alpha-agonists can be useful, but are avoided in small children and sickling disorders.
• Topical steroids should be used since they reduce inflammation and possibly the risk of secondary haemorrhage.
• Atropine is recommended by some authorities to achieve constant mydriasis and reduce the chance of secondary haemorrhage, but clear evidence is lacking.

Laser photocoagulation of angle bleeding points via a gonioprism has been described, though gonioscopy should probably be deferred for 5–6 days post-injury.

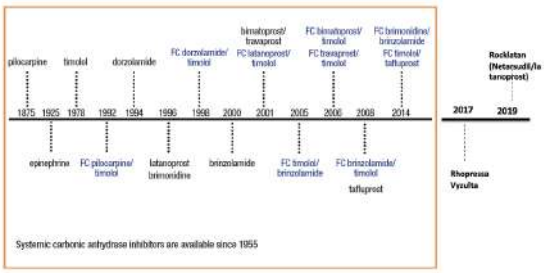
Surgical evacuation of blood is required in around 5%. If a total hyphaema or persistently intolerable IOP lasts for more than 5 days surgery should be considered to reduce the risk of permanent corneal staining (rare) and optic atrophy, and to prevent the occult development of peripheral anterior synechiae and chronic secondary glaucoma; a lower threshold is required in haemoglobinopathy patients (even moderate pressure elevation can lead to optic atrophy), patients with prior glaucomatous optic neuropathy and in young children with a risk of amblyopia. A glaucoma filtration procedure may be necessary in some cases. A bleeding point should be cauterized if possible.

On discharge the patient should be advised to avoid any activity with a risk of even minor eye trauma for several weeks; symptoms of a rebleed should prompt immediate review.</td>
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<tr>
<td>Angle Recession</td>
<td>• Long-term regular review of patients at risk is required.
• Medical treatment is as for other types of secondary open-angle glaucoma but is frequently unsatisfactory; laser trabeculoplasty is likely to be of little benefit.
• Trabeculectomy with adjunctive antimetabolite is generally effective.
• An artificial filtering shunt or cyclodiode should be considered if trabeculectomy fails.</td>
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WEEK 4: GLAUCOMA MEDICATIONS

LECTURE 5: Glaucoma Medications

History of Glaucoma Medications



- Glaucoma medications have been around since the late 1800s. The introduction of Timolol changed Glaucoma treatment in the 1970s, becoming the gold standard against which all glaucoma medications are compared to. Prostaglandin analogues revolutionized glaucoma treatment in the 1990s.
- The mid 2000s saw the introduction of fixed combination medications that helped improve compliance.
- In 2017, we saw the first new classes of glaucoma medications introduced (Rhopressa, Vyzulta) and Rolatan in 2019.

Theory Behind Glaucoma Therapy

- There are 5 major classes of topical glaucoma medications. These operate on the aqueous humour by either decreasing production or increasing outflow :

Prostaglandin Analogs (PGA)	Beta Blockers	Alpha Agonist	Carbonic Anhydrase Inhibitors (CAI)	Cholinergic Agonists (Miotics)

Topical Glaucoma Drugs MOAs and Adverse Affects

- All glaucoma medications have side effects, but beta blockers have the main systemic side effects.

Class	MOA	Significant/Common Side Effects
Alpha Agonists	↓ aqueous production ↓ outflow resistance (increases outflow)	Conjunctival hyperaemia, headache
Beta Blockers	↓ aqueous production	Bronchospasm, bradycardia, CHF (congestive heart failure)
Carbonic Anhydrase Inhibitors	↓ aqueous production	Paraesthesia, depression, anorexia
Mitotics	↑ aqueous outflow	Eye/brow pain, myopia, visual effects
Prostaglandin Analogs	↑ aqueous outflow (uveoscleral outflow)	Conjunctival irritation, hyperaemia

Trade Name	Generic Name	Concentration Bottle size	Formulation	Schedule	Pregnancy	PBS
Iopidine	Apraclonidine	0.5% [10mL]		S4	B3;	MP only
Betaquin Betoptic	Betaxolol	0.5% [5mL]	Solution	S4	C	MP, OP
Lumigan Lumigan PF	Bimatoprost	0.03% [3mL] (30 x 0.4mL)	Solution Unit dose available	S4	B3	MP, OP
Alphagan Alphagan P	Brimonidine	0.2% [5mL] 0.15% [5mL)	Solution	S4	B3;	MP, OP
Azopt BrinzoQuin	Brinzolamide	1% [5mL]	Suspension	S4	B3	MP, OP
Trusopt	Dorzolamide	2% [5mL]	Solution	S4	B3;	MP, OP
Xalatan	Latanoprost	0.005% [2.5mL]	Solution	S4	B3;	MP, OP
Isopto Carpine	Pilocarpine	1%, 2%, 4% [15mL]	Solution	S4	B3;	MP, OP
Saflutan	Tafluprost	0.0015% [30 x 0.3mL}	Solution Unit dose available	S4	B3;	MP, OP
Timoptol	Timolol	0.25% [2.5mL] 0.5% [5mL]	Solution	S4	C	MP, OP
Travatan	Travoprost	0.004% [2.5mL]	Solution	S4	B3	MP, OP

(1) Prostaglandin Analogs

General Overview:

Mechanism of Action (MOA)

MOA: Increase uveoscleral outflow of AH.
Prostaglandins are a group of lipid compounds derived from arachidonic acid. Within the eye, prostaglandins lower IOP by allowing for enhanced uveoscleral outflow.

Possible mechanisms include relaxation of the ciliary muscle and remodeling of extracellular matrix tissue within the ciliary body leading to increased aqueous outflow via this route.

1. Increase matrix metalloproteinases (MMP)
2. Decrease collagen and other ECM constituents
3. MMPs degrade ECM
4. MMPs reduce hydraulic resistance around muscle fibres

✅ Excellent 24-hour control of IOP
✅ Once daily dosage
✅ Few (if any) systemic side effects

Counselling
💬 These eye drops may slowly change the colour of your eye making the iris appear darker. This change is permanent and may be more noticeable when only one eye is being treated.

CATEGORY	TGA DESCRIPTION
A	Drugs which have been taken by a large number of pregnant women and women of childbearing age without any proven increase in the frequency of malformations or other direct or indirect harmful effects on the foetus having been observed.
B1	Drugs which have been taken by only a limited number of pregnant women and women of childbearing age, without an increase in the frequency of malformation or other direct or indirect harmful effects on the human foetus having been observed. Studies in animals have not shown evidence of an increased occurrence of foetal damage.
B2	Drugs which have been taken by only a limited number of pregnant women and women of childbearing age, without an increase in the frequency of malformation or other direct or indirect harmful effects on the human foetus having been observed. Studies in animals are inadequate or may be lacking, but available data show no evidence of an increased occurrence of foetal damage.
B3	Drugs which have been taken by only a limited number of pregnant women and women of childbearing age, without an increase in the frequency of malformation or other direct or indirect harmful effects on the human foetus having been observed. Studies in animals have shown evidence of an increased occurrence of foetal damage, the significance of which is considered uncertain in humans.
C	Drugs which, owing to their pharmacological effects, have caused or may be suspected of causing, harmful effects on the human foetus or neonate without causing malformations. These effects may be reversible. Accompanying texts should be consulted for further details.
D	Drugs which have caused, are suspected to have caused or may be expected to cause, an increased incidence of human foetal malformations or irreversible damage. These drugs may also have adverse pharmacological effects. Accompanying texts should be consulted for further details.
X	Drugs which have such a high risk of causing permanent damage to the foetus that they should not be used in pregnancy or when there is a possibility of pregnancy.

Side Effects

1. Conjunctival Hyperaemia:

- Have several ocular side effects. The most common one is conjunctival hyperemia (5-50% of patients).
- Caused by release of nitric oxide and neuropeptides causing vasodilation.
- Minimal - mild hyperemia is normally seen.
- Latanoprost is associated with a lower incidence of conjunctival hyperaemia when compared to travoprost and bimatoprost

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2. Hypertrichosis of eyelashes

- Can get an increase in the number of lashes, may become longer, thicker and darker.
- Probably due to vasodilation via opening the potassium channels in the peri-follicular vessels, thus stimulating follicular hair growth
- Usually apparent by 3rd month of treatment
- Less than <1% of patients complain about this side effect
- May also get **occasional trichasis**

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3. Perioribital darkening

- Most notable on the skin of eyelids.
- Fairly uncommon side effect that becomes apparent in a matter of months – although can take as long as 3 years to manifest.
- Is **reversible** within a few months of discontinuation of the PGA.

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4. Iris Pigmentation

- Risk varies depending on eye colour ; hazel and mixed patchy irides tend to have the highest risk. If you have homogenous green and brown eye there is less risk. Homogeneous grey or blue eyes have very little risk of this happening.
- Onset begins within a few months of commencing this treatment.
- Is **irreversible** and hence patients must be counseled of this risk (esp when we are only treating one eye).

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5. PGA Periorbitopathy

- Deepening of upper lid sulcus
- Might have upper eyelid ptosis.
- Involution of dermatochalasis (excess skin on upper or lower eyelid)
- Periorbital fat atrophy which can lead to mild enophthalmos (eye sunk into orbit)
- Inferior scleral show
- Increase prominence of lid vessels
- Tight eyelids
- Easy to miss if not looked at closely as often eyes are treated bilaterally.
- **Reversible** on discontinuation of PGA – partial to complete – within 4-6 weeks of discontinuation

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6. Inflammatory Complications

- Clinically detectable inflammatory side effects (onset or recurrence of uveitis, cystoid macular oedema or recurrence of herpetic keratitis) are complications that occur only rarely with PGA use.
- These side effects usually occur on eyes that are susceptible to inflammation because of previous or low-grade ongoing uveitis, herpetic keratitis or macular diseases associated with cystoid macular oedema (which can occur after complicated cataract surgery, or epiretinal membrane, retinal vein occlusion in the presence of an open or missing posterior lens capsule after cataract surgery)
- The time of onset of these side effects varies from one day to several months after starting treatment.
- All these side effects are **reversible** if the PGA analogue therapy is discontinued soon after the development of the symptoms
- To promote recovery topical corticosteroid or nonsteroid anti-inflammatory therapy is recommended in the short-term
- Use carefully with prior history of anterior uveitis, herpetic keratitis or macular disease associated with CMO.

DRUG NAME	OVERVIEW DETAILS	INDICATIONS/CONTRAINDICATIONS	SAFETY PROFILE (PRECAUTIONS/INTERACTIONS)	ADVERSE EFFECTS
Active Ingredient: Bimatoprost **Trade Names:** Lumigan, Lumigan PF ***SCHEDULE 4*** ✅On the OBA ✅ **PBS-UNRESTRICTED**, **B3**	**Concentrations:** 0.03% [3mL] **Preservative:** BKC or PF **Multi-Dose Solution, Unit dose available** **Dosage:** Adult, child 1 drop once daily at night	✅ OHT ✅ OAG (usually as 1st line therapy as a , monotherapy or as part of combination therapy) ❌Active intraocular inflammation (uveitis)	**Precautions:** • Intraocular inflammation (uveitis, iritis) • History of herpetic keratitis (may cause reactivation) • Aphakia, pseudophakia • Torn lens/capsule, • Known risk factors for macular oedema (as you will have an increased risk of developing macular oedema) 💊**Interactions:** Other PGA, NSAIDs eyedrops (NSAIDs will decrease the efficacy of the prostaglandin e.g ketorolac,diclofenac) ✅ Can be used with children ❌Avoid risk with pregnant women if possible **B3=** Taken by a limited number of pregnant women WITHOUT increased risk of malformations. AHM avoid use if possible.	**Common:** • Iris hyperpigmentation • Eyelash changes • Periorbital changes • Conjunctival hyperaemia • Eye irritation • Allergic conjunctivitis • Blurred vision • SPK • Blepharitis • HAs **Infrequent:** • Reversible macular oedema • Eyelid & conjunctival oedema • Uveitis **Rare:** Reactivation of herpetic keratitis ★ 3 – 36% chance of eyelash changes ★ 2.9% chance of periorbital darkening ★ 1.1-1.5% chance of iris pigmentation changes ★ More frequent chance of PGA periorbitopathy
Active Ingredient: Latanoprost **Trade Names:** Xalatan ***SCHEDULE 4*** ✅On the OBA ✅ Most commonly prescribed **PBS-UNRESTRICTED**, **B3** **ADVANTAGES:** • Fewer systemic side effects	**Concentrations:** 0.005% [2.5mL] **Preservative:** BKC **Multi-Dose Solution** **Dosage:** Adult, child 1 drop once daily at night	✅ OHT ✅ OAG (usually as 1st line therapy as a , monotherapy or as part of combination therapy) ❌Active intraocular inflammation (uveitis)	**SAME AS ABOVE**	**SAME AS ABOVE** ★ Less likely to have conjunctival hyperemia than **bimatoprost** & **travoprost** ★ 0 – 25% chance of eyelash changes ★ 1.5% chance of periorbital darkening ★ 5.1-10.1% chance of iris pigmentation changes
Active Ingredient: Tafluprost **Trade Names:** Saflutan ***SCHEDULE 4*** ✅On the OBA **PBS-UNRESTRICTED**, **B3**	**Concentrations:** 0.0015% [0.3mL single use] **Preservative FREE** **Dosage:** **Adult** 1 drop once daily at night **Solution, Unit dose available**	✅ OHT ✅ OAG (usually as 1st line therapy as a , monotherapy or as part of combination therapy) ❌Active intraocular inflammation (uveitis)	**Precautions:** • Intraocular inflammation (uveitis) • History of herpetic keratitis (may cause reactivation) • Aphakia, pseudophakia • Torn lens/capsule, • Known risk factors for macular oedema (as you will have an increased risk of developing macular oedema) 💊**Interactions:** Other PGA, NSAIDs eyedrops (NSAIDs will decrease the efficacy of the prostaglandin) ❌Avoid risk with pregnant women if possible **B3=** Taken by a limited number of pregnant women WITHOUT increased risk of malformations. AHM avoid use if possible	**SAME AS ABOVE**
Active Ingredient: Travoprost **Trade Names:** Travatan ***SCHEDULE 4*** ✅On the OBA ✅ **PBS-UNRESTRICTED**, **B3**	**Concentrations:** 0.004% [2.5mL] **Preservative:** Polyquaternium **Dosage:** Adult, child 1 drop once daily at night	✅ OHT ✅ OAG (usually as 1st line therapy as a , monotherapy or as part of combination therapy) ❌Active intraocular inflammation (uveitis)	**Precautions:** • Intraocular inflammation (uveitis) • History of herpetic keratitis (may cause reactivation) • Aphakia, pseudophakia • Torn lens/capsule, • Known risk factors for macular oedema (as you will have an increased risk of developing macular oedema) 💊**Interactions:** Other PGA, NSAIDs eyedrops ✅ Can be used with children ❌Avoid risk with pregnant women if possible	**SAME AS ABOVE** ★ 0.7 – 52% chance of eyelash changes ★ 2.9% chance of periorbital darkening ★ 1.0-3.1% chance of iris pigmentation changes

(2) Beta Blockers

- Beta-blockers reduce IOP by decreasing aqueous production, mediated by an effect on the ciliary epithelium. In approximately 10% of cases the response decreases with time (tachyphylaxis), sometimes within only a few days. There may be limited supplementary effect if a topical beta-blocker is added when a patient already takes a systemic beta-blocker; the combination may also involve a relatively high risk of systemic side effects.
- Beta-blockers should **not** be instilled at bedtime as they may cause a profound drop in blood pressure while the individual is asleep, thus reducing optic disc perfusion and potentially causing visual field deterioration; the IOP-lowering effect is also believed to be less marked during sleep, as nocturnal aqueous production is normally less than half the daytime rate.
- However, a beta- blocker may be preferred under some circumstances such as monocular treatment to avoid the cosmetic disadvantage of the asymmetrical periocular skin darkening and/or conjunctival hyperaemia with prostaglandins.
- Beta-blockers are also preferred in conditions such as ocular inflammation and cystoid macular oedema, or where there is a history of herpes simplex keratitis.

MOA: Reduce aqueous humour formation, probably by blockade of beta receptors on the ciliary epithelium. There are 2 types of beta blockers:

1. Non-Selective (**Timolol**)
2. Selective (**Betaxolol**)

NOTE: Remember the other systemic beta-blockers as many of your patients will be on this medications

DRUG NAME	OVERVIEW DETAILS	INDICATIONS/CONTRAINDICATIONS	SAFETY PROFILE (PRECAUTIONS/INTERACTIONS)	ADVERSE EFFECTS
Active Ingredient: Timolol Maleate **Trade Names:** Timoptol ***SCHEDULE 4*** ✅On the OBA ✅👨‍👧 as long as child >12 months **PBS-UNRESTRICTED, C** **NON-SELECTIVE BETA BLOCKER** ***NOTE:*** Treatment with a systemic beta-blocker reduces intraocular pressure (but less than with a topical beta-blocker). Adding a topical beta-blocker further reduces intraocular pressure (but this may be less effective than using an alternative class of topical agent). Risk of systemic adverse effects is increased. **ADVANTAGES:** • Good efficacy **DISADVANTAGES:** • Lots of systemic side effects (as non-selective) • Timolol lowers IOP during the diurnal/wake period but has little IOP-lowering efficacy during the nocturnal/sleep period (better to use **Brinzolamide** in combo with **Latanoprost**)	**Concentrations:** 0.25% [2.5mL], 0.5% [5mL] **Preservative:** BKC ***NOTE:*** There is no evidence of a clinically significant difference in efficacy between the two solution concentrations. **Solution** **Dosage:** Adult, child > 12months: 1 drop twice daily **NON-SELECTIVE** = Affects all beta receptors. • B1 receptors = Found in heart • B2 receptors = Found in pulmonary system **Timolol** is an extremely potent beta adrenergic blocking agent and is considered the gold standard of topical beta blockers as far as IOP control is concerned. Even nowadays FDA approval of a new glaucoma medication requires that its efficacy be measured against **Timolol's** efficacy. ***NOTE:*** Timolol binds to ocular melanin therefore, individuals with dark irises may need higher concentrations and ocular hypotensive action of the drug persists for at least 2 weeks after withdrawal in patients with chronic twice-daily use ➔ Need to do a 4 week washout if you discontinue chronic use.	✅ OHT ✅ Chronic OAG ✅ Aphakia with glaucoma ❌Bronchial asthma ❌Bronchospasm ❌Bradarrhythmia ❌2nd/3rd degree atrioventricular block ❌Reactive airways disease ❌Severe COPD ❌Sinus bradycardia ❌Overt cardiac failure ❌Cardiogenic shock	**Precautions:** • Diabetes • Hyperthyroidism • Cardiac failure • Depression • **Elderly** (systemic adverse effects more common) • **Children** (may cause apnoea in neonates and bradycardia in children) ***NOTE:*** A retrospective study of all glaucoma patients with obstructive pulmonary disease (OPD) who were hospitalised showed that 78.5% were tx with topical beta-blockers with majority being on non-selective beta-blockers These findings indicate that glaucoma patients with OPD are more likely to be either hospitalized or require an emergency room visit when treated with topical beta-blockers compared with other IOP-lowering agents Owing to the potential occurrence of these adverse events, caution must be exercised while treating patients with coexisting underlying medical conditions that may be aggravated by topical beta-blockers. ✅Breastfeeding (although adverse effects are unlikely, consider monitoring the infant) **Interactions:** 💊Adrenaline (epinephrine) 💊Catecholamine depletors 💊Other systemic beta-blockers; clonidine – high risk of systemic side effects 💊Ca channel blockers (**Verapamil** – use under specialist supervision only) ➔ Has an ADDITIVE effect with other beta-blockers, exercise caution with **Verapamil** (increase in effect) ➔ Avoid treatment with **Verapamil** due to the potential for profound bradycardia. **C =** Could be harmful, but AHM says can be used but need to monitor fetus (BUT FINE FOR PREGNANCY)	Allergy, punctate keratitis Non-selective beta-blockers have several side effects that affect the following systems: **General** • Lethargy • Weakness **Pulmonary** • **Bronchospasm** • Reduced pulmonary function **CNS** • Anxiety • Confusion • Decreased libido • Depression • Emotional lability • Hallucination • Psychosis **Cardiovascular** • **Bradycardia** • Inability to increase heart rate during exercise • Hypotension • **Worsening of congestive heart failure** • Heart block • Syncope **Common:** Stinging on instillation (VERY COMMON); bradycardia; blurred vision **Infrequent:** Decrease corneal sensation; hypotension; fainting; fatigue; confusion; hallucinations; bronchospasm

Active Ingredient: Betaxolol hydrochloride **Trade Names:** Betaquin, Betoptic ***SCHEDULE 4*** ✅On the OBA ✅ 👶 **PBS-UNRESTRICTED, C** Betaxolol hydrochloride was introduced in the early 1980s as the first topical cardioselective β1-adrenergic antagonist for the treatment of glaucoma **ADVANTAGES:** • By having a greater affinity for cardiac (β1) than pulmonary (β2) receptors, it has been reported to offer a reduced potential for pulmonary side effects, particularly in patients with pre-existing pulmonary diseases (e.g for COPD it is better to use **Betaxolol**) • Potential neuroprotective and blood flow properties **DISADVANTAGES:** • Reduced efficacy compared to **Timolol**	**Concentrations:** 0.25% suspension NOT ON PBS, 0.5% solution **Preservative:** BKC **Solution (0.5%)** **PRACTICE POINTS: Betaxolol 0.25% suspension** (which may be available through the SAS) **is as effective as betaxolol 0.5% solution**; use of suspension reduces local stinging, which may help compliance **Dosage:** Adult, child 1 drop twice daily ***NOTE:*** Betaxolol twice daily has a lower hypotensive effect than timolol. However, optic nerve blood flow may be increased due to a calcium-channel blocking effect, so that visual field preservation may be superior. Betaxolol is relatively cardioselective (beta-1 receptors), so causes less bronchoconstriction.	✅ OHT ✅ Chronic OAG (either as mono or combination therapy) ❌Sinus bradycardia > 1st degree block ❌Cardiogenic shock ❌Overt cardiac failure history	**Precautions:** • Cardiac failure • Heartblock history • Severe cardiac disease • Cerebrovascular insufficiency • CV disease **Interactions (SIMILAR TO ABOVE)** 💊Ca channel blockers (**Verapamil** – use under specialist supervision only) 💊Antiarrhythmics 💊Digitalis 💊Adrenaline 💊Catecholamine depletors 💊Other beta-blockers	**Common:** Stinging on instillation; bradycardia, blurred vision **Infrequent:** Decreased corneal sensitivity; hypotension; fainting; fatigue; confusion; hallucinations; bronchospasm

(3) Adrenergic Agonists

Non-Selective Adrenergic Agonist

DRUG NAMES	OVERVIEW DETAILS + MOA
Epinephrine **Propine**	**MOA:** Had various MOAS, either decreasing aqueous production or increasing outflow through the trabecular or uveoscleral outflow. • Rarely seen anymore (have been superceded by different drugs) • The relatively high frequency of adverse events associated with this agent limit its use in routine clinical care (makes it super unappealing) • Weak efficacy

Alpha Agonist: Side Effects

General	• Lethargy • Fatigue • Drowsiness • Dry mouth • Dry nose
Cardiovascular	Mild decrease in blood pressure
Cardiovascular	Allergic blepharoconjunctivitis – red eye with follicular reaction (less with Brimonidine)

Lethargy – More common in:
- Children
- Frail individuals
- Elderly
- Small individuals

Alpha Agonist

- Ocular alpha-2 receptor stimulation decreases aqueous synthesis via an effect on the ciliary epithelium, and increases uveoscleral outflow. There is probably a neuroprotective effect.
- They cross the blood–brain barrier and should be used with great caution in young children, in whom severe central nervous system (CNS) depression and hypotension been reported, and are contraindicated under the age of 2 years.
- They may potentiate vascular insufficiency. They should not be given with oral monoamine oxidase inhibitor antidepressants due to the risk of hypertensive crisis.

MOA: Suppress production of AH and increase uveoscleral outflow of AH
- **Alpha-1 receptors =** Smooth muscle, heart & liver
- **Alpha-2 receptors =** Iris, CB, retina, RPE, platelets, vascular smooth muscle, nerve termini, pancreatic islets

DRUG NAME	OVERVIEW DETAILS	INDICATIONS/CONTRAINDICATIONS	SAFETY PROFILE (PRECAUTIONS/INTERACTIONS)	ADVERSE EFFECTS
Active Ingredient: Apraclonidine hydrochloride **Trade Names:** Iopidine **SLIGHT SELECTIVITY FOR ALPHA-2 (Selective alpha-2-receptor sympathomimetics)** **PBS-RESTRICTED, B3** ✅On the OBA, but only MP ❌👶 **CHILDREN, PREGNANCY** **DISADVANTAGES:** • Limited use in clinical care of glaucoma	**Concentrations:** 0.5% [5mL or 10mL] solution or 1% ***NOTE:*** 1% (or 0.5%) is used principally to prevent or treat an acute rise in IOP following laser surgery on the anterior segment. The 0.5% concentration is typically used as a temporizing measure over the course of several	✅ Used during ocular surgery to prevent acute rise in IOP (MAIN USE) ✅ Short-term control of IOP (for ≤ 3 months) in glaucoma patients on max tolerated glaucoma therapy ❌Systemic clonidine hypersensitivity ❌ Concurrent monoamine oxidase inhibitors (MAOIs), tricyclic antidepressants (TCAs), systemic sympathomimetics	**Precautions:** • Severe cardiovascular disease • **AVOID IN PREGNANCY** **Interactions** 💊Additive effect with beta-blockers (GOOD THING WITH GLAUCOMA THERAPY) 💊Cardiac glycosides 💊CNS depressants (alcohol, opiates, sedatives, anaesthetics, TCAs)	High frequency (48%) of allergic reactions **Common:** Ocular irritation; dry mouth & nose; taste disturbances **Infrequent:** Mydriasis, conjunctival blanching, lid retraction **Rare:** Fainting, chest tightness, reduction in BP, HA

	weeks, such as whilst a patient is awaiting glaucoma surgery. It is generally not suitable for long-term use because of a loss of therapeutic effect over weeks to months and a high incidence of local side effects. **Solution** **Dosage:** Adult, 1 drop 2 or 3 times daily			
Active Ingredient: Brimonidine Tartrate **Trade Names:** Alphagan, Alphagan P **HIGHLY SELECTIVE FOR ALPHA-2** ❌ **CHILDREN** (ONLY GREATER THAN 12 YEARS) **PBS-UNRESTRICTED, B3** **ADVANTAGES:** • Favourable ocular & systemic safety profile • Additive with other glaucoma medications • Has neuroprotective effects – ability to preserve anatomic and functional integrity of RGCs other than just by reduction of IOP, thus preserving VF. ➔ Patients treated with **Brimonidine** significantly lower rate of VF defect progression compared to timolol (9% vs 30%) – **MAY BE INCONCLUSIVE** **DISADVANTAGES:** • **Brimonidine** is a relatively safe long-term IOP-reducing agent, although ocular allergy may lead to discontinuation in some individuals • Peak and trough effects. Has a reduced half-life: require tid (X3) dosage if sole agent ➔ The peak and trough effect is because **Brimonidine** has poorer efficacy during the nocturnal/sleep period. Research has found that while effective during the diurnal/wake period, it did not significantly lower IOP during the nocturnal/sleep period – thus exhibiting a lack of IOP lowering effect at night-time. The lack of IOP effect at night can be explained by failure to overcome a normal nighttime reduction of uveoscleral outflow.	**Concentrations:** 0.15% [5mL] or 0.2% [5mL] solution **Preservative:** BKC **Solution** **Dosage:** Adult, child > 12y: 1 drop twice a day (of either strength) BUT due to reduced half life, most likely dosed at x3 a day	✅ Used as a 2nd/3rd line agent for glaucoma (IOP reduction ~ 22%) ✅ Lowering IOP in chronic OAG or OHT (monotherapy/combination with beta-blocker) ❌Concomitant MAOI ❌Children younger than 2 years	**Precautions:** • Severe cardiovascular disease • **AVOID USE IN CHILDREN UNDER 12** **Interactions:** 💊Additive effect with beta-blockers (GOOD THING WITH GLAUCOMA THERAPY) 💊Cardiac glycosides 💊CNS depressants (alcohol, opiates, sedatives, anaesthetics, TCAs)	Allergic conjunctivitis is common. **Common:** Ocular irritation, ocular allergic reaction, conjunctival blanching, lid retraction, blepharitis, dry mouth & nose, taste disturbances, fatigue, HA, drowsiness, dizziness **Infrequent:** Systemic allergic reactions, depression, palpitations, systemic hypotension

(4) Carbonic Anhydrase Inhibitors (CAI)

- The carbonic anhydrase inhibitors (CAI) are chemically related to sulfonamide antibiotics. They lower IOP by inhibiting aqueous secretion, and via the topical route are used three times daily as monotherapy or twice daily as adjunctive treatment.
- In general, they are slightly less effective than beta-blockers but are hypothesized to have a supplementary neuroprotective effect.
- They may precipitate corneal decompensation in patients with corneal endothelial dysfunction, but some benefit has been reported in the treatment of cystoid macular oedema. Idiosyncratic bone marrow suppression can occur.
- Though cross-reaction is uncommon, topical (and systemic) CAI are relatively contraindicated in patients allergic to sulfonamide antibiotics. Research suggests that concomitant treatment with a topical and systemic CAI does not usually give an additive effect.

General Overview:

Mechanism of Action (MOA)	Side Effects (ORAL)	TOPICAL CAI
Carbonic anhydrase generates Na+ and HCO3- (bicarbonate) ions, allowing water to enter the ciliary epithelial cells, leading to aqueous humor production. Carbonic anhydrase inhibitors are small chemical compounds that inhibit the production of aqueous humor production by inhibiting the action of the enzyme carbonic anhydrase. ***1.*** Inhibit carbonic anhydrase II which reduces AH formation. AH secretion depends on the production of bicarbonate (from carbonic anhydrase II) – an isoenzyme found in non-pigmented ciliary body epithelium. Bicarbonate formation influences fluid transport so carbonic anhydrase II inhibitors reduce bicarbonate ion formation thus reducing fluid transport and IOP. ***2.*** Also proposed that CAI cause vascular dilation and increase blood flow by increasing tissue carbon dioxide concentrations and/or tissue pH. ***3.*** Suggested to increase ocular perfusion parameters. There are oral + topical forms of CAI: • Oral = **Acetazolamide, Methazolamide** • Topical = **Brinzolamide, Dorzalamide**	Most common side effects include fatigue , paresthesias of the face and extremities , metallic taste in the mouth, nausea, vomiting (associated with drug induced metabolic acidosis). Other side effects may include weight loss, dizziness or intestinal colic. **Pulmomonary**: Respiratory decompensation in COPD. **Gastrointestinal**: • Cramps • Diarrhoea • Epigastric burning • Metallic taste in mouth • Nausea • Vomiting **Renal**: • Nephrolithiasis • Renal failure **Hematologic**: • Acute leukopenia • **Agranulocytosis** (RARE – SUDDEN + IRREVERSIBLE) • **Aplastic anaemia** (RARE – SUDDEN + IRREVERSIBLE) • Neutropenia **Other**: • Hypokalemia (low level of potassium in blood serum) • Metabolic acidosis • Steven-Johnson syndrome Patients on oral Acetazolamide should be co-managed by a primary care provider. Patients should have occasional complete blood count with differentials done (especially if worrisome signs present like new onset fatigue or fever).	Considered a safe medication as they have fewer systemic side effects compared to oral. ✅ 2nd/3rd line drug for glaucoma treatment, but can be used as monotherapy ✅ Excellent efficacy when used in combination with other glaucoma medications ✅ Minimal side effects: • **Ocular side effects =** Hyperemia • **Systemic side effects =** Metallic taste ✅ Have demonstrated nocturnal IOP-lowering efficacy **Lotanoprost monotherapy VS Lotanoprost + Timolol VS Lotanoprost + Brinzolamide Study** **Results =** IOP in Brinzolamide treatment were lower than the other 2 treatments. This is because PGA derivatives improve the uveoscleral outflow but also increase the activity of carbonic anhydrase in the ciliary epithelium causing a secondary increase in AH secretion, thus slightly reducing the efficacy of PGA. Therefore, the topical CAI will decrease IOP by inhibiting the carbonic anhydrase and thus improving PGA efficacy.

DRUG NAME	OVERVIEW DETAILS	INDICATIONS/CONTRAINDICATIONS	SAFETY PROFILE (PRECAUTIONS/INTERACTIONS)	ADVERSE EFFECTS
Active Ingredient: Acetazolamide **Trade Names:** Diamox **B3** ❌On the OBA	**Concentrations:** 250mg (tablet) **ORAL** **Dosage:** 250-1000mg/24 hrs in divided doses	✅ Chronic OAG ✅ Acute ACG ✅ Idiopthaic intra-cranial hypertension ❌Adrenal or respiratory failure or metabolic acidosis ❌Sodium or potassium depletion ❌Renal disease ❌Hepatic impairment/cirrhosis	**Precautions:** • Gout – may worsen • Allergy to sulfonamides • **Elderly** (increased risk of adverse events) **Interactions:** 💊High dose **Aspirin** – Severe adverse effects including coma, anorexia, tachypnea and death 💊Lithium 💊Cyclosporine 💊Diuretics 💊Digoxin 💊Comittent **Metformin** – May have increased risk of poor glucose control and lactic acidosis	Up to 50% of patients do not tolerate it. BITTER TASTE + TINGLING FINGERS **Common:** Paraesthesia; fatigue; drowsiness; depression; decreased libido; bitter/metallic taste; nausea; vomiting; abdominal cramps; diarrhoea; black faeces; polyuria; renal stones; metabolic acidosis; electrolyte changes **Infrequent:** Transient myopia; bullous skin eruptions **Rare:** Aplastic anaemia; thrombocytopenia; agranulocytosis, neutropenia; anaphylaxis
Active Ingredient: Brinzolamide **Trade Names:** Azopt, BrinzoQuin **PBS-UNRESTRICTED, B3** ✅On the OBA ✅🤰 **ADVANTAGES:** • Brinzolamide better tolerated than dorzolamide (in terms of comfort) • Brinzolamide is similar to dorzolamide, but with a lower incidence of stinging and local allergy. It is a suspension, and a white residue may be left on the eyelids after instillation if excess is not	**Concentrations:** 1% [5ml] **Preservative:** BKC **TOPICAL (Suspension)** **SHAKE BOTTLE WELL!** **Dosage:** Adult, child, 1 drop twice daily	✅ OHT ✅ OAG ❌**PREGNANCY**	**Precautions:** • Corneal grafts → May cause corneal oedema and precipitate corneal decompensation. • Endothelial dystrophy → Corneal oedema and precipitate corneal decompensation. • Sulphonamides allergy → May increase risk of allergy to carbonic anhydrase inhibitors. **Interactions:** 💊Miotics 💊Oral CAI 💊Adrenergic agonists	**Common:** Ocular irritation; FB sensation; bitter taste **Infrequent:** Blepharitis; vision changes; corneal staining; GI disturbances; HA; paraesthesia; dizziness; dermatitis **Rare:** Allergic reactions (urticaria, angioedema, bronchospasm)

wiped away. • Add on IOP lowering efficacy during nocturnal period. • When considering an add-on treatment for patients already receiving latanoprost monotherapy, brinzolamide provides a more sustained lowering of IOP than timolol gel throughout the 24-hour day. • Brinzolamide lowers both daytime and nighttime IOP in these patients.				
Active Ingredient: Dorzolamide **Trade Names:** Trusopt PBS-UNRESTRICTED, B3 ✅On the OBA ✅👶	**Concentrations:** 2% [5ml] solution **Preservative:** BKC **TOPICAL (Solution)** **Dosage:** • iAdult, child, 1 drop three times a day (IF monotherapy) • 1 drop twice daily (IF adjunct to beta-blocker)	✅ OHT ✅ OAG ❌**PREGNANCY**	**Precautions:** • Corneal grafts → May cause corneal oedema and precipitate corneal decompensation. • Endothelial dystrophy → Corneal oedema • Sulphonamides allergy → May increase risk of allergy to carbonic anhydrase inhibitors. **Interactions:** 💊Oral CAI 💊Adrenergic agonists	The main adverse effects are stinging and a transient bitter taste following administration **Common:** Ocular irritation; FB sensation; bitter taste **Infrequent:** Blepharitis; vision changes; corneal staining; GI disturbances; HA; paraesthesia; dizziness; dermatitis **Rare:** Allergic reactions (urticaria, angioedema, bronchospasm)

(5) Miotics (Cholinergic Agonist)

- Miotics are cholinergic agonists that are now predominantly used in the treatment of angle closure, though were formerly a mainstay of the treatment of open-angle glaucoma. In angle-closure glaucoma, miotic-induced contraction of the sphincter pupillae pulls the peripheral iris away from the trabeculum, opening the angle.
- Miotics also reduce IOP by contraction of the ciliary muscle, which increases the facility of aqueous outflow through the trabecular meshwork.
- Local side effects include miosis, brow ache, myopic shift and exacerbation of the symptoms of cataract.
- Visual field defects appear denser and larger.
- Systemic side effects are rare but include confusion, bradycardia, bronchospasm, gastrointestinal symptoms and urinary frequency.

DRUG NAME	MOA	OVERVIEW DETAILS	INDICATIONS/ CONTRAINDICATIONS	SAFETY PROFILE (PRECAUTIONS/INTERACTIONS)	ADVERSE EFFECTS
Active Ingredient: Pilocarpine hydrochloride **Trade Names:** Isopto Carpine PBS-UNRESTRICTED, B3 ✅On the OBA ✅👶 **DISADVANTAGES:** • Due to topical side effects, drugs are RARELY used now. The use of cholinergic agents has declined in recent years with the availability of newer medications that have comparable efficacy and fewer side effects.	These drugs cause contraction of the ciliary muscle and scleral spur leading to mechanical opening of the trabecular meshwork and facilitation of aqueous humor outflow. The changes in the anterior chamber angle results from a dual responses to pilocarpine: 1. One is a forward movement of the iris plane, which results from increased resistance to the flow from the posterior chamber to the anterior chamber, accommodation, and increased outflow. 2. The other is contraction of the sphincter muscle of the iris, which leads to a decrease in total thickness and a pull on the root of the iris. The net response of the angle opening to pilocarpine is the combination of these opposite reactions: the **narrowing by forward movement of the iris plane** and **widening by contraction of the sphincter muscle of the iris.** ***NOTE:*** Meiosis itself has no effect on reducing the IOP.	**Concentrations:** 1%, 2%, 4% [15mL] **Preservative:** BKC ★ Usual concentration for glaucoma **1-4%** ★ Dark irides may benefit from **6%** ★ Higher concentration = greater side effects ★ **4%** can be used bid with beta-blocker **ORAL** ***NOTE:*** Heavily pigmented eyes; pigment can reduce pilocarpine's effect; a higher concentration or more frequent doses may be required. **Dosage:** • **Adult, child > 2 years:** 1 drop 1% pilocarpine 3 or 4 times daily • **Child 1 month – 2 years:** 1 drop 1% pilocarpine 3 times daily Pilocarpine 0.5%, 1%, 2%, or 4% solution as four times daily monotherapy is equal in efficacy to beta-blockers. Pilocarpine gel (Pilogel®) 4% is instilled once daily at bedtime so that induced myopia and miosis are predominantly confined to sleep. Gel, or drops twice daily, may be used to prevent angle closure following laser iridotomy in the presence of a substantial non-pupillary block element.	✅ Chronic OAG ✅ Acute angle-closure (narrow angle) Contraindications include acute inflammatory disease of the anterior segment and conditions where pupillary constriction is not recommended: ❌Anterior Uveitis ❌Iritis ❌Some forms of secondary glaucoma ❌ Previous retinal detachment ❌Pre-existing retinal damage ❌Young patients with myopia (increases risk of retinal detachment).	**Precautions:** • Previous retinal detachment	**Common:** HA, blurred vision; myopia; reduced vision in low light; constriction of VF; ocular irritation ***NOTE:*** Frontal headache (brow ache) due to ciliary spasm is common initially but usually improves after a few weeks; paracetamol may reduce pain. **Infrequent:** Lacrimation **Rare:** Retinal detachment; systemic effects (increased sweating; diarrhoea; bronchospasm; bradycardia)

(6) Fixed Combo Drugs

- Several patients do not respond to one type of medication and need a combination therapy for that fixed combination agents reduce IOP more effectively than either of their component medications used separately as monotherapy. You will use these after a single option doesn't work.
- Fixed combination drugs (whilst on the PBS) have restricted benefits – which means they have to meet certain clinical criteria before you can prescribe them on PBS.

FIXED COMBINATION	BRAND NAME	DOSAGE	PBS, PREGNANCY
Timolol + PGA	**Ganfort and Ganfort PF 0.3/5** **Timolol** 0.5% + **Bimatoprost** 0.03% **Xalacom, Xalamol** **Timolol** 0.5% + **Latanoprost** 0.005% **DuoTrav** **Timolol** 0.5% + **Travoprost** 0.004%	1 drop **once** daily	MP, OP C+ **PBS-RESTRICTED**; • Elevated IOP • Inadequately controlled with monotherapy • OAG/OHT
Timolol + CAI	**Azarga** **Timolol** 0.5% + **Brinzolamide** 1% **Cosdor, Cosopt → COSPOT PF NOT ON PBS** **Timolol** 0.5% + **Dorzolamide** 2%	1 drop **twice** daily	MP, OP C+ **PBS-RESTRICTED** (SAME AS ABOVE) • Elevated IOP • Inadequately controlled with monotherapy • OAG/OHT
Timolol + ALPHA AGONIST	**Combigan** **Timolol** 0.5% + **Brimonidine** 0.2%	1 drop **twice** daily	MP, OP C+ **PBS-RESTRICTED** (SAME AS ABOVE) • Elevated IOP • Inadequately controlled with monotherapy • OAG/OHT
ALPHA AGONIST + CAI	**Simbrinza** **Brimonidine** 0.2% + **Brinzolamide** 1%	1 drop **twice** daily	MP, OP B3 **PBS-RESTRICTED** (SAME AS ABOVE) • Elevated IOP • Inadequately controlled with monotherapy • OAG/OHT

KEY POINTS:

- Prostaglandin analogues (e.g. **latanoprost** 0.005% - **MOST COMMONLY PRESCRIBED**, **travoprost** 0.004%, **bimatoprost** 0.03%) increase uveoscleral outflow and are typically first-line due to high efficacy and once-daily dosing. ALSO less systemic side effects.
- Beta-blockers (e.g. **timolol** 0.5%, **betaxolol** 0.5%): Decrease aqueous production; dosed once or twice daily
- Alpha2-agonists (e.g. **brimonidine** 0.15%, **apraclonidine** 0.5%): Dual action i.e., reduce production and increase uveoscleral outflow; dosed two to three times dally
- Carbonic anhydrase inhibitors (e.g. **dorzolamide** 2%, **brinzolamide** 1%): Decrease aqueous production; dosed two to three times dally.
- Oral **acetozalmide** can also be used in acute angle closure management (CAI)
- Miotics (e.g. **pilocarpine** 1%, 2%, 4%): Increase trabecular outflow; less commonly used due to side effects; dosed three to four times daily

Table 7: IOP medications available in Australia for the management of glaucoma

Preparations by Class	Mechanism of action	Efficacy	Daily dosage	Order of treatment choices	Ocular side effects	Systemic side effects	Contraindications
Prostaglandin analogues • Latanoprost 0.005% (e.g. Xalatan) • Travoprost 0.004% (e.g. Travatan) • Bimatoprost 0.03% (e.g. Lumigan#) • Tafluprost 0.0015% (e.g. Saflutan#)	Increase aqueous outflow	**25-35%** Maximum effect: 8-12 hours	Once daily (night)	First	• Increase in iris pigmentation • Darkening • Thickening & lengthening of eyelashes • Conjunctival hyperaemia • Periorbital pigmentation	Uncommon - may cause respiratory symptoms in susceptible individuals	No contraindications **Precautions:** • Intraocular inflammation (iritis, uveitis) • History of herpetic keratitis • Aphakia or pseudophakia (potential for macular oedema)
Beta-blockers Non-selective agents: • Timolol 0.25%, 0.5%, 1% (e.g. Timoptol, Nyogel, Timoptic#) • Selective agents: • Betaxolol 0.25%, 0.5% (e.g. Betoptic)	Decrease aqueous production	**20-25%** Maximum effect: 2 hours	One to two times daily	First	• Transient ocular discomfort • Blurred vision • Increased lacrimation • Foreign body (FB) sensation	• Headache • Bradycardia • Decreased libido • Bronchospasm • Nausea	• Sinus bradycardia • Overt cardiac failure history • Cardiogenic shock **Precautions:** • Asthma • Severe COPD (selective agents, i.e. betaxolol preferred)
Alpha2-agonists • Brimonidine 0.2%, 0.15% (e.g. Alphagan) • Apraclonidine* 0.5% (e.g. Iopidine)	Increase aqueous outflow and decrease aqueous production	**10-25%** Maximum effect: 1-4 hours	Two to three times daily	Second	• Allergic reactions • Hyperaemia • Burning/stinging • Foreign body (FB) sensation • Blurring	• Headache • Bradycardia • Decreased libido • Bronchospasm • Nausea	Patients receiving MAOIs **Precautions:** • Severe cardiovascular disease • May have loss of effect over time
Carbonic anhydrase inhibitors Topical: • Dorzolamide 2% (e.g. Trusopt) • Brinzolamide 1% (e.g. Azopt)	Decrease aqueous production	**15-25%** Maximum effect: 2 hours	Two to three times daily	Second	• Allergic reactions • Burning/stinging	• Headache • Bitter taste • Dry mouth • Nausea • Fatigue	• Allergy to sulfonamides • Severe renal impairment **Precautions:** • Corneal grafts • Endothelial dystrophy (may cause corneal oedema)
Systemic: • Acetazolamide 250mg (e.g. Diamox)	Decrease aqueous production	**25-30%**	Two to four times daily	Third		• CNS depression • Lactic acidosis • (N.B. Up to 50% of patients do not tolerate acetazolamide)	• Adrenal or respiratory failure • Sodium or potassium depletion
Cholinergics (miotics) • Pilocarpine 1%, 2% (e.g. Isopto Carpine, Pilocarpine minims#)	Increase aqueous outflow	**15-20%** Maximum effect: 3-4 hours	Three to four times daily	Third	• Eye ache/pain • Blurred vision • Myopic shift • Miosis • Rare - retinal detachment	• Headache • Nausea • Dizziness	• Uveitis • Iritis • Secondary glaucoma
Combination therapies • Combigan (brimonidine 0.2%/timolol 0.5%) • Cosopt (dorzolamide 2%/timolol 0.5%) • DuoTrav (travoprost 0.004%/timolol 0.5%) • Xalacom (latanoprost 0.005%/timolol 0.5%) • Ganfort (bimatoprost 0.03%/timolol 0.5%) • Azarga (brinzolamide 1%/timolol 0.5%) • Simbrinza (brinzolamide 1%/brimonidine 0.2%)	As for individual components	**20-35%**	Combigan: Twice daily Cosopt: Twice daily DuoTrav: Once daily Xalacom: Once daily Ganfort: Once daily Azarga: Twice daily Simbrinza: Twice daily	Second	As for individual components		As for individual components

Preservative-free option
†Currently not available on the PBS
‡Restrictive benefit: the condition must have been inadequately controlled with monotherapy

1. NHMRC. Guidelines for the Screening, Prognosis, Diagnosis, Management and Prevention of Glaucoma. Canberra, Australia; 2010.
2. MIMS Online [Internet]. Medical Information Management System. [cited 2018 December 12]. Available from https://www.mimsonline.com.au.

Preservative Free Therapy

- Ocular surface disease (OSD) is relatively prevalent among the elderly population and glaucoma patients in particular.
- The preservative agents contained in most topical IOP-lowering medical formulations may exacerbate pre-existing OSD leading to ocular symptoms as well as decreased medication compliance
- BKC is the most common ophthalmic preservative and also has been most implicated as a causative agent in exacerbation of OSD.
- Recent studies have suggested that BAK may also be neurotoxic to corneal tissues
- The recent availability of preservative-free topical glaucoma therapy has offered an important treatment option in this regard.
- Preservative free: IOP-lowering medicines include: **Tafluprost**, **Bimatoprost**, **Latanoprost**, **Timolol** and fixed combination **Dorzolamide/Timolol** (**Cosopt PF**)

Class	Active Ingredient	Brand Name
Prostaglandin Analogues	Bimatoprost 0.03%	Lumigan PF
	Tafluprost 0.015%	Saflutan
Beta-blockers	Timolol 0.25 or 0.5%	Timoptic in Ocudose*†
Miotics	Pilocarpine 2%	Pilocarpine minims*
Fixed Combinations	Bimatoprost 0.03% + Timolol 0.5%	Ganfort PF
	Dorzolamide 2% + Timolol 0.5%	Cosopt PF*

†Not available in Australia

* Not available on the PBS

New Drugs for Glaucoma

- No access in Australia yet, not on OBA.

DETAILS	MOA + INDICATIONS	SAFETY PROFILE + SIDE EFFECTS
Rho-Kinase Inhibitor		
❌On the OBA **Rhopressa** (**Netarsudil** 0.02%) **TOPICAL** **Dosage:** Once daily	✅ OAG ✅ OHT ✅May have role in secondary glaucoma like steroid-induced glaucoma (as steroids increase resistance to outflow through TM but Rhopressa works to decrease it, **MOA:** Has two mechanisms of action and two targets – targeting rho-kinase (ROCK) and a norepinephrine transporter (NET). 1. Rhopressa decreases actin-myosin contraction and reduces actin stress fibres and focal adhesions in the TM to improve the outflow of AH 2. Increase aqueous outflow through TM and decrease episcleral venous pressure **Netarsudil** increases aqueous outflow through the trabecular meshwork and decreases episcleral venous pressure by inhibiting the effect of Rho kinase on actin and myosin contraction. In the ROCKET clinical trials, once daily **Netarsudil** was found to be **non-inferior** to twice daily timolol, reducing IOP by an average of about 4mmHg. **ADVANTAGES:** ● Netarsudil may have a particularly important role in **treating patients with lower starting IOPs.** ● In addition to its action on the trabecular meshwork to increase outflow, **Netarsudil's** unique ability to lower episcleral venous pressure can achieve a target beyond the low teens, which is otherwise difficult to achieve with a venous back pressure in the 8-12mmHg range. HENCE **may be better for normal tensive glaucoma patients.** ● Likely second-line treatment	Has a favorable safety profile: ● Conjunctival haemorrhage **~15-20%** ● Conjunctival hyperemia **~ 20%** ● Corneal verticillate **~ 20%** **AKA vortex keratopathy,** is a whorl-like pattern of golden-brown or gray opacities in the cornea's inner layer. While clinical evidence on biomicroscopic exams is present within the first 4 weeks of treatment, corneal verticillata have not been found to be functionally or visually significant. Most corneal verticillata are resolved upon discontinuation of treatment.
Rocklanten ❌On the OBA (**Netarsudil** 0.02%**+ Latanoprost** 0.005%) **Dosage:** Once daily	✅ OAG ✅ OHT	**SAME AS ABOVE**
Nitric Oxide (NO)		
Vyzulta (**Latanoprostene bunod** 0.024%) ❌On the OBA	**MOA:** Dual action,combining the effects of a prostaglandin analog with the action of nitric oxide on the trabecular meshwork outflow. NO induces cell relaxation in the TM by activating the NO-cyclic guanosine monophosphate signalling pathway, which is thought to lead to a widening of the intercellular spaces in the TM, thereby increasing the conventional outflow Nitric oxide (NO) levels are known to be lower in glaucomatous eyes, and NO deficiency may lead to trabecular contraction and decreased outflow facility. Additionally, when instilled into the eye, NO diffuses to the trabecular meshwork to promote cell relaxation and increase outflow to lower IOP. ***NOTE:*** ● LUNAR & APOLLO trials: non-inferior to Timolol ● In the VOYAGER study, patients receiving latanoprostene bunod had an average of 1.23mmHg lower reduction in IOP compared to patients receiving latanoprost alone - significant as epidemiologic studies have shown that for every 1mmHg incremental decrease in IOP you can reduce the risk of VF loss by about 10% **Latanoprostene bunod** leverages these effects of NO by releasing the molecule upon metabolizing in the eye.	The safety profile of latanoprostene bunod is similar to that of other prostaglandin analogs, including increase iris and periorbital skin pigmentation and eyelash growth. **Caution** should be used in patients with a history of **intraocular inflammation or macular edema** (same effects as other PGAs). The most common adverse effects reported are conjunctival hyperemia (6%), eye irritation (4%), eye pain (3%), and pain at the instillation site (2%).

LECTURE 6: Neuro-opthalmology

Pupil Anomalies

- Diseases that affect connections between nerves and the brain, affect the visual pathway.
- Are sight + life saving conditions.

REFRESHER

The pupillary light reflex is a four neuron pathway:

Retina → Pretectal nucleus → Edinger-Westphal nuclei → Ciliary ganglion → Short ciliary nerve → Sphincter pupillae

Sensory Nerve: Connects each retina with both pretectal nuclei in the midbrain at the level of the superior colliculi

- **Nasal retina:** Fibres cross over at the optic chiasm and go to the opposite (*contralateral*) pretactal nucleus
- **Temporal retina:** Fibres stay on the same side (*ipsilateral*) and go to the same side pretectal nucleus

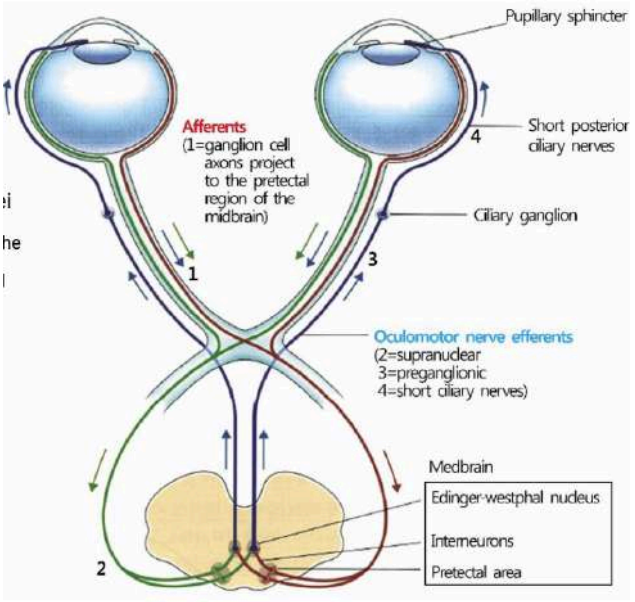
Internuncial Nerve : Each pretectal nucleus sends signals to both Edinger-Westphal (EW) nuclei in the brainstem. Connect pretectal nucleus to EW.

- Light in one eye causes both pupils to construct equally (direct + consensual pupillary reflex – routine test)
- Damage here can cause light-near dissociation e.g Argyll Robertson

Preganglionic Motor Nerve: Connects the Edinger-Westphal nucleus to the ciliary ganglion.

- Parasympathetic fibres pass through the oculomotor nerve, enter its inferior division to reach the ciliary ganglion.
- Hence why we can have pupillary involvement in CNIII palsy.

Postganglionic Motor Nerve: Leaves the ciliary ganglion and passes in the short ciliary nerves to innervate the sphincter pupillae, causing pupillary construction.



Near triad Near reflex

Combination of involuntary movements

1. Accommodation
2. Convergence
3. Miosis

- Vision is not a prerequisite, meaning that this near reflex is still initiated even if the patient cannot see the stimulus.
- There's **no known condition** where the light reflex is normal but the near reflex is absent
- Both the light reflex and near reflex share the same final pathway: CN III → ciliary ganglion → short ciliary nerves but the brain centre that controls the near reflex isn't well defined (likely to involve higher cortical areas + midbrain)
- Likely involves higher cortical areas (frontal and occipital lobes) and the midbrain centre

Identify common conditions that present with anisocoria:

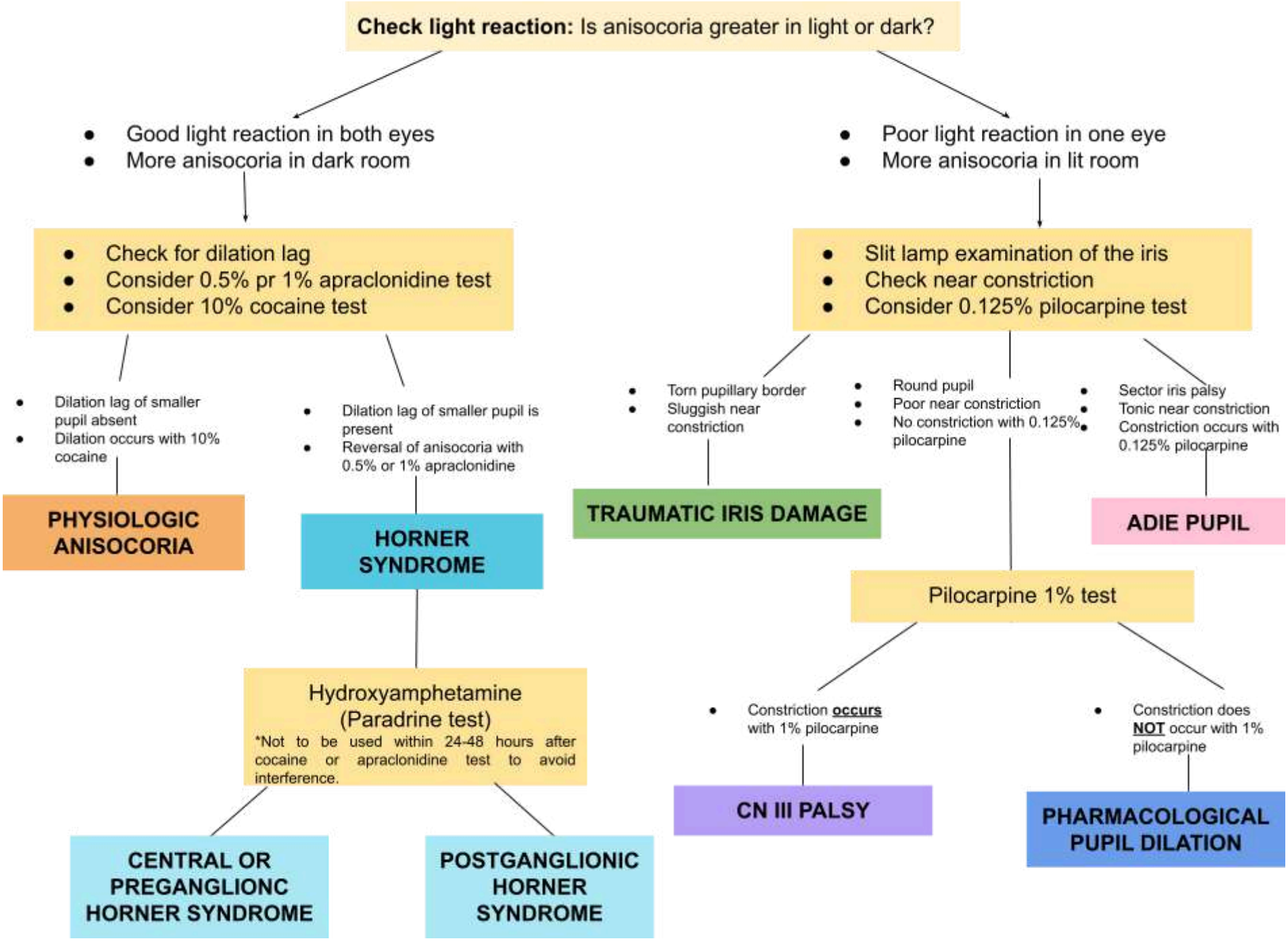
Anisocoria greater in LIGHT = Large pupil is abnormal

Anisocoria greater in DIM = Small pupil is abnormal

Abnormal pupil is constricted	Abnormal pupil is dilated
• Horner's Syndrome (mild ptosis on the side of the small pupil) • Argyll Robertson (syphilitic - always bilateral, irregularly round miotic pupils) • Iritis • Long-standing Adie's (pupil is initially dilated but over time constricts) • Unilateral exposure to a miotic agent	• Adie's (starts off dilated because constricted chronic) • CN III palsy (ALWAYS has associated ptosis) • Iris sphincter muscle damage from trauma or surgery (will see torn pupillary margin or iris transillumination effects on SLE). • Unilateral exposure to a mydriatic agent (**atropine**, **scoplamine** patch for motion sickness, ill fitting mask on nebulizer using **ipatropium bromide**, use of sympathetic medications e.g **pseudoephedrine**) • Physiological anisocoria (pupil size disparity sane in light + dark, pupil reacts normally to light, size difference is usually <2 mm in diameter). • Unilateral exposure to a mydriatic agent

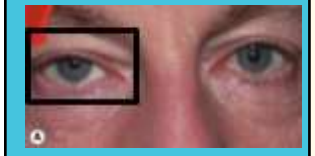
NOTE: Both = pharmacologically related

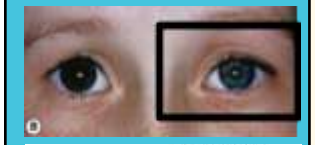
Anisocoria Workup



CONDITION	SIGNS/SYMPTOMS	WORKUP	MANAGEMENT + FOLLOW UP

Horner Syndrome
🚩NEW ONSET🚩







Is an **ocular sympathetic palsy** – deficiency in the sympathetic system (stress/fight and flight response). Losing the ability for pupil to dilate.

Symptoms: Ptosis and anisocoria. May have anhydrosis. Often asymptomatic.

KEY = Ptosis, miosis and anhydrosis

Mild ptosis (2mm) and lower eyelid elevation ("reverse ptosis") occur on the side of the small pupil.

NOTE: The upper eyelid drops due to the superior tarsal connection.

Other signs:
- Low IOP
- Anhydrosis (lack of sweating)
- Lighter iris in congenital cases
- Transient increase in accommodation (older patients hold card closer in Horner eye)
- Involved eye may have conjunctival hyperemia due to increased episcleral vascular tone

Pupil is 3 neuron pathway, which beginss at the posterior hypothalmus in the brain → goes down spine in the cilispinal centre of Budge → goes up the superior cervical ganglion (passing structures like the lung) and then runs parallel to the internal carotid back into the eye).

Anything along that pathway can cause Horner's syndrome (described as 1st, 2nd or 3rd order neuron cause)

Causes:
1. **First Order (Central):** Stroke (vertebrobasilar artery insufficiency or infarct) , tumour, multiple sclerosis
   + Uncommon (~10-15% with identifiable cause)
   + Usually associated with neurologic findings
2. **Second Order (Pre-ganglionic):** Trauma, tumour (e.g lung carcinoma metastasis, thyroid adenoma, neurofibroma), aortic aneurysm
   + Common (~44%)
   + Usually Hx of trauma/procedures OR lung/mediastinal tumour (as passing lung) – may have other symptoms if affecting the lungs
   + May have arm/shoulder pain
3. **Third Order (Post-ganglionic):** Headache syndrome, **internal carotid dissection**, VZV.
   + Common (~44%)
   + Wide range of causes (anything that runs parallel to internal carotid)
   + Ask re HA
   + Look for other neural involvement suggesting localisation to cavernous sinus, orbit etc
4. **Congenital** (may also be caused by birth trauma)

NOTE: MS is autoimmune, demyelinated nerve fibres. Immune system mistakenly attacks the myelination.

New-onset Horner syndrome requires diagnostic workup **ASAP** (URGENT, within 1-2 days at most) to exclude life-threatening causes e.g., internal carotid artery dissection which can present with: • Transient visual loss • Head/face/neck pain • Pulsatile tinnitus • Foul taste in mouth (dysgeusia)	Chronic Horner syndrome is more likely to be benign so less urgent (like congenial PIC)

History (direct to the underlying cause):
- Determine duration of Horner Syndrome from the patient's history and an examination of old photographs.
- Headaches, ipsilateral neck pain? – classic signs associated with carotid artery dissection
- Arm pain (goes hand in hand with lung cancer or tumour)?
- Previous stroke? – if had one before more likely to have it again
- Previous surgery, head or neck trauma that may have damaged the sympathetic chain, including cardiac, thoracic, thyroid or neck surgery?
- Neurodegenerative conditions – multiple sclerosis, tumours

PHYSICAL WORKUP
- Physical examination: check for supraclavicular nodes (above collar bone) looking for thyroid enlargement, or a neck mass
- Complete blood count (CBC) with differential (to help identify metastasis or tumours, provides you with indirect blood markers)
- CT or MRI of the chest to evaluate for possible lung mass e.g., Pancoast tumor
- MRI of the brain and neck
- Angiography of the head/ neck (AKA MRA/CTA) to evaluate for carotid artery dissection (especially with neck pain)
- Lymph node biopsy when lymphadenopathy (if abnormal lymph is present)

PHARMACOLOGICAL TESTS FOR HORNER'S SYNDROME

1. **Apraclonidine** 0.5% or 1% → **Reversal** of anisocoria (miotic pupil appears larger than the normal pupil) after 40-60 mins. MOST ACCESSIBLE ONE (more likely a specialist clinic)

2. **Cocaine** 10%
   - One drop in both eyes
   - Check after 15 mins
   - If no change in either pupil, repeat and recheck in 15 mins
   - Repeat until normal pupil dilates, can take up to 60 mins
   - Positive result if >1mm anisocoria (Horner pupil dilates less than the normal pupil)

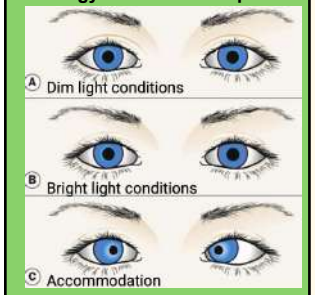
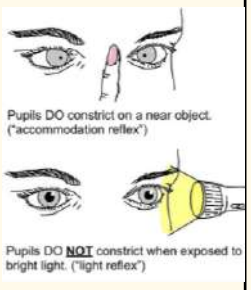
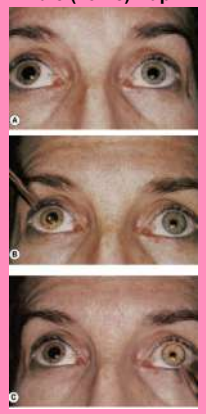
3. **Hydroxyamphetamine** 1%
   - Used to distinguish third order lesions from first and second order
   - Place one drop, check after 15 mins. If no change, repeat and recheck in 15 mins
   - Failure of the Horner pupil to dilate to an equivalent degree as the fellow eye indicates a third order lesion,
   - Sensitivity 93%, specificity 83% for a third order neuron BUT difficult to get from compounding pharmacies.

*Not to be used within 24-48 hours after cocaine or apraclonidine test to avoid interference.
NOTE: No anaesthetic drops beforehand + make sure corneal epithelium is intact.

- Treat underlying disorder
- Elective ptosis surgery
- No ocular complications requiring follow up

Refer to neuro-opthalmologist ideally

1. Workup acute Horner syndromes as soon as possible to rule out life-threatening causes. Neuroimaging should be performed relatively immediately for dissection. Refer urgently to neuro-opthalmologist. Remaining workup may be performed within 1-2 days.
2. Chronic Horner syndrome can be evaluated with less urgency. There are **no ocular complications that necessitate close follow up.**

Argyll Robertson Pupils  	**Symptoms:** Usually asymptomatic **Signs: Small, bilateral irregular pupils** that exhibit "light-near" dissociation (react poorly or not at all to light but constrict normally during accommodation/convergence). By definition, vision must be intact. Pupils dilate POORLY in darkness. ALWAYS bilateral but may be asymmetric. **Prostitute's pupil – Tertiary Syphilis**	• **Pupils:** Test the pupillary reaction to light and convergence. + To test the reaction to convergence, patients are asked to look first at a distance target and then at their own finger, which the examiner holds in front of them and slowly brings in towards their face. • SLE, DFE for accompanying interstitial keratitis, chorioretinitis, papillitis and uveitis – looking for accompanying signs of Syphilis. • Blood workup is to look for signs of Syphilis • Fluorescent treponemal antibody absorption or treponemal-specific assay and rapid plasma reagin or Venereal Disease Research Laboratory test for Syphilis  ***NOTE:*** • **Cocaine** 10% should dilate • With **pilocaprine** 0.125% neither pupil should constrict	Based on presence of active disease and previous treatment (penicillin therapy, cycloplegic, oral steroids – if syphilis related) Pupil findings alone are not an emergency and workup can be undertaken within a few days.
Adie (Tonic) Pupil  Considered a stable anomaly.	**Symptoms:** Difference in size of pupils, blurred near vision and photophobia. May be asymptomatic. **Signs:** An irregularly dilated pupil that has minimal or no reactivity to light. Slow, tonic constriction with convergence and slow redilation. May have vermiform iris movements and/or sectoral iris sphincter paresis. May have an acute onset and become bilateral. The involved pupil may become smaller than the normal pupil over time. **Causes:** • Idiopathic (MOST COMMON • Orbital trauma • VZV infection • Botulism • GCA **Dilated large pupil, slow + lazy (doesn't react), typically UNILATERAL + younger woman** **Chronic Adie's becomes little Old Ladies = Small pupil over time**	• Pupils and SLE observing for slow constriction or abnormal movements • Constricts 15 min after instilling **pilocarpine** 0.125% or lower concentrations (pilocarpine hypersensitivity – checking for cholinergic hypersensitivity) + An Adie pupil constricts, a normal pupil does NOT. • If bilateral, refer to GP to rule out other causes e.g., orbital trauma, varicella zoster, early syphilis etc	**Pilocarpine** 0.125% 2-4X a day may be considered for cosmesis and to aid in accommodation. Constriction of pupil can also help with depth of focus. If diagnosis is certain, routine follow up is sufficient (as stable anomaly).

REFRESHER

CN 1: Olfactory sensory-smell
CN 2: Optic sensory-vision
CN 3: Oculomotor motor-moves EOM (except SO and LR) and upper lids, pupil constriction and accommodation
CN 4: Trochlear motor-superior oblique
CN 5: Trigeminal mixed -chewing and head/face sensory
CN 6: Abducens motor-lateral rectus
CN 7: Facial mixed-facial muscles, salivation, lacrimal gland and taste CN 8: Vestibulocochlear sensory-balance and hearing
CN 9: Glossopharyngeal mixed -taste posterior tongue, swallowing, monitors carotid body
CN 10: Vagus mixed - swallowing, voice, elevates palate, communication with thorax area and abdomen viscera
CN 11: Accessory motor-shoulder lift/shrug and head turn
CN 12: Hypoglossal motor-tongue movement

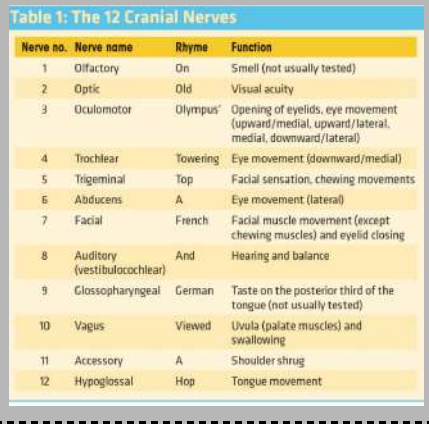
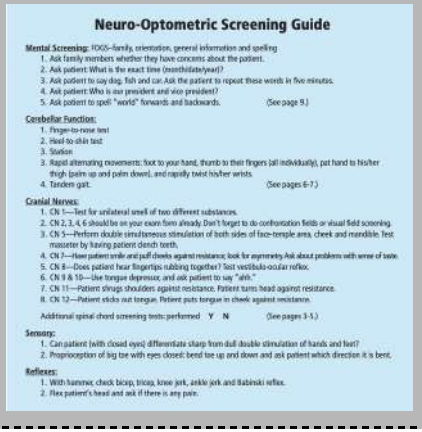


Table 1: The 12 Cranial Nerves

Nerve no.	Nerve name	Rhyme	Function
1	Olfactory	On	Smell (not usually tested)
2	Optic	Old	Visual acuity
3	Oculomotor	Olympus'	Opening of eyelids, eye movement (upward/medial, upward/lateral, medial, downward/lateral)
4	Trochlear	Towering	Eye movement (downward/medial)
5	Trigeminal	Top	Facial sensation, chewing movements
6	Abducens	A	Eye movement (lateral)
7	Facial	French	Facial muscle movement (except chewing muscles) and eyelid closing
8	Auditory (vestibulocochlear)	And	Hearing and balance
9	Glossopharyngeal	German	Taste on the posterior third of the tongue (not usually tested)
10	Vagus	Viewed	Uvula (palate muscles) and swallowing
11	Accessory	A	Shoulder shrug
12	Hypoglossal	Hop	Tongue movement



Neuro-Optometric Screening Guide

Mental Screening: FOGS–family, orientation, general information and spelling
1. Ask family members whether they have concerns about the patient.
2. Ask patient: What is the exact time (month/date/year)?
3. Ask patient to say dog, fish and car. Ask the patient to repeat these words in five minutes.
4. Ask patient: Who is our president and vice president?
5. Ask patient to spell "world" forwards and backwards. (See page 9.)

Cerebellar Function:
1. Finger-to-nose test
2. Heel-to-shin test
3. Station
3. Rapid alternating movements: fast to your hand, thumb to their fingers (all individually), pat hand to his/her thigh (palm up and palm down), and rapidly twist his/her wrists
4. Tandem gait. (See pages 6-7.)

Cranial Nerves:
1. CN 1—Test for unilateral smell of two different substances.
2. CN 2, 3, 4, 6 should be on your exam form already. Don't forget to do confrontation fields or visual field screening.
3. CN 5—Perform double simultaneous stimulation of both sides of face–temple area, cheek and mandible. Test masseter by having patient clench teeth.
4. CN 7—Have patient smile and puff cheeks against resistance; look for asymmetry. Ask about problems with sense of taste.
5. CN 8—Does patient hear fingertips rubbing together? Test vestibulo-ocular reflex.
6. CN 9 & 10—Use tongue depressor, and ask patient to say "ahh."
7. CN 11—Patient shrugs shoulders against resistance. Patient turns head against resistance.
8. CN 12—Patient sticks out tongue. Patient puts tongue in cheek against resistance.

Additional spinal chord screening tests: performed Y N (See pages 3-5.)

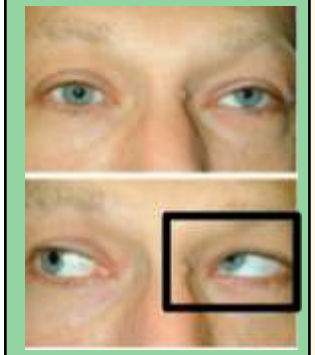
Sensory:
1. Can patient (with closed eyes) differentiate sharp from dull; double stimulation of hands and feet?
2. Proprioception of big toe with eyes closed: bend toe up and down and ask patient which direction it is bent.

Reflexes:
1. With hammer, check bicep, tricep, knee jerk, ankle jerk and Babinski reflex.
2. Flex patient's head and ask if there is any pain.

Ocular Motility Disorders

CONDITION	SIGNS/SYMPTOMS	WORKUP	MANAGEMENT + FOLLOW UP
Isolated CN3 Palsy **AKA OCCLUMOTOR** 🚩URGENT REFERRAL🚩   Ptosis, small pupil, exotropia, hypotropia	**Symptoms:** Binocular diplopia and ptosis; with or without pain. Pain does not distinguish between microvascular infarction and compression. **DOWN + OUT EYE in primary gaze** **Signs:** **1. External Ophthalmoplegia** • **Complete palsy:** Limitation of ocular movements in all fields of gaze except abduction • **Incomplete palsy:** Partial limitation of ocular movement • **Superior division palsy:** Ptosis and poor eye elevation • **Inferior division palsy:** Inability to abduct + depress the eye **2. Internal Ophthalmoplegia** • **Pupil involving (MORE COMMON):** A fixed, dilated, poorly reactive pupil • **Pupil sparing:** Pupil not dilated and normally reactive to light • **Reactive pupil sparing:** Pupil partially dilated and sluggishly reactive to light. Exotropia OR Hypotropia Ptosis, small pupil, exotropia, hypotropia Caused by: • Pupil involving + Common = Aneurysm + Less common = Tumour, trauma, congenital, cavernous sinus mass lesion, VZV, GCA, leukemia • Pupil sparing: **Ischaemic microvascular disease** (like DM), rarely cavernous sinus syndrome or **GCA** • Relative pupil sparing: Ischemic microvascular disease, less likely compressive Ischemic causes of CN3 palsy are primarily due to microvascular disease, often linked to systemic conditions like diabetes mellitus and hypertension. These conditions can damage the blood vessels supplying the nerve, leading to a lack of blood flow that causes damage to the central motor fibers of the nerve.	**History:** • Onset and duration of diplopia? • Recent trauma? • Pertinent medical history (e.g., diabetes, hypertension, known cancer or central nervous system [CNS] mass, recent infections) • If ≥55 years old, ask specifically about GCA symptoms **Complete ocular examination** Check for: • Pupillary involvement • Directions of motility restriction (in both eyes) • Ptosis • A visual field defect (visual fields by confrontation) – to look for ON involvement, suggesting it is NOT just an isolated CN3 palsy • Proptosis – resistance to retropulsion (start to suspect accompanying orbital disease e.g TED) • Orbicularis muscle weakness • Eyelid fatigue with sustained upgaze • Signs of aberrant regeneration Assess the other cranial nerves on both sides • Immediate CNS imaging to rule out mass/aneurysm is indicated for all third cranial nerve palsies whether it is pupil-involving or pupil-sparing. • **For suspected ischemic disease:** Check blood pressure, fasting blood sugar, and HbA1c (refer to GP) • Complete Blood Count (CBC) with differential in children • Ice pack test, rest test, or edrophonium chloride test if myasthenia gravis is suspected • Immediate ESR, CRP, and platelet count if GCA is suspected (would all be elevated).	**1.** Treat the underlying abnormality **2.** If causing diplopia, an occlusion patch or prism may be placed over the involved eye. ***NOTE:*** Patching is usually not performed in children <11 years old for risk of amblyopia. **3.** Strabismus surgery may be considered for persistent significant misalignment Varies depending on underlying disorder and stability of examination findings. • Co-management with GP, neurosurgery, and/or neurology necessary. • If secondary to ischemia: + Function should return within 3 months + Refer to GP for long-term management of vasculopathic disease risk factors • If pupil-involving and imaging/angiography are negative, lumbar puncture should be considered.
Aberrant Regeneration of CN3 	**Symptoms:** Binocular diplopia and ptosis; with or without pain **Signs:** The most common signs of aberrant CNIII nerve regeneration include: • Eyelid gaze dyskinesis: Elevation of involved eyelid on downgaze or adduction (Pseudo-von Graefe sign) • Pupil gaze dyskinesis: Pupil constricts on downgaze or adduction. Other signs may include limitation of elevation and depression of eye, adduction of involved eye on attempted elevation or depression. **Cause:** Thought to result from misdirection of the 3rd cranial nerve fibres from their original destination to alternate 3rd cranial nerve controlled fibres (e.g IR to the pupil). X3 possible etiologies: **1. Abberancy from congenital CN3 palsies (⅔ of patients)** **2. Abercancy from prior acquired CN3 palsies** (seen in patients recovering from CN3 palsy damage by trauma/compression by a posterior communicating artery aneurysm). **3. Primary aberrant regeneration:** Presence of condition without any primary history of CN3 palsy. Indicates presence of a progressively enlarging parasellar lesion like a carotid aneurysm or meningioma within the cavernous sinus. ***NOTE:*** Ischemic CN3 palsies DO NOT produce aberrancy, If aberrant regeneration develops in a presumed ischemic palsy, neuroimaging should be performed.	Workup depends on etiology: **(1) Abercancy from congenital CN3 palsies =** NONE **(2) Abercancy from prior acquired CN3 palsies =** Same as CN3 palsy **(3) Primary aberrant regeneration =** Neuroimaging to rule out slowly compressive lesion or aneurysm.	1. Treat the underlying disorder 2. Consider strabismus surgery if significant symptoms are present. Follow-up depends on etiology: **(1) Abercancy from congenital CN3 palsies =** Routine **(2) Abercancy from prior acquired CN3 palsies =** As per the underlying disorder identified in the workup **(3) Primary aberrant regeneration =** As per the neuroimaging + clinical examination findings. Patients instructed to return IMMEDIATELY for any changes (e.g. ptosis, diplopia, sensory abnormality).

Isolated CN4 PALSY
AKA TROCHLEAR



[PIC] Top: Left eye is hypertrophic, Bottom: On right gaze it gets worse (3 steps parks test is indicated)

Symptoms: Binocular **VERTICAL** (or oblique) diplopia, difficulty reading, sensation that objects appear tilted, may be asymptomatic.

Signs: Deficient inferior movement of an eye when attempting to look down + in. The 3 step test isolates a palsy of the SO muscle.

The involved eye is higher (HYPERTROPIC) in primary gaze. The hypertropia increases when looking in the direction of the uninvolved eye or tilt the head towards the ipsilateral shoulder. The patient maintains a head tilt TOWARDS the contralateral shoulder to eliminate diplopia.

Many of these will be long-standing.

NOTE: Actions of SO include intorsion, depression and abduction.

Causes:
- **Common:** Trauma, vascular infarct (often from DM or HTN), congenital or demyelinating disease.
- **Rare:** Tumor, hydrocephalus, aneurysm, GCA

Most common causes of CN4 palsy are trauma + stroke.

History:
- Onset and duration of the diplopia?
- Misaligned eyes or head tilt since early childhood?
- Trauma?
- Stroke?

(1) Examine **old photographs** to determine whether the head tilt is long-standing, indicating a chronic or congenital fourth cranial nerve palsy.
(2) Perform the double **Maddox rod test** if bilateral fourth cranial nerve palsies are suspected.
(3) Measure **vertical fusional amplitudes** with a vertical prism bar to distinguish a congenital from an acquired palsy.
(4) Ice test, rest test, or less commonly edrophonium chloride test if myasthenia gravis is suspected.
(5) CT of head and orbits (axial, coronal, and parasagittal views) for suspected orbital disease.
(6) Blood pressure measurement, fasting blood sugar, and HbA1c.
(7) Immediate ESR, CRP, and platelets if GCA is suspected.
(8) Brain MRI for:
+ A fourth cranial nerve palsy accompanied by other cranial nerve or neurologic abnormalities.
+ All patients <45 years of age with no history of significant head trauma, and patients aged 45 to 55 years with no vasculopathic risk factors or trauma

Perform the **three-step test:**
1. Perform the cover-uncover test to determine which eye is deviated upward in primary gaze. The higher eye comes down after being uncovered.
2. Determine whether the upward deviation is greater when the patient looks to the left or to the right.
3. Determine whether the upward deviation is greater when tilting the head to the left shoulder or right shoulder.

Expect hyperdeviation that is worse:
- On contralateral gaze (OPPOSITE gaze)
- When tilting the head toward the ipsilateral shoulder (SAME shoulder)
- In downgaze than in upgaze

Treat the underlying disorder.
- If causing diplopia, an occlusion patch or prism may be placed over the involved eye.
- Strabismus surgery may be considered for persistent significant misalignment.
  + Defer for at least 6 months after onset to allow for deviation stabilisation or for possible spontaneous resolution/ head turn modifications.

- **Congenital fourth cranial nerve palsy:** Routine.
- **Acquired fourth cranial nerve palsy:** As per the underlying disorder. If the workup is negative, the lesion is presumed vascular or idiopathic and the patient is reexamined in **1 to 3 months.**
- If **non-resolving in 3 months** or if an additional neurologic abnormality develops, appropriate neuroimaging is indicated.
- Instruct patients to return immediately for any changes e.g., ptosis, worsening diplopia, sensory abnormality, pupil abnormality.

Isolated CN6 Palsy
AKA ABDUCENS



[PIC] RE is not abducting fully

Symptoms: Binocular **HORIZONTAL** diplopia,worse for distance than near, most pronounced in the direction of the paretic lateral rectus muscle.

Signs: Deficient LATERAL movement of an eye with negative forced duction testing. NO proptosis.

Causes:

ADULT	CHILDREN
Common: • Vasculopathic (e.g., diabetes • Hypertension • Other atherosclerotic risk factors) • Trauma • Idiopathic.	Benign and usually self-limited after viral infection or vaccination, trauma, increased intracranial pressure (e.g., obstructive hydrocephalus), pontine glioma, and Gradenigo syndrome (petrositis causing sixth and often seventh cranial nerve involvement, with or without eighth and fifth cranial nerve involvement on the same side; associated with complicated otitis media).

(ADULT) History:
- Do the symptoms fluctuate during the day? – hints at MG rather than CN6 palsy
- Cancer, diabetes, or thyroid disease?
- Symptoms of GCA (in the appropriate age group)?

Complete neurologic and ophthalmic examinations:
➔ Because of the risk of corneal damage, it is especially important to evaluate the fifth cranial nerve. Corneal sensation (supplied by the first division) can be tested by touching a wisp of cotton or a tissue to the corneas before applying topical anesthetic.
➔ Ophthalmoscopy looking for papilledema is required because increased intracranial pressure from any cause can result in unilateral or bilateral sixth cranial nerve palsies.

Check blood pressure, fasting blood sugar, and HbA1c.

Brain MRI
Indicated for patients:
+ Younger than 45 years of age (if MRI is negative, consider LP).
+ Patients aged 45 to 55 years with no vasculopathic risk factors.
+ Sixth cranial nerve palsy accompanied by severe pain or any other neurologic or neuro-ophthalmic signs.
+ Any history of cancer.
+ Bilateral sixth cranial nerve palsies.
+ Papilledema is present.
- **Immediate** ESR, CRP, and platelet count if GCA is suspected.
- Consider Lyme antibody.

(CHILD) Workup:
- History: Recent illness or trauma? Neurologic symptoms, lethargy, or behavioral changes? Chronic ear infections?
- Complete neurologic and ophthalmic examinations as described for adults.
- Otoscopic examination to rule out complicated otitis media.
- MRI of the brain in all children.

Treat the underlying disorder.
- If causing diplopia, an occlusion patch or prism may be placed over the involved eye.

NOTE: In patients <11 years, patching is avoided, and these patients are monitored closely for the development of amblyopia.
- Strabismus surgery may be considered for persistent significant misalignment.

As with CN4 palsy:
- Reexamine **every 6 weeks** after the onset of the palsy until it resolves.
- If non-resolving in 3 months or if an additional neurologic abnormality develops, appropriate neuroimaging is indicated (MRI)
- Instruct patients to return **immediately** for any changes (e.g., worsening diplopia).
- MRI of the head is indicated if any new neurologic signs or symptoms develop, the abduction deficit increases, or the isolated sixth cranial nerve palsy does not resolve in 3 to 6 months.

Isolated CN7 Palsy



[PIC] Demonstrates lagopthalmos

Symptoms: Weakness or paralysis of one side of the face, inability to close one eye, excessive drooling.

Signs: Unilateral weakness or paralysis of the facial musculature.

- **Central lesion:** Weakness or paralysis of lower facial musculature only. Upper eyelid closure and forehead wrinkling intact.
- **Peripheral lesion:** Weakness or paralysis of upper and lower facial musculature.

Other signs include flattened nasolabial fold, droop of corner of the mouth, ectropion, and lagophthalmos (inability to close eyelids fully). May have ipsilateral decreased taste on anterior two-thirds of tongue, decreased basic tear production, or hyperacusis.

Causes: Also includes stroke, tumour

History

- Onset, duration, first episode or recurrence
- Associated symptoms: facial/ear pain, trauma, stroke, recent infection, hearing loss, tinnitus, vertigo
- PMHx: sarcoidosis, cancer

Examination

- General/Neuro: Review old photos for chronicity. Distinguish central vs peripheral palsy; assess other cranial nerves (V, VI, VII), motor/cerebellar signs, and taste (ant. ⅔ tongue).
- Ocular: Check motility, nystagmus, orbicularis strength, ectropion, Bell's phenomenon, cornea for exposure signs; perform Schirmer test; assess for uveitis.
- ENT: Inspect ear/oropharynx for vesicles or lesions; palpate parotid; assess hearing.

Investigations

- Imaging:
  + CT temporal bone if trauma
  + MRI/CT brain if neuro signs, cancer history, or symptoms >3 months

- Lab tests (as indicated):
  + Chest imaging + ACE for sarcoidosis
  + CBC, Lyme serology, syphilis tests if infectious cause suspected
  + Autoimmune markers (RF, ESR, ANA, ANCA) for collagen-vascular disease

Depends on etiology, treat underlying disease (mostly referral to neuro, e.g stroke, Ramsay-Hunt syndrome)..

For Bell's palsy, 86% of patients recover completely with observation alone within 2 months. Options for treatment include the following:

- Physical therapy with facial massage and/or electrical stimulation of facial musculature.
- For new-onset Bell palsy, steroids (e.g., **prednisone** 60 mg p.o. daily for 7 days, followed by a taper of 5 to 10 mg per day) have been shown to increase the likelihood of facial nerve recovery. Antiviral agents, in combination with prednisone, may be offered to patients although the benefit has not been well established.

The primary ocular complication of facial palsy is corneal exposure, which is managed as follows:

- **Mild exposure keratitis:** Artificial tears q.i.d. with lubricating ointment every night at bedtime (q.hs.)
- **Moderate exposure keratitis:** Preservative-free artificial tears, gel or ointment q1-2h or moisture chamber with lubricating ointment during the day; moisture chamber with lubricating ointment or tape tarsorrhaphy q.h.s. Consider a temporary tarsorrhaphy.
- **Severe exposure keratitis:** Temporary or permanent tarsorrhaphy. For expected chronic facial palsy, consider eyelid gold weight to facilitate eyelid closure.

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1. Recheck all patients at 1 and 3 months and more frequently if corneal complications arise.
2. If not resolved after 3 months, order MRI of the brain to rule out mass lesion.

In nonresolving facial palsy with repeatedly negative workup, consider referral to neurosurgeon or plastic surgeon for facial nerve graft, cranial nerve reanastomosis, or temporalis muscle transposition for patients who desire improved facial motor function.

Myasthenia Gravis





Hallmark is variability + fatigability feature.

Symptoms: Painless, droopy eyelid or double vision that is variable throughout the day or worse when the individual is fatigued; may have weakness of facial muscles, proximal limb muscles, and difficulty swallowing or breathing.

Signs: Worsening of ptosis with sustained upgaze or diplopia with continued eye movements, weakness of the orbicularis muscle (cannot close the eyelids forcefully to resist examiner's opening them). No pupillary abnormalities or pain.

Etiology: Autoimmune antibody-mediated disease; sometimes associated with underlying thyroid dysfunction. All age groups may be affected.

History:

- Do the signs **fluctuate** throughout the day and **worsen with fatigue**?
- Any systemic weakness?
- Difficulty swallowing, chewing, or breathing? – Recognise it is a systemic disorder (not just eyes– will deteriorate quite quickly)
- Medications (worsened by betablockers, macrolides)?

Assess for presence of fatigability:

- Measure the degree of ptosis in primary gaze.
  1. Have the patient focus on your finger in upgaze for 1 minute.
  2. Observe whether the ptosis worsens.
- **Assess orbicularis strength** by asking the patient to squeeze the eyelids shut while you attempt to force them open.
- Test pupillary function = **ALWAYS** normal pupils in myasthenia gravis

Blood test for acetylcholine receptor antibodies

In adults, **ice test**, **rest test**, or, if cardiac monitoring present, edrophonium chloride/neostigmine test may confirm diagnosis:

Ice test	An ice pack is placed over closed ptotic eye(s) for 2 minutes

Refer to a neurologist familiar with this disease.

Difficulty swallowing or breathing may require **urgent hospitalisation.**

- If the condition is mild, purely ocular, and is not disturbing to the patient, patient may simply patch one eye as needed (if need to concentrate or something) .
- If the condition is disturbing or more severe, an oral anticholinesterase agent such as pyridostigmine should be given.
  + Systemic steroids if symptoms persist.
  + **Azathioprine** (2-3 mg/kg/d) may be helpful in older patients.
  + Other medications to consider include **mycophenolate mofetil** and **cyclosporine**.
- **Treat any underlying thyroid disease** or infection.
- Surgical removal of the thymus can be performed.

Patients who have had their isolated ocular abnormality for an extended time (e.g., months) should be **seen every 4 to 6 months** and if proven to be stable, **every 6 to 12 months**.

Patients should always be warned to **return immediately if swallowing or breathing difficulties arise.**

After isolated ocular myasthenia has been present for **2 to 3** years, **progression** to **systemic involvement** is **unlikely**

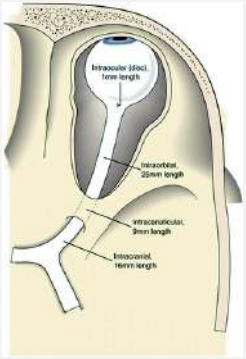
		<table><tr><td></td><td>**Improvement of ptosis** by at least 2 mm is a **positive test** for myasthenia gravis</td></tr><tr><td>**Rest test**</td><td>In children, observation for improvement immediately after a 1- to 2- hour nap is a safe alternative. A similar rest test (keeping eyes closed) for 30 minutes in adults may be similarly diagnostic.</td></tr></table> **Other workups:** • Check **swallowing and breathing function** as well as proximal limb muscle strength to rule out systemic involvement (GP will do this) • **Thyroid function** tests (including thyroidstimulating hormone) • **CT scan of the chest** to rule out thymoma • **Consider ANA,** rheumatoid factor, and other tests to rule out other autoimmune disease (refer immunologist or rheumatologist) • A **single-fibre EMG** including the orbicularis muscle may be performed **if other testing is negative and the diagnosis is still suspected**. May be the most sensitive test for involvement of the ocular muscles.	
Cavernous sinus and associated syndromes (Multiple Ocular Nerve Palsies)	**Anatomy** • Cavernous Simus are located on either side of the sella turcica. • Contains CN III, IV, V1, V2, VI, sympathetic fibres, and the internal carotid artery. • Close relationship explains why lesions here often affect multiple cranial nerves simultaneously. **Symptoms:** Double vision, eyelid droop, pain in the distribution of the V-1 and V-2 branches of the ipsilateral trigeminal nerve, or numbness. **Signs:** • Multiple ocular motor palsies: CN III, IV, VI → ophthalmoplegia, ptosis, diplopia. • Facial pain or numbness or both corresponding to first or second branches of the fifth cranial nerve (V1/V2) • Horner syndrome may occur (due to sympathetic involvement). • Proptosis, chemosis, venous congestion (due to impaired venous drainage). • Often unilateral, but may be bilateral if lesion crosses midline. ***NOTE:*** All signs involve the same side of the face when one cavernous sinus/superior orbital fissure is involved. **Causes** • **Vascular:** Cavernous sinus thrombosis, carotid–cavernous fistula, aneurysm. • **Inflammatory:** Tolosa–Hunt syndrome (idiopathic granulomatous inflammation). • **Neoplastic:** Meningioma, pituitary tumour extension, metastasis. • **Infectious:** Sinusitis (esp. sphenoid, ethmoid), facial infections (via venous spread).	• **History:** Diabetes? Hypertension? Recent significant head trauma? Prior cancer (including skin cancer)? Weight loss? Ocular bruit? Recent infection? Severe headache? Diurnal variation of symptoms? • CT scan (axial, coronal, and parasagittal views) or MRI of the sinuses, orbit, and brain. • Bloods / cultures if infectious. • Lumbar puncture if inflammatory suspected (e.g carcinomatous meningitis). • Lymph node biopsy when lymphadenopathy is present.	Treat underlying cause **1. Arteriovenous Fistula (Dural)** • Many close spontaneously or after arteriography; others need interventional neuroradiology. • IOP usually normalizes after closure. • Secondary glaucoma: Use aqueous suppressants (↑ episcleral venous pressure → outflow drugs less effective). **2. Metastatic Disease to Cavernous Sinus** • Oncology referral → systemic chemo ± radiotherapy. **3. Intracavernous Aneurysm** • Neurosurgical referral for work-up ± treatment

Chronic Progressive External Ophthalmoplegia	Chronic Progressive External Ophthalmoplegia happens because of problems with the mitochondria (the "energy factories" of cells), usually due to mitochondrial DNA mutations. ● The extraocular muscles rely heavily on energy, so when mitochondria don't work properly, these muscles gradually weaken over time. ● Result: Slowly progressive bilateral ptosis (droopy eyelids) and limitation of eye movements, without much pain or double vision early on. **Symptoms:** Slowly progressive, symmetric ophthalmoplegia (the weakness or paralysis of the muscles that move the eye and droopy eyelids) Almost never have diplopia. Usually bilateral; there is no diurnal variation; there may be a family history. **Signs:** Ptosis, limitation of ocular motility (sometimes complete limitation), normal pupils, and orthophoric.	**(1)** Careful history: Determine the rate of onset (gradual versus sudden, as in cranial nerve disease). **(2)** Family history. **(3)** Carefully examine the pupils and ocular motility. **(4)** Test orbicularis oculi strength. **(5)** Fundus examination: Look for diffuse pigmentary changes. **(6)** Check swallowing function. **(7)** Ice test, rest test, or edrophonium chloride test to check for myasthenia gravis. **(8)** Prompt referral to a cardiologist for full cardiac workup (including yearly electrocardiograms) if Kearns-Sayre syndrome is suspected. **(9)** If neurologic signs and symptoms develop, consult a neurologist for workup (including possible LP). **(10)** Lipoprotein electrophoresis and peripheral blood smear if abetalipoproteinemia suspected. **(11)** Serum phytanic acid level if Refsum disease suspected. **(12)** Consider genetic testing.	NO CURE! Associated abnormalities treated as follows: **1.** Treat exposure keratopathy with lubricants at night and artificial tears during the day. See 4.5, EXPOSURE KERATOPATHY. **2.** Single vision reading glasses or base-down prisms within reading glasses may help reading when downward gaze is restricted. **3.** In Kearns-Sayre syndrome, a pacemaker may be required. **4.** In oculopharyngeal dystrophy, dysphagia and aspirations may require cricopharyngeal surgery. **5.** In severe ptosis, consider ptosis crutches or surgical repair, but watch for worsening exposure keratopathy. **6.** Genetic counseling as needed. Follow up depends on ocular and systemic findings
Internuclear Ophthalmoplegia	Defined as ophthalmoplegia secondary to lesion in the medial longitudinal fasciculus (MLF). **Symptoms:** Double vision, blurry vision, or vague visual complaints. **Signs:** Weakness or paralysis of adduction, with horizontal jerk nystagmus of the abducting eye. **Causes:** MS, brainstem stroke, brainstem mass lesions	(1) History: Age? Are symptoms constant or only toward the end of the day with fatigue? Prior optic neuritis, urinary incontinence, numbness or paralysis of an extremity, or another unexplained neurologic event (concerning for MS)? (2) Complete evaluation of eye movement to rule out other eye movement disorders (e.g., sixth cranial nerve palsy, skew deviation). (3) Ice test, rest test, or edrophonium chloride test when the diagnosis of myasthenia gravis cannot be ruled out. (4) MRI of the brainstem and midbrain.	**1.** If an acute stroke is diagnosed, admit to the hospital for neurologic evaluation and observation. **2.** If concern for demyelinating disease, consider treatment recommendations as for optic neuritis **3.** Patients are managed by physicians familiar with the underlying disease.

Optic Nerve Disorders Other than Glaucoma

REFRESHER:
Entire optic nerve is **5cm** in total in length
Subdivisions of the optic nerve:
- **Intraocular (1mm) –** Only visible portion to us in fundoscopy (normally seeing 1.5-1.8 mm diameter of ONH)
- **Intraorbital (25-30mm) –** Extending from ONH to apex of the globe
- **Intracanlicular (5-9mm) –** Has myelin which flattens it up
- **Intracranial (5-16mm) –** Goes to optic chiasm

Classification of optic neuropathy by cause:
- Glaucomatous (recognise it is only 1 type of neuropathy – more common)
- Inflammatory (e.g. optic neuritis)
- **Ischaemic:** AION
- **Rarer:** Hereditary, nutritional and toxic causes
- Papilloedematous
- Traumatic
- Compressive
- Infiltrative (from tumours/infections e.g meningitis)



CONDITION	SIGNS/SYMPTOMS	WORKUP	MANAGEMENT + FOLLOW UP
Optic Neuritis  **TYPICALLY UNILATERAL** **Associated with MS** Typically young women, UNILATERAL vision loss, not supposed to see disc oedmea , YOUNG WHITE WOMEN Atypical is BILATERAL Hiccups is a sign of atypical, narcolepsy, nausea "Do you have uncontrollable hiccups?" More common to be retro	Optic neuritis is an inflammation of the optic nerve that can cause temporary vision loss, blurred vision, dimmed vision, and pain with eye movement. It occurs when the body's immune system mistakenly attacks the optic nerve, often triggered by infections, autoimmune diseases, or as a symptom of **multiple sclerosis (MS)** Immune mediated disorders of the CNS like neuromyelitis optica spectrum disorders (NMSOD) have also been described as causes of atypical optic neuritis. **Typical (MS-associated) Symptoms:** • Onset: Vision loss over hours–days; worst at ~1 week. • Severity: Mild → profound. • Usually unilateral, age 18–45. • Symptoms: Retro-orbital pain (↑ with eye movement), ↓ color vision, ↓ light perception. • May have neurologic symptoms (e.g., weakness, numbness), viral prodrome, Pulfrich phenomenon (when moving side-side seems like there is depth), or Uthoff sign (worse with heat/exercise e.g hot bath). • Common in young caucasian women **Atypical (e.g., NMOSD)** • Older age, more severe vision loss, poor recovery. • Often bilateral (simultaneous or rapid sequential). **Signs:** RAPD in unilateral/asymmetric causes, decreased CV, central, cecocentral, or arcuate visual field defects. Other signs include swollen disc (in one-third of patients) usually without peripapillary hemorrhages (papillitis most commonly seen in children and young adults) or a normal disc (in two-thirds of patients; retrobulbar optic neuritis more common in adults). Posterior vitreous cells possible.	**History** Enquire about: • Patient's age: younger 18-45 • UNILATERAL + rapid onset of visual loss • Previous episodes • Many will not have visual symptoms esp if unilateral • Motilities – **pain with eye movement**, double vision (hence why ask with Broad H) • **Uhthoff** (pain worse with exercise) and **Pulfrich** signs (sideways moving object tends to have depth) • Antecedent flu-like viral symptoms or MS symptoms – weakness, numbness, tingling in extremities **Check for:** • Reduced VA, CV depth changes, **red cap test**, motilities, pupils – RAPD in unilateral cases • VF testing – SAP (static automated perimetry) 30-2 — **central** field defect most common BUT anything can happen • **DFE including vitreous + ONH assessment** (as well as signs of swelling) **Workup performed by neuro-ophthalmologist:** • MRI of the brain + orbits • Check blood pressure • Consider blood tests including CBC, ESR, ACE level, Lyme antibody, FTAABS or treponemal-specific assay and RPR or VDRL tests and chest x-ray or CT ***NOTE:*** The **Optic Neuritis Treatment Trial (ONTT)** found steroid treatment reduced initial progression to clinically definite multiple sclerosis (CDMS) for 3 years. Steroid therapy only increases the rapidity of visual return but does not improve final visual outcome	ONTT – Optic Neuritis Treatment Trial **IF patient seen acutely with no prior history of MS or optic neuritis AND MRI shows 1+ typical area of demyelination: pulsed iv steroids within 14 days of decreased vision:** • **Methylprednisolone** 1 g/day i.v. for 3 days, then • **Prednisone** (orally) 1 mg/kg/day p.o. for 11 days, then • Taper **Prednisone** over 4 days (20 mg on day 1, 10 mg on days 2 through 4). • In addition, while being on steroids need prophylactic anti-ulcer medication (e.g., **Omeprazole** 20 mg p.o. daily or **Ranitidine** 150 mg p.o. b.i.d.) for gastric prophylaxis The reasoning behind this treatment is not to improve/regain vision but it does hasten the return of the vision. Does NOT improve final visual outcome, but does delta progression to systemic disease + hastens visual recovery. **If patient seen acutely with no prior history of MS or optic neuritis BUT MRI shows 2+ characteristic demyelinating lesions, TREATMENT AS ABOVE + possible systemic Tx of MS e.g.,:** • **Interferon-Beta** • **Glatiramer Acetate** • **Fingolimod** • **Dimethyl Fumarate** • **Terifluonomide** • **Alemtuzumab** • **Natalizumab** • **Ocrelizumab** • **Dalfampridine** The reasoning behind this treatment is that these patients have high risk of developing multiple sclerosis in the near future ***NOTE:*** **NEVER** use oral **prednisone** – needs to be IV on onset to avoid the rebound effect (recurrence) as found in the ONTT. NEEDS TO BE IV at the onset. **If patient seen acutely with no prior history of MS or optic neuritis AND MRI is negative (no white matter lesions lesions, risk of MS in next 5-10 years is low): still use pulsed iv steroids to hasten visual recovery or compromised fellow eye AND consider mimicking disorders like NMOSD:** NMOSD – *Neuromyelitis optica spectrum disorders* – spectrum of autoimmune diseases characterized by acute inflammation of the optic nerve and the spinal cord. Typical features include • "Non-recovering optic neuritis" • **Bilateral** • Spinal cord white matter lesions • Test aquaporin antibodies **Patient with a prior diagnosis of MS or optic neuritis are observed OR treated with pulsed iv steroids** • Re-examine at **4-6 weeks, 3-6 months thereafte**r (typically done by the neurologist/ neuro-ophthalmologist) • Repeat MRI • Evaluation and management of possible MS + other systemic causes
Papilloedema  **BILATERAL** [PIC] Blurred margins, peripapillary haemorrhages + CWS, no cupping, spontaneous venous pulsation, obscuration of small vessels, peyton;s folds. Flame shaped haemorrhages may occur. ***NOTE:*** In chronic papilledema, the hemorrhages and cotton-wool spots resolve, disc hyperemia disappears, and the disc becomes gray in color. VA fine UNLESS we are atrophy stage.	Papilledema is a term that is exclusively used when a disc swelling is secondary to increased intracranial pressure (ICP). **Symptoms:** Episodes of transient, usually bilateral visual loss (lasting seconds), often precipitated after rising from a lying or sitting position (altering intracranial pressure); headache; double vision; nausea; vomiting; and, rarely, a decrease in visual acuity (a mild decrease in visual acuity can occur in the acute setting if associated with a macular disturbance). Visual field defects and severe loss of central visual acuity occur more often with chronic papilledema. **Signs:** Bilaterally swollen, hyperemic discs (in early papilledema, disc swelling may be asymmetric) with nerve fiber layer edema causing blurring of the disc margin, often obscuring the blood vessels. Other signs include papillary or peripapillary retinal hemorrhages (often flame shaped); loss of venous pulsations (20% of the normal population do not have venous pulsations); dilated, tortuous retinal veins; normal pupillary response and color vision; an enlarged physiologic blind spot or other visual field defects by formal visual field testing. Causes include: • Primary metastatic intracranial tumours • Idiopathic intracranial hypertension • Meningitis • Brain abscess	• Enquire about related vision loss, neuro symptoms including HA, double vision, nausea, vomiting • VA, CV, red cap test, motilities, pupils, SAP, vitreous and DFE including ONH assessment • Physical examination and BP (BP to rule out hypertensive or severe hypertensive retinopathy since hallmark is optic disc swelling) • **Emergency (within 24hrs) MRI or CT** • Lumbar puncture with CSF analysis and opening pressure measurement (as it is optic disc swelling coming from increased intracranial pressure) • Consider FAF, B-scan US to rule out pseudopapilloedema ***NOTE:*** FAF helpful in workup, ONH drusen is common DDx will show HYPERfluoresce (glow) on FAF	Treatment directed at the underlying cause of the increased intracranial pressure: • Primary and metastatic intracranial tumors • Hydrocephalus • Idiopathic intracranial hypertension • Subdural and epidural hematomas • *Subarachnoid hemorrhage:* Severe headache, may have preretinal hemorrhages (Terson syndrome) • Arteriovenous malformation • *Brain abscess:* Often produces high fever and mental status changes. • *Meningitis:* Fever, stiff neck, headache (e.g., syphilis, tuberculosis, Lyme disease, bacterial, inflammatory, neoplastic) • *Encephalitis:* Often produces mental status abnormalities • Cerebral venous sinus thrombosis

Idiopathic Intracranial Hypertension (IIH)/ Benign intracranial hypertension



Will see same as above condition BILATERAL

A syndrome in which patients present with symptoms and signs of elevated intracranial pressure, the nature of which may be either idiopathic or due to various causative factors.

Symptoms: Headache, transient episodes of visual loss (typically lasting seconds) often precipitated by changes in posture, double vision, pulsatile tinnitus, nausea, or vomiting accompanying the headache. Occurs predominantly in obese women.

WIll have transient visual obsecuration (TVOs) - specific for increased intracranial pressure. Worse with bending, coughing, changing position (anything strenuous). Also associated with papilloedema. AMas can last longer (~ 30 mins) + uni/bilateral. TVO is bilateral .

The typical profile is **obese females.**

Signs:

- Papilledema due to increased intracranial pressure.
- Negative MRI/MRV of the brain.
- Increased opening pressure on LP with normal CSF composition.
- Unilateral/bilateral CN6 palsy

AS WITH PAPILLOEDEMA AND:

- Enquire also about recent weight gain, pregnancy, medications that can cause weight gain:
  - \+ OCP (oral contraceptives)
  - \+ Tetracycline
  - \+ Cyclosporins
  - \+ Vitamin A derivatives (accutane)
  - \+ Amiodarone
  - \+ Sulfa antibiotics
  - \+ Lithium
  - \+ Nalidixic acid
  - \+ Systemic steroids

WORKUP SAME AS ABOVE – EMERGENCY MRI (assuming tumour of brain bleeding).

In pseudopapilloedmea = No leakage
In papilloedema always leakage , which is why in MRI papilloedema shows enhancement/looks brighter as there is leakage. IVFA gold standard for optic disc oedema.

Idiopathic intracranial hypertension may be a self-limited process. Treatment is indicated in the following situations:

- ★ Severe, intractable HA
- ★ Progressive VA/VF loss
- ★ Presence of papilloedema

Ophthalmologist guides the workup of such treatment plans:
(1) Weight loss if overweight (since part of typical profile)
(2) **Acetazolamide** 250mg po qid, building up to 500-1000mg qid if tolerated
(3) Discontinue causative medications
(4) Consider short course of systemic steroids, or surgical interventions

If treatment by these methods is unsuccessful, consider a surgical intervention (mostly for rapidly progressive vision loss – fulminant):

- A **neurosurgical shunt** (ventriculoperitoneal or lumboperitoneal) or venous sinus stenting procedure should be considered if intractable headache is a prominent symptom.
- **Optic nerve sheath decompression surgery** or **neurosurgical shunt** (ventriculoperitoneal or lumboperitoneal) is often effective if vision is threatened.

- Depends on symptoms, severity of disc oedema and visual loss
- Every 3-4 weeks to every 3 months (documenting resolution of the disc swelling)
- VF test is the **most important method** for following these patients

1. If acute, patients can be monitored every 3 months in the absence of visual field loss. If chronic, initially follow patient every 3 to 4 weeks to monitor visual acuity and visual fields and then every 3 months depending on the response to treatment.
2. In general, the frequency of follow up depends on the severity of visual loss. The more severe, the more frequent the follow up.

Arteritic ischaemic optic neuropathy (Giant Cell Arteritis) – AAION



UNILATERAL

Different kind of swollen disc appearance tend to be chalky/white. It is not as red as NAION.

AAION are typically >55 years and will often have other systemic symptoms: HA, jaw claudication, scalp tenderness (when they brush their hair – along temporal artery), muscle and joint aches, anorexia, weight loss or fever. WHITE PEOPLE

Symptoms: Sudden, painless visual loss. Initially unilateral, but may rapidly become bilateral. Occurs in patients ≥55 years of age. Antecedent or simultaneous headache, jaw claudication (pain with chewing), scalp tenderness especially over the superficial temporal arteries (e.g., tenderness with hair combing), proximal muscle and joint aches (polymyalgia rheumatica), anorexia, weight loss, or fever may occur.

Signs: Afferent pupillary defect; visual loss (often counting fingers or worse); pale, swollen disc, at times with flame-shaped hemorrhages. Later, optic atrophy and cupping occur as the edema resolves. The ESR, CRP, and platelet count may be markedly increased.

Other: Visual field defect (commonly altitudinal or involving the central field); a palpable, tender, and often nonpulsatile temporal artery; and a central retinal artery occlusion or a cranial nerve palsy (especially a sixth cranial nerve palsy) may occur.

Polymyalgia rheumatica shoulder + neck stiffness and hip weakness. “Do you have trouble getting out of a chair?”. Especially in the morning – 50% of patients with GCA have this but 15% of people with PR have GCA

- Note age and enquire about:
  - \+ Pattern of vision loss
  - \+ Associated systemic symptoms
- Sudden dramatic, drop in VA, pupils (**RAPD**), DFE including ONH assessment
- Immediate ESR, CRP and platelets: Elevated in AAION
- Orbital ultrasound and brain MRI: to rule out any other presentations (not for work up of GCA, just trying to work out other causes of optic disc oedema)
- **Temporal artery biopsy (gold standard)** if GCA is suspected, within 1 week of starting treatment

Alternatives to temporal artery biopsy:

- **PET-CT scan:** sens 71% spec 91% NPV 98%
- **MRI/MRA:** sens 80%, spec 97%
- **Scalp doppler/colour doppler ultrasound:** looking for halo sign, sens 87%, spec 96%

Cupping is specific to **glaucoma**, compressive optic neuropathy (tumour) + GCA

RED FLAG PRESENTATION = Steroids started before a temporal artery biopsy is done.

A lot of the management for AAION is about risk mitigation (since such a red flag presentation). Without steroids (and occasionally on adequate steroids), the contralateral eye can become involved within 1-7 days.

✅ Immediate systemic/oral steroids as soon as GCA is suspected:
- ★ **Methylprednisolone** 250 mg IV q6h for 12 doses (AKA 3 days) followed by **Prednisone** 80-100mg orally daily

✅ If biopsy is positive for GCA:
- ★ Maintain **Prednisone** 1mg/kg po daily initially, tapered on an individual basis (lots of trial and error)

✅ If biopsy is negative for GCA:
- ★ **Steroids are usually discontinued** unless the clinical presentation is classic and a response to Tx has occurred
- ★ Biopsy of the contralateral artery may be performed to ENSURE there is no GCA.

ALSO CONSIDER while being on steroids need prophylactic anti-ulcer medication (e.g., **Omeprazole** 20 mg p.o. daily or **Ranitidine** 150 mg p.o. b.i.d.) for gastric prophylaxis

NOTE: The treatment for optic neuritis typically only lasts a few weeks, but for AAION patients, they are on therapy for a long time ~ 6-12 months. Will experience side effects (must be aware of this)

GCA IS AN OCULAR EMERGENCY = TREAT IMMEDIATELY (seen same day)

Oral steroids are maintained **until signs+ symptoms resolve and lab values normalise** then tapered repeating bloodwork with each dosage change and/or monthly to ensure adequate disease suppression

1. Patients suspected of having GCA must be evaluated and treated immediately.
2. After the diagnosis is confirmed by biopsy, the initial oral steroid dosage is maintained until the symptoms resolve and lab values normalize. The dosage is then tapered slowly, repeating bloodwork with each dosage change and/or monthly (whichever appropriate based on disease stability) to ensure that the new steroid dosage is enough to suppress the disease.
3. If the ESR or CRP increase or symptoms return, the dosage must be increased.
4. Treatment should last at least 6 to 12 months. The smallest steroid dose that suppresses disease is used.

Non-arteritic Ischaemic Optic Neuropathy (NAION)  **UNILATERAL**	Assumed that there is some sort of nocturnal dip in the blood pressure (which often the patient will have some predisposing vasculopathic risk factors as listed in next column) + fellow disc at risk (so small disc/small cup) Worry about the risk to fellow eye especially if there is a crowded disc. Unilateral but can be bilateral (mostly unilateral) Patients may be **younger** cf to AION. **Visual loss is often less severe**, does not have the accompanying symptoms of GCA listed previously, and usually has a **normal ESR and CRP**. **Signs:** Afferent pupillary defect, pale disc swelling (often segmental), flame-shaped hemorrhages, normal ESR and CRP. More likely in caucasian people as they have small ONH = more likely disc at risk. RARELY RECURS IN THE SAME EYE. 15% in the other eye in 5 years.	As with AAION i.e., **immediate** referral. GP to rule out • Cardiovascular disease • Diabetes • HTN • Sleep apnoea Ask about any **phosphodiesterase-5 inhibitors** e.g **sildenafil (Viagra)** –AFIL ending eg Tadalfil HTN is diurnal , BP lower at night. If you take BP medications at night = hypoperfusion of ON. People will notice that their vision is reduced in one eye in the morning. 1. Do you snore really loudly at night? 2. When do you take your BP medication?	Ischemic Optic Neuropathy Decompression Trial **(IONDT)** • Observation • OR cardiovascular risk factor modification to avoid that nocturnal hypotension episode that predisposes to the acute event. Includes strategies like: + Avoiding blood pressure medication at bedtime (liaise with GP), **avoiding phosphodiesterase-5 inhibitors** e.g **sildenafil (Viagra)** Follow up monthly for 3-6 months + advise patient about risk to the fellow eye
Posterior ischaemic optic neuropathy	Uncommon. **Symptoms:** Painless visual loss. Most commonly occurs in the postoperative setting at any time from upon awakening from anesthesia to 4 to 7 days thereafter. May be unilateral or bilateral, with a partial or complete deficit. **Signs:** Same as NAION, optic discs may appear normal initially in acute posterior ischemic optic neuropathy, but eventually pale disc edema, followed by pallor, develops. **Etiology:** Had predisposing surgery, obvious symptoms but no signs (as superior posterior and we can only see some of the ON)	**AS WITH NAION +** perimetric VF to document any VF loss	• Although no controlled studies exist for postoperative posterior ischemic optic neuropathy, it has been suggested that prompt blood transfusion with correction of hypotension and anemia may be beneficial and should be considered. • Treat any associated inflammatory or infectious causes • Monthly for **3-6 months**

KEY POINTS:

- Urgent referral saves sight: Sudden vision loss or disc swelling needs immediate action to prevent irreversible vision loss or systemic complications
- Treat the cause, not just the swelling: Management depends on the underlying diagnosis e.g., steroids for optic neuritis/aAION, weight loss and acetazolamide for IIH
- Think beyond the eye: Many optic nerve disorders need systemic workup and multidisciplinary shared care

Miscellaneous Optic Neuropathies

- Conditions that are far less common:

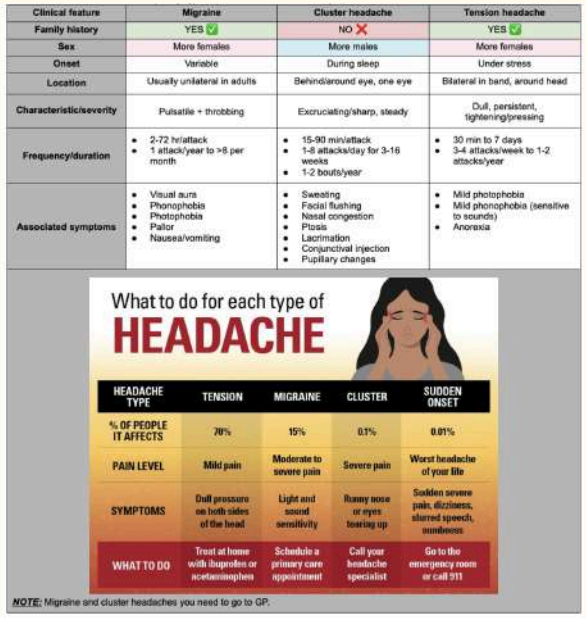
Toxic/Metabolic Optic Neuropathy **Signs:** Bilateral cecocentral or central visual field defects, signs of alcoholism, tobacco/substance use, certain medication use, heavy metal exposure, or poor nutrition. **Symptoms:** Painless, progressive, bilateral loss of vision. Visual acuity of 20/50 to 20/200, reduced color vision, temporal disc pallor, optic atrophy, or normal-appearing disc initially.	**Compressive Optic Neuropathy** **Signs:** Central visual field defect, relative afferent pupillary defect. The optic nerve can be normal, pale, or, occasionally, swollen; proptosis; optociliary (collateral) shunt vessels. Collateral vessels occur only with intrinsic lesions of the nerve (never with extrinsic lesions). **Symptoms:** Slowly progressive visual loss, although occasionally acute or noticed acutely. **Etiology:**	**Leber Hereditary Optic Neuropathy** **Signs:** Mild swelling of optic disc progressing over weeks to optic atrophy; small, telangiectatic blood vessels near the disc that do not leak on i.v. fluorescein angiography often present acutely; usually occurs in young men aged 15 to 30 years, and less commonly in women who are often older. Visual acuity 20/200 to counting fingers, cecocentral visual field defect.

Etiology: Associated with tobacco/alcohol or other substance abuse, severe malnutrition with vitamin B1, pernicious anemia (B12 malabsorption), toxic causes associated with medications e.g chloramphenicol, methanol, lead. **Workup:** **(1)** **History:** Drug or substance abuse? Medications? Diet? **(2)** Complete ocular examination, including pupillary assessment, dyschromatopsia evaluation, and optic nerve examination. **(3)** Formal visual field test. **(4)** CBC with differential and peripheral smear. **(5)** Serum vitamin B1, B12, and folate levels. **(6)** Consider a heavy metal (e.g., lead, thallium) screen. **(7)** If disc is swollen, consider blood test for Leber hereditary optic neuropath **Treatment:** ● **Thiamine** 100 mg p.o. b.i.d. ● **Folate** 1.0 mg p.o. daily. ● Multivitamin tablet daily. ● Eliminate any causative agent (e.g., alcohol, medication). ● Coordinated care with an internist, including vitamin B12 1,000 mg intramuscularly every month for pernicious anemia. **Follow Up:** Every month at first and then every 6 to 12 months	● Optic nerve glioma (age >20 years), associated with neurofibromatosis ● Optic nerve meningioma: Usually adult women. Orbital imaging may show an optic nerve mass, diffuse optic nerve thickening, or a railroad-track sign (increased contrast of the periphery of the nerve). ● Any intraorbital or intracranial mass (e.g., hemangioma, schwannoma). **Workup:** All patients with progressive visual loss and optic nerve dysfunction should have an MRI of the orbit and brain. **Treatment:** (1) Depends on the etiology. (2) Treatment for optic nerve glioma is controversial. These lesions are often monitored unless there is evidence of intracranial involvement, at which point surgical excision may be indicated. Most of these patients are young children, who are very susceptible to cognitive complications of radiotherapy. Chemotherapy may be considered if there is progressive visual loss. (3) For optic nerve sheath meningiomas, fractionated stereotactic radiotherapy treatment should be considered. Serial MRIs may be used to monitor tumor control.	**Symptoms:** Painless progressive visual loss in one and then the other eye within days to months of each other. Visual loss is bilateral at onset in approximately 25% of cases. **Transmission:** By mitochondrial DNA (transmitted by mothers to all offspring). However, 50% to 70% of sons and 10% to 15% of daughters manifest the disease. All daughters are carriers, and none of the sons can transmit the disease. **Workup:** Genetic testing **Treatment:** 1. Idebenone is used in Canada and the United Kingdom but is not available in the United States. Studies thus far have shown variable results regarding its effectiveness. Phase 3 clinical trials of gene replacement are ongoing. 2. Tobacco (or exposure to smoke) and alcohol avoidance are recommended. 3. Genetic counseling should be offered. 4. Consider cardiology consult because of increased incidence of cardiac conduction defects.
Dominant Optic Atrophy Mild-to-moderate bilateral visual loss (20/40 to 20/200) usually presenting at approximately age 4 years old. Slow progression, temporal disc pallor, cecocentral visual field defect, tritanopic (blue-yellow) color defect on Farnsworth-Munsell 100-hue test, strong family history, and no nystagmus. OPA1 and other mutations are responsible.	**Complicated Hereditary Optic Atrophy** Bilateral optic atrophy with spinocerebellar degenerations (e.g., Friedreich, Marie, Behr), polyneuropathy (e.g., Charcot-Marie-Tooth), or inborn errors of metabolism.	**Radiation Optic Neuropathy** Delayed effect (usually 1 to 5 years) after radiation therapy to the eye, orbit, sinus, nasopharynx, and brain with acute or gradual stepwise visual loss that is commonly severe. Disc swelling, radiation retinopathy, or both may be present. Enhancement of optic nerve or chiasm on MRI.

Headaches

- Most are not dangerous, but others may be life or vision threatening:
  - GCA
  - Acute ACG
  - Severe malignant hypertension
  - Increased ICP/papilloedema

CONDITION		WORKUP	MANAGEMENT + FOLLOW UP
Headache	**KEY =** Think about what is suggestive of a worse etiology VS common **Etiology:** **(1) Life Threatening:** ● **GCA:** Age ≥55 years. May have high ESR, CRP, and platelet count. Acute angle closure glaucoma: Decreased vision, painful eye, fixed mid-dilated pupil, and high intraocular pressure. ● **Ocular ischemic syndrome:** Periorbital eye pain. ● **Malignant hypertension:** Marked increase of blood pressure, often accompanied by retinal cotton-wool spots, hemorrhages, and, when severe, optic nerve swelling. Increased intracranial pressure: May have papilledema and/or a sixth cranial nerve palsy. Headaches usually worse in the morning and with Valsalva. ● **Infectious CNS disorder (meningitis or brain abscess):** Fever, stiff neck, mental status changes, photophobia, and neurologic signs. ● **Structural abnormality of the brain (e.g., tumor, aneurysm, arteriovenous malformation):** Mental status change, signs of increased intracranial pressure, or neurologic signs during, and often after, the headache episode.	**History** ● FOLDARQ ● Enquire specifically about symptoms of a serious disorder and other concerning symptoms ● Trauma ● Medications including OCP (oral contraceptive pills) ● FHx of migraines **Suggestive symptoms** ● 'First or worst' HA – "worst headache I've ever had in my life!" ● Change in pattern of the HA – had headaches for years but these are different in nature ● Always in the same location ● Awakens the person from sleep ● Varies with position/posture change (i.e getting up makes it worse) ● Non-responsive to pain medications that previously relieved it ● Associated nausea and vomiting ● HA followed by migraine-like visual symptoms	Neurologic examination + Blood pressure monitoring

	• **Subarachnoid hemorrhage:** Extremely severe headache, stiff neck, mental status change; rarely, subhyaloid hemorrhages seen on fundus examination, usually from a ruptured aneurysm. • **Epidural or subdural hematoma:** Follows head trauma; altered level of consciousness; may produce anisocoria or cranial neuropathy. **(2) Others:** • Migraine • Cluster headache • Tension headache • VZV • Sinus disease	**Physical work up** • Pupils • Motility • VFs • IOP • Cycloplegic refraction • DFE including ONH assessment to look for any posterior eye symptoms • Refer to GP for full workup	
Cluster Headache	**Symptoms:** Typically unilateral, very painful (stabbing), periorbital, frontal, or temporal headache associated with ipsilateral tearing, rhinorrhea, sweating, nasal stuffiness, and/or a droopy eyelid. Usually lasts for minutes to hours. Typically recurs once or twice daily for several weeks, followed by a headache-free interval of months to years. The cycle may repeat. Predominantly affects men. Headache awakens patients, whereas migraine does not. **Signs:** Ipsilateral conjunctival injection, facial flush, or Horner syndrome (third-order neuron etiology) may be present. Ptosis may become permanent. Precipitating factors include alcohol, nitroglycerin	**1.** History and complete ocular examination. **2.** Neurologic examination, particularly a cranial nerve evaluation. **3.** If Horner syndrome present, consider imaging studies to eliminate other causes. **4.** Obtain an MRI of the brain when the history is atypical or a neurologic abnormality is present.	Avoid alcoholic beverages or cigarette smoking during a cluster cycle. Refer patients to neurologists to help coordinate pharmacologic therapy. **Abortive (Acute) Treatment:** • Oxygen 5–8 L/min by face mask × 10 min at onset → relief in ~70 %. **Triptans:** • Sumatriptan 6 mg SC or 20 mg intranasal. • Zolmitriptan 5–10 mg intranasal. **Alternatives:** Ergotamine inhalation, dihydroergotamine, corticosteroids. **Severe acute attack:** IV diazepam may be used. **Prophylaxis (During Cluster Periods):** • **Verapamil** 360–480 mg/day (divided). • **Lithium** 600–900 mg/day (monitor renal + thyroid function; avoid certain drugs). • **Ergotamine** 1–2 mg/day. • **Methysergide** 2 mg b.i.d. (max 3–4 mo; many contraindications). • Oral steroids (e.g., **prednisone** 40–80 mg/day × 1 week, taper) + antiulcer agent. • **Galcanezumab** (anti-CGRP mAb) – FDA approved for prevention. **1.** Patients started on systemic steroids are seen within a few days and then every several weeks to evaluate the effects of treatment and monitor intraocular pressure. **2.** Patients taking methysergide or lithium are reevaluated in 7 to 10 days. Plasma lithium levels are monitored in patients taking this agent.
Migraine	Migraine prevalence is highest between ages 30 and 39 years and progressively declines after age 40. Migraine attacks may be shorter and less typical with	**WORKUP SAME AS ABOVE**	Avoid precipating factors (may keep diary to identify triggers)

	advancing age. New-onset migraines are uncommon after the age of 50, and these patients should be worked up for secondary causes such as vascular lesions, intracranial hemorrhages, infarcts, masses, or GCA. Occurs in 16% (1 in 5) of the population, present to optometrist because of associated visual symptoms. At least 2 of: ● Unilateral ● Throbbing ● Worse with movement ● Moderate to severe intensity Plus at least one of: ● Nausea, vomiting ● Photophobia (light), Usually no signs. ***International Classification*** Consult International Classification of Headache Disorders, 3rd edition, for further information. • Migraine without aura (common migraine; 80%): Lasts 4 to 72 hours. Unilateral location, pulsating quality, moderate-to-severe pain, and/or aggravation by physical activity. Nausea, vomiting, photophobia, and/or phonophobia are the characteristics. • Migraine with typical aura (classic migraine; 10%): Fully reversible binocular visual symptoms that may be perceived as monocular (e.g., flickering lights, spots, lines, loss of vision) or fully reversible unilateral sensory symptoms (e.g., numbness, "pins and needles"). Symptoms gradually develop over 5 minutes and last between 5 and 60 minutes. No motor symptoms are present. • Typical aura without headache (acephalgic migraine): Visual or sensory symptoms as above without accompanying or subsequent headache. • Familial hemiplegic and sporadic hemiplegic migraine: Migraine with aura as above with accompanying motor weakness with (familial) or without (sporadic) history in a first-degree or second-degree relative. Sporadic cases always require neuroimaging. • Retinal migraine: Fully reversible monocular visual phenomenon (e.g., scintillations, scotoma, blindness) accompanied by headache fulfilling migraine definition. Appropriate investigations to exclude other causes of transient monocular blindness should be completed. Existence of retinal migraine is controversial. • Basilar-type migraine: Aura symptoms mimic vertebrobasilar artery insufficiency in a patient with migraine. See 10.23, VERTEBROBASILAR ARTERY INSUFFICIENCY. • Ophthalmoplegic migraine: Onset in childhood. Headache with third cranial nerve palsy. Likely an inflammation, rather than migraine, as MRI shows enhancement of CNIII.	**Complicated migraines may have a permanent neurologic or ocular deficit,** documenting VF may be helpful; ● History may establish the diagnosis + determine time sequence + "How long does it last for?" + "When does the headache start relative to the visual symptoms?" ● Educate patient on the possibility of it worsening over time. ● Ocular and neurologic examinations, including refraction. ● CT scan or MRI of the head is indicated for: + Atypical migraines: New onset migraines in patients over 50 years old. Migraines that are always on the same side of the head or those with an unusual sequence such as visual disturbances persisting into or occurring after the headache phase. + Complicated migraines. ● Consider checking for uncontrolled blood pressure or low blood sugar (hypoglycemic headaches are almost always precipitated by stress or fatigue). <u>Typical visual aura:</u> In a typical migraine, visual aura **precedes the headache** (AKA incoming migraine). If the headache comes first and the aura follows, that's atypical and should prompt consideration of other causes such as: ● Arteriovenous malformations (AVMs) ● Mass lesions (e.g., tumours) ● Cerebral swelling (e.g., raised ICP) ● Seizure foci (especially occipital seizures producing visual phenomena) ● **Fortification spectra:** Zigzag figure appearing near fixation, gradually spreading left or right, convex shape with angulated scintillating edge, leaves rel/absolute scotoma behind. ● **Other less common include;** micropsia, macropsia, metamorphopsia, distorted body image, palinopsia, cereb polyopia, achromatopsia ● Homonymous, positive (flickering lights spots, lines) or negative (vision loss), lasts 5-60mins	 Refer to GP ● **Abortive therapy**: medications used at the onset of therapy, best if infrequent ● **Initial therapy**: Aspirin or NSAID **IF that fails**: Ergotamines or selective serotonin receptor agonists (triptans), calcitonin gene-related peptide antagonists **Prophylaxis in patients with frequent or severe HAs or neurologic changes**: Beta blockers, calcium channel blockers, antidepressants, anti-CGRP monoclonal antibiotics, and others Antinausea medication as needed during an acute episode. Reevaluate in 4 to 6 weeks to assess the efficacy of the therapy.
🚩**Occipital seizure**🚩	*<u>Can be mistaken for migraine:</u>* Flashing lights, (but **only couple of minutes** compared to migraine >10min), nausea, postictal headache *<u>Other symptoms:</u>* Shapes, bilateral loss of vision (black out or white out), gaze deviation, eye pain *<u>Focal cognitive seizures if extra-striate visual areas:</u>* Complex hallucinations/palinopsia, nystagmus, lid fluttering, impaired awareness, focal atonic motor seizures		

<u>KEY POINTS:</u>

- Most serious or life-threatening headaches present with acute onset of pain
- A person with their first or worst headache has a much greater risk of harbouring an ominous cause than someone with a 5-year history of headache which hasn't changed in pattern.

Non-Physiologic Visual Loss

Signs: Loss of vision. Malingerers frequently are involved with an insurance claim or are looking for other forms of financial gain. Those with psychogenic visual loss truly believe they have lost vision.

Symptoms: Loss of vision. Malingerers frequently are involved with an insurance claim or are looking for other forms of financial gain. Those with psychogenic visual loss truly believe they have lost vision.

Workup: Try to get rid of malingerers (pupils, mirror test, base-out prism test, W4DT test)

Treatment:

1. Patients are usually told that no ocular abnormality can be found that accounts for their decreased vision. In general, they should not be told that they are faking the visual loss.
2. Hysterical patients often benefit from being told that their vision can be expected to return to normal by their next visit. Psychiatric referral is sometimes indicated.

Follow Up:

- If nonphysiologic visual loss is highly suspected but cannot be proven, reexamine in 1 to 2 weeks.
- Consider obtaining an ERG, visual-evoked response, macular OCT, or an MRI of the brain.
- If functional visual loss can be documented, have the patient return as needed.

Transient Vision Loss/ Amaurois Fugax

Signs: May see an embolus within an arteriole or the ocular examination may be normal.

Symptoms: Monocular visual loss that usually lasts seconds to minutes, but may last up to 1 to 2 hours. Vision returns to normal.

Etiology:

- Embolus from the carotid artery (most common), heart, or aorta.
- Vascular insufficiency as a result of arteriosclerotic disease of vessels anywhere along the path from the aorta to the globe causing hypoperfusion often precipitated by a postural change or cardiac arrhythmia.
- Hypercoagulable/hyperviscosity state.
- Rarely, an intraorbital mass may compress the optic nerve or a nourishing vessel in certain gaze positions, causing gaze-evoked transient visual loss.
- Vasospasm.

Workup:

(1) Amaurosis fugax is considered by the American Heart Association to be a form of TIA. Current guidelines recommend MRI with diffusion weighted imaging, urgent carotid and cardiac studies, and neurology consultation. If patient is seen within days of the event, emergent referral to a stroke center or emergency room is required.
(2) Immediate ESR, CRP, and platelet count when GCA is suspected.
(3) History: Monocular visual loss or homonymous hemianopsia (verified by covering each eye)? Duration of visual loss? Previous episodes of transient visual loss or TIA? Cardiovascular disease risk factors? Oral contraceptive use? Smoker? Vascular surgeries?
(4) Ocular examination, including a confrontation visual field examination and a dilated retinal evaluation. Look for an embolus or signs of other aforementioned disorders.
(5) Medical examination: Cardiac and carotid auscultation.
(6) Noninvasive carotid artery evaluation (e.g., duplex Doppler US). Consider orbital color Doppler US, if available, which may reveal a retrolaminar central retinal artery stenosis or embolus proximal to the lamina cribrosa. MRA or CTA may also be considered, but cannot evaluate flow like a duplex Doppler US.
(7) CBC with differential, fasting blood sugar, hemoglobin A1c, and lipid profile (to rule out polycythemia, thrombocytosis, diabetes, and hyperlipidemia).
(8) Cardiac evaluation including an echocardiogram.

Treatment:

1. **Carotid disease.**
   - Consider aspirin 81 mg or 325 mg p.o. Daily.
   - Consult vascular surgery in select patients if a surgically accessible, high-grade carotid stenosis is present for consideration of carotid endarterectomy or endovascular stent.
   - Control hypertension, diabetes, and dyslipidemia (follow up with a medical internist).
   - Lifestyle modification (e.g., smoking cessation).
2. **Cardiac disease.**
   - Consider aspirin 325 mg p.o. Daily.
   - In presence of a thrombus, coordinate care with internal medicine and/or cardiology with likely hospitalization and anticoagulation (e.g., heparin therapy).
   - Consider referral to cardiac surgery as needed.
   - Control arteriosclerotic risk factors (follow up with medical internist).
3. If **carotid and cardiac diseases are ruled out**, a vasospastic etiology can be considered (extremely rare). Treatment with a calcium channel blocker may be beneficial.

Pituitary Adenous

Pituitary adenomas cause progressive bitemporal hemianopia through optic chiasm compression, often requiring combined neurosurgical and endocrine management.

Signs:

- Visual field defects, classically bitemporal hemianopia (compression of the optic chiasm).
- Optic disc pallor or optic atrophy if chronic.
- Decreased visual acuity if longstanding compression.
- Possible signs of endocrine dysfunction (e.g., acromegaly, Cushing's, hypopituitarism).
- Ocular motility deficits if cavernous sinus is involved (less common).

Symptoms:

- Painless, progressive vision loss, often noticed peripherally first.
- Visual field defects (usually bitemporal).
- Headache (due to mass effect).

Etiology:

- Benign tumour of anterior pituitary cells (can be functioning or non-functioning).
- Causes compression of the optic chiasm (especially from suprasellar extension).
- May invade cavernous sinus or surrounding structures.

Workup:

- MRI brain with pituitary protocol – gold standard for diagnosis.
- Formal visual field testing (e.g., Humphrey 24-2) to detect and monitor defects.
- Endocrine workup (prolactin, GH, ACTH, TSH, LH/FSH, cortisol, etc.).
- Ophthalmic exam for VA, pupils, optic discs.

Treatment:

- Endocrine management depending on tumour type (e.g., dopamine agonists for prolactinoma).
- Surgical removal (e.g., transsphenoidal resection) for non-functioning, large, or vision-threatening adenomas.
- Radiotherapy if residual or recurrent tumour.
- Treat hormonal deficiencies as needed.

Follow Up:

- Regular VF + optic disc exams to monitor recovery or progression.
- MRI surveillance for recurrence or residual tumour.
- Endocrine review for ongoing hormonal control.
- Multidisciplinary follow-up (endocrinology, neurosurgery, ophthalmology).

Cortical Blindness

Signs: Markedly decreased vision and visual field in both eyes (sometimes no light perception) with normal pupillary responses.

Symptoms: Bilateral complete or severe loss of vision. Patients may deny they are blind (Anton syndrome) or may perceive moving targets but not stationary ones (Riddoch phenomenon).

Etiology:

- **Most common:** Bilateral occipital lobe

Retrograde Degeneration

Signs:

- Optic disc pallor (develops weeks–months after the primary lesion)
- Usually normal initial fundus, then progressive pallor
- May show afferent pupillary defect (RAPD) if unilateral/asymmetric
- No active disc swelling (unlike papilledema or optic neuritis)

Symptoms:

- Painless, gradual vision loss following damage to the post-retinal visual pathway (e.g., optic tract, LGN, occipital cortex)
- May not be noticed early if the fellow eye is normal
- Visual field defects correspond to site of original lesion (e.g., homonymous hemianopia)

Vertebrobasilar Artery Insufficiency

Signs: May have a hemianopsia, ocular motility deficits, or nystagmus, but often presents with a normal ocular examination.

Symptoms: Transient, bilateral blurred vision, sometimes accompanied by flashing lights. Ataxia, vertigo, dysarthria or dysphagia, perioral numbness, and hemiparesis or hemisensory loss may accompany the visual symptoms. History of drop attacks (the patient suddenly falls to the ground without warning or loss of consciousness). Recurrent attacks are common.

Workup:

(1) History: Associated symptoms of vertebrobasilar insufficiency? History of

infarctions.
- **Other:** Toxic, postpartum (amniotic embolus), posterior reversible encephalopathy syndromes.
- **Rare:** Neoplasm (e.g., metastasis, meningioma), incontinentia pigmenti.
-

Workup:

(1) Test vision at distance (patients with bilateral occipital lobe infarcts may appear completely blind, but actually have a very small residual visual field). Patients will do much worse with near-card testing than distance if only a small island remains.
(2) Complete ocular and neurologic examinations.
(3) MRI of the brain.
(4) Rule out nonphysiologic visual loss by appropriate testing (see 10.25, NONPHYSIOLOGIC VISUAL LOSS).
(5) Cardiac auscultation and electrocardiography to rule out arrhythmia.
(6) Check blood pressure.
(7) Consult neurologist or internist for evaluation of stroke risk factors.

Treatment:

1. Patients diagnosed with a stroke within 72 hours of symptom onset are admitted to the hospital for neurologic evaluation and observation.
2. If possible, treat the underlying condition.
3. Arrange for services to help the patient optimize function at home and their surrounding environment.

Follow Up: As per the internist or neurologist.

Etiology: Post-retinal lesions causing loss of axonal input to retinal ganglion cells → retrograde degeneration

- Optic tract lesions
- Lateral geniculate body lesions
- Occipital lobe infarcts or trauma
- Tumours compressing post-retinal pathways

Workup:

- Neuroimaging (MRI brain ± orbits) to localize primary lesion
- Visual field testing to map defects
- OCT / fundus exam to document RNFL and ganglion cell loss over time

Treatment:

- Treat the underlying cause (e.g., stroke management, tumour removal, addressing trauma)
- No direct treatment reverses degeneration of optic nerve fibers

Follow Up:

- Regular vision and visual field monitoring
- OCT / optic disc exams to track progression of pallor
- Manage underlying neurologic condition and provide low vision support if needed

carsickness or migraine? Symptoms of GCA? Smoker?
(2) Dilated fundus examination to rule out retinal emboli or papilledema.
(3) Blood pressure in each arm to rule out subclavian steal syndrome.
(4) Cardiac auscultation to rule out arrhythmia.
(5) CBC to rule out anemia and polycythemia, with immediate ESR, CRP, and platelet count if GCA is considered.
(6) Electrocardiography, echocardiography, and cardiac monitoring to rule out dysrhythmia.
(7) Consider noninvasive carotid flow studies.
(8) MRA, CTA, or transcranial/vertebral artery Doppler US to evaluate posterior cerebral blood flow.

Treatment:

1. Coordinated care with internal medicine or neurology with initiation of antiplatelet and/or anticoagulation therapy.
2. Consult internist for hypertension, diabetes, and dyslipidemia control if present.
3. Lifestyle modification (e.g., smoking cessation).
4. Correct any underlying problem revealed by the workup.

Follow Up: If outpatient, 1 week to check test results. Thereafter follow-up time is dictated by identified underlying causative condition(s).

WEEK 7: MULTIDISCIPLINARY AND COLLABORATIVE CARE

LECTURE 7: Multidisciplinary and Collaborative care

What is collaborative care?

- Synonymous with shared care and team-based care:
  "*When the care of a patient is provided by **two or more health practitioners**, each **practising within their sphere of expertise in consultation** with the patient".* – Optometry Board of Australia
- Collaborative care goes by lots of different terms like "shared care" and "team care". The term indicates that the patient is involved in the decision.
- With collaborative care, it is important to recognise your own limitations in terms of experience, skills, knowledge, expertise and experience. For e.g ophthalmologists have been trained to cut, laser and inject. We need to refer to their expertise whenever that level of care is required.
- As a primary care provider, a big part of what we should do is coordinate care with multiple practitioners. This can be stifled by medicare reimbursement for referral.

Related Terms

Interdisciplinary care	Team members from different disciplines working collaboratively, with a common purpose, to set goals, make decisions and share resources and responsibilities. ➔ Will have interdisciplinary/interprofessional group discussions where a single case is shared and different professions bring in their expertise (e.g optometry and audiology, optometry and pharmacy). ➔ Shared discussion, co-educating each other, all working towards the same goal.
Multidisciplinary care	Team members from different disciplines working independently to create discipline-specific care plans that are implemented simultaneously, but without explicit regard to their interaction. ➔ Multiple disciplines work together for the patient but work more independently in their areas of expertise. ➔ Less cross talk, shared learning, shared talking, growth together.
Integrated care	The provision of seamless, effective, and efficient care that reflects the whole of a person's health needs ➔ Wrap around care, putting patient's priorities front and centre and making the system work for PATIENTS (rather than the standard other way around).

WHY collaborate?

- Principles to keep in the back of your minds:
  - All health practitioners have a shared purpose – want to help patients with their health.
  - "Right care, right time, right place, right professional, right patient" – appropriating care to patients and their needs.
  - Integration is a means to improve services in relation to access, quality, user satisfaction and efficiency – WHO

What is quality health care?🍎
It is EFFECTIVE care.
- Thinking about red flags
- Care that expedites recovery
- Care that prolongs visual outcomes

We collaborate because it is:

1. **Effective**
2. **Safe**
3. **Appropriate** → Doing tests that are only on indication. As you become more experienced, you start to think more about the appropriateness of care.
4. **Provides continuity of care** → VERY IMPORTANT. There is so much value in record keeping (like long-term monitoring of CSCR), helps provide seamless care.
5. **Accessible** → Healthcare access is concerned with affordability and whether the care is going to work for the patient. How accommodating is the care – is it being offered and is available at a time that works for the patient?
6. **Efficient and sustainable** → Consider the urgency of referrals for e.g.. Ensure that you triage well so the right person gets the right care at the right time.

We can collaborate because it;
- Provides us with our own professional satisfaction
- Patients love it!
- Forms a point of difference
- Helps optimise health care delivery i.e. to solve a "problem":
  + Ageing population
  + Extended wait times for specialty care
  + Higher patient expectations
  + The "burden" of early detection

Statistics
- The prevalence of self-reported eye conditions in patients over 55 years is above 90% - AIHW (2017)
- The median wait time for cataract surgery in NSW is 235 days – Bureau of Health Information (2018)
- Around 22 patients per month in the UK are permanently losing vision while sitting on hospital wait lists – Foot & Macewan (2017)

- Collaborative care delivers:
  - "…safe as well as cost-effective" – Devarajan et al (2011)
  - "… significantly more likely to be satisfied" – Gray et al (1997)
  - "…reduction in delay between visits" – Botha et al (2015)
  - "…reductions in cases without a diagnosis" – Ly et al (2016) – significantly reduces patient anxiety of not knowing what is happening
  - "Healthcare costs were significantly lower" when we use multiple members of a healthcare team – Holtzer-Goor et al (2010)

Perks of Collaborative Care:
- ★ Cool cases!
- ★ Relationship building
- ★ Intrinsically rewarding
- ★ Opportunities for growth
- ★ Recognition Work satisfaction
- ★ Diversity of opinion

- "With ageing populations and rising chronic disease burden the demand for eye care services will also grow, and integrated models of care such as C-EYE-C are necessary to make healthcare sustainable." – Ford et al (2019)

WHO can I collaborate with?

- Can collaborate with anyone and everyone! – the key is to be proactive (would rather have more information than less)
  1. **Horizontal integration –** across professional teams
  2. **Vertical integration –** across primary/secondary/tertiary care boundaries:
     - Primary = Optometrist, GP (first care of entry)
     - Secondary = Ophthalmologist/Specialist
     - Tertiary = Hospital, ED

Ophthalmologists	Teachers
General practitioners	Aged-care nurses
Neurologists	Psychologists
Endocrinologists	Psychiatrists
Nurses / Diabetes educators	Orthoptists
Cardiologists	Occupational therapists
Nephrologists	Social workers
Rheumatologists / Immunologists	O&M specialists
Pharmacists	Health care educators

★ Other optometrists within or in a different practice:
+ Additional imaging
+ Specialty contact lens fits inc. orthokeratology
+ Dry eye management
+ Therapeutics
+ Binocular vision, behavioural/paediatric optometry
+ Low vision
+ Sports vision

Examples include grad-mentoring programs, or peer supervising consults – you can gain a lot from immediate feedback.

★ Ophthalmologists
★ Doctors e.g., GPs, medical specialists – neurologists, rheumatologists, immunologists
★ Allied health practitioners: physiotherapists, dietitians etc.
★ Other professionals: social workers, educators etc.
★ Family and carers

Collaborative Eye Care has 4 different flavours 🍨:
- The same condition
- Different aspects of ocular health
- Systemic disease with ocular sequelae
- Ocular disease with systemic sequelae

Collaborative Eye Care - Example Conditions
- Glaucoma, Diabetic retinopathy, AMD, Cataract
- Paediatrics (e.g teachers, social workers)
- Emergency (referrals if urgent)
- Corneal Disease / Contact Lenses
- Dry Eye
- Low Vision
- Medication Toxicity screening (Deferoxamine, Plaquenil, Ethambutol)
- Neuro
- IRDs = Inherited Retinal Dystrophies

Collaborative Eye Care Cases

(1) For the same condition e.g glaucoma

CASE 1

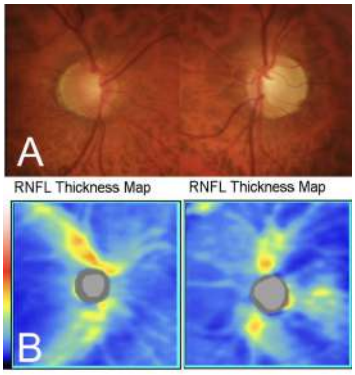
57-year-old asymptomatic male. Mother with NTG. Denied symptoms of blurry vision, red sore eyes or halos around lights.

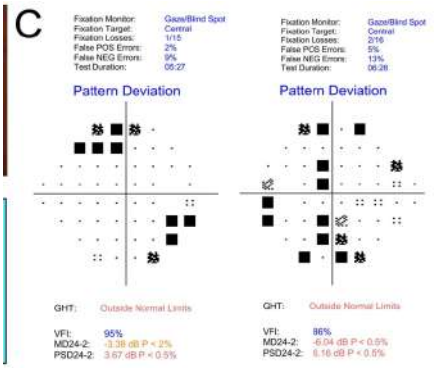
	OD	OS
Refraction	-3.75/-0.75x75 (6/6^{-1})	-3.75/-1.00x111 (6/7.5^{-2})
Applanation tonometry @11am	15 mmHg	19 mmHg
Pachymetry	616 microns	604 microns
Slitlamp examination and gonioscopy	• Bilateral Krukenberg's spindles and subtle peripheral iris transillumination defects • CBB all quadrants with heavily pigmented trabecular meshwork and posterior iris bowing	

In this case, we are suspecting: pigment dispersion syndrome → pigmentary glaucoma (secondary glaucoma)
- Affects >50years for glaucoma.
- Higher myopic males have a higher risk (also have higher risk of RD).

<u>DIAGNOSIS:</u> Pigmentary glaucoma (burnout phase)

<u>MANAGEMENT:</u> Long term surveillance every 6 months + Ancillary testing
- Can manage collaboratively with an ophthalmologist
- Can refer to an ophthalmologist, not expecting to see the patient again unless directed by the ophthal
  + Can manage independently
  + 'Damage has been done' - okay to monitor long term since it is non-progressive





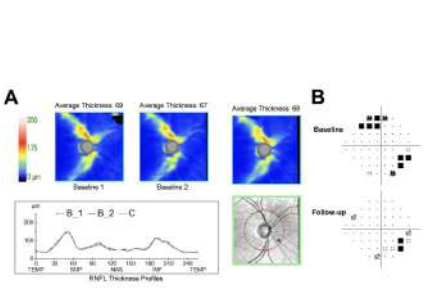


IMAGE A = CD 0.6 (but ambiguous), asymmetry in cup size OS>OD
- OS has sloped cupping + also has

Partial arcuate defects superiorly and inferiorly extending to blindspot in RE
- Worse in the LE than the RE which

Progression analysis showed stability. As patient got better at VF , they still have an inferior arcuate defect extending to

more cupping • Seeing rim loss • Structural evidence of glaucoma **IMAGE B =** Inferior RNFL loss OS > OD in the thickness map. Presence of RNFL thinning (wedge like) • OD inferior temporal RNFL thinning • OS is more focal supero-temporal RNFL wedge defect.	is consistent with structural results. • Defects are more edge-like, so start to think of it being more artifactual, like clover leaf. ***NOTE:*** Glaucoma respects midline Clover leaf from poor patient concentration in LE (is showing concordance but VFs need to be repeated)	blindspot inferiorly which probably corresponds to possible superior loss. No evidence of glaucoma progression over the time that the patient was monitored.





Shows pigmentary dispersion (heavily pigmented angle)

Other considerations for workup: Gonio photo, IOP phasing (measuring at different times of days to capture diurnal variation), repeat perimetry, provocative testing (to see dynamic of angle — in bright or dim light).

Clinical Pearls:
- Collaborative care is effective especially for chronic disease requiring further data to guide management.
- Consider proactively how you plan to co-manage glaucoma.

Glaucoma Example

- It is the most common opportunity for collaborative care.

(1) Optometry independently managed	Optometry Independently Managed: Managed competently and safely in the hands of community optometrists **without input from ophthalmology** Indicated for: (1) Glaucoma suspects: Risk factors but no definitive evidence of glaucoma + telehealth review option (F/U every 6 months for high risk glaucoma suspects) (2) Treated high risk glaucoma suspects: • Stable patients, SLT (maybe had SLT done recently, are stable now) vs topical tx • To treat or not to treat (discuss individual preferences with ophthal) + IOP cut off? + Family History / Genetics + High Myopia
(2) Optometry led	Optometry Led: Optometrist diagnosed patient with glaucoma and **commenced on treatment in conjunction with ophthalmology** input as per OBA guidelines; **long-term care returned to the optometrist** OBA Guidelines ★ **Initial ophthalmology input within 4 months of starting treatment** ★ Indicated for: • Pre-Perimetric glaucoma: Defined structural damage to the nerve without field loss (requires exclusion of alternate pathology). Almost like an early form of glaucoma. • Early Glaucoma: Structural and functional evidence of glaucoma + Mean Deviation (MD) better than -6 dB +/- Field loss limited to one hemisphere + Treated with Xalatan, repeat script every time they are seen. Adherent, routine patients patients
(3) Ophthalmology led	Ophthalmology Led: Patients whose care is coordinated by Ophthalmology with optometry input as directed • **Moderate** glaucoma: **Field loss** with MD **between -6 dB and -12 dB** +/- affecting either both hemispheres AND field loss **not** within 10 degrees of fixation • **Advanced** glaucoma: MD worse than -12 dB or central visual field loss within 5 degrees either side of fixation. Central visual field loss within 10 degrees of fixation ***NOTE:*** A secondary glaucoma is more likely to progress, so will be considered advanced form.

RECALL:

The optometrist MUST provide a referral for ophthalmological assessment and advice after making an initial diagnosis and initiating treatment for chronic glaucoma IF:
- The anti-glaucoma treatment **does not** stabilise the patient's condition
- A patient needs assessment by an ophthalmologist or ophthalmology service for possible surgical intervention or laser treatment, or
- A patient experiences side effects of initial treatment

When you first initiate treatment for glaucoma (whether new or high risk), ensure there is a follow up plan in place, review appointment at least every 4-6 weeks, minimally. Checking response to therapy, if they are using drug, checking adherence, side effects etc.
NOTE: IN ANY EVENT **MUST** provide the patient with a referral to an ophthalmologist or ophthalmology service **WITHIN 4 MONTHS** of starting treatment for chronic glaucoma

- Things to think about if you are getting involved:

Initial Dataset
This is a list of data you can put in your referrals:
- Diagnosis and severity grading
- Glaucoma risk factors and a general health history including medications (especially thinking about potential drug interactions or any overlaps e.g. systemic beta blockers, allergies to sulphur medications).
- Baseline Applanation IOP (highest recorded pre-treatment IOP and even previous records of IOP)
- Central Corneal Thickness (pachymetry)
- Gonioscopy results
- Relevant disc and field findings with associated images
- Copy of OCT optic nerve and Ganglion Cell analyses (progression analysis if available)
- Treatment plan initiated including Target IOP/ proposed treatment plan
- Confirmation that the patient is interested in collaborative care
- Suggested review schedule and which management classification this patient is likely to fall under e.g optometry/opthalmology led.
- +/- Structure Function Report, IOP monitoring / phasing performed

NOTE: Glaucoma is a diagnosis of exclusion. You can follow and better manage the case if there is more data.

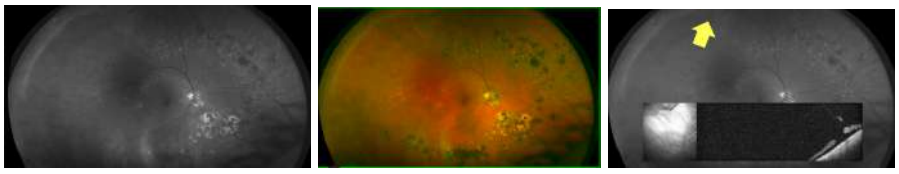
Subsequent Visits
- Comments on stability/compliance and drug compatibility, treatment efficacy (if it is reaching treatment target or not)
- Applanation IOP
- OCT Report of the days visit
- GPA reports of OCT and VFT
- Comparative photos (high level of care — unexpected).
- Management Plan including next appointment

(2) For different aspects of ocular health

CASE 2

- 69 year old female with history of choroiditis
- **PH-VA** 6/7.5 R&L
- **Refraction**: OD +0.75/-0.75x90 (6/6-), OS +0.75/-0.50x90 (6/7.5)
- **Reason for referral**: recent REO (right eye orbital) pain that has been recovered
- **Pertinent exam findings**: had dry eyes, maybe sinus issues? Also long-standing choroiditis – not reviewed for years

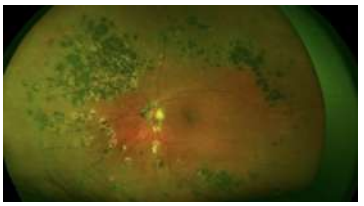
RE



Can see a very shallow retina detachment superiorly in image 1.
The yellow arrow in image 3 shows a rhegmatogenous retinal break with associated shallow RD (superior-temporal RD).
Subtle elevation

DIAGNOSIS: Macular ON retinal detachment

LE



Chorioretinal scarring (associated with chorioritis), affecting midperiphery spreading outward to the periphery, PPA (larger blind spot) , stable

MANAGEMENT: Next day referral to Sydney Eye Hospital (requiring urgent collaboration)
- Get it sealed at SEH (i.e. vitrectomy, scleral buckle)
- Come back to optometrist for follow-up

Clinical Pearls:
- Collaborative care is safe
- Think of using collaborative care especially in sight or life threatening disease

HOW to collaborate?



Person centred care: Care that is respectful of, and responsive to, the preferences, needs and values of the individual patient

- Seeking out and understanding what is important to the patient
- Shared decision making, with patient consent
- Some patients don't want tx
- Document findings if px doesn't want treatment (have the conversation → document it → follow-up)

Role understanding: Formal vs informal arrangements, Define and redefine as needed over time

Continual quality improvement

- Correlates with less isolation, stress and dissatisfaction
- Reflective practice
- Peer-review. Mentoring

Evidence-based practice: guidelines

Each party must have a clear understanding of

- The diagnoses, treatment(s) and ongoing recommendations to the patient of the other treating practitioners
- The information to provide to other parties involved in the care of the patient
- Timeframes and format in which this information should be provided
- Who is responsible for ongoing patient care, and follow up of patients who miss scheduled appointments
- Roles and responsibilities of each person participating in the shared care.

Examples: OBA for use of scheduled medicines. Examination and management of patients with diabetes (OBA). 2019 Clinical practice guide for the diagnosis, treatment and management of AMD (OBA)

Communications: Interprofessional, Across team

Medicare

10905 = Medicare item no for referred comprehensive consultations

- Not enough referrals to other optoms

Requested Medicare items processed from July 2019 to June 2020

Click on the item hyperlinks below to get the patient age/gender breakdown of that particular item

	State								Total
	NSW	VIC	QLD	SA	WA	TAS	ACT	NT	
	Services	Services	Services	Services	Services	Services	Services	Services	Services
Item									
10905	6,338	1,638	2,227	514	369	123	121	20	11,350
10907	76,546	89,305	59,169	21,869	34,845	5,476	4,567	2,769	303,547
10910	617,495	453,783	384,251	128,668	192,457	36,984	34,330	14,708	1,862,686
10911	382,827	256,457	250,068	97,255	119,373	28,707	17,797	4,875	1,157,359
10913	201,517	162,209	130,202	45,002	65,010	14,146	10,610	5,518	634,496
10914	303,845	264,597	250,609	64,987	81,145	22,563	14,234	8,883	1,010,864
10915	80,710	75,535	43,886	19,867	13,282	5,133	4,166	3,715	247,282
10916	680,555	514,489	453,869	146,650	170,712	48,339	31,163	12,693	2,058,470
Total	2,349,839	1,818,013	1,583,283	524,892	677,195	162,473	117,188	53,181	7,286,054

Made up just 0.15% of all the major Medicare optometry billed consultations in the last financial year

Existing models of collaborative care

Varies according to location of the pation + skill-base of local health care practitioners

- **Ad-hoc**:communication through reports and referrals
- **Semi-formalised**: discussed arrangements and agreed upon methods
- **Formalised**: Structured criteria, templates and programs, often with published outcomes

Wide range of technology uptake and utilisation: formalised image sharing platforms, oculo, referral letters etc

Optometrists employed within the public system	*E.g. Townsville, Kununurra, Womens and Childrens, Hobart* ● Triage new patients / referrals ● Scheduled review appointments ● Specialist clinics (LV / CL etc.) ● Collaborative clinics (glaucoma, paeds etc.) e.g. SEH Glaucoma, Westmead Cataract, Fremantle Glaucoma
Optometry clinic subcontracts on premises	*E.g. Former Sutherland Hospital* ● Similar model to IVIs – see public Px privately but bulkbilled
Hub and spoke model	*E.g. C-Eye-C, Flinders -CEPOS* – Triage new referrals + scheduled review appointments
Community models	*E.g. GC3P, POAP* – Scheduled Review appointments
Alternative pathway model	*E.g. CFEH –POWH, Karratha Eyecare* – External alternative Private Optometry (bulkbilled) referral to public system

Collaborative care progams

Establishing a collaborative care program

- ★ The diagnoses, treatment(s) and ongoing recommendations to the patient of the other treating practitioners
- ★ The information to provide to other parties involved in the care of the patient

- ★ Timeframes in which this information should be provided
- ★ The preferred format for this information
- ★ Who is responsible for ongoing patient care and the follow up of patients who miss scheduled appointments
- ★ The roles and responsibilities of each person participating in the shared care

CFEH: centre for eye health	A community initiative to provide state of the art imaging, diagnostic and treatment services staffed by optometrists and consultant ophthalmologist • Joint initiative of UNSW Sydney and Guide Dogs NSW/ACT • Main clinic at UNSW, affiliated with Prince of Wales Hospital • Satellite clinic in Western Sydney • We examined ~10,000 patients per year at no charge • **10+ year history of working closely with ophthalmology**
HTH: health translation hub	A UNSW initiative that will bring together educational and medical researchers, clinicians, educators, industry partners and public health officials - To drive excellence, and support the rapid translation of research, innovation and education into improved patient care, delivering better health outcomes to the community

Communication

Telemedicine care models:

- Synchronous (real-time)
- Asynchronous (store-and-forward)
- Hybrid

Methods of communication:

- Postal mail, face-to-face, fax, phone
- Encrypted Email *e.g. ProtonMail, PreVeil*
- Direct entry websites
- Web-based communications tools *e.g. WhatsApp, Zoom*
- e-communication networks *e.g. Healthlink, Telstra Health, Oculo, MyHealth record*
- File transfer *e.g. Citrix Sharefile, Dropbox*
- Direct network log in for external parties *e.g. Forum, Heyex2, Citrix receiver, Remote desktop services*

Use protocols, templates or forms (make sure there is all the necessary information e.g. for cataract referral

HOW TO COLLABORATE:

1. Discuss Concept with Local Ophthalmologist
2. Face to Face Meeting: Practice vs Individual
3. One condition at a time
4. Characteristics of suitable patients
5. When to refer / communicate
6. How to Communicate
7. What to include in reports / communication: referral guidelines
8. Funding Model

Eye hospital can reject referrals if no OCT is provided
If in doubt, check with the website of the practice / hospital you're referring to for structure/criteria

8 C's of communication

- ★ **Conciseness** – Referral letters should be short and to the point
- ★ **Correctness** – Accurately state the facts; check your grammar
- ★ **Clarity** – Make it easy to understand
- ★ **Completeness** – Cover the essentials to get your intended response
- ★ **Consideration** – Put yourself in the place of the receiver
- ★ **Concreteness** – Be specific to reinforce confidence in your skills
- ★ **Courtesy** – Strengthen your relations
- ★ **Consistency** – Make it a habit in your daily practice

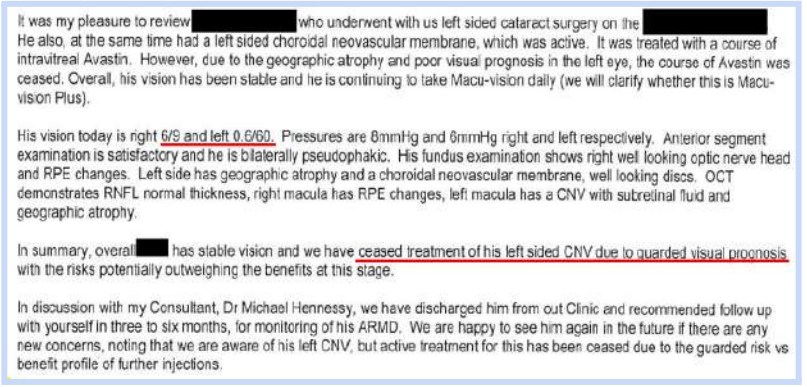
Practice vs individual collaboration

Practice has same patient base: different Interests / Experience / Knowledge / Therapeutic qualification

Solutions for consistency	• Internal Case Conferencing • REVIEW REPORTS and effect change • Organisational conferencing • External support: telehealth, collaborating colleagues/ophthalmologists
Considerations	• Grading scales (keep up to date with new scales) • Degree of urgency: ability to secure timely appointments (esp for red flags), appropriate equipment • Patient physical characteristics: location/language etc • Practice equipment – minimum dataset: available at only one location?

Funding

- Public vs private
- Consultation fee vs bulk billed
- Imaging costs
- Medical model of optometry



It was my pleasure to review [redacted] who underwent with us left sided cataract surgery on the [redacted]. He also, at the same time had a left sided choroidal neovascular membrane, which was active. It was treated with a course of intravitreal Avastin. However, due to the geographic atrophy and poor visual prognosis in the left eye, the course of Avastin was ceased. Overall, his vision has been stable and he is continuing to take Macu-vision daily (we will clarify whether this is Macu-vision Plus).

His vision today is right 6/9 and left 0.6/60. Pressures are 8mmHg and 6mmHg right and left respectively. Anterior segment examination is satisfactory and he is bilaterally pseudophakic. His fundus examination shows right well looking optic nerve head and RPE changes. Left side has geographic atrophy and a choroidal neovascular membrane, well looking discs. OCT demonstrates RNFL normal thickness, right macula has RPE changes, left macula has a CNV with subretinal fluid and geographic atrophy.

In summary, overall [redacted] has stable vision and we have ceased treatment of his left sided CNV due to guarded visual prognosis with the risks potentially outweighing the benefits at this stage.

In discussion with my Consultant, Dr Michael Hennessy, we have discharged him from out Clinic and recommended follow up with yourself in three to six months, for monitoring of his ARMD. We are happy to see him again in the future if there are any new concerns, noting that we are aware of his left CNV, but active treatment for this has been ceased due to the guarded risk vs benefit profile of further injections.

LECTURE 8: Vitreous and Posterior Review

Approach to Uveitis

- Uveitis is not a single disease but a collection of 30 to 40 different disorders that may be characterized by clinical symptoms, anatomic location, morphology, presence or absence of key anatomic findings (such as ciliary flush, keratic precipitates [KP], iris nodules, synechiae, snowballs, snowbanks, retinal vasculitis, macular edema, and optic neuropathy), and response to treatment.
- A thorough history and review of systems with a complete examination will typically narrow the differential diagnosis to a much more manageable number of possibilities on which the workup should be based.
- The history defines the course of the disease:
  - ★ Onset → Sudden versus insidious
  - ★ Duration → Limited (<3 months) versus persistent (>3 months).
  - ★ Course:
    - + Acute (sudden onset and limited duration).
    - + Recurrent (flare-ups occurring >3 months after stopping therapy).
    - + Chronic (persistent or flaring up <3 months after stopping therapy).

Predisposing factors/ causes of uveitis
- ★ Infectious.
- ★ Systemic inflammatory disease.
- ★ Vascular disorders.
- ★ Host immune status.
- ★ Genetic.
- ★ Drug-induced.
- ★ Trauma.
- ★ Environmental.

NOTE: That by these criteria, the term “acute or chronic” has no meaning, and uveitis controlled with medication should be considered “suppressed” and not “in remission.”

- ★ Laterality:
  - + Unilateral.
  - + Unilateral/alternating (bilateral and nonsimultaneous).
  - + Bilateral and simultaneous.
- ★ Anatomic location:

Anterior uveitis: Cells limited to the anterior chamber (iritis) or with some anterior vitreous involvement (iridocyclitis) ● Anterior chamber cells must be greater than vitreous cells. ● Isolated anterior uveitis should NEVER be diagnosed without assessment of the retina.	**Intermediate uveitis:** Cells in the vitreous cavity (vitritis) without chorioretinal involvement; may have anterior chamber cells. ● Vitreous cells must be greater than anterior chamber cells. **Panuveitis:** Concurrent anterior, intermediate, and posterior uveitis.	**Posterior:** Isolated retinitis, choroiditis, or both ● **Retinochoroiditis:** Primarily retinal with secondary choroidal involvement. ● **Chorioretinitis:** Primarily choroidal with secondary retinal involvement. ● There may be posterior vitreous cells. ● Optic disc edema and hyperemia may be present. ● Neuroretinitis. ● Retinal vasculitis: + Venous. + Arterial. + Mixed venous/arterial.	**Lesions** in posterior uveitis should be characterized by: ● Colour: + White lesions are usually retinal. + Yellow lesions are usually choroidal. + Pigmented lesions usually indicate long-standing disease. ● Presence or absence of hemorrhage ● Presence or absence of retinal vasculitis ● Border appearance: creamy versus granular versus sharply defined ● Pattern (focal or paucifocal, multifocal) ● Morphology + Placoid. + Punched-out. + Serpiginous. + Amoeboid. + Ovoid. + Punctate.

- ★ Hypopyn (layering of WBC in A.C)
  - ● Usually white, flat-topped.
  - ● Bloody hypopyon often suggests herpetic uveitis.
  - ● Shifting hypopyon (changes with head position) suggests Behçet disease.

Principles of uveitis workup:

1. Distinguish infectious from noninfectious uveitis.
2. Distinguish purely ocular disease from uveitis associated with systemic conditions.
3. Consider masquerade syndromes (e.g., retained intraocular foreign body, tumors, chronic retinal detachment, etc.).
4. Obtain additional testing only if the results will influence the differential diagnosis, medical or surgical management, prognosis, or referral patterns.
5. Recognize that up to 40% of uveitis is undifferentiated (the term preferred to “idiopathic,” since nearly all noninfectious uveitis has no known cause) and explain this to patients.
6. Imaging, like laboratory testing, should be based on the disease and not used indiscriminately. Testing may include:
   - ● Optical coherence tomography (OCT).
   - ● Intravenous fluorescein angiography (IVFA).
   - ● Indocyanine green angiography (ICGA).
   - ● Fundus autofluorescence (FAF).
   - ● OCT angiography.
   - ● Visual field testing.
   - ● Electrophysiology

Overall uveitis tx:

- Refer to internist or pulmonologist → for systemic evaluation & medical managmenet
- Early referral to uveitis specialist → for complicated case
- Anterior uveitis:
    - Cyclopletic (Cyclopentolate 1% tid & atropine 1% bid)
    - Topical steroid (Prednisolong acetate 1% q1-6h)
- Posterior uveitis
    - Supplement calcium with vitamin D (600mg with 400iU) 1-2x daily + histamine type 2 receptor blocker (Ranitidine 150mg po bid) OR proton pump inhibitor (Pantoprazole 40mg daily)
    - Periocular steroid (triamcinolong 40mg/mL 0.5-1.0mL) → instead of systemic in unilateral/asymmetric case
        - Repeat every 3-4 weeks
    - Topical steroid alone → inadequate tx
- Prediposing factors management :
    - Retinal NV → panretinal photocoagulation
    - Orbital disease → systemic steroid
    - ON granulomas → Consultation with neuro-ophthalmologist & systemic steroid
    - Pulmonary disease, facial nerve palsy, CNS disease & renal disease → systemic steroid + refer to internist /neurologist

Overall follow time

- Anterior uveitis:
    - Initially: Every 1-7 days (depend on inflammation)
        - Aimed: adjust steroid dosage, Monitor IOP & re-evaulate fundus + slowly taper
    - Quiescent/stable: Every 3-6 months
- Intermediate uveitis
    - If treated with steroid or systemic immunosuppression: Every 2-6 weeks
    - If poor response to steroid → prompt workup for other cause OR referral to subspecialist

Revision

Section 1: 50 Multiple choice questions, 1 mark each, 50 marks total

Topic area	Number of Qs
Ophthalmic drug delivery, prescribing, OBA guidelines	8
Collaborative care	3
Vitreous & Posterior Uvea	5
Macular/retinal disease	8
Glaucoma & Glaucoma medications	10
Retinal vascular disease	8
Neuro-ophthalmology	8

What to expect: multiple choice questions

Section 1: 50 Multiple choice questions, 1 mark each, 50 marks total

- Assessing your understanding of lectures and core reading material
  - Workup
  - Management and treatment options
  - Follow up
- Underlying specific management considerations e.g.,
  - Risk of future/related complications
  - Referral/treatment time frames and urgency
  - Treatment goals/indications/High-risk disease features requiring intervention
- Glaucoma medications
  - Essential prescribing knowledge

Section 2: 22 Short answer, organised into 6 topic areas, 50 marks

- Prescription writing (1Q, 2 marks)
- Ophthalmic drug delivery, OBA guidelines, collaborative care (3Qs, 8 marks)
- Macular/retinal disease (4Qs, 10 marks)
- Glaucoma (5Qs, 10 marks)
- Retinal vascular disease (5Qs, 10 marks)
- Neuro-ophthalmology (4Qs, 10 marks)

What to expect: short answer questions

- Common clinical scenarios and diseases
  - E.g., initiating glaucoma treatment
  - Adapting management plans to common scenarios
- Red-flag situations
- Drugs/treatments that optometrists can prescribe or recommend

Describe / Outline / List / Explain / Justify / Discuss / Relate

- methods or a process e.g., streamlined authority prescribing, QUM
- core / additional /alternative additional test/s and expected result
- first line / first steps
- management plan / pharma. and non-pharma. options
- follow up strategy
- referral urgency / review schedule
- treatment goals / purpose / influencing factors
- disease prognosis

Identify / name / list:

- drugs / treatment
- route of administration
- concentration
- alternative test/s
- disease complications

WEEK 1 = 18 marks
WEEK 2 = 18 marks
WEEK 3/4 = 20 marks
WEEK 5 = 18 marks
WEEK 7= 3 marks

RETINAL DISEASES (8 CONDITIONS)

1. Posterior Vitreous Detachment (PVD)

A **62-year-old hyperopic woman** presents complaining of "**sudden onset flashes** in her right eye for the past two days" and "**a few new floaters**," including what she describes as a "**spider-web-like shadow**."

- She denies any curtain or veil over vision. She has no recent trauma.
- History of mild age-related cataracts and no previous retinal issues.
- VA: **6/6 R**, **6/6 L**.
- IOP normal.
- Anterior segment unremarkable.

The patient has been diagnosed with **posterior vitreous detachment.**

Describe how PVD can lead to retinal breaks/tears.

PVD is when the posterior vitreous cortex mechanically separates from the retina, usually starting at the optic disc then peeling forward. A PVD occurs when age-related vitreous liquefaction (syneresis) causes collapse of the vitreous gel, leading to separation of the posterior vitreous cortex from the internal limiting membrane of the retina.

When the vitreous separates, it ideally detaches smoothly from the retina.
But if it doesn't detach cleanly, it stays attached at strong adhesion points:

- Vitreous base
- Around retinal vessels
- At lattice degeneration
- Around areas of previous inflammation
- ONH

At these points, the vitreous pulls which causes persistent traction. This can create:

1. **Retinal breaks**
   + Flap tears (horseshoe) – most common with acute traction
   + Operculated holes – if a piece of the retina is pulled off
2. **Retinal detachment**

NOTE: A retinal HOLE can be caused by **atrophy** NOT traction. There are 2 types of retinal holes, atrophic and tractional.

What history and symptom questions should be asked to someone who has a PVD?

- How long have you been experiencing these symptoms (of flashes/floaters, curtain over vision)?
- Distinguish retinal photopsias from visual distortion of migraines, which may be accompanied by new floaters.

NOTE: Visual aura is described more as a zigzag pattern that advances across the visual field as time passes. Retinal photopsia is described as a very quick sudden flash, like a lightning flash in the periphery, often temporal

- Also important to ask about risk factors for retinal break:
  ★ Age 46-65 year olds (if a younger onset than it is more likely to be a retinal tear)
  ★ Associated symptoms
  ★ Myopic (especially if high)
  ★ Trauma/surgeries (already existing vulnerability)
  ★ FOHx of retinal tears/ detachments

Discuss the workup for a patient with PVD. Explain the reasoning behind each step and which signs you are looking for.

The patient will need to undergo a complete ocular examination. It is important that BOTH eyes are assessed for presence of PVD and peripheral retinal pathology.

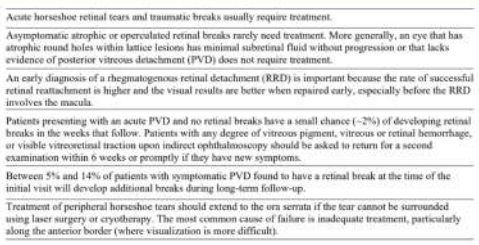
(1) **DFE WITH scleral indentation** → This is a mandatory part of the workup. This is to rule out RD and tears. If you do not do scleral indentation, you will miss ~11% of retinal breaks/detachments. Can also look for the presence of Weiss ring over the optic disc.
(2) **Evaluate the anterior vitreous** → Signs of pigmented cells (tobacco dust). Tobacco dust has a high correlation with peripheral retinal breaks. Ask patient to look around as pigment settles by gravity.
(3) **OCT** → Can help confirm presence or absence of PVD
(4) **SLE** → Can try to visualise PVD with 60D or 90D lens. Patient makes rapid saccades then looks straight to float PVD in view if not visible (looks like a grey-black strand).
(5) If VH obscures view then perform **ultrasonography** to rule out a retinal break, RD, or other causes of VH. Inferior layering VH may mimic a retinal break on US.

NOTE: Pseudophakic patients may have smaller anterior breaks than phakic patients.

What is the follow up period if:
i) a MILD VH is found/mild peripheral punctate retinal haemorrhages
ii) a SIGNIFICANT VH is found or anterior pigmented cells (tobacco dust - scheffer's sign)

i) If no retinal break is found, but mild VH or peripheral punctate retinal hemorrhages are present (indicating increased vitreous traction), **repeat examinations are performed in 2 weeks.**

ii) If no retinal break is found but significant VH or anterior pigmented vitreous cells are present, repeat examination should be performed **within 24 hours** by a retina specialist because of the high likelihood of a retinal break.



Acute horseshoe retinal tears and traumatic breaks usually require treatment.

Asymptomatic atrophic or operculated retinal breaks rarely need treatment. More generally, an eye that has atrophic round holes within lattice lesions has minimal subretinal fluid without progression or that lacks evidence of posterior vitreous detachment (PVD) does not require treatment.

An early diagnosis of a rhegmatogenous retinal detachment (RRD) is important because the rate of successful retinal reattachment is higher and the visual results are better when repaired early, especially before the RRD involves the macula.

Patients presenting with an acute PVD and no retinal breaks have a small chance (~2%) of developing retinal breaks in the weeks that follow. Patients with any degree of vitreous pigment, vitreous or retinal hemorrhage, or visible vitreoretinal traction upon indirect ophthalmoscopy should be asked to return for a second examination within 6 weeks or promptly if they have new symptoms.

Between 5% and 14% of patients with symptomatic PVD found to have a retinal break at the time of the initial visit will develop additional breaks during long-term follow-up.

Treatment of peripheral horseshoe tears should extend to the ora serrata if the tear cannot be surrounded using laser surgery or cryotherapy. The most common cause of failure is inadequate treatment, particularly along the anterior border (where visualization is more difficult).

How do you treat PVD? What would be the treatment indicated for this patient? Justify the follow up period.

Normally, no treatment is indicated for PVD unless an acute retinal break or dense vitreous hemorrhage is found.

In the presence of **symptomatic PVD**, chronic retinal breaks or lattice degeneration usually warrant Tx.

Same day/ within 24 hours referral to a SURGICAL retinal specialist if you suspect a retinal break.

In this patient's case, given she has not observed a curtain in her vision, her VA is good and we are operating on the assumption that no retinal tear was observed, this patient should still be referred urgently in case of a retinal break (erring on the side of caution). If a patient is getting flashes of light, this indicates that the patient is getting some sort of traction happening, so it is best to seal those off.

Educate patient on symptoms that require urgent review (curtain, sudden increase in floaters, new flashes, loss of vision or worsening of it, shadow in vision). The timing of these symptoms can be from days to years later. Patient should also be informed of the likelihood of PVD in fellow eye,.

Routine follow-up is a dilated fundus exam WITH scleral depression in 4–6 weeks to check for delayed retinal tears. 2-5% risk of developing new retinal breaks in patients with PVD with no retinal break at presentation.

NOTE: In the setting of acute PVD symptoms, chronic retinal breaks (pigmented) or lattice degeneration usually warrant treatment.

2. Retinal break (tear)

A 62-year-old **moderately myopic woman** presents with a 2-day history of **sudden onset flashes** in her right eye, described as "brief sparkles in the peripheral vision," especially with eye movements. She also reports **a new large floater that appeared suddenly yesterday**. No curtain or shadow in her vision. She denies headache or history of migraine. No trauma. She underwent uncomplicated cataract surgery in the same eye 6 months ago.

The patient has been diagnosed with **a retinal break**

What are the predisposing conditions to a retinal break?	List the workup procedures for a retinal break.	Discuss which treatment options are recommended and how urgent this is.
• Lattice degeneration • High myopia • Aphakia + pseudoaphkia • Age-related retinoschisis • Vitreoretinal tufts • Meridional folds • Hx of previous retinal breaks or detachments in fellow eye • Family Hx of retinal breaks or detachments • Collagen vascular disorders (systemic lupus erythematosus AKA lupus, rheumatoid arthritis,, Sjogren's syndrome) • Trauma	**SAME AS PVD** (1) Complete ocular examination with slit lamp and indirect ophthalmoscopy of both eyes (DFE) (2) SLE = Looking for tobacco dust (3) B-scan US to be used in cases where retina is not visible (dense VH or cataract) ***NOTE:*** After trauma, scleral depression may be performed gently once open globe injury has been ruled out.	**Laser photocoagulation or cryotherapy** required **within 24hrs** especially for: • Acute symptomatic break/new onset • Acute traumatic break (including a dialysis) • Acute symptoms and lattice degeneration Treatment to be considered: • Asymptomatic retinal break that is large (e.g., ≥1.5 mm), above the horizontal meridian, or both, particularly in the absence of PVD. • Asymptomatic retinal break in an aphakic or pseudophakic eye, a highly myopic eye, or an eye in which the involved or contralateral eye has had an RD. **Asymptomatic** cases are not treated unless it is **superior and large ≥1.5 mm** (since gravity will pull it down). Can also consider treating in absence of PVD and if associated with other risk factors like pseudophakia (cataract surgery), high myopia. Treatment may be less urgent for chronic breaks. ENSURE that when you see a retinal break/tear document the absence or presence of a PVD. Can also document the area of rub-retinal fluid or detachment to see if it is advancing over time.

Justify the follow up periods for retinal breaks.

1. Patients with predisposing conditions or retinal breaks that do not require treatment are followed at 3 months and then every 6 to 12 months if stable.
2. Patients treated for a retinal break are reexamined in 2 weeks (as scars take that long to form), 6 weeks, 3 months, and then every 6 to 12 months.

RD symptoms (a dramatic increase in floaters or flashing lights, worsening visual acuity, or the appearance of a curtain, shadow, or bubble anywhere in the field of vision) are explained and patients are told to return immediately if these symptoms develop.

3. Retinal detachments

- **Exudative/serous**
- **Rhegmatogenous**
- **Tractional**

List the workup procedures for a retinal detachment of exudative nature	List the workup procedures for a retinal detachment of rhegmategnous nature	List the workup procedures for a retinal detachment of tractional nature
(1) Intravenous fluorescein angiography (IVFA) → Will show leakage or pooling identifying source of SRF (2) OCT → Helps with identifying source of SRF (e.g CNV) (3) B-scan US → Helps identify underlying cause (4) Systemic workup → Rule out underlying cause (eg. neoplastic, inflammatory, vascular)	(1) SLE to assess signs of uveitis, lens status and presence of pigment/VH and PVD (2) DFE+ SLE of periphery helps find small breaks. (3) B-scan US helpful if media opacities are present. ***NOTE:*** We look for signs of uveitis because inflammatory cells in the vitreous can mimic Shafer's sign, uveitis can both mask and predispose to retinal tears, and the management pathways differ completely. We check lens status because we want to assess lens status to identify pseudophakia (a risk factor for retinal tears) and to check for media opacities. If the view to the retina is compromised, perform a B-scan to rule out a tear or detachment."	(1) SLE to assess lens status, sequelae of retinal vascular disease (e.g vascularization of the iris) and PVD (2) DFE+ SLE of periphery helps find small breaks. (3) B-scan US helpful if media opacities are present. (4) OCT is useful in identifying tractional membranes and can be useful in differentiating tractional membranes from detached retina.

Outline management and follow ups for all 3 types of retinal detachments:

RD TYPE	MANAGEMENT	FOLLOW UP
Exudative/Serous (Exudative RD's)	Treat underlying condition	Depends on cause and individual patient factors
Rhegmategonous (RRD)	**Macula on but threatened:** Should have URGENT surgical repair (scleral buckle, vitrectomy or combination) performed urgently. Also photocoagulation, cryotherapy and pneumatic retinopexy. **Macula OFF:** Should be repaired but not necessarily urgently, as it has been shown there is no difference in visual outcomes if performed within 7-10 days. Chronic **macula OFF** RD's are treated within 1 week if possible.	All patients with RRD require urgent follow up with a retina specialist. Follow up after surgery is typically 1 day, 1 week, 1 month, 2-3 months then every 6-12 months thereafter.

Tractional (TRD)	May or may not require intervention depending on etiology, status of the fellow eye, and the extent/location of retinal traction. Mostly associated with PDR	Depends on cause and patient factors.

4. Retinoschisis (Age related AKA Degenerative)

A 65-year-old mildly hyperopic woman presents for a routine eye examination.
- She reports no visual symptoms: no flashes, no floaters, and no visual field defects.
- Her vision is 6/6 in both eyes with normal IOP.
- She has no history of trauma, retinal disease, or surgery.
- On dilated fundus exam with a 90D lens, in the inferotemporal periphery of the right eye, you observe a smooth, immobile, dome-shaped retinal elevation.

The patient has been diagnosed with **retinoschisis (age-related).**

List the workup procedures for this condition.	Discuss which treatment options are recommended and how urgent this is. Also justify a follow up plan.
(1) Slit lamp evaluation for anterior chamber inflammation and pigmented anterior vitreous cells; neither should be present in isolated retinoschisis. (2) DFE with scleral depression to rule out a concomitant RD or an outer layer retinal hole, which may lead to an RD. (3) SLE to aid in recognizing outer layer retinal breaks. (4) OCT can help determine which layer of the retina is split. (5) Visual field testing will reveal an **absolute scotoma** in the area of schisis.	Surgery is indicated when a clinically significant RD develops. A small RD walled off by a demarcation line is usually not treated. This may take the form of pigmentation at the posterior border of outer layer breaks. Follow up every **6 months.** Educate patient on RD symptoms (an increase in floaters or flashing lights, blurry vision, or the appearance of a curtain or shadow anywhere in the field of vision) are explained to all patients, and patients are told to return immediately if these symptoms develop.

5. Retinoschisis (X linked - Juvenillie)

A 9-year-old boy is brought in by his parents because he has been struggling to see the board at school and recently failed a vision screening. He reports blurred central vision in both eyes but denies pain, flashes, or floaters. Parents note that he has always been "clumsy" and sometimes has difficulty with ball sports.
There is no history of trauma, and he is otherwise healthy. Mother mentions that her maternal uncle had "poor vision from childhood.". DFE of macula reveals 'spoke-wheel' pattern foveal schisis in both eyes
The patient has been diagnosed with **retinoschisis (Juvenile).**

List the workup procedures for this condition.	Discuss which treatment options are recommended and how urgent this is. Also justify a follow up plan.
(1) Family Hx (2) DFE with scleral depression to rule out retinal break or detachment (3) OCT can help determine the layer of schisis and help differentiate schisis from RD (4) Fundus autofluorescence (FAF) may help delineate areas of schisis. (5) Electroretinography (ERG) is **NOT** necessary for diagnosis but can show a reduced b wave with a preserved a wave.	1. No definitive treatment for stellate maculopathy. Topical carbonic anhydrase inhibitors (CAI) have been shown to decrease foveal thickness and improve visual acuity in some patients. 2. For nonclearing VH, consider vitrectomy. 3. Surgical repair of an RD should be performed. 4. Superimposed amblyopia may be present in children younger than 11 years of age when one eye is more severely affected, and a trial of patching should be considered. Follow up every **6 months.** This should be more frequent if treating amblyopia.

VASCULAR DISEASES (14 CONDITIONS)

6. Angioid Streaks

Describe what is angioid streaks	Outline the workup for angioid streaks	How do you treat angioid streaks? Outline a follow up period. What would you do in the case that the patient plays high-impact sports?
They are bilateral reddish-brown or gray bands located deep to the retina. Occur due to **breaks in Bruch's membrane**, which may occur spontaneously or secondary to blunt trauma. Fifty percent of cases are associated with systemic diseases; the rest are idiopathic. ● Sx: Asymptomatic, but can cause visual impairment if macula involved (CNV/SR haemorrhage)	**(1)** History: Any known systemic disorders? Previous ocular trauma? **(2)** DFE = Looking at macula for CNV. **(3)** FAF if diagnosis uncertain or IVFA or OCTA if CNV suspected. **(4)** Physical examination: Look for clinical signs of etiologic diseases. **(5)** Systemic workup (eg. serum alkaline phosphatase/urine calcium levels for Paget disease, sickle cell preparation in patients of African descent, skin or scar biopsy if PXE suspected, CBC if hematologic etiology suspected etc)	**1.** Management of any underlying systemic disease by an internist. **2.** Treat any secondary ocular complications ➔ CNV → Anti-VEGF for angioid streak-associated CNV (focal laser photocoagulation and PDT have discouraging outcomes) Patients who play sport should be recommended to wear one-piece polycarbonate safety glasses due to an increased risk of subretinal hemorrhage and choroidal rupture from minor trauma. **Follow up:** Fundus examination every 6 months, monitoring for CNV. Instruct patient to **check Amsler grid daily** and return immediately if changes are noted.

7. Cotton Wool Spot

Describe the etiology of a CWS.	Outline the workup in the presence of a CWS.	How does it take for a CWS to fade? What is the treatment?
Thought to be an acute obstruction of a precapillary retinal arteriole causing **blockage of axoplasmic flow** and subsequent buildup of axoplasmic debris in the NFL. Found commonly in: ● DM ● Chronic or acute HTN ● Retinal vein occlusions ● Retinal emboli ● Collagen vascular disease (lupus) ● GCA ● Cancer ● HIV retinopathy ***NOTE:*** Even 1 CWS is **NOT** normal.	(1) History: Diabetes or HTN? Prior ocular or periocular radiation? GCA symptoms in appropriate age group? Symptoms of collagen vascular disease including joint pain, rashes, etc.? HIV risk factors? Hematologic abnormalities? (2) Complete ocular examination, including DFE + SLE = Look for concurrent hemorrhages, vascular occlusion, vasculitis, hard exudates. (3) Check blood pressure. (4) Check fasting blood sugar and hemoglobin A1c. (5) Consider erythrocyte sedimentation rate (ESR), C-reactive protein (CRP), and platelets if GCA is suspected. (6) Consider blood and urine cultures, chest x-ray, carotid and orbital Doppler examination, chest CT, and echocardiography if emboli are suspected. (7) Consider HIV testing. (8) Fluorescein angiography is **generally not helpful** for an isolated CWS **without concomitant pathology**. IVFA reveals areas of capillary nonperfusion adjacent to CWS location (not very useful for isolated CWS).	Takes **5-7 weeks to fade but can remain longer if associated with DR.** Treatment and follow up depends on underlying etiology. If concern for infectious processes, serial dilated examinations are recommended.

8. CRAO (Central Retinal Artery Occlusion)

A 74-year-old female with no previous ocular pathology presented with acute vision loss OD. Vision was LP OD and 20/25 OS. There was a right RAPD. She denied any family history of ocular disease. Her medications included rosuvastatin, ramipril, metformin and sliding scale insulin.

The patient has been diagnosed with **CRAO**.

How would you inquire about a patient's visual symptoms? What other tests would be part of your workup? What would these tests show?	Describe the blood tests you would carry out and why.	What is the point in assessing risk factors? Which tests are these?
Enquire about **amaurosis fugax** (phrase as "Any history of transient vision loss, any recent loss of vision that lasted seconds or more, in one or both eyes?" Workup will be treated as an acute stroke. Tests will include: ● Pupils = RAPD ● VA = Will be lowered ● DFE= Superficial opacification or whitening of the retina in the posterior pole and a cherry-red spot in the center of the macula (may be subtle).	To exclude arteritis causes (GCA) – which is an ocular emergency: : ● Erythrocyte sedimentation rate (ESR - will be elevated in GCA) ● CRP (c-reactive protein) ● Platelet Count All will be elevated in GCA.	BP, fasting glucose, and fasting lipids are obtained in CRAO because it is an ocular stroke; these tests identify systemic vascular risk factors such as hypertension, diabetes, and dyslipidaemia, which contribute to **emboli** formation and predict risk of further cerebrovascular events.

Why do we investigate embolic sources and how do we do this?	If the patient was less than 50 years old and had no risk factors, what would you do additionally as part of the workup? Justify why.	Summarise the workup.
● MRA = Helps identify patients at risk for brain stroke ● Echocardiogram = Looks for cardiac emboli ● Carotid doppler = Looks for carotid artery plaques MRA, Carotid Doppler, and Echocardiogram are performed in CRAO workup to identify embolic sources—most commonly carotid atherosclerotic plaques and cardiac emboli—because CRAO is an ocular stroke with significant risk of subsequent cerebral stroke. ***NOTE:*** Emboli from the heart can travel to the brain and give patient a stroke.	CRAO is an arterial occlusion. In patients under 50, a major cause is abnormal clotting tendencies rather than atherosclerosis. We would perform hypercoagulable and Vasculitic Screens, Blood Film & take a detailed MHx. **Hypercoagulable screen** — Rule out clotting disorders like spontaneous arterial thrombosis **Vasculitic screen** — Rule out autoimmune/inflammatory vasculitis damaging retinal arteries **Blood film** — Detect malignant, haematologic, or hyperviscosity problems **Detailed MHx** — Identify lifestyle/drug/medication or systemic causes not seen on labs In patients younger than 50 years or with appropriate risk factors or positive review of systems, consider lipid profile, antinuclear antibody (ANA), rheumatoid factor, syphilis testing (RPR or VDRL and FTA-ABS or	(1) Should be treated as an acute stroke. The American Academy of Ophthalmology (AAO) 2018 guidelines suggest that all these patients should be sent **immediately** to an emergency department, preferably affiliated with a stroke center, for evaluation and workup. ***WORKUP DONE WITH GP:*** (2) GCA suspect: Immediate ESR, CRP, and platelets to rule out GCA if the patient is 55 years of age or older and no embolus seen on examination. Query GCA review of systems. **If the patient's history, laboratories, or both are consistent with GCA, start high-dose systemic steroids.** (3) Check the blood pressure. (4) Other blood tests: Fasting blood sugar and hemoglobin A1c, complete blood count (CBC) with differential, prothrombin time/activated partial thromboplastin time (PT/PTT). (5) Carotid artery evaluation by duplex Doppler US. (6) Cardiac evaluation with electrocardiography (ECG),

	treponemal-specific assay), serum protein electrophoresis, hemoglobin electrophoresis, and further evaluation for hypercoagulable state. **Test** / **What it detects** / **Why it matters** **Lipid profile** — Familial hyperlipidaemia — Early atherosclerosis/emboli **ANA** — Autoimmune diseases — Autoimmune vasculitis affecting retinal artery **RF** — Autoimmune inflammatory conditions — Vasculitic occlusion **Syphilis tests** — Infectious vasculitis — Can mimic CRAO **SPEP** — Plasma cell disorders, hyperviscosity — Thick blood → occlusion **Hb electrophoresis** — Sickle cell/hemoglobinopathies — Vessel obstruction **Hypercoag panel** — Thrombophilia — Clots causing CRAO	echocardiography, and possibly Holter monitoring or bubble study. (7) OCT (8) Can also consider IVFA. **Less commonly ERG is used.**

Outline a management plan. Describe what can cause <u>anecdotal reports</u> of improvement. Justify the use for these anecdotal methods	**What is the follow up period for CRAO?**
RAO is treated as an **acute stroke** and **immediate referral** to **an emergency department with an affiliated stroke center** is warranted. For **specific management of ocular signs and symptoms,** there are anecdotal reports of improvement after the following treatments, if instituted within 90 to 120 minutes of the occlusive event. None of these treatments have been proven effective in randomized, controlled clinical trials and **<u>should not be considered standard of care.</u>** 1. Immediate ocular massage with fundus contact lens or digital massage. 2. Anterior chamber paracentesis 3. IOP reduction with **acetazolamide**, 500 mg i.v. or two 250-mg tablets p.o. or a topical beta-blocker (e.g., timolol or levobunolol, 0.5% daily or b.i.d.).	**1.** Follow as directed by managing **internist and/or neurologist.** **2.** **Repeat eye examination in 1 to 4 weeks**, checking for neovascularization of the iris/disc/angle/retina (NVI/NVD/NVA/NVE), which develops in up to 20% of patients at a mean of 4 weeks after onset. **3.** If neovascularization develops, perform panretinal photocoagulation (PRP) and/or administer an anti-vascular endothelial growth factor (anti-VEGF) agent

9. BRAO (Branch Retinal Artery Occlusion)

A 72-year-old man has been diagnosed with a **Branch Retinal Artery Occlusion (BRAO)** in his right eye. He reports sudden onset superior visual field loss that started 3 hours ago. He describes it as “a dark patch at the top of my vision.”. He denies pain, flashes, or floaters.
Past medical history:

- Hypertension
- Hyperlipidaemia
- Smoker

Describe the workup for this patient. What is the exception between CRAO and BRAO workup?	**What do you do in the case that a patient’s symptoms are consistent with GCA?**	**What do the AAO 2018 guidelines suggest for BRAO treatment? Justify why the follow up period differs between CRAO and BRAO.**
Should be treated the same as a CRAO.With BRAO, ERG is not helpful. ERG is not useful in BRAO because it measures global retinal function and BRAO causes a localized sectoral ischemia that is too small to affect full-field ERG results. Workup for BRAO will include: (1) Immediate referral to ED ideally with an affiliated stroke centre for workup (2) If a patient is over 55 years of age, then ask about GCA symptoms. Perform ESR, CRP and platelet counts to rule out GCA. (3) Check blood pressure (4) Other blood tests: Fasting blood sugar and hemoglobin A1c, complete blood count (CBC) with differential, prothrombin time/activated partial thromboplastin time (PT/PTT). (5) Carotid artery evaluation by duplex Doppler US. (6) Cardiac evaluation with electrocardiography (ECG), echocardiography, and possibly Holter monitoring or bubble study. (7) OCT can be very helpful in making the diagnosis. Can also consider IVFA.	If the patient’s history, laboratories, or both are consistent with GCA, start high-dose systemic steroids.	The AAO 2018 guidelines suggest that all these patients should be sent immediately to an emergency department, preferably affiliated with a stroke center, for evaluation and workup. BRAO follow up period is longer (3-6 months) because NV is rarer (as compared to CRAO).

10. CRVO (Central Retinal Vein Occlusion)

Describe the two types of CRVO. Why does DFE help in making this distinction in CRVO?	Describe the SYSTEMIC workup for this patient + Describe the OCULAR workup for this patient.	
1. **Ischemic CRVO:** Vision typically worse (<20/200) with RAPD and visual field defects. Extensive retinal hemorrhage, CWSs, venous tortuosity, and widespread capillary nonperfusion on IVFA (often >10 disc diameters). ERG shows decreased b-wave amplitude. Higher risk of neovascularization. 2. **Nonischemic CRVO:** Vision often better than 20/200, mild or no RAPD, mild fundus changes. Lower risk of neovascularization.	(1) History: Medical problems, medications (especially antihypertensive medications, oral contraceptives, diuretics), eye diseases? (2) Check blood pressure. (3) Blood tests: Fasting blood sugar and hemoglobin A1c, CBC with differential, platelets, PT/PTT, lipid profile. (4) If clinically indicated, particularly in younger patients, consider hemoglobin electrophoresis, RPR or VDRL, FTA-ABS or treponemal-specific assay, ANA, cryoglobulins, antiphospholipid antibodies, factor V Leiden mutation, protein C and S levels, antithrombin III mutation, prothrombin mutation, homocysteine levels, serum protein electrophoresis, and chest radiograph. (5) Complete medical evaluation, with careful attention to cardiovascular disease or hypercoagulability.	(1) Complete ocular examination, including IOP measurement, careful slit lamp examination and gonioscopy to rule out NVI and NVA (both of which are best observed before dilation), and dilated fundus examination. (2) IVFA: Risk of neovascularization proportional to degree of capillary nonperfusion. (3) OCT: Used to help detect presence and extent of ME as well as to monitor response to therapy. (4) If the diagnosis is uncertain, oculopneumo-plethysmography or ophthalmodynamometry may help to distinguish CRVO from carotid disease (but are infrequently performed). Ophthalmic artery pressure is low in carotid disease but is normal to increased in CRVO.

Describe the management plan for CRVO	Justify the use of Aspirin in this case.	How do you treat CRVO related macular oedema? What are the associated complications?
1. Discontinued related medicines if possible (e.g contraceptives & diuretics – HTN meds) 2. Treat underlying medical cause (for ischemic type) 3. Aspirin 81-325mg (po/oral) daily → however, could increase risk of haemorrhage 4. If IOP is increased, reduce it → reduce using anti-glaucoma drugs 5. If NV developed → Anti-VEGF as adjunct to PRP — panretinal photocoagulation	Aspirin 81 to 325 mg po daily often recommended but no clinical trials have demonstrated efficacy and it may increase risk of haemorrhage	• **Intravitreal anti-VEGF (1st line)** — **bevacizumab,** 0.5 mg **ranibizumab ,** 2 mg **aflibercept, or faricimab** • Intravitreal corticosteroid implant (40 mg, injecting 1-4 mg dose **triamcinolone** and **dexamethasone**) • May be combined with **laser photocoagulation**

Intra-vitreal Ranibizumab (FDA), Bevacizumab (Off-Label)	Dexamethasone (FDA)
Risks of intravitreal injections are low but include VH and endophthalmitis, among others.	Complications include cataract formation and elevated IOP.

Justify follow up period for CRVO

With CRVO, we are concerned about risk of stroke, neovascularisation (especially NVI) and macular oedema. 50% risk of developing Rubeosis iridis (NVI), resulting in high risk of neovascular glaucoma.

Hence with follow up, you will want to see the patient every month initially, gradual interval taper based on vision, presence/absence of macular oedema and response to treatment. Advise patient of 8-10% risk of fellow eye BRVO/CRVO.

11. BRVO (Branch Retinal Vein Occlusion)

Outline workup of BRVO	List treatment of BRVO	What is the gold standard treatment for BRVO?	Why do we not use focal retinal laser photocoagulation anymore?
(1) History: Systemic disease, particularly HTN or diabetes? (2) Complete ocular examination, including dilated retinal examination with indirect ophthalmoscopy to look for retinal neovascularization and macular oedema (ME). (3) OCT: Used to help detect presence and extent of ME as well as monitor response to therapy. (4) Check blood pressure. (5) Blood tests: Fasting blood sugar and hemoglobin A1c, lipid profile, CBC with differential and platelets, PT/PTT. If clinically	• Retinal neovascularization: Sector PRP to the ischemic area, which corresponds to area of capillary nonperfusion on IVFA. • Prompt and appropriate treatment of underlying medical conditions (e.g., HTN).	1. Anti-VEGF treatment is now the gold standard. Intravitreal **ranibizumab 0.5 mg** and **aflibercept 2 mg** are FDA-approved for treating RVO-associated ME. Intravitreal bevacizumab has also been used off-label. Risks of intravitreal injection are low but include VH and endophthalmitis. 2. **Dexamethasone** intravitreal implant or off-label intravitreal steroid (e.g., triamcinolone 40 mg/mL, inject 1 to 4 mg). **FOLLOW UP SAME AS CRVO**	Focal retinal laser photocoagulation has historically been the gold-standard treatment if edema is present for 3 to 6 months duration, and visual acuity is below 20/40 with macular capillary perfusion. However, anti-VEGF treatment is now largely favored. Limitations of focal laser include length of time before effect (often several months) and the need to wait until retinal hemorrhages clear to perform the surgery.

indicated, consider a more comprehensive workup. (6) Medical examination: Performed by an internist to check for cardiovascular disease. (7) An IVFA is obtained after the hemorrhages clear or sooner if neovascularization is suspected.			

12. Diabetic Retinopathy

List some risk factors of DR	What should the discussion of a diabetic patient's medical history include?	What is the most frequent loss of diabetes-related vision loss?	What is the recommended examination procedure for examining patients with diabetes?
• Hyperglycaemia • Systemic hypertension (>130mmHg) • Elevated HbA1c (>7%) • Hyperlipidaemia • Indigenous Australians or those of NESB • Dependence on insulin/Type of diabetes • Retinal arteriolar tortuosity • Other factors ○ Smoking ○ Pregnancy ○ Renal disease ○ Nutritional factors ○ Genetic factors	• Type and duration of diabetes • Blood pressure and cholesterol/lipid status • Current treatment (diet, oral medications, insulin type and dosage, other injectable medications such as exenatide) • Glycaemic control (including recent laboratory values for HbA1C) • Renal (kidney) function: presence of microalbuminuria29 or overt proteinuria • Presence of peripheral neuropathy (tingling or numbness in extremities) • Adherence, method, frequency, and results of self-monitoring of blood glucose • Practitioners managing the patient's diabetes care (GP, endocrinologist and/or renal specialist)	DMO	

Procedure	Comments
Visual acuity (with correction)	Distance and Near, monocularly, including pinhole acuity if indicated
Pupil reactions	Direct/Consensual and Near pupillary responses
Ocular motility and cover test	Extent, fluency and symmetry of ocular movements in all directions of gaze, ruling out eye movement anomalies. Relevant history taking regarding the onset and direction of diplopia is necessary. If diplopia is manifest, cover test and prism neutralisation is also indicated.
Visual field screening	- Confrontation - Note: visual field screening and baseline testing in the absence of actual or suspected pathology of the visual pathways or brain will not attract a Medicare rebate for item 10940 or 10941.
Refraction	- On indication - Where patient reports a change in vision or visual function (e.g. increased glare sensitivity) or where a change in habitual visual acuity is measured.
Slitlamp biomicroscopy	- Recommended at every visit - Examination for iris neovascularisation (NVI), diabetic cataract, corneal integrity
Tonometry	Pre-dilation, post-dilation as indicated
Stereoscopic fundus examination with pupil dilation	- NHMRC guidelines describe pupil dilation using 0.5 to 1.0% tropicamide as safe (noting dilated ocular fundus examination increases the sensitivity of DR screening) and so consider it mandatory in performing ophthalmoscopy or slitlamp biomicroscopy[2] unless contraindicated by the presence of potentially occludable anterior chamber angles. - Gonioscopy may be necessary to evaluate whether narrow angles are potentially occludable. Where potentially occludable angles are detected, the patient should be referred to an ophthalmologist for an opinion on management e.g. suitability for PI. - People with diabetes can show a poor response to mydriatic agents. 2.5% phenylephrine hydrochloride, unless contraindicated, may help achieve maximum pupillary dilation in patients with DM, particularly those of Aboriginal and Torres Strait Islander descent and patients with heavy iris pigmentation - Optometry Australia pupil dilation guidelines can be viewed here - Symptoms of VA reduction or distortion of vision or a significant change of DM control should always prompt dilated pupil examination

What does supplementary testing of DR include?	What is the recommended frequency of eye examination according to patient classification?
• Colour vision assessment (MONOCULARLY) • Contrast sensitivity testing • Gonioscopy • Macular function assessments • OCT • Ocular fundus photography • Wide-field fundus imaging	

Classification	One year risk of PDR	Review / referral timeframe*	Ophthamologist referral
No apparent DR, no DMO	N/A	1–2 yrs	Not required
Mild NPDR	1-5%	6–9 mos	Not required
Moderate NPDR	12-27%	4 mos (3–6 mos)	Required
Severe NPDR	~52%	4 weeks (<3 mos)	Required
Proliferative DR		Within 1 week (<1 mo)	Required
Non-center-involving DMO		3 mos	Required
Center-involving DMO		1 mo	Required

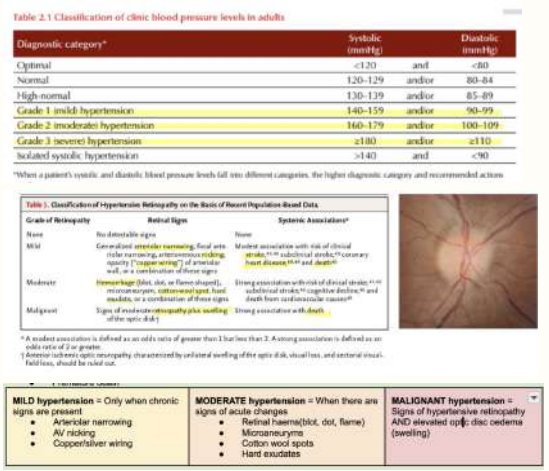
Patient Classifications	Frequency of Examination
All patients with diabetes (T1DM or T2DM)	At diagnosis and 2 yearly thereafter. Annually for higher-risk patients: Aboriginal and Torres Strait Islander patients, longer duration of diabetes, poor glycaemic control, blood pressure or blood lipid control. For children with pre-pubescent type 1 diabetes, Optometry Australia advises introducing eye examinations early in the course of diabetes, to assess patient's ocular fundus, measure baseline VA and perform other tests appropriate for child's age and stage of development. Otherwise, children with pre-pubertal diabetes should undergo annual examinations from puberty.
Patients with non-proliferative diabetic retinopathy (NPDR) or mild NPDR	Depending on level of DR present, 6–12 monthly or annually. • **Mild NPDR without DMO**: 6–12 months. • **Mild NPDR with DMO:** Ophthalmology referral
Moderate NPDR	**Moderate NPDR WITHOUT DMO:** 3-6 monthly or referral if at high risk of progression **Moderate NPDR <u>WITH</u>** DMO: Ophthalmology referral within 4 Wk
Pregnant women with diabetes	Pregnancy in females with T1DM or T2DM may accelerate the development and progression of DR. Hence, women with diabetes who become pregnant should: • Have a comprehensive eye exam prior to conception if planning pregnancy • Have a comprehensive eye examination in the first trimester

	• Base the frequency of examinations during the pregnancy should be based on first trimester exam results and blood glucose control during pregnancy • Have a comprehensive eye exam 6–12 weeks post-partum.
Women with gestational diabetes	Women with gestational diabetes do not need ophthalmic surveillance during or after pregnancy, unless diabetes persists.

NOTE: Referral to an ophthalmologist should be urgent if diabetic macular oedema (DMO) or proliferative diabetic retinopathy (PDR) is suspected. If an unexplained reduction in visual acuity is observed, OCT and/or OCT-A may be useful tools to confirm or rule out oedema or macular ischaemia.

What does Optometry Australia recommend that optometrists inform the patient's GP and endocrinologist of?	**Outline the management and follow up plan for diabetic retinopathy.**
Correspondence (at a minimum) should include: **A.** VA **B.** Whether a DFE was performed **C.** Whether there is presence of retinopathy and if so, its classification **D.** The next recommended eye review	1. Patient education → Diabetic related (like weight, HbA1c) & how this needs to be closely managed with GP 2. Optimisation of vascular risk factors 3. Communicate findings with GP & Endocrinologist (treat with hormones and endocrine glands) 4. Management with GP a. Statins → 1 st line (Lower # in liver production) b. **Fenofibrate** therapy → only reduce the progression of DR by lowering free fatty acids i. Indication: T2DM & mild-moderate DR) <u>Management with DMO:</u> (1) If better than 6/7.5 → Observation (2) If VA is 6/9 or worse → Refer to ophthalmologist within 4 wks (a) Intravitreal Anti-VEGF (1st line) : Ranibizumb & Aflibercept, Bevacizumb (off-label) (b) Intravitreal corticosteroid: Dexamethasone, fluocinolone acetonide (long-lasting)Triamcinolone 40mg/mL injecting 1-4mg (off-label) <u>Management of PDR</u> (1) Panretinal Photocoagulation (PRP) if (a) NVD >¼ disc area (b) Any NVD with pre-retinal or VH (c) NVE >½ disc area with pre-retinal or VH (d) Any NVI & NVA (2) ± Intravitreal Anti-VEGF (3) ± Pars Plana vitrectomy

13. Hypertensive Retinopathy

Outline workup	**Classification of hypertension and hypertensive retinopathy**	**How do we manage and follow up for hypertensive retinopathy?**	**What are the risk factors for hypertensive retinopathy**
(1) History: Known HTN, diabetes, or adnexal radiation? (2) Check blood pressure. (3) Complete ocular examination, particularly dilated fundus examination. (4) Refer patient to a medical internist or an emergency department. The urgency depends on the blood pressure reading and whether the patient is symptomatic. A systolic blood pressure ≥180 mm Hg, a diastolic blood pressure ≥110 mm Hg or the presence of chest pain, difficulty breathing, headache, change in mental status, or blurred vision with optic disc swelling requires immediate medical attention. (5) Patients may need workup for secondary causes of hypertension such as those listed above.		• Refer to GP /internist → control HTN • Refer urgently to ED if symptomatic, malignant signs, or if BP ≥180/110 (severe HTN) <u>Follow up:</u> • Initially every 2-3 months, 6-12 month is stable	• Age (>60) • Smoking • Overweight • Diet: high salt/saturated fat • Lack of physical exercise • Alcohol (>1-2/day) • Stress • FoHx • Chronic kidney disease • Pregnancy

14. Ocular histoplasmosis

Outline workup	List treatment	Develop a follow up treatment plan
(1) History: Time spent in the Ohio-Mississippi River Valley or endemic area? Prior exposure to fowl? (2) Amsler grid test = To evaluate the central visual field of each eye. (3) Slit lamp examination: Anterior chamber or vitreous cells and flare should not be present. (4) DFE= Look for signs of CNV and vitreous cells. (5) IVFA and OCT to help detect CNV and monitor response to treatment.	**1.** Antifungal treatment is not helpful — the disease is thought to be a scarring process from an old fungal infection, not a current one. **2.** Intravitreal anti-VEGF therapy for CNV is the mainstay of treatment. PDT for subfoveal CNV and focal laser photocoagulation for extrafoveal CNV may rarely be used as well.	• Instruct all patients to use an Amsler grid daily and to return immediately if any sudden visual change is noted. • Patients treated with anti-VEGF injections are seen every 4 to 6 weeks, depending on clinical response to therapy. Generally, a more complete treatment response is achieved with fewer injections than in AMD. • Patients are often able to stop injections rather than undergoing lifelong treatment. Patients treated with PDT or focal laser are typically seen at 2 to 3 weeks, 4 to 6 weeks, 3 months, and 6 months after treatment, and then every 6 months thereafter. • A careful macular examination and OCT is performed at each visit. IVFA may be repeated whenever renewed neovascular activity is suspected. • Patients without CNV are seen every 6 months when macular changes are present in one or both eyes and yearly when no macular disease is present in either eye.

15. Ocular Ischemic Syndrome/Carotid Occlusive Disease (OIS)

What is ocular ischemic ischemic syndrome ?	List out workup	Outline the management plan for OIS	Develop a follow up plan
• Vision threatening condition with severe carotid artery occlusive disease leading to ocular hypo-perfusion • Commonly in old males	(1) History: Previous episodes of transient monocular visual loss? Cold hands or arm muscle spasms with exercise? (2) Complete ocular examination: Search carefully for anterior chamber flare, asymmetric cataract, and NVI/NVA/NVD/NVE. (3) Medical examination: Evaluate for HTN, diabetes, and atherosclerotic disease. Check pulses. Cardiac and carotid auscultation. (4) Laboratory workup for GCA in the appropriate settings. (5) Consider IVFA for diagnostic purposes. (6) Noninvasive carotid artery evaluation: Duplex Doppler US, oculoplethysmography, magnetic resonance angiography, others. (7) Consider orbital color Doppler US. (8) Consider ophthalmodynamometry if CRVO diagnosis cannot be excluded. (9) Carotid arteriography is reserved for patients in whom surgery is to be performed. (10) Consider cardiology consultation, given the high association with cardiac disease.	(1) Lifestyle modifications (a) Smoke cessation (b) Weight reduction (2) Refer to neurologist for full medical and neurological assessment → treat underlying systemic disease (3) Topical steroid & cycloplegic (a) Stabilise blood-aqueous barrier and limit iris movement to decrease the likelihood of hypothermia (4) Surgical treatment: (a) Carotid artery endarterectomy (b) Carotid artery stenting <u>Complication management:</u> (1) If nAMD developed → IOP control (2) If NV developed → Panretinal photocoagulation (PRP) / cryotherapy (3) If NVI and CMO developed → Intravitreal anti-VEGF (e.g bevacizumab & ranibizumab and triamcinolone)	Depends on the age, general health of the patient, and the symptoms and signs of disease. Surgical candidates should be evaluated urgently.

16. Retinal Arterial Macroanerysm

Describe etiology	Outline workup	List treatment + Develop a follow up treatment plan
Acquired vascular dilation of retinal artery or arteriole usually at the site of arteriolar bifurcation or arteriovenous crossing. Usually related to systemic HTN and general atherosclerotic disease.	(1) History: Systemic disease, particularly HTN or diabetes? (2) DFE= Look for concurrent retinal venous obstruction (present in one-third of cases) and signs of hypertensive retinopathy (visible in fellow eye as well). (3) Check blood pressure. (4) Consider checking lipid panel as well as fasting or random blood sugar and hemoglobin A1c. (5) IVFA may demonstrate early hyperfluorescence if there is no blockage from hemorrhage. Late frames may show leakage or staining of vessel wall. (6) OCT is helpful in demonstrating and following any ME.	**<u>Treatment:</u>** • Consider laser treatment if edema and/or exudate threatens central vision. • Caution must be taken when treating arterioles that supply the central macula since distal thrombosis and obstruction with resultant ischemia can occur. • Laser can also cause aneurysmal rupture resulting in retinal and vitreous hemorrhage. • Anti-VEGF agents may be beneficial in patients with macroaneurysm-associated ME. Dense or nonclearing vitreous hemorrhage, sub-internal limiting membrane (ILM) hemorrhage, or thick submacular hemorrhage may benefit from vitrectomy. **<u>Follow Up:</u>** Frequency based on the amount and location of exudate and hemorrhage.

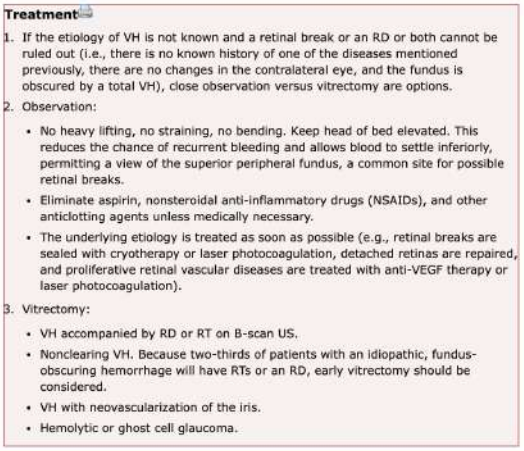
17. Sickle Cell Retinopathy

Outline workup	List treatment + Develop a follow up treatment plan
(1) Medical history and family history: Sickle cell disease, diabetes, or known medical problems? Intravenous drug abuse? (2) Dilated fundus examination using indirect ophthalmoscopy. (3) Sickledex, sickle cell preparation, and hemoglobin electrophoresis. (4) Consider IVFA (particularly widefield) to aid in diagnostic and therapeutic considerations.	**Treatment:** There are no well-established indications or guidelines for treatment. Isolated retinal neovascularization itself does not require treatment, as there may be a high probability of autoinfarction. Neovascularization with associated VH should receive PRP to the avascular area (anterior to the neovascularization). RD and VH may be best treated with vitrectomy. Anti-VEGF agents may be beneficial, but caution should be used in cases with significant traction. **Follow Up:** 1. **No retinopathy:** Annual dilated fundus examinations. 2. **Retinopathy present:** Repeat dilated fundus examination every 3 to 6 months, depending on severity.

18. Valsalva Retinopathy

What is the etiology for Valsalva retinopathy?	Describe treatment options	Describe treatment options + follow up plan
• Pre-retinal haemorrhage caused by a sudden increase in intrathoracic or intraabdominal pressure • Coughing, vomiting • Lifting • Straining for a bowel movement	(1) History: History of Valsalva including any recent heavy lifting, straining during bowel movement, coughing, sneezing, vomiting, etc.? The patient may not remember the incident. (2) DFE+ SLE = Look for findings suggestive of a different etiology including microaneurysms, dot-blot hemorrhages, CWSs, RT, PVD. (3) If dense VH is present, perform a B-scan US to rule out RT or RD. (4) IVFA may be helpful to rule out other causes including RAM or diabetic retinopathy. (5) Usually not required lab test	(1) Observation → self-resolved within weeks to months (2) Advised px to avoid anticoagulants drugs and strenuous physical activity (3) Surgical intervention for large haemorrhages (a) Neodymium:YAG laser (b) Laser membranotomy → disrupts the ILM & promote drainage of blood **Prognosis:** Prognosis is excellent. Most patients are observed, as sub-ILM hemorrhage usually resolves after a few days to weeks. Occasionally a laser is used to permit the blood to drain into the vitreous cavity, thereby uncovering the macula. Vitrectomy is rarely considered, typically only for nonclearing VH. **Follow Up:** May follow up **every 2 weeks for the initial visits** to monitor for resolution, then **follow up routinely.**

19. Vitreous Haemorrhage

Outline workup	List treatment + Develop a follow up treatment plan
(1) History: Any ocular or systemic diseases? Trauma? (2) Complete ocular examination, including slit lamp examination with undilated pupils to check for iris neovascularization, IOP measurement, and dilated fundus examination of both eyes by using indirect ophthalmoscopy. In cases of spontaneous VH, scleral depression is performed if a retinal view can be obtained. (3) When no retinal view can be obtained, B-scan US is performed to detect an associated RD or intraocular tumor. Flap RTs may be detected with scleral depression and may be seen on B-scan US (elevated flap). (4) IVFA may aid in defining the etiology, although the quality of the angiogram depends on the density of the hemorrhage. Additionally, it may be useful to highlight abnormalities in the contralateral eye.	 **Treatment** 1. If the etiology of VH is not known and a retinal break or an RD or both cannot be ruled out (i.e., there is no known history of one of the diseases mentioned previously, there are no changes in the contralateral eye, and the fundus is obscured by a total VH), close observation versus vitrectomy are options. 2. Observation: • No heavy lifting, no straining, no bending. Keep head of bed elevated. This reduces the chance of recurrent bleeding and allows blood to settle inferiorly, permitting a view of the superior peripheral fundus, a common site for possible retinal breaks. • Eliminate aspirin, nonsteroidal anti-inflammatory drugs (NSAIDs), and other anticlotting agents unless medically necessary. • The underlying etiology is treated as soon as possible (e.g., retinal breaks are sealed with cryotherapy or laser photocoagulation, detached retinas are repaired, and proliferative retinal vascular diseases are treated with anti-VEGF therapy or laser photocoagulation). 3. Vitrectomy: • VH accompanied by RD or RT on B-scan US. • Nonclearing VH. Because two-thirds of patients with an idiopathic, fundus-obscuring hemorrhage will have RTs or an RD, early vitrectomy should be considered. • VH with neovascularization of the iris. • Hemolytic or ghost cell glaucoma. **Follow Up:** If observation is elected, the patient is evaluated daily for the first 2 to 3 days. If a total, dense VH persists, and the etiology remains unknown, vitrectomy should be considered.

20. AMD

- **Exudative**
- **Non-Exudative**

Outline the workup for AMD. What classification system should optometrists use for patients with AMD?

The Beckman Clinical Classification System. (OBA guideline) Work up is as follows:

(1) Hx & Sx
- Risk factors; include FoHx, smoking status and history of smoking
- Nutritional supplement use
- Driving status

(2) VA: Especially monocular BCVA

(3) DFE: To evaluate the macula

(4) Amsler grid: To monitor progression and help with diagnosis of nAMD.

(5) Screen for new symptoms suggestive of new-onset neovascular AMD/other progression during history-taking:
- Scotoma
- Scintillating areas
- Colour vision abnormalities

Table 5: Optometric assessment of a patient with AMD

Clinical test	Notes
History	Screen for new symptoms suggestive of AMD (see Table 4). Establish risk factors (see Table 1), including family history of AMD, smoking history and status, as well as documentation of nutritional supplement use and driving status.
Visual acuity	Monocular best-corrected visual acuity.
Fundus examination	Dilated fundus exam (DFE), including stereoscopic biomicroscopic evaluation of the macula, is recommended at least annually for those exhibiting signs of AMD.
Amsler grid	Presentation of the grid at 30cm leads to a retinal projection of 20°, with each square representing a 1° angle.[1] It has been shown that white lines on the black background is more sensitive and reliable than the black lines on the white background.[42] The sensitivity and specificity of the Amsler grid to diagnose neovascular AMD is only moderate in at-risk patients, so patients should be encouraged to attend regular eye examinations, regardless of the results of their Amsler grid self-assessment.[43]

List the risk factors for AMD

Strong: (risk of **developing**)
- Older age (>60yrs, x3 in those older than 75)
- Smoking (strongest → could double the risk)
- FoHx of AMD (genetic)

Moderate: (risk of **progressing to late AMD**)
- Hypertension
- Cardiovascular disease
- BMI ≥ 30kg/m2

Weak:
- Poor diet
  - \+ Low in Omega 3, vitamins,
  - \+ Low macular carotenoid (Zeaxanthin & lutein) and mineral
  - \+ High in fat
- Lack of exercise

NOTE:
Mediterranean diet associated with lower risk of developing/progression in AMD → where fat is unsaturated

Melbourne study: Lower prevalence of advanced AMD with diet high in fruits, vegetable, chicken and nuts & less red meat (if you eat higher amounts of red meat = higher prevalence).

List the imaging recommendations for late atrophic AMD (GA) patients.

- Colour fundus photography = Helps for monitoring drusen number, size, presence of pigmentary abnormalities and signs of late disease
- OCT = Same as above + crucial for late AMD (such as SRF) or early signs of atrophic AMD. Helpful for looking at drusen in a 3D manner.
- FAF = Helps demarcate **GA** better, helpful for measuring size and extent of atrophic lesions

How do we estimate the risk of progression to late AMD?

Risk factors:
- Each eye with large drusen
- Each eye with pigmentary abnormalities associated with medium drusen
- If neither eye has large drusen and both eyes have medium drusen (early AMD)

Table 3: Five-year risk of progression to late AMD[30]

Risk factors	Risk of progression for patients without late AMD in either eye at baseline[a,b]
0	0.5%
1	3%
2	12%
3	25%
4	50%

What do we define AMD pigmentary abnormalities as?

AMD pigmentary abnormalities are defined as any hyper-pigmentary or hypo-pigmentary abnormalities associated with medium or large drusen, **but not associated with** other known disease entities (e.g., inherited retinal degenerations).

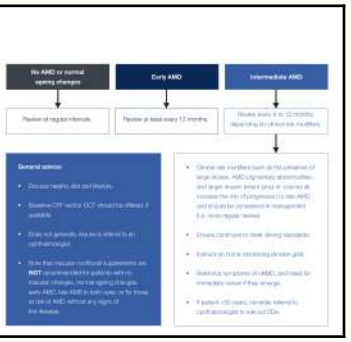
List the AMD clinical biomarkers to consider when determining review timeframes.

- Presence of large drusen (>125 um → average central retinal vein is 125um)
- Presence of pigmentary abnormalities
- Drusen volume – consider the total extent of the drusen in the macular region

NOTE: **Reticular pseudodrusen** (RPD) has also been associated with an increased risk of progression to late AMD when present in the fellow eyes of those with unilateral neovascular AMD. Evidence is equivocal for those with non-late AMD (e.g intermediate)

Current evidence indicates that **hyper-reflective foci (HRF)** that do not correspond with hyperpigmentary abnormalities does not improve the prediction of late AMD development than considering hyperpigmentary abnormalities alone.

Summarise review and follow up pathways for all forms of AMD.

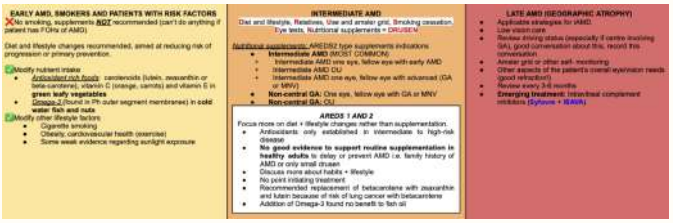
TYPE OF AMD	REVIEW PERIOD	REFERRAL?	Risk of Progression to Late AMD	
Early AMD	At least annually	Patients with early or intermediate AMD do not require referral to an ophthalmologist, unless they have abnormal structural, functional or historical clinical findings that require a second opinion, including a younger, atypical age of AMD onset (<50 years).	**0.9%** (medium drusen)	
Intermediate AMD	Reviewed 6-12 monthly, depending on risk factors for progression		**20.5%**	

Late AMD (GA OR non-active neovascular AMD)	Every 3-6 months GA/n	Patients with symptomatic, new-onset late AMD (neovascular) should be referred to an ophthalmologist urgently (within 1 week).	Can progress to Neovascular AMD	

★ Patients with asymptomatic AMD (drusen), with macular fluid detected on OCT, should be referred to an ophthalmologist within 2 weeks.
★ Optometrists should refer patients with geographic atrophy (GA) interested in further advice or considering treatment (when it becomes available in Australia) to an ophthalmologist.
★ Patients with neovascular AMD (nAMD) should be advised of the typical treatment regime (i.e. anti-VEGF injections) and the importance of adherence to the regimen.

Outline Management for NON-exudative and EXUDATIVE AMD. Describe use of AREDS 1 & 2 supplements

NON-EXUDATIVE



Recommendations:
- Patient-centred care is paramount and AMD referral pathways need to be managed to maximise patient access to the practitioner who can provide the care that is required.
- Patients with AMD should be offered a home Amsler grid for self-monitoring.
- Optometrists should counsel patients on the appropriate course of action if they notice a change in their vision.
- Optometrists should offer patients a referral to low vision support services as soon as their AMD causes visual impairment.

AREDS 2 formula:
★ 500mg of **Vitamin C**
★ 400 international units of **Vitamin E**
★ 80mg **Zinc** (Zinc Oxide → dosage exceeds the upper level of intake guidelines of Australia and NZ)
★ 2mg copper as **cupric oxide**
★ 10mg **lutein**
★ 2mg **zeaxanthin**

Nutritional supplements: AREDS2 type supplements indications
- **Intermediate AMD** (MOST COMMON)
  + Intermediate AMD one eye, fellow eye with early AMD
  + Intermediate AMD OU
  + Intermediate AMD one eye, fellow eye with advanced (GA or MNV)
- **Non-central GA:** One eye, fellow eye with GA or MNV
- **Non-central GA:** OU

AREDS 2 finding:
The original AREDS supplements had beta carotene 15mg . The replacement of beta carotene with lutein/zeaxanthin (carotenoid) reduces the incidence of lung cancer in past and current smokers in addition to helping halt disease progression.

It was found that adding omega-3 adding omega-3 fatty acids to the AREDS2 formula did not provide any additional benefit in slowing the progression of age-related macular degeneration (AMD)

There is also no good evidence to support routine supplementation in healthy adults to delay or prevent AMD i.e. family history of AMD or only small drusen. No prophylactic effect.

AREDS 1/2
- Reduces the risk of progression from intermediate to advanced AMD by ~25%
- No effect on cataract

EXUDATIVE

15 years ago AMD was largely untreatable, until the rise of anti-VEGF therapy. BUT the problem with VEGF is that:
1. Anti VEGF blocks permeability but does not cause the MNV to regress so when treatment is stopped the MNV often becomes active again
2. Patients therefore need ongoing therapy which is a huge burden

It is important to consider other aspects of a patient's overall eye and vision needs (in collaboration with an ophthalmologist).
- LV care, reviewing driving status
- Amsler grid or other self monitoring to help detect progression of disease.

Management options (ALL ANTI-VEGF)
★ **Ranibizumab** 0.5 mg (Lucentis)
★ **Aflibercept** 2 mg (Eylea)
★ **Bevacizumab** (Avastin) "off label"
★ **Brolucizumab** (Beovu) – extended dosing (3 monthly)
★ **Faricimab** (Vabysmo) – extended dosing (3 monthly)

A lot of these drugs (when compared), none have shown a significant improvement ; they are non-inferior.

- Guided by RCTs; however, most RCTs have not been designed for head-to-head comparison
- Efficacy: No consistent meaningful difference has been demonstrated
- Side effects: Safety is so good with existing drugs that new drugs need to be just as safe
- Systemic side effects: meta-analyses suggest little difference
- Treatment frequency: No difference between Eylea and Lucentis when used monthly or treat and extend

Ultimately, drug of choice comes down to individual doctor preference.

Usual treatment regimen is typically monthly until MNV is inactive based on FFA and/or OCT
- PRN = As needed
- Fixed monthly
- Treat and extend (MOST common treatment regimen used in Australia) – warn these patients about signs and symptoms of endophthalmitis and RD.

REFERRAL PERIOD
Refer to ophthalmologist (from Optometry Australia guideline)
- Within 1 week - if symptomatic
- Within 2 week - if asymptomatic
- Urgent [within 1 week] referral nAMD:
  - Recent (within 3 months) Hx of vision loss, spontaneously report distortion or onset of missing patchy/blurring in central vision
  - Suspected or definite new-onset macular haemorrhage
  - Macular fluid (SRF/intra-retinal) → OCT

AMD in 1 eye , risk of getting in 2nd is 10-12%

Review as what opthal suggests

21. Central Serous Chorioretinopathy (CSCR)

Differentiate between acute and chronic CSCR?		**Outline workup for CSCR. List the extra workup that needs to be undertaken in a chronic case**
ACUTE	**CHRONIC**	(1) History: Ask about **corticosteroid use** (topical cream or nasal spray), sleep patterns, **stress**, anxiety, pregnancy, systemic lupus erythematosus (SLE), h.pylori (stomach gut ulcers). (2) Amsler grid test to document the area of field involved. Can also do a visual field. You will see

• Usually resolved by 2-3 months	• > 3 months • Gravitational tract • Drusen on affected area??

scotoma and distortion of straight lines.
(3) SLE of the macula to rule out concomitant CNV. In addition, search for an optic pit of the disc.
(4) DFE to rule out a choroidal tumor, concomitant CNV or RRD.
(5) OCT is helpful in demonstrating subretinal fluid or PEDs and for monitoring purposes. Enhanced-depth imaging OCT often demonstrates choroidal thickening and may be a useful adjunct in diagnosis
(6) IVFA and ICGA if the diagnosis is uncertain or presentation atypical, CNV is suspected, or laser treatment is to be considered. IVFA shows the nearly pathognomonic "smoke-stack" pattern of dye leakage in 10% to 20% of cases. ICGA shows choroidal artery and choriocapillaris filling delays and characteristic multifocal hyperfluorescent patches in the early phase.
(7) In cases of **chronic CSCR**, a systemic workup including cortisol levels and renal function should be considered

Describe the etiology of CSCR?

- **Idiopathic:** Most common in young adult to middle-aged men. In women, CSCR has an association with pregnancy. Increased incidence in patients with lupus.
- **Increased endogenous cortisol:** This might explain a possible association with psychological or physiologic stress (type A personality). Rare cases exist with cortisol producing adrenal adenomas or Cushing syndrome.
- **Exogenous cortisol:** Corticosteroid use, including nasal corticosteroid sprays and topical creams.

Management for CSCR (Acute VS Chronic)

ACUTE	CHRONIC
• **Observation:** Acute CSCR is usually self-limited (within 2-3 months) with good visual prognosis. Worse prognosis for patients with recurrent disease, multiple areas of detachment, bilateral cases or prolonged course. • Stress management • Stop corticosteroids, if possible, including topical skin and nasal spray preparations. • If CNV develops, consider anti-VEGF therapy.	• Photodynamic therapy (PDT): May be helpful for chronic CSCR. • Mineralocorticoid receptor antagonists: **Eplerenone** and **spironolactone** have been associated with improved anatomic and visual outcomes in **chronic CSCR.** • Laser photocoagulation: May accelerate visual recovery, but long-term benefit and safety are unclear. May increase risk of CNV formation. Given the CNV risk, use low laser intensity. Consider laser for: + Persistence of a serous detachment for several months. + Recurrence of the condition in an eye that sustained a permanent visual deficit from a previous episode. + Occurrence in the contralateral eye after a permanent visual deficit resulted from a previous episode. + The patient requires prompt restoration of vision (e.g., occupational necessity).
Observe every 6-8 weeks until resolved (acute). For chronic will be managed by opthal.	

22. Chloroquine/Hydroxychloroquine Toxicity (Drug Induced Bull's Eye Maculopathy)

List the risk factors with Chloroquine/Hydroxychloroquine toxicity	Describe workup for a patient with Chloroquine/Hydroxychloroquine Toxicity	Does ocular toxicity regress?
★ Chloroquine daily >2.3mg/kg ★ Hydroxychloroquine daily >5mg/kg ★ Duration >5 year ★ Renal disease, comorbid macula disease	(1) **Enquire about risk factors:** Daily dosage, weight, duration of use, total cumulative dose • **Chloroquine** >300g, >2.3mg/kg • **Hydroxychloroquine** >1000g, >5mg/kg • Duration of use > 5years • Renal disease, tamoxifen use, comorbid macular disease (2) Best corrected visual acuity. (3) DFE → observed any pigmentary alterations. (4) Consider posterior pole fundus photographs. (5) Consider visual fields and OCT if maculopathy is present.	Once ocular toxicity develops, it usually does not regress even if the drug is withdrawn. In fact, new toxic effects may develop, and old ones may progress even after the chloroquine/hydroxychloroquine has been discontinued.

Patient has just started taking chloroquine, what tests should be completed in his first year at baseline evaluation?	Management and follow plan for Chloroquine/Hydroxychloroquine Toxicity + Follow Up
• BCVA • SLE (slit lamp exam) • DFE • CFP (colour fundus photograph) • 10-2 standard automated perimetry (SAP) • One of mfERG, FAF or SD-OCT	1. Discontinue medication in conjunction with the prescribing physician/GP 2. **Annual screening AFTER 5 years** after medication use (sooner for higher-risk px) a. Automated VF i. 10-2 for **Non-Asian** ii. 24-2 or 30-2 for **Asian** b. Spectral Domain OCT i. Observing any parafoveal photoreceptor thinning, disruption of ORL, RPE atrophy, loss of foveal contours 3. Additional tools that may be used as available or in suspect cases include multifocal ERG and FAF

23. Choroidal Effusion/ Detachment

What causes choroid effusion?	List the workup process.
• Abnormal accumulation of fluid in the suprachoroidal space • Causes: + Common complication of glaucoma surgery + Inflammatory and infectious disease + Trauma + Neoplasms (excessive growth of tissue/cancer) etc. **EXCRUCIATINGLY PAINFUL!**	(1) History: Recent ocular surgery or trauma? Known eye or medical problem? (2) Slit lamp examination: Check for presence of a filtering bleb and perform Seidel test to rule out a wound leak. (3) Gonioscopy of the anterior chamber angle: Look for a cyclodialysis cleft (a detachment of the ciliary body from the scleral spur in the eye → usually due to trauma and surgery). (4) Dilated retinal examination: Determine whether there is SRF, indicating a concomitant RD, and whether an underlying choroidal disease or tumor is present. Examination of the contralateral eye may be helpful in diagnosis. (5) In cases suggestive of melanoma, B-scan US and transillumination of the globe are helpful in making a diagnosis. B-scan US is also useful in distinguishing between serous and hemorrhagic choroidal detachment and in determining if hemorrhage is mobile or coagulated.

Discuss treatment for choroidal detachments. What are the associated complications?

General Treatment	Specific Treatment (treating underlying cause)
1. Cycloplegic (e.g., **atropine 1%** t.i.d.). 2. Topical steroid (e.g., **prednisolone acetate 1%** four to six times per day). 3. Consider oral steroids. 4. Surgical drainage of the suprachoroidal fluid may be indicated for a flat or progressively shallow anterior chamber, particularly in the presence of inflammation	1. Serous: • Wound or filtering bleb leak: Patch for 24 hours, decrease steroids and add aqueous suppressants, suture the site, use cyanoacrylate glue, place a bandage contact lens on the eye, or a combination of these. • Cyclodialysis cleft: Laser therapy, diathermy, cryotherapy, or suture the cleft to close it. • Uveitis: Topical cycloplegic and steroid as discussed previously. • Inflammatory disease: See the specific entity. • RD: Surgical repair. Proliferative vitreoretinopathy after repair is common. 2. Hemorrhagic: Drainage of the choroidal detachment using intraocular infusion with or without vitrectomy is performed for severe cases. More successful if hemorrhage is liquefied, which occurs 7 to 10 days after the initial event. Otherwise use general treatment.

24. Choroidal (Naevus)

Describe risk factors for malignant transformation.	Outline workup for choroidal naevus	What is the treatment and prognosis?	Justify follow up period.
TFSOM-DIM • T: Thickness >2 mm (US). • F: Fluid subretinal (OCT). • S: Symptoms (vision loss, <20/50 on Snellen visual acuity). • O: Orange pigment hyperautofluorescence (autofluorescence). • M: Melanoma hollow (US). • DIM: Diameter >5 mm (photography). If **four or more factors are present,** the lesion has a >50% chance to show growth and is likely to be a small choroidal melanoma.	(1) Complete ophthalmologic examination including dilated fundus examination (2) Detailed clinical drawing of the lesion with careful attention to location and size. (3) Baseline color photos of the lesion to assist in diameter assessment and in documenting growth. (4) OCT to document the overlying retinal features, SRF, and the lesion itself using enhanced-depth imaging. (5) Autofluorescence to document the presence of lipofuscin or RPE disturbance. (6) US for tumor thickness measurement and internal acoustic qualities.	Usually benign and asymptomatic. **Observation** = First examination should be in 3 to 4 months to confirm stability and then one to two times yearly to document lack of change. **Risk of progress to melanoma**: Very rare (1 in 5000)	Low-risk lesions can be followed with annual dilated fundus examination. High-risk lesions should be followed every 3 to 6 months.

25. Choroidal (Malignant Melanoma)

Identified the risk features for Naeve in progressing to melanoma	Outline workup for a malignant melanoma
TFSOM-UHHD: To **F**ind **S**mall **O**cular **M**elanoma-**U**sing **H**elp **H**ints **D**aily + **T**hickness >2 mm + **F**luid subretinal + **Symptoms** of flashes/floaters + **O**range pigment + **M**argin within 3mm on the optic nerve head + **U**ltrasound Hollow + **H**alo absent: A halo of depigmentation surrounding a pigmented choroidal lesion is more commonly seen in nevi and indicates stability. + **D**rusen absent: Drusen overlying	(1) History: Ocular surgery or trauma, cancer, anorexia, weight loss, or systemic illness? (2) Dilated fundus examination using indirect ophthalmoscopy. (3) IVFA: Can rule out lesions that simulate melanoma, but may not differentiate melanoma from large nevus, metastases, or hemangioma. (4) A-scan and B-scan US: Documents thickness and confirms clinical impression. With choroidal melanoma, US usually shows low-to-moderate reflectivity with choroidal excavation. Thickness is often >2 mm. May show a mushroom appearance. (5) OCT: Often documents fresh SRF. (6) Autofluorescence: Often shows prominent overlying lipofuscin (orange pigment). (7) ICGA: Can show double circulation with prominent blood vessels within the melanoma. (8) Consider fine-needle aspiration biopsy for genetic analysis of the tumor for prognostication and in selected cases for cytologic confirmation. (9) Consider CT scan or MRI of the orbit and brain (useful in patients with opaque media): (10) If melanoma is confirmed, systemic follow up is done according to the risk level for metastatic disease. Blood work: Lactate dehydrogenase, gamma-glutamyl transferase, aspartate and alanine

choroidal lesions suggest chronicity.	aminotransferases, and alkaline phosphatase twice yearly. If liver enzymes are elevated, consider an MRI or liver scan to rule out liver metastases: → likely to grow in other part of the body ● Annual chest x-ray ● Annual MRI of the **liver** ● Annual complete physical examination by a medical internist or oncologist (11) Referral to an internist or oncologist for breast examination, full skin examination, chest CT, and possibly a carcinoembryonic antigen assay if choroidal metastases are suspected.
Justify follow up.	**Outline treatment plan**
~~After first diagnosis, this should be repeated at a **3-4 month** follow-up.~~ ~~If no change in the lesions has been identified, the minimum follow-up should be fundus photography **every 6-12** months for the remainder of the patient's life.~~	Depending on the results of the metastatic workup, the tumor characteristics, the status of the contralateral eye, and the age and general health of the patient, melanoma of the choroid may be managed by: ● Observation ● Photocoagulation ● Transpupillary thermotherapy ● Radiation therapy, local resection ● Enucleation — surgical removal of the entire eyeball, leaving the eye muscles and other orbital tissues intact ● Exenteration — A radical surgical procedure that involves the complete removal of all organs within a specific body cavity, such as the pelvis or orbit Most cases are managed with **plaque radiotherapy.** **Anti-VEGF injections every 4 months for the first 2 years** have been shown to minimize ultimate vision loss from radiation retinopathy. In some cases, sector photocoagulation is added.

26. Crystalline Retinopathy

Describe etiology for crystalline retinopathy	**Describe workup**	**Outline treatment and follow up.**
● Canthaxanthin toxicity ● Tamoxifen – breast cancer ● Retinal arterial emboli ● Talc (IV drug use)	(1) Complete past medical and medication history: Intravenous drug use? Cardiovascular risk factors such as HTN, elevated cholesterol? Breast cancer? Use of oral tanning agents? History of anesthesia in patients with renal failure? IV drug use? (2) Complete ocular examination, including DFE Carefully assess the location, depth, color, and morphology of crystals as well as the potential presence of ME, neovascularisation of the disc and peripheral retina, or retinal infarction. Examine the cornea for crystals. (3) Consider carotid Doppler US and echocardiography in older patients and those with cardiovascular risk factors (measure blood flow) (4) Examine patients for evidence of intravenous drug abuse. (5) Consider testing for diabetes if idiopathic juxtafoveal/parafoveal telangiectasis suspected. (6) Optical: ○ IVFA may be helpful to demonstrate areas of nonperfusion distal to an intravascular crystal. ○ OCT may be helpful to determine depth.	**1. Discontinue associated drug use** ● Stop **tamoxifen** or **canthaxanthin** use if responsible for toxicity ● Stop intravenous drug use. 2. If there is peripheral nonperfusion or neovascularization, **consider PRP or anti-VEGF agents**. a. Prognosis: Visual loss may be permanent if there has been vascular nonperfusion in the macula secondary to blockage from intraretinal crystals. Follow-up: depends on underlying aetiology

27. Cystoid macular oedema (CMO)

What do you need to ask about during H/S for CMO?	**Outline workup for CMO**
Any of these can cause CMO (take history): ● Recent intraocular surgery? ● Diabetes? HTN? ● BRVO/CRVO ● Uveitis or ocular inflammation? ● Night blindness or FOHx (like RP)? ● Medicines including topical epinephrine, dipivefrin or prostaglandin analogs (especially in patients that have undergone cataract surgery)	(1) Take complete medical history = Looking for causes for CMO (2) Complete ocular examination including gonio = Rule out the presence of retained lens fragments following cataract surgery and haptic malposition of implanted anterior chamber intraocular lens. (3) DFE = Looking for pars planitis (4) IVFA/FFA = Shows early leakage of perifoveal capillaries and late macular staining, classically in a petaloid or spoke-wheel pattern. ***NOTE:*** Fluorescein leakage does not occur in select cases of pseudo-CME (5) OCT = Documents presence of CME and demonstrates efficacy of therapy. (6) Other diagnostic tests when indicated: Fasting blood sugar and hemoglobin A1c, ERG (associated with CRVO) , others.
Describe the management and follow up plan for CMO	
(1) Treat the underlying cause: DR, CRVO, BRVO, uveitis, RP, retinal vasculitis etc (2) Consider discontinuing associated medications → **Medications containing nicotinic acid**, **PGA**, **epinephrine**, **dipivefrin**	

(3) Postsurgical = NOT emergency, other forms of CMO may benefit from early treatment
(4) Topical NSAID (**Ketorolac 0.5%** q.i.d, **bromfenac 0.09%**, or **nepafenac 0.3%** daily) in conjunction with topical steroid (**Prednisolone acetate 1%** q.i.d)
(5) Systemic steroid, Intravitreal steroid/Anti-VEGF

- Subtenon steroid (e.g., triamcinolone 40 mg/mL, inject 0.5 to 1.0 mL).
- Intravitreal steroid (e.g., triamcinolone 40 mg/mL, inject 1 to 4 mg).
- Intravitreal anti-VEGF therapy (e.g., bevacizumab 1.25 mg in 0.05 mL).
- Systemic steroids (e.g., prednisone 40 mg p.o. daily for 5 days and then taper over 2 weeks).
- Systemic NSAIDs (e.g., indomethacin 25 mg p.o. t.i.d. for 6 weeks).
- Topical or systemic carbonic anhydrase inhibitors (e.g., dorzolamide 2% t.i.d. or acetazolamide 500 mg p.o. daily starting dose) in cases of RP-associated CME.
- CME with or without vitreous incarceration in a surgical wound may be improved by vitrectomy or YAG laser lysis of the vitreous strand.

Follow-Up
Postsurgical: 4-6 weeks (determine response to therapy – which is topical NSAID and topical steroid)
Other CME : Similar time frame

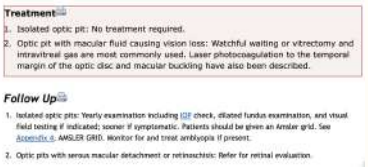
28. Epiretinal membrane

Describe workup	Outline follow up	Justify the management and follow up time for ERM
(1) History: Previous eye surgery or eye disease? Diabetes? (2) Complete ocular examination, including DFE. A careful peripheral examination should be performed to rule out a retinal break. (3) OCT is helpful in evaluating ERMs	This is not an emergency condition, and treatment may be instituted at any time. Often does not progress over time. A small percentage of ERMs recur after surgical removal. Annual observation	Treat the underlying disorder → RD, post-surgical, DR, inflammatory (1) Monitoring with Amsler grid: for macula involving (2) Optimise VA (3) If poor VA & photopsia → Vitrectomy with membrane peeling (GOLD STANDARD)

29. Idiopathic Polypoidal Choroidal Vascuolopathy (IPCV)

Describe workup	Justify follow up – how does the prognosis compare to nAMD?	Describe the management for IPCV
(1) Fundo = Detect signs of exudation (2) ICGA = **GOLD STANDARD,** used to confirm the presence of a branching network of vessels arising from the inner choroidal circulation with terminal aneurysmal dilations (popcorn lesions). Unlike occult CNV, the IPCV lesions **do not stain late unless active leakage is present.** (3) IVFA = To evaluate for other causes of CNV. (4) OCT = Used to assess for ME, SRF, and PEDs and can detect polyps in some eyes.	The prognosis of IPCV is generally **better** than for neovascular AMD. Symptomatic or macular IPCV is followed **every 1 to 2 months** with periodic OCT, IVFA, and ICGA as needed for disease progression. Consider treatment, or retreatment, if symptomatic, persistent, or new leakage occurs.	(1) Asymptomatic lesions observed and may resolve spontaneously. (2) If absence of NV and vision is not threatened → Observation (3) Refer within 1 week to retinal specialist for PDT, Anti-VEGF mono/combo therapy (if NV present) ***NOTE:*** EVEREST-II and PLANET trials supports anti-VEGF mono or combination therapy Review every **1-2 months** with repeat OCT, FFA and ICGA as needed

30. Optic Pit

Differentiate between acquired and congenital optic pits	Outline workup for optic pit	Justify treatment and follow up plan	Describe the management for optic disc pit <u>with</u> maculopathy.
<table><tr><th>Congenital</th><th>Acquired</th></tr><tr><td>• Etiology: Incomplete closure of the fetal fissure during embryogenesis • Appearance: Pit is typically located at the OD margin (temporal) • Developed maculopathy at a younger age & exhibited visual symptoms more frequently</td><td>• Etiology: complication of advance glaucoma or ocular trauma due to complete ON avulsion • Typically no visible ODP • Higher C/D ratio • Lacked NFL retinoschisis & Outer retinal layer defect • Spared the foveal center</td></tr></table>	(1) Complete ophthalmologic examination including SLE/DFE to evaluate for macular fluid. (2) Measure IOP. (3) Obtain baseline automated visual field testing. (4) If macular fluid is present, consider OCT to characterize precisely and IVFA to rule out CSCR or CNV.	 There is no clear consensus regarding treatment of optic pit maculopathy. There is no evidence for anti-VEGF therapy for treatment of optic pits. Conservative management would not be appropriate given the vision of 20/200 and the associated risk of amblyopia in the **pediatric population**. The most commonly used intervention for the treatment of optic pit maculopathy is pars plana vitrectomy (PPV). **Annual follow up.**	(1) Oral acetazolamide/corticosteroids (aims to reduce macular fluid) (2) Pars plana vitrectomy (3) Macular buckling (4) Laser photocoagulation to temporal margin of optic disc. Follow up: • Every 6 months with Amsler grid • Repeat test for VA, VF and OCT

31. Pathologic/Degenerative Myopia

How is pathological myopia different from myopia in terms of structural changing?	Outline the work up for pathological myopia	Describe the management and followup plan for pathological myopia
• >-6.00D, AXL>26.5mm • Risk of developing ○ CNV/RRD ○ Posterior staphyloma ○ Spots of subretinal hemorrhage ○ Choroidal sclerosis ○ Peripheral retinal thinning → VF defect	(1) Baseline glaucoma (POAG) assessment (IOP, CCT, Gonioscopy, DFE, VF) (2) DFE on the macula and peripheral retina (3) Optional: • OCT: Check for foveal schesis or full thickness macula hole • FFA, IVFA	No need to treat if nothing is wrong. Treatment depends on specific presentation: (1) Symptomatic retinal break → laser photocoagulation/cryotherapy (2) NV due to high myopia → Anti-VEGGF (3) RD prophylaxis → polycarbonate safety googles for sports Also if you suspect glaucoma (remember one of the 4 types was myopic, PPA, tilted) <u>**Follow up:**</u> Every **6-12 months** with repeat DFE

32. Solar or Photic Retinopathy

Outline Workup	Justify treatment options	Describe follow up + prognosis
(1) History: Eclipse viewing or sungazing? Work exposure? Photosensitizing medications? (2) Complete ocular examination, including DFE (3) Amsler grid testing = May identify central or **paracentral scotomata.** (4) IVFA = Shows a window defect **late in the disease course.** (5) OCT findings in the: • **Acute setting include:** Hyporeflectivity at the level of the RPE and occasional hyperreflectivity of the injured neurosensory retina. • **Chronic stage:** A central hyporeflective defect at the level of the photoreceptor inner segment-outer segment junction is seen	**Observation + patient counselling**. Eyes with better visual acuities on initial examination tend to recover more vision. Long-term significant reduction in visual acuity is rare. However, central or paracentral scotomata may persist despite improvement in visual acuity.	Follow-up for solar retinopathy typically involves an initial check within 1-3 days, with regular checkups continuing over the next 3-6 months. Recovery is spontaneous in nature and occurs over the course of 3-6 months after the inciting event, though visual recovery may be incomplete, and patients may suffer from permanent visual acuity deficits and central or paracentral scotomata. The bulk of visual acuity improvement has been noted to take place in the first month and 6 months after the photoxic event.

33. Vitreomacular Adhesion/Vitreomacular Traction/Macular Hole

Outline Workup	What is the gold standard treatment for VMH?	What treatment can you consider for symptomatic VMA, VMT and macular hole?	How many cases resolve spontaneously? Justify observation as a management option.
(1) History: Previous trauma? Previous eye surgery? Sun gazer? (2) Complete ocular examination with DFE. If a PVD is present, careful examination of the peripheral fundus to rule out peripheral breaks is important. (3) A true macular hole can be differentiated from a pseudo- or lamellar hole by directing a thin, vertical slit beam across the area in question using a 60- or 90-diopter lens with the slit lamp biomicroscope. The patient with a true hole will report a break in the line (Watzke-Allen test). A pseudohole or lamellar hole may cause distortion of the line, but it should not be broken. (4) IVFA may be helpful in identifying exudative retinal vascular disease (i.e., diabetic retinopathy, vein occlusion, pseudophakic CME) in cases which also have VMA/VMT. (5) OCT is critical for evaluating the vitreoretinal interface and determining the degree of traction from vitreous or ERMs. It is also useful in staging macular holes, differentiating from pseudo- or lamellar holes, and evaluating for progression	Pars plana vitrectomy with membrane peel **within 6 months** of onset (GOLD STANDARD)→ Removes vitreous gel from posterior eye	Can consider: **Ocriplasmin** for **symptomatic** VMA, VMT, and macular hole → *Enzyme medication injected into eye (recombinant version of plasmin → breaks down protein matrix responsible for abnormal adhesions between vitreous gel and retina)*	50% of cases resolve spontaneously (STAGE 1 will resolve, other 2 refer for pars plana vitrectomy). It is okay to watch, but if progressing significantly + lots of risk factors (maybe hole in fellow eye)

SUMMARY OF ALL TREATMENTS:

1. Stage 1 macular holes can be observed, as 50% resolve spontaneously.
2. Ocriplasmin is a recombinant protease with activity against components of the vitreoretinal interface (fibronectin and laminin). It is FDA-approved for the treatment of symptomatic VMA, VMT, and macular hole. It does not consistently work and rare, but important side effects such as ERG changes, lens subluxation, and dyschromatopsias have limited acceptance of this drug.
3. For symptomatic macular holes, pars plana vitrectomy with ILM peel and gas tamponade remains the gold standard for treatment. It is preferable to operate within the first 6 months of onset for highest chance of visual recovery. Serious complications are rare, but cataract progression in phakic patients is almost universal.

RETINAL DYSTROPHIES (4 DISEASES)

34. Best Disease (Vitelliform macular dystrophy - Egg Yolk)

List the risk factor of best disease	Outline the workup	Outline treatment + follow up
• Genetic: BEST1 gene (AD) • Common in hyperopia with angle closure glaucoma	(1) Family history. It is often helpful to examine family members. (2) Complete ocular examination, including a DFE. (3) EOG is highly specific and can be used to confirm the diagnosis or to detect the carrier state of the disease. (4) Genetic testing to confirm mutation of BEST1 gene. (5) Consider IVFA and OCT to detect the presence of or to delineate CNV.	There is no effective treatment for the underlying disease. In the era of intravitreal anti-VEGF agents, these are now typically first-line agents for CNV, but PDT and focal laser may also play a potential role. Additionally, due to development of subretinal hemorrhage with mild trauma, polycarbonate lenses are recommended at all times and especially while playing sports. **Follow Up:** Annually with DFE & Amsler grid monitoring (RTC if change noted) ***NOTE:*** Patients with CNV should be treated promptly. Otherwise, there is no urgency in seeing patients with this disease.

35. Cone Dystrophies

What is the usual onset?	Describe inheritance pattern of cone dystrophies	Outline the workup	Outline treatment + follow up
First 3 decades of life	Usually sporadic. Hereditary forms are usually autosomal dominant or less often X-linked.	(1) Family history. (2) Complete ophthalmic examination, including formal assessment for dyschromatopsia (e.g., Farnsworth-Munsell 100-hue test). (3) Formal visual field test. (4) Full-field ERG: abnormal photopic response with normal rod-isolated response. (5) OCT can show disruption of outer retinal layers but may be normal even in patients with electrophysical abnormalities. (6) FAF can be useful in the diagnosis (particularly sensitive to disturbances in the RPE), as well as for monitoring these diseases.	**NO PROVEN CURE. MEASURES ARE PURELY PALLIATIVE** (1) Palliative treatment • Tinted lens → maximise vision • Miotic drops (Pilocarpine 0.5/1% q.i.d) → manage photophobia (2) Genetic counselling (3) Low vision **Follow Up:** Yearly

36. Retinitis Pigmentosa and Inherited Chorioretinal Dystrophies

Outline workup,management and follow up

CONDITION	WORKUP	TREATMENT	FOLLOW UP
RP RP65	• VA/Contrast sensitivity = Will be reduced • VF = Loss of peripheral vision (tunnel vision) • ERG= reduced amplitude of the wave, absent a & b wave	**1. VitA supplementation** Some evidence supports that diets high in vitamin A (15,000 IU/d) and lutein (12 mg/d) may help slow midperipheral visual field loss in nonsmokers with RP, but the literature remains inconclusive. If high-dose vitamin A therapy is initiated, **monitor liver function tests and vitamin A levels.** **2. Epiretinal and Subretinal Microchip Implants** Both epiretinal and subretinal microchip implants have been used with some success to improve vision in patients with very advanced RP .Clinical trials are ongoing with a variety of designs to determine the safety and efficacy of retinal implant technology. In addition, research in gene therapy for specific types of RP is underway. **3. Low Vision** **4. Luxturna therapy (gene therapy) → RP65**	**Annual** ocular examinations usually are sufficient to measure visual acuity and visual field. If medical treatment is initiated, more frequent visits and laboratory blood work may be indicated.
Systemic Diseases Associated With Hereditary Retinal Degeneration	(1) Guided medical history (pertaining to each cause) (2) Drug history. (3) Family history with pedigree and genetic testing for diagnostic and counseling purposes (see above). (4) Ophthalmoscopic examination. (5) Formal visual field testing (e.g., Humphrey). (6) Dark-adaption studies and ERG: May help distinguish stationary rod dysfunction (congenital stationary night blindness) from RP (a progressive disease). (7) Fundus photographs. (8) Consider syphilis testing (RPR or VDRL and FTA-ABS or treponemal-specific assay). (9) If the patient is male and the inheritance pattern is unknown, examine his mother and perform an ERG on her. Women carriers of X-linked disease often have abnormal pigmentation in the midperiphery and abnormal results on dark-adapted ERGs. (10) If neurologic abnormalities such as ataxia, polyneuropathy, deafness, or anosmia are present, obtain a fasting (at least 14 hours) serum phytanic acid level to rule out Refsum disease. (11) If hereditary abetalipoproteinemia is suspected, obtain serum cholesterol and triglyceride levels (levels are low), a serum protein and lipoprotein electrophoresis (lipoprotein deficiency is detected), and peripheral blood smears (acanthocytosis is seen). (12) If Kearns-Sayre syndrome is suspected, the patient must be examined by a cardiologist with sequential EKGs; patients can die of complete heart block. All family members should be evaluated.	**FOR Hereditary Abetalipoproteinemia (Bassen-Kornzweig Syndrome):** 1. Water-miscible vitamin A, 10,000 to 15,000 IU p.o. daily. 2. Vitamin E, 200 to 300 IU/kg p.o. daily. 3. Vitamin K, 5 mg p.o. weekly. 4. Restrict dietary fat to 15% of caloric intake. 5. Biannual serum levels of vitamins A and E; yearly ERG, and dark adaptometry. 6. Consider supplementing the patient's diet with zinc. OTHERWISE general treatment is: **Treatment** For syphilis, see 12.12, SYPHILIS. For vitamin A deficiency, see 13.7, VITAMIN A DEFICIENCY. No definitive treatment for RP is currently known. See above, RETINITIS PIGMENTOSA. Cataract surgery may improve central visual acuity. Topical or oral carbonic anhydrase inhibitors (e.g., acetazolamide 500 mg/d) may be effective for CME. All patients benefit from genetic counseling and instruction on how to deal with their visual handicaps. Tinted lenses may provide comfort outdoors and may provide better contrast enhancement. In advanced cases, low-vision aids and vocational rehabilitation are helpful.	Depends on underlying etiology.
Hereditary chorioretinal dystrophies (other causes of nyctalopia)	Similar approach	For Gyrate atrophy: (1) Reduce dietary protein consumption and monitor serum ammonia levels (optimal in between 0.15-0.2mmol/L) (2) Supplemental vitamin B6 (Pyridoxine) 20mg/d po initial and increase up to 500mg/d po → change dosage according to ammonia levels For Choroideremia: (1) Darkly tinted sunglasses → relieve symptoms (2) Genetic counseling → X-lined recessive	Depend on the underlying etiology

37. Stargardt Disease (Fundus Flavimaculatus)

Describe the inheritance pattern.	Outline the workup	Outline treatment + follow up
Usually autosomal recessive, but occasionally autosomal dominant (dominant cases may be asymptomatic into middle age). ABCA4	Indicated when the diagnosis is uncertain or must be confirmed. (1) History: Age at onset, medications, family history? (2) Dilated fundus examination. (3) OCT may show photoreceptor disorganization, outer retinal atrophy, and RPE changes, which may appear early even with a normal fundus examination. (4) IVFA often shows blockage of choroidal fluorescence producing a "silent choroid" or "midnight fundus" as a result of increased lipofuscin in RPE cells. (5) FAF can be helpful in diagnosis and in monitoring disease progression. (6) ERG and EOG. (7) Formal visual field examination (e.g., Octopus, Humphrey). (8) Consider genetic testing: Sequencing of the ABCA4 gene (found to be abnormal in many cases of Stargardt disease and other related maculopathies).	Avoid vitamin A supplementation Ultraviolet light-blocking glasses when outdoors may be beneficial. Low-vision aids, services dedicated to helping the visually handicapped, and genetic counseling are helpful. ✅Palliative, as with cone dystrophy ❌Avoid VitA supplementation → cause lipofuscin buildup which worsen the condition **Follow Up =** Annual

PUPIL ANOMALIES (3 CONDITIONS)

38. Posterior Vitreous Detachment (PVD)

PALSIES (3 CONDITIONS)

39. Posterior Vitreous Detachment (PVD)

Section 2: 22 Short answer, organised into 6 topic areas, 50 marks

- Prescription writing (1Q, 2 marks)
- Ophthalmic drug delivery, OBA guidelines, collaborative care (3Qs, 8 marks)
- Macular/retinal disease (4Qs, 10 marks)
- Glaucoma (5Qs, 10 marks)
- Retinal vascular disease (5Qs, 10 marks)
- Neuro-ophthalmology (4Qs, 10 marks)

TOPIC 1: OPTOMETRIC PRESCRIBING OPHTHALMIC OPTHALMIC DRUG DELIVERY, OBA GUIDELINES, COLLABORATIVE CARE

List three responsibilities optometrists have when prescribing scheduled medicines. ● Follow OBA standards, guidelines, code of conduct ● Know TGA-approved indications ● Ensure prescriptions meet legal requirements (name, dose, repeats, etc.) ● Store scheduled medicines securely ● Record all in-practice administration ● Report theft/loss of scheduled medicines	**Optometrists who prescribe scheduled medicines should observe the Quality Use of Medicines principles as they apply to the scope of the endorsement. List four factors that may be taken into account when choosing a suitable medicine.** ● The individual ● The clinical condition ● Risks and benefits ● Dosage and length of treatment ● Any coexisting conditions ● Other therapies ● Monitoring considerations, and costs for the ● individual, the community and the health system as a whole

Midterm Questions

MULTIPLE CHOICE QUESTIONS

A patient presents with papilloedema and BP of 210/120 mmHg. What should a practising optometrist do?

- **A.** Start topical treatment
- **B.** Refer to GP
- **C.** Refer urgently to hospital
- **D.** Review in 24 hours

ANSWER = C

In a patient over 50 with diabetes and hypertension presenting with third nerve palsy with complete pupil-sparing and complete paresis of extraocular muscles innervated by the third cranial nerve, what is the most likely management plan?

- **A.** Immediate neurosurgical referral
- **B.** Observation with outpatient follow-up
- **C.** Intravenous antibiotics
- **D.** Emergency CT angiography

ANSWER = B
More likely to be an ischemic cause rather than a mass. If secondary to ischemia, function should return in 3 months.

Which of the following findings in a patient with retinoschisis MOST warrants consideration of surgical intervention?

- **A.** Bilateral inferotemporal age-related retinoschisis
- **B.** Outer retinal hole with progressive subretinal fluid
- **C.** Inner retinal hole without subretinal fluid
- **D.** Stable posterior retinoschisis for 5 years

ANSWER = B

A patient with moderate POAG has a baseline IOP of 24 mmHg and progressive VF loss despite treatment. What is the most appropriate target IOP?

- **A.** 18 mmHg
- **B.** 21 mmHg
- **C.** 14 mmHg
- **D.** Any IOP below 24 mmHg

ANSWER = C

Original target IOP was 25% from baseline = 24-6 = 18 mmHg. Given progression, reduce by a further 20% = 18-4 = 14 mmHg.

Which of the following is a TGA approved therapeutic indication of Xalatan (latanoprost)?

- **A.** Reduction of elevated IOP in patients with open-angle glaucoma
- **B.** Reduction of elevated IOP in acute angle closure glaucoma
- **C.** Treatment of neovascular glaucoma
- **D.** Prevention of macular oedema in patients with pseudoaphakia

ANSWER = A
❌B because with AACG, there is a closed angle. If we try to increase uveoscleral outflow, it won't work. Peak effect is also ~ 8 hours, so it would also take too long
❌D because Xalatan increases risk of macular oedema in those with pseudoaphakia

Which of the following statements about glaucoma by optometrists is FALSE? Optometrists must:

- **A.** Assess the retinal nerve fibre layer using OCT
- **B.** Have equipment to measure central corneal thickness
- **C.** Perform threshold visual field testing
- **D.** Refer patients experiencing side effects from anti-glaucoma medication

ANSWER = A

OBA legally requires optometrists:

- **Minimum:** IOP, Gonioscopy (AC angle), Threshold VF, DFE (OD examination)
- **Optional:** OD imaging & RNFL images

OCT is RECOMMENDED but NOT legally required for optometric management.

What is the recommended first-line treatment for typical optic neuritis associated with multiple sclerosis?

- **A.** Oral corticosteroids alone
- **B.** Intravenous methylprednisolone followed by taper
- **C.** Antibiotics and antivirals
- **D.** Observation only, regardless of severity

ANSWER = B
❌A as oral corticosteroids worsen recurrence risk
❌C as irrelevant to an inflammatory condition like optic neuritis

What features define a molecule as a steroid?

- **A.** Characteristic ring structure
- **B.** Biological site of synthesis
- **C.** Cholesterol as parent molecule
- **D.** Ant-inflamamtory action

ANSWER = A

Which of the following is a valid indication for surgical peeling of an epiretinal membrane?

- **A.** Presence of mild metamorphopsia without VA loss
- **B.** Significant reduction on vision attributable to the membrane
- **C.** Any stage 3 ERM detected on OCT
- **D.** ERM recurrence after previous surgery

ANSWER = B
❌C as structure does not influence function

In a patient with exudative retinal detachment secondary to CSCR, which of the following is a potential pharmacologic treatment?

- **A.** Methotrexate
- **B.** Eplerenone
- **C.** Bevacizumab
- **D.** Prednisolone

ANSWER = B

A 68-year-old male presents to your clinic with a history of gradually worsening vision in his right eye over the past year. His medical history includes hypertension and type 2 diabetes, for which he is taking metoprolol and metformin. On examination, his intraocular pressure (IOP) is 28mmg in the right eye and 22 mmg in the left eye. Gonioscopy reveals open angles with pseudoexfoliative material on the lens capsule and pupillary margin in the right eye. Fundoscopy shows optic nerve cupping with a cup-to-disc ratio of 0.7 in the right eye and 0.5 in the left eye. Visual field testing indicates superior arcuate scotoma in the right eye.

Write a prescription for the most appropriate initial pharmacological treatment to lower his elevated IOP (filling in the four text boxes below). Include all aspects of the prescription, ensuring it is compliant with the Optometry Board of Australia standards and relevant jurisdictional medicines/poisons acts. (2 Marks)	
Describe two ocular features of pseudoexfoliative syndrome that may complicate cataract surgery. (2 Marks) (From case above)	
What is the prognosis of pseudoexfoliative glaucoma if left untreated? Explain your answer. (2 Marks) (From case above)	
How should the effectiveness of the management plan be monitored over time? Include specific tests and intervals for follow-up over the 12 months following initiation of treatment. (3 Marks) (From case above)	
Considering the patient's systemic health history, what additional factors should be taken into account when formulating his management plan? Describe how these factors might influence your choice of treatment. (1 Mark) (From case above)	
List three indications for neuro-imaging in migraine. What additional case history questions would you ask the patient? (3 Marks)	
Describe the typical visual aura associated with migraine. Include in your answer when the headache is likely to start relative to the visual symptoms. (2 Marks)	
List an indication for initiating migraine prophylaxis and name two classes of medication commonly used for this purpose. (2 Marks)	
Outline three key clinical features of non-arteritic ischemic optic neuropathy that help differentiate it from arteritic ischemic optic neuropathy. (3 Marks)	

SHORT ANSWER QUESTIONS

TOPIC 1: SCRIPT WRITING

A patient has high IOPs and is currently taking topical steroids to treat their uveitis. What drug would you prescribe?

Rx: Timolol Maleate 0.5% eyedrops

Dosage: Instil 1 drop in **both** eyes 2 times a day

Quantity: 1, Repeats: 0

❌ Common mistakes:

- Not specifying the eye – glaucoma is bilateral
- Misspelt drug names – just needs to be semi-recognisable
- 2x drops a day (in a single administration) VS 2x times a day (morning and evening)

Uveitis = contraindication to PGAs (which can increase ocular inflammation)

✅ Correct scoring answers were:

- 1x drop in **both** eyes daily
- 1x drop in **both** eyes 2x times a day
- 1x drop in **each eye** twice daily

0.5 marks each for drug name, drug concentration, correct dosage, quantity (1), number of repeats (0 – new glaucoma px + need to review for 4-6 weeks first, max repeats of 5 at follow-up visits)

TOPIC 2: PROFESSIONAL OPTOMETRY

1. **Identify 3x methods of ophthalmic drug delivery used to treat posterior eye disease. Give an example and a posterior condition. (3)**
   - Topical administration (topical eyedrops, solutions, suspension) → E.g topical Latanoprost eyedrops for treatment of POAG
   - Intravitreal injection →E.g anti-VEGF (Ranbizumab) for CNV
   - Oral tablet →E.g Acetazolamide tablets for acute angle closure attack

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2. **Define "Quality Use of Medicines" and describe how it might be applied when initiating treatment for a patient newly diagnosed with ocular hypertension at high risk of glaucoma. (3 marks for 3 principles)**
   - Recognising the role of non-pharmacological options
   - Recognising the individual risks, side effects, efficacy if selecting pharmacological options
   - Monitoring the patient's drug use

Optometrists who prescribe scheduled medicines should observe the Quality Use of Medicines (QUM) 4 principles as they apply to the scope of the endorsement. Quality use of medicine means:

A. Selecting management options wisely by:

- Considering the place of medicines in treating illness and maintaining health, and
- Recognising there may be better ways than medicine to manage many disorders.

In terms of OHT, we should see if the patient actually needs to be treated. In general, only those at higher risk should be treated, although patient preference may be a decisive factor.

B. Choosing suitable medicines (if a medicine is considered necessary) so that the best available option is selected by taking into account:

- The individual
- The clinical condition
- Risks and benefits
- Dosage and length of treatment
- Any coexisting conditions
- Other therapies
- Monitoring considerations, and
- Costs for the individual, the community and the health system as a whole.

If we decide to treat OHT, need to consider the above factors.

If we decide to treat OHT, need to ensure:

C. Using medicines safely and effectively to get the best possible results by:

- Monitoring outcomes
- Minimising misuse, over-use and under-use
- Improving people's ability to solve problems related to medication, such as negative effects, and
- Managing multiple medications.

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3. Describe the role of optometrists in the management of patients requiring intra-vitreal injections in Australia. Include 2x key responsibilities.

- Educating
- Emphasising importance of referral
- Patient adherence
- Thinking of timing of referral
- Communication with ophthalmologists

TOPIC 3: RETINAL + MACULAR DISEASE

1. Outline an evidence-based assessment and monitoring strategy for an ASIAN patient at risk of hydroxychloroquine-induced retinal toxicity. Include what signs you would look for.

List the risk factors with Chloroquine/Hydroxychloroquine toxicity	Describe workup for a patient with Chloroquine/Hydroxychloroquine Toxicity	Does ocular toxicity regress?
★ Chloroquine daily >2.3mg/kg ★ Hydroxychloroquine daily >5mg/kg ★ Duration >5 year ★ Renal disease, comorbid macula disease	(1) **Enquire about risk factors:** Daily dosage, weight, duration of use, total cumulative dose • **Chloroquine** >300g, >2.3mg/kg • **Hydroxychloroquine** >1000g, >5mg/kg • Duration of use > 5years • Renal disease, tamoxifen use, comorbid macular disease (2) Best corrected visual acuity. (3) DFE → observed any pigmentary alterations. (4) Consider posterior pole fundus photographs. (5) Consider visual fields and OCT if maculopathy is present.	Once ocular toxicity develops, it usually does not regress even if the drug is withdrawn. In fact, new toxic effects may develop, and old ones may progress even after the chloroquine/hydroxychloroquine has been discontinued.
Patient has just started taking chloroquine, what tests should be completed in his first year at baseline evaluation?	**Management and follow plan for Chloroquine/Hydroxychloroquine Toxicity + Follow Up**	
• BCVA • SLE (slit lamp exam) • DFE • CFP (colour fundus photograph) • 10-2 standard automated perimetry (SAP) • One of mfERG, FAF or SD-OCT	1. Discontinue medication in conjunction with the prescribing physician/GP 2. **Annual screening AFTER 5 years** after medication use (sooner for higher-risk px) a. Automated VF i. 10-2 for **Non-Asian** ii. 24-2 or 30-2 for **Asian** b. Spectral Domain OCT i. Observing any parafoveal photoreceptor thinning, disruption of ORL, RPE atrophy, loss of foveal contours 3. Additional tools that may be used as available or in suspect cases include multifocal ERG and FAF	

- Core test for screening is visual fields or spectral OCT
- Workup: VFs (asians 24-2 or 30-2 because maculopathy + toxicity goes wider in peripheral retina and adopts a broader pattern, 10-2 for non-asians)
- Additional tools: ERG, FAF
- Recognising risk factors for toxicity

2. List two therapeutic options for managing polypoidal choroidal vasculopathy (PCV). Briefly describe the rationale for each.

(1) Asymptomatic lesions observed and may resolve spontaneously.
(2) If absence of NV and vision is not threatened → Observation
(3) Refer within 1 week to retinal specialist for PDT, Anti-VEGT mono/combo therapy (if NV present)

3. Explain why is ocriplasmin (Jetrea) is not initiated by optometrists? Describe how an optometrist contributes to the decision-making process for its use.

Ocriplasmin for symptomatic VMA, VMT, and macular hole → Enzyme medication injected into eye (recombinant version of plasmin → breaks down protein matrix responsible for abnormal adhesions between vitreous gel and retina)

It is an injection that is why optometrists cannot initiate it.

4. Explain why vitamin A supplementation may be discussed in retinal dystrophies. Outline the optometrist's responsibility when this is raised.

- **Avoid VitA supplementation in Stargardt disease** → Stargardts has an abnormal accumulation of lipofuscin, so if you use VitA it will make more lipofuscin which will make things worse.
- **RP** → Some evidence supports that diets high in vitamin A (15,000 IU/d) and lutein (12 mg/d) may help slow midperipheral visual field loss in nonsmokers with RP, but the literature remains inconclusive. If high-dose vitamin A therapy is initiated, monitor liver function tests and vitamin A levels.

TOPIC 4: VASCULAR DISEASE

1. **What are the likely treatment options for CRVO with macular oedema seen on OCT. Include in your answer, key risks and expected clinical outcomes of the treatment. (3)**
2. **List 4x systemic blood tests should an optometrist recommend following diagnosis and explain its purpose. (4)**
3. **What treatment, if any, is a patient with BRVO likely to receive and why? Include in your answer to the specific factors that will influence treatment choice. (3)**

1. What are the likely treatment options for CRVO with macular oedema seen on OCT. Include in your answer, key risks and expected clinical outcomes of the treatment. (3)
    - *Talk about the treatment options for CRVO – core treatment guidelines*
    - *Discuss key risks to intravitreal injections: endophthalmitis, steroids (increased IOP, cataracts, increased risk of infection) macular oedema, RDs*
    - *Clinical outcomes: recover VA, try to stabilise VA, avoid reduction of VA, reduce macular thickness*
2. List 4x systemic blood tests should an optometrist recommend following diagnosis and explain its purpose. (4)
    - *Hypercholesterolameia*
    - *Total lipid counts (LDL vs HDL)*
    - *Blood glucose*
    - *Fasting blood glucose*
    - *HbA1ct*
    - ***BP IS <u>NOT</u> A SYSTEMIC BLOOD TEST** → because it is a non-invasive clinical measurement, not a blood test that analyses blood in a lab*
    - *<50yo – Blood clotting factors, hyperclotting, hyperviscosity factors, platelet counts*
3. What treatment, if any, is a patient with BRVO likely to receive and why? Inlcude in your answer to the specific factors that will influence treatment choice. (3)
    - *Answer required conditional reasoning – tx depends on severity*
    - *BRVO should typically be observed but **need to look for ischaemia, neovas, macular oedema** (required to get the mark)*
    - *Needed to state that many cases are observed initially*
    - *But anti-VEGF, sector PRP or intravitreal corticosteroids are required for specific complications*

TOPIC 5: GLAUCOMA

1. **Describe the initial first step for a patient presenting wit h acute primary angle closure including purpose, drug, concentration. Role of optometrist if you cannot prescribe this drug? (2 marks)**
2. **What topical medication should be subsequently applied as a second step, including drug(s), concentration(s) and dosage(s) (3m)**
3. **If the eye is red and inflamed, what additional treatment should be administered and when?**
4. **Why should pilocarpine be withheld ?**
5. **Once the angle closure attack is broken, what is the final management option for this patient why?**
6. **What is the final management to keep the angle open?**

GLAUCOMA (10 marks) – OBA guidelines Appendix C

1. Describe the initial first step for a patient presenting wit h acute primary angle closure including purpose, drug, concentration. Role of optometrist if you cannot prescribe this drug? (2 marks)
    - *Acetazolamide (Diamox) 500mg*
2. What topical medication should be subsequently applied as a second step, including drug(s), concentration(s) and dosage(s) (3m)
3. If the eye is red and inflamed, what additional treatment should be administered and when?
4. Why should pilocarpine be withheld ?
5. Once the angle closure attack is broken, what is the final management option for this patient why?
6. What is the final management to keep the angle open?
    - *AKA asking how do we open the angle permanently to prevent recurrence?*
    - *ANS Laser peripheral iridotomy!!*

TOPIC 6: NEURO-OPTHALMOLOGY

The patient has AAION. Fundoscopy shows swollen, pale optic discs. Pupils show R RAPD. Symptoms of jaw claudication, scalp tenderness, ... VA was 6/30?

1. **List 3x additional tests that would need to be performed in the workup of the patient? Include rationale (3 marks)**
2. **What was the initial step of optometric management/treatment on suspicion of AAION?**
3. **What treatment would they be put on?**
4. **Explain the goal(s) of this treatment strategy?**
5. **What are the potetial complications of untreated AAION? (1 mark)**
6. **Outline a follow-up plan. (3 marks)**

NEURO-OPHTHALMOLOGY (10 marks)

The patient has AAION. Fundoscopy shows swollen, pale optic discs. Pupils show R RAPD. Symptoms of jaw claudication, scalp tenderness, ... VA was 6/30?

7. List 3x additional tests that would need to be performed in the workup of the patient? Include rationale (3 marks)
8. What was the initial step of optometric management/treatment on suspicion of AAION?
9. What treatment would they be put on?
10. Explain the goal(s) of this treatment strategy?
11. What are the potetial complications of untreated AAION? (1 mark)
12. Outline a follow-up plan. (3 marks)
    - *Following up the temporal artery biopsy results after 1 week*
    - *Recognising this patient would need long-term mx – co-management with rheumatologist, immunologist, neurologist…*
    - *Taper steroids over time depending on clinical response*
    - *Follow-up the optic nerve (easy mark – optometrist role!)*
    - *Did not need to see a set time regime for the review*